

**GREEN
CLIMATE
FUND**

Meeting of the Board
9 – 13 November 2020
Virtual meeting
Provisional Agenda Item 11

GCF/B.27/02/Add.10/Rev.01

3 November 2020

Consideration of funding proposals – Addendum X

Funding proposal package for FP150

Summary

This addendum contains the following six parts:

- a) A funding proposal summary titled “Promoting private sector investment through large scale adoption of energy saving technologies and equipment for Textile and Readymade Garment (RMG) sectors of Bangladesh” submitted by the Infrastructure Development Company Limited (IDCOL);
- b) No-objection letter issued by the national designated authority or focal point(s);
- c) Environmental and social report(s) disclosure;
- d) Independent Technical Advisory Panel’s assessment;
- e) Response from the accredited entity to the independent Technical Advisory Panel’s assessment; and
- f) Gender documentation of the funding proposal.

These documents are presented as submitted by the accredited entity and the national designated authority(ies) or focal point(s), respectively. Pursuant to the Comprehensive Information Disclosure Policy of the Fund, the funding proposal titled titled “Promoting private sector investment through large scale adoption of energy saving technologies and equipment for Textile and Readymade Garment (RMG) sectors of Bangladesh” submitted by the Infrastructure Development Company Limited (IDCOL) is being circulated on a limited distribution basis only to Board Members and Alternate Board Members to ensure confidentiality of certain proprietary, legally privileged or commercially sensitive information of the entity.

Table of Contents

Funding proposal submitted by the accredited entity	1
No-objection letter issued by the national designated authority or focal point(s)	156
Environmental and social report(s) disclosure	157
Independent Technical Advisory Panel's assessment	159
Response from the accredited entity to the independent Technical Advisory Panel's assessment	169
Gender documentation	171

Funding Proposal

Project/Programme title:	<i>Promoting private sector investment through large scale adoption of energy saving technologies and equipment for Textile and Readymade Garment (RMG) sectors of Bangladesh</i>
Country(ies):	<i>Bangladesh</i>
Accredited Entity:	<i>Infrastructure Development Company Limited (IDCOL)</i>
Date of first submission:	<i>2020/01/23</i>
Date of current submission	<u><i>2020/10/18</i></u>
Version number	<i>V.04</i>



Contents

Section A	PROJECT / PROGRAMME SUMMARY
Section B	PROJECT / PROGRAMME INFORMATION
Section C	FINANCING INFORMATION
Section D	EXPECTED PERFORMANCE AGAINST INVESTMENT CRITERIA
Section E	LOGICAL FRAMEWORK
Section F	RISK ASSESSMENT AND MANAGEMENT
Section G	GCF POLICIES AND STANDARDS
Section H	ANNEXES

Note to Accredited Entities on the use of the funding proposal template

- Accredited Entities should provide summary information in the proposal with cross-reference to annexes such as feasibility studies, gender action plan, term sheet, etc.
- Accredited Entities should ensure that annexes provided are consistent with the details provided in the funding proposal. Updates to the funding proposal and/or annexes must be reflected in all relevant documents.
- The total number of pages for the funding proposal (excluding annexes) **should not exceed 60**. Proposals exceeding the prescribed length will not be assessed within the usual service standard time.
- The recommended font is Arial, size 11.
- Under the [GCF Information Disclosure Policy](#), project and programme funding proposals will be disclosed on the GCF website, simultaneous with the submission to the Board, subject to the redaction of any information that may not be disclosed pursuant to the IDP. Accredited Entities are asked to fill out information on disclosure in section G.4.

Please submit the completed proposal to:

fundingproposal@gcfund.org

Please use the following name convention for the file name:

"FP-[Accredited Entity Short Name]-[Country/Region]-[YYYY/MM/DD]"

A. PROJECT/PROGRAMME SUMMARY															
A.1. Project or programme	Programme	A.2. Public or private sector	Private												
A.3. Request for Proposals (RFP)	Not applicable Not applicable														
A.4. Result area(s)	<u>Mitigation:</u> Reduced emissions from: ☐ Energy access and power generation: ☐ Low-emission transport: ☒ Buildings, cities, industries and appliances: ☐ Forestry and land use: <u>Adaptation:</u> Increased resilience of: ☐ Most vulnerable people, communities and regions: ☐ Health and well-being, and food and water security: ☐ Infrastructure and built environment: ☐ Ecosystem and ecosystem services:		<u>GCF contribution:</u> <u>Enter number</u>% <u>Enter number</u>% <u>100</u>% <u>Enter number</u>% <u>Enter number</u>% <u>Enter number</u>% <u>Enter number</u>%												
A.5. Expected mitigation impact	14.53 million t CO ₂ eq	A.6. Expected adaptation impact	Not applicable												
A.7. Total financing (GCF + co-finance)	340.50 million USD	A.9. Project size	Not applicable												
A.8. Total GCF funding requested	256.48 million USD		Medium (Upto USD 250 million) As per GCF policy, each sub-project will be within the IDCOL accreditation scope.												
A.10. Financial instrument(s) requested for the GCF funding	Mark all that apply and provide total amounts. The sum of all total amounts should be consistent with A.8. <table border="0"> <tr> <td>☒ Grant</td> <td>6.48 million</td> <td>☐ Equity</td> <td><u>Enter number</u></td> </tr> <tr> <td>☒ Loan</td> <td>250 million</td> <td>☐ Results-based payment</td> <td><u>Enter number</u></td> </tr> <tr> <td>☐ Guarantee</td> <td><u>Enter number</u></td> <td></td> <td></td> </tr> </table>			☒ Grant	6.48 million	☐ Equity	<u>Enter number</u>	☒ Loan	250 million	☐ Results-based payment	<u>Enter number</u>	☐ Guarantee	<u>Enter number</u>		
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☒ Loan	250 million	☐ Results-based payment	<u>Enter number</u>												
☐ Guarantee	<u>Enter number</u>														
A.11. Implementation period	12 years ; considering last disbursement for any sub-project loan under this programme will take place at Year 12 as per project implementation timetable	A.12. Total lifespan	21 Years												
A.13. Expected date of AE internal approval	Within 03 months after signing of FAA.	A.14. ESS category	B												
A.15. Has this FP been submitted as a CN before?	Yes ☒ No ☐	A.16. Has Readiness or PPF support been used to prepare this FP?	Yes ☒ No ☐												
A.17. Is this FP included in the entity work programme?	Yes ☒ No ☐	A.18. Is this FP included in the country programme?	Yes ☒ No ☐												
A.19. Complementarity and coherence	Yes ☒ No ☐														

A.20. Executing Entity information	Component 1, 2 & 4 : AE i.e. Infrastructure Development Company Limited (IDCOL) is also the EE Component 3: Local Financial Institution (LFIs) will be the EEs namely- South East Bank Limited, City Bank Limited, BRAC Bank Limited and IDLC Finance Limited or any other eligible LFIs as per selection criteria. IDCOL may also be the EE in some cases when it lends directly to the RMG manufacturers. Component 5: Infrastructure Development Company Limited (IDCOL) as EE. Sustainable Renewable Energy Development Authority (SREDA), GoB will be the Implementing Partner for this component and a beneficiary, although not an EE.
A.21. Executive summary (max. 750 words, approximately 1.5 pages)	
CO₂ emissions per capita of Bangladesh increased from 0.18 metric tons in 1997 to 0.47 metric tons in 2014, having an average annual growth rate of 6%.¹ The rising trend is a ramification of current economic growth due to industrial expansion. In order to support the present industrial and economic growth, energy production is soaring in the country, causing a higher fossil fuel emission². The industrial sector in Bangladesh accounts for 47.8%³ of the commercial energy consumption, predominantly in the form of natural gas and electricity. Among this the textile sector is the second largest energy consuming industrial sub-sector with 12.40%⁴ consumption after Garment sector (15.40%). The GoB's Energy Efficiency and Conservation Master Plan states that manufacturing industries in Bangladesh are not efficient in energy use because of continuous usage of old/ mal-maintained machines and poor energy management. The impetus towards efficient end use of energy in the industrial sector is lagging mostly due to inadequate financial incentives and lack of technical expertise. If the current industrial energy intensity persists then considering the economic growth outlook in the medium to long term, it can be assured that GoB will face difficulties to manage the rising energy demand, and subsequently achieve GHG emission reduction targets under the UN Paris climate accord. Hence, the proposed programme aims to support business establishments in the textile and RMG sector with financial and market resources to avail investment opportunities for energy saving technology upgrades. The intervention will create a favorable market environment and scalable business model for investment in energy efficiency improvements, leading to sizeable energy savings and accompanying with greenhouse gas (GHG) emissions reductions. It will substantially contribute to achieving Bangladesh's Nationally Determined Contribution (NDC); i.e. 15% GHG emission reduction compared to the Business-As-Usual scenario by 2030 with international support, under the Paris Agreement. Furthermore, the program also contributes to the GoB target to achieve 10% of energy consumption reduction in the industry sector by 2030 and 20% improved energy intensity by 2030. Overall the proposed program will comprise an integrated package of concessional financing for textile & RMG borrowers, and technical assistance for creation of enabling environment by covering areas such as capacity building, awareness, support in loan disbursement and monitoring and evaluation of the program parameters. The Programme comprises of five interrelated and closely coordinated components:	

¹ <https://data.worldbank.org/indicator/EN.ATM.CO2E.PC?locations=BD>

² <https://tradingeconomics.com/bangladesh/fossil-fuel-energy-consumption-percent-of-total-wb-data.html>

³ Energy Efficiency and Conservation Master Plan up to 2030 by SREDA, Power Division

⁴ Energy Efficiency and Conservation Master Plan up to 2030 by SREDA, Power Division

- **Component 1: USD 133.00 million financing for Energy Saving Equipment & Technology for Textile sector**
- **Component 2: USD 3.05 million GCF Technical Assistance (TA) to develop enabling environment for EE investment in textile sector**
- **Component 3: USD 200.00 million financing for Energy Saving Equipment & Technology for RMG sector**
- **Component 4: USD 2.30 million GCF Technical Assistance (TA) to develop enabling environment for EE investment in RMG sector**
- **Component 5: USD 1.15 million GCF Technical Assistance (TA) to strengthen regulatory & institutional framework at the national level to overcome the operational constraints related to implementing EE&C in the country**

In case of Component 1, 2 & 4: IDCOL will be the Executing Entity for Component 1, 2 & 4. Under Component 1, IDCOL will make GCF funds available in the form of concessional loans to the end borrower who are in this case the private textile manufacturing entities. The end borrower will seek disbursements from IDCOL in both USD and BDT for energy efficient equipment & technology in line with the capacity proposed to be deployed every year over the programme implementation period. Under Component 2, IDCOL will also be responsible for execution of activities under Technical Assistance (TA).

Just like component 2, component 04 will be executed by IDCOL which consists of activities proposed as Technical Assistance to promote access to information & resources and to build capacity for successful implementation of Component 03 to bring about a systemic change in the RMG sector.

In case of Component 3, IDCOL will channelize funds from GCF in the form of loan, to the Executing Entities (EEs) i.e. LFI's viz. BRAC Bank, City Bank, South East Bank and IDLC or any other eligible LFI's as per selection criteria to reach the ultimate end borrowers who are in this case the private garments manufacturing entities. In some instances, IDCOL will lend directly to the RMG manufacturers. The Accredited Entity (AE), IDCOL will receive fund from GCF and channel it to Executing Entities (EEs). The EEs will seek disbursements from IDCOL for energy efficient equipment & technology according to their loan application pipeline every year over the program implementation period. The transaction throughout the channel will be in USD.

Component 05: This component will address the policy level issues for creation of enabling environment for energy efficiency financing of the country. It will target to strengthen regulatory & institutional framework at the national level to overcome the operational constraints related to implementing EE&C in the country. Sustainable and Renewable Energy Development Authority (SREDA) under Ministry of Power, Energy & Mineral Resources, GoB, will be the implementing partner of this component.

Under Component 01 and 03, GCF funding in the form of concessional senior loans will support the end borrowers of textile and RMG sector respectively, by deploying a credit line to ensure that long term concessional financing is available in the market to overcome the barriers of high up-front costs, and longer payback periods, to promote private sector investment in EE equipment. Even though the implementation modality and the addressed sectors of Component 01 and 03 will be different as mentioned above, but the overall objective will be the same which is to progress investment opportunities for energy saving technology

upgrades in the country. Furthermore, GCF non-reimbursable resources (technical assistance), comprising under Component 02 and 04 aims to improve market perception about energy efficiency and provide support to the relevant stakeholders at all stages of implementing energy efficient measures. It is envisaged to create an interface between energy efficiency equipment suppliers or manufacturers and the decision makers behind the investment for energy efficiency i.e. Textile & RMG end borrowers, thereby helping disseminate the information related to energy efficiency and resolve any queries that the unit owners may have. Policy makers will also be apprised the barriers to implementing energy efficiency measures and provide guidance in developing enabling policies, regulations to help overcome these barriers. It is envisaged that without such enabling activities, a concessional line of financing would not help in generating interest among the stakeholders in undertaking energy efficiency measures. Also, given that the energy efficiency market in Bangladesh is in its infancy, such support activities would be critical to enable market transformation and galvanize the equipment procurement decisions of Textile & RMG units. Hence, the TA is viewed to be a critical component to ensure greater uptake of energy efficiency measures during the programme duration in the two sectors, as well as ensure sustainability of energy efficiency market for both the sector beyond the programme duration.

The programme overall is expected to reduce or avoid about 14.53 MtCO₂eq of GHG emission reductions over the lifetime of the investments in both the sectors. The programme is first of its kind for the country considering specific focus on mitigation thorough energy efficiency of two largest energy intensive industries i.e. textile & RMG. The other ongoing and past programmes on EE have a broader focus targeting various energy intensive industries of the country. The paradigm shift potential for the proposed programme lies in the project's focus on overcoming technical & financial barriers for the private sector (including the Local Financial Institutions), by providing know-how & capacity building and finally by creating an enabling environment to promote EE&C investments in textile and RMG sector of Bangladesh. Nationwide impact will also be achieved by implementing component 05, by Government owned institution SREDA, to strengthen regulatory & institutional framework at the national level to overcome the operational constraints related to implementing EE&C in the country.

B. PROJECT/PROGRAMME INFORMATION

B.1. Climate rationale and context (max. 1000 words, approximately 2 pages)

Climate rationale: Describe the climate change problem the proposal is expected to address. Describe the mitigation needs (GHG emissions profile) and/or adaptation needs (climate hazards and associated risks based on impacts, exposure, and vulnerabilities) that the proposed interventions are expected to address.

Climate rationale:

Global warming is the most alarming phenomenon of the present time and to combat it, it is high time to take all possible measures in every sector. Bangladesh submitted its INDC to the United Nations Framework Convention on Climate Change (UNFCCC) in September 2015. Bangladesh ratified the Paris Agreement on 21 September 2016 and its NDC can now be found on the UNFCCC's interim NDC Registry⁵. Bangladesh's NDC describes its plans for tackling greenhouse gas (GHG) emissions through a progressive approach to developing its economy on a low carbon pathway. In the NDC, Bangladesh committed to reduce GHG emissions in the power, industry and transport sectors by 5% below 'business-as-usual' GHG emissions by 2030 using only domestic resources, or by 15% below 'business-as-usual' GHG emissions by 2030 if sufficient and appropriate support is received from developed countries.

Energy consumption, especially in the industrial sector is a significant factor for the GHG emissions which makes it imperative to help in the adoption of environment friendly practices by the industries on a priority basis. In Bangladesh, the industry sector accounts for 47.8%⁶ of the commercial energy demand, predominantly in the form of natural gas and electricity. However, as per GoB's Energy Efficiency & Conservation Master Plan 2030, most of this commercial energy is inefficiently consumed and utilised for economic activities. Implementation of energy efficiency and conservation (EE&C) measures within the industrial sector can help reduce the energy consumption in the sector by 30%. This is exemplified by the diagram below which gives the break-up of energy consumption in various industrial sub-sectors under business-as-usual (BAU) scenario and an EE&C scenario which take in to account maximum energy efficiency potential for the industrial sector⁷:

⁵https://www4.unfccc.int/sites/ndcstaging/PublishedDocuments/Bangladesh%20First/INDC_2015_of_Bangladesh.pdf

⁶ Bangladesh's EE&C Master Plan 2030 (March 2015)

⁷ Source: [Energy Efficiency and Conservation Master Plan up to 2030 \(JICA website, 2015\)](#)

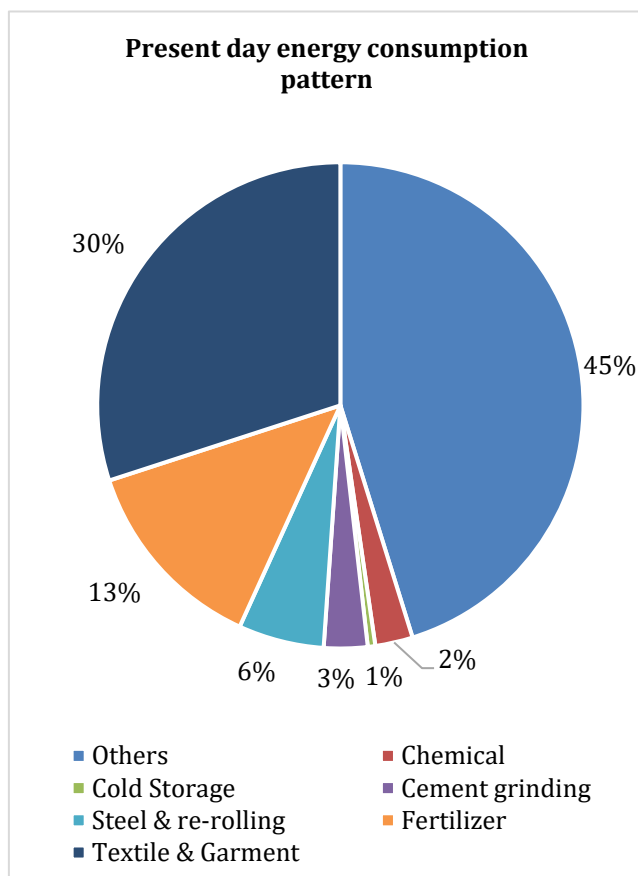


Figure 1: Energy consumption pattern under baseline scenario

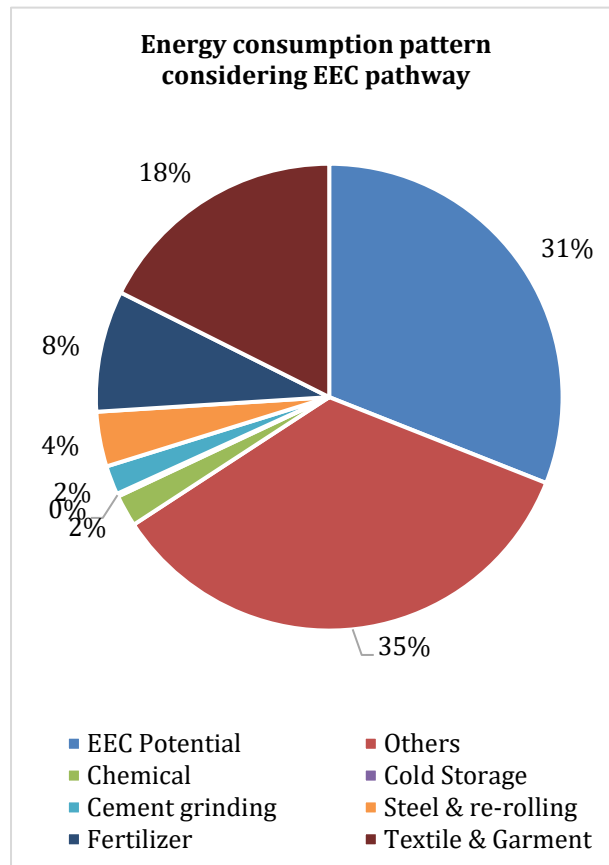


Figure 2: Energy consumption under EE&C scenario

While the energy savings opportunity is present across the various industrial sub-sectors, the Textile and RMG sector in particular provides substantial scope of savings. The textile and RMG sectors combined accounts for approximately 27.80% of the energy consumption in the industrial sector.

The RMG sector being the most energy intensive consumes 15.4% and textile sector being the second most energy intensive consumes 12.4%⁸ of the total industrial energy demand (1,701 kTOE/year).

⁸ SREDA Energy Balance 2016-17

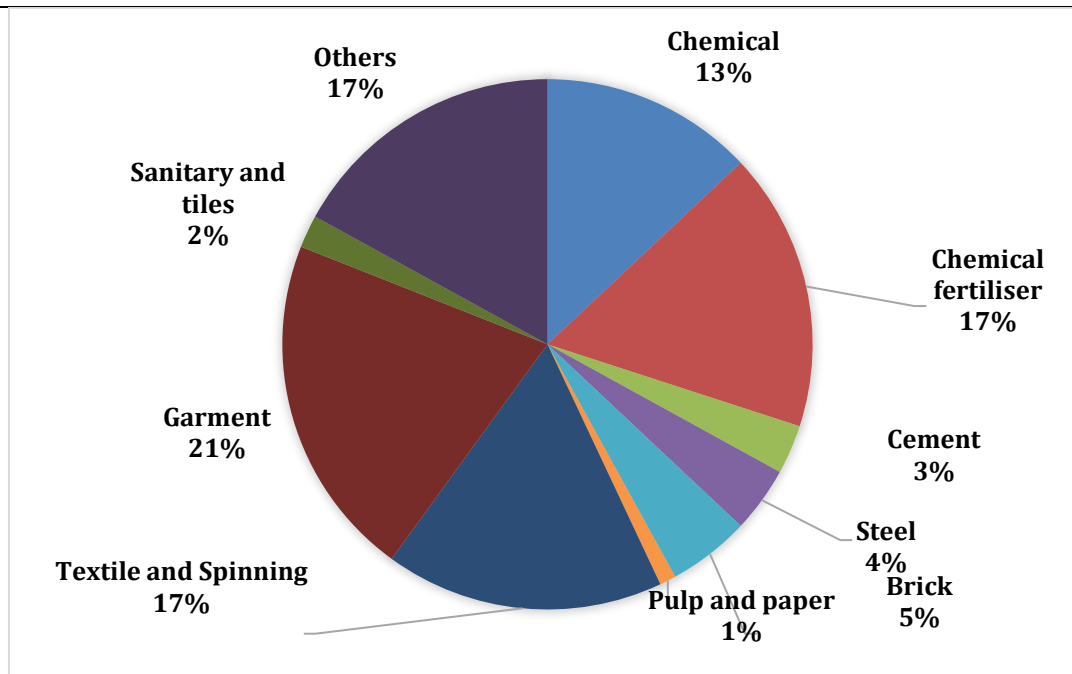


Figure 3: Energy consumption by sectors (source: SREDA energy balance 2016-17)

The feasibility study conducted for the textile sector suggests, the larger textile units are typically 10 to 15% more energy efficient than the 'small and medium' units with efficiency being as high as 50% in some cases. Hence, under Component 01, IDCOL based on successful due diligence may consider financing a segment of small and medium units as part of the programme.

The RMG sector also showed similar trend, in terms of energy efficiency, small factories having small-scale investment are more oriented towards increasing revenues and short-term profits, rather than undertaking initiatives which will help improve productivity and increase profits in the long run. Expenses on energy efficiency is seen as an additional financial burden and is rarely undertaken. Therefore, under Component 03, IDCOL will encourage the LFI (Executing Entities) to increase awareness among the RMG unit through consultation and to create access to concessional finance for small and medium units on its successful due diligence.

The proposed programme aims to support entrepreneurs and business establishments in the textile and RMG sector, with financial and market resources to avail investment opportunities for energy saving technology upgrades. The programme will create a favorable market environment and scalable business model for investment in energy efficiency improvements, leading to sizeable energy savings and accompanying greenhouse gas (GHG) emissions reductions of 14.53 million tCO₂ over the 20-year lifetime of the project investments.

Also describe the most likely scenario (prevailing conditions or other alternative) that would remain or continue in the absence of the proposed interventions. Include baseline information. The methodologies used to derive the climate rationale should be included in the feasibility study.

Snapshot of Textile Industry in Bangladesh

The modern story of the growth of textile industry in Bangladesh starts with the establishment of the Bangladesh Textile Mills Corporation (BTMC) on March 26, 1972 set up with the point of observing, planning and controlling the exercises of nationalized factories of Bangladesh and building up and growing new enterprises. BTMC began its formal program with 74 nationalized factories from 1 July, 1972.

In addition to BTMC, Bangladesh Textile Mills Association (BTMA) is the prime national trade organization representing Yarn, Fabric Manufacturers and Textile Product Processors mills of the country under private sector and private sectors are registered under BTMA. BTMA has been registered in 1983 and at present, 1476 Member Mills are now listed under BTMA.

Table 1: Composition of Bangladesh's textile sector⁹

Types of Factory	Numbers	Installed Capacity	Annual Capacity
Yarn Manufacturing Mills	430	Spindles 13.17 Million Rotors 0.231 Million	2943 Million Kgs
Fabric Manufacturing Mills	802	Shuttle-less Loom & Shuttle Loom: 52169 Million	3707 Million Mtrs.
Dyeing-Printing-Finishing Mills	244	N/A	Fabric Processing: 3448 Million Mtrs Woven Dyeing: 2480 Million Mtrs Yarn Dyeing: 396 Million Kgs Knit Dyeing: 572 Million Kgs
Total	1476		

Textile industries are the one of the fastest growing and amongst the most important industries of Bangladesh and contributes to a significant portion of the nation's economy, employment and global profile. Textile sector contributes more than 13% of Bangladesh's GDP while also supporting 86% of the country's export in the form of the final clothing/garment (from Textiles & Textile related products¹⁰), which is a testament to the huge importance of the sector in the country's growth. The export of clothing/garment in Bangladesh based on the textile production has increased from USD 24 bn to USD 33 billion from 2014-2019 with an average annual growth of ~8%.

Even though textile sector is a growing industry in Bangladesh, however it is still not self-sufficient in terms of serving the demand of the local garments sector. The local textile millers can meet 85 percent of the demand from the knitwear sector and 35 percent from the woven sector, according to data from the BTMA. Bangladesh needs USD 1.17 billion additional investments in the primary textile sector to reduce its import dependence for fabrics for the export-oriented garment sector¹¹.

The industry (along with the garment industry) is one of the largest employers in the country, generating more than 65% of the country's industrial employment and employing over 4 million. Out of this a significant proportion of employees is women, with textile and garment overall having around equal number of men and women employed but textile in particular has around 20-30% female employment according to the feasibility study. Thus any improvement in the conditions in this sector will play an important role in the

⁹ <http://www.btmadhaka.com/wp-content/uploads/2019/01/Basic-Data-of-BTMA-2018.pdf>

¹⁰ Ibid

¹¹ <https://www.thedailystar.net/business/news/textile-sector-needs-tk-10000cr-investment-1700014>

gender empowerment as well. In general, the state of the textile industry impacts indirectly the well-being of 10-12 million people in Bangladesh¹².

Textile sector energy baseline

Energy in the textile industry is mostly used in the forms of electricity, as a common power source for machinery, cooling and temperature control systems, lighting, office equipment, etc.; diesel oil as a fuel for boilers which generate steam; liquefied petroleum gas; coal; and natural gas.

A breakdown of the processes is done to study the various process deployed in textile units and examine energy consumed in each process.

Table 2: Break down of processes in textile units

Spinning	Weaving	Knitting	Dyeing and drying
•Yarn Fabrication •Blending •Cleaning, carding and combing	•Drawing •Roving •Twisting and Spinning	•Starching and Chemical treatment for tensile strength	•Chemical treatment and coloring •Drying and steaming

Each of the listed processes are energy consuming. The textile units comprise a large number of machines which together consume a significant amount of energy from the grid, diesel and natural gas.

The energy consumption for each process of the sub processes is presented in the graph below.

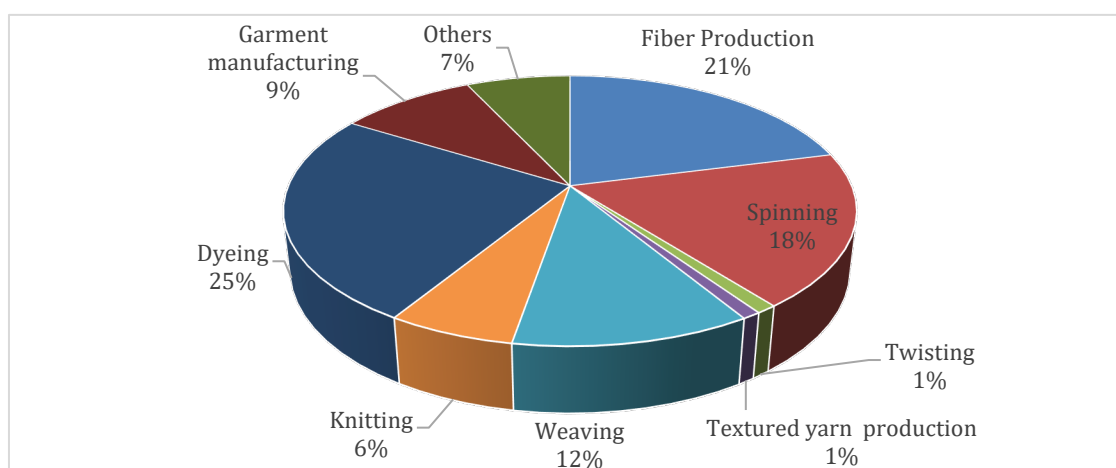


Figure 4: Energy consumption per deployed process in textile units

Feasibility study shows that spinning and weaving are quite energy intensive. However, largest share of energy is consumed in dyeing and drying which uses energy for boiling, cooling and steaming of water.

¹² Ibid (6)

EE&C Master Plan for 2030, indicates in the energy blends for the textile sector that the natural gas is the highest consumed fuel in textile in comparison to other sources i.e. Grid/REB¹³ and diesel remain negligible. However, the actual consumption of electricity from Grid is 92 toe/annum and that of natural gas is 1,586 toe/annum¹⁴. Some of the small yarn producing textile units solely rely on electricity from the REB. As the feasibility study conducted under the PPF also includes the non-registered small units, the energy blend ratio is a representation of the actual scenario.

The feasibility study¹⁵ indicates that a number of small and medium units utilize diesel and oils which isn't registered in national statistics while natural gas continues to be the most consumed out of the rest of the energy sources. Figure 02 compares the energy mix indicated in SREDA's Energy Efficiency and Conservation Master Plan 2030, prepared in 2015 with the data identified during the feasibility study of this programme. Even though Natural Gas remains the dominant source of energy but its share has declined by almost 22%, whereas diesel share has increased by 2440 % and share of grid electricity has declined by 75%. This indicates that the textile industry is increasing its carbon footprint (diesel has the highest emission factor) in order to maximize production with the available and affordable sources of energy and thus energy efficiency transformation of the sector becomes imperative.

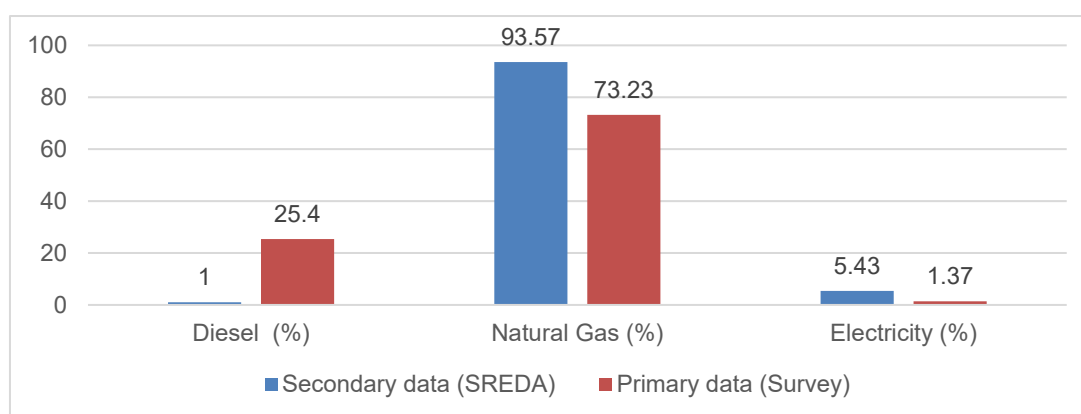


Figure 5: Comparative analysis of energy blends as emerged from National statistics and Survey conducted

Growth of Textile sector and GHG emission:

As of 2018, Bangladesh holds second place¹⁶ after China in apparel export. The expansion of the textile industry in Bangladesh is conceivable mainly due to cheaper labor. However, other factors are also closely linked, such as supportive business conditions, freightage facilities, and so on. There are organizations such as the Bangladesh Textile Mills Association (BTMA), Bangladesh Garment Manufacturers and Exporters Association (BGMEA) and Bangladesh Knitwear Manufacturers and Exporters Association (BKMEA) which focus on reinforcing and promoting the entire apparel industry sector. They take measures including ensuring an amicable atmosphere for business, educating the laborers, making an effort for the betterment of the laborers' social compliance, and building relationships among the concerned stakeholders, with the focus of

¹³ Rural Electrification Board

¹⁴ Energy Efficiency and Conservation Master Plan up to 2030
(http://sreda.gov.bd/files/EEC_Master_Plan_SREDA.pdf)

¹⁵ Annex

¹⁶ RMG Sector towards a Thriving Future. The Daily Star. 2 February 2016. Available online:
<https://www.thedailystar.net/supplements/25th-anniversary-special-part-2/rmg-sector-towards-thriving-future-210886> (accessed on 10 October 2018).

expanding the trade income of Bangladesh. The country is aiming now to achieve \$50 billion in RMG exports by 2021¹⁷. Figure 6 illustrates the yearly apparel export of Bangladesh in United States dollars (USD).

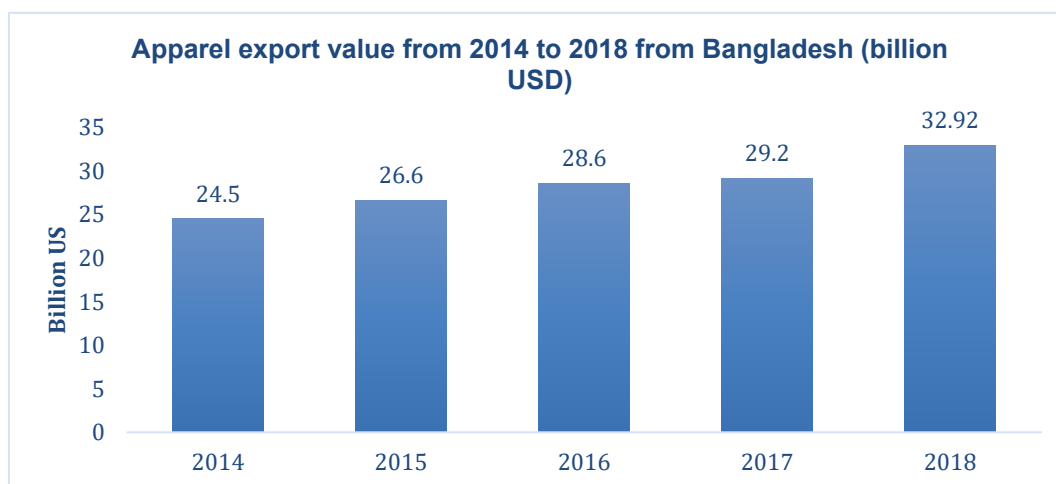


Figure 6: Yearly apparel export of Bangladesh in United States dollars (USD) from 2014-2018

With a yearly average growth of 8% in the textile export, energy demand of this sector is exponentially rising. The historical trend analyzed in the feasibility study identified that the total industrial energy consumption (KTOE) is progressing at a CAGR of 8%, which is mainly driven by the rise in the energy demand in the textile sector. In such case improved energy efficiency will play a critical role, along with sustainable energy, thereby minimizing the anthropogenic greenhouse gas (GHG) emission by the industry as a major energy user.

Snapshot of RMG Industry in Bangladesh

The roots of the RMG sector in Bangladesh, lies in late 1970s with the success of garment manufacturers in catering to export market. The sector gradually became fragmented and formation and institutionalization of the Bangladesh Garment Manufacturers and Exporters Association (BGMEA), one of the largest trade associations in the country, in 1983 helped in structuring the sector and increase its contribution to the country's economy, as well as ensure all legitimate rights and privileges of the garment workers leading to the betterment of the society and environment.

The sector, which comprises over 4500 factories, has been a leading contributor to the economy, accounting for 11% of the country's GDP in 2017-18¹⁸. It is the largest exporting industry in Bangladesh and has experienced phenomenal growth over the last three decades – contribution in total export increased from 4% in 1983-84 to 84% in 2017-18², amounting to USD 30.62 billion. The sector has witnessed a growth of nearly 7.3% CAGR in last 5 years i.e. 2013 to 2018.

¹⁷ Finance Division, Ministry of Finance-Government of the People's Republic of Bangladesh. Available online: <https://mof.gov.bd/> (accessed on 15 October 2018).

¹⁸ Source: [Statement by BGMEA \(Textile Today, 2019\)](#)

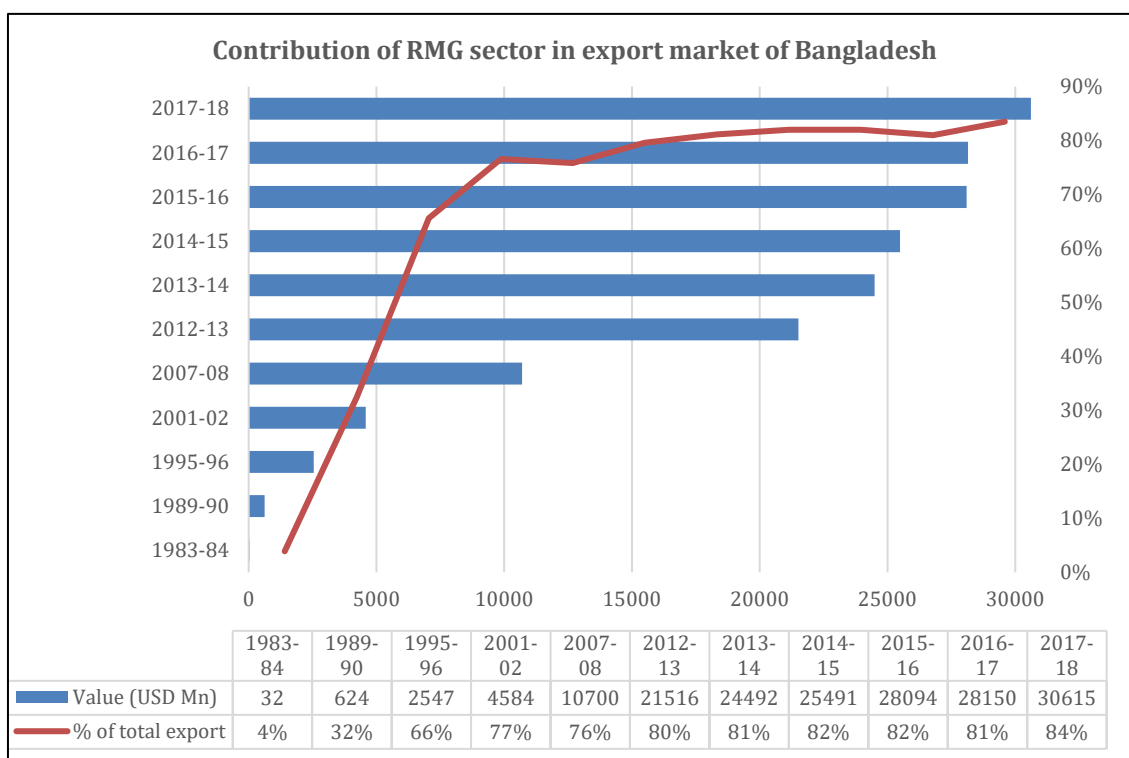


Figure 7: Bangladesh RMG sector export contribution

While the sector continues in its growth trajectory, there are a number of hurdles in the horizon which need to be overcome to reach USD 50 billion export by 2021 set by the national body for garments manufacturers in Bangladesh - Bangladesh Garment Manufacturers and Exporters Association (BGMEA). These impediments include rising energy costs, increasing worker wages, low worker productivity, graduation from LDC status of Bangladesh and increased demand of sustainable production practices from customers.

The sector plays a key role in employment generation and in the provision of income to the poor – it employs nearly 4.4 million workers¹⁹. The sector has also played a significant role in the socio-economic development of the country with women comprising nearly 80% of the total workforce in the sector²⁰. It also plays a pivotal role to promote the development of linkage small scale industries, since 99% of the accessories like dyeing, printing, zippers, labels are manufactured in Bangladesh²¹. This had helped open up employment opportunities through indirect economic activities, thereby contributing to the country's overall socio-economic development, woman empowerment and poverty alleviation.

¹⁹ Source: Information disseminated by BGMEA

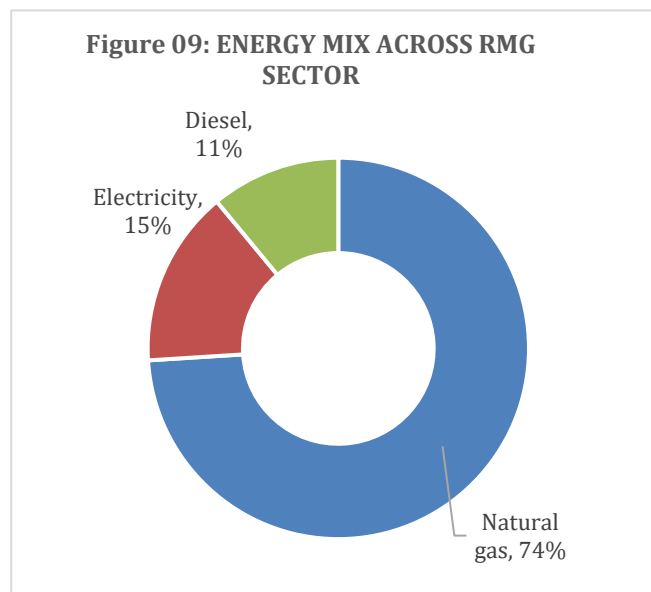
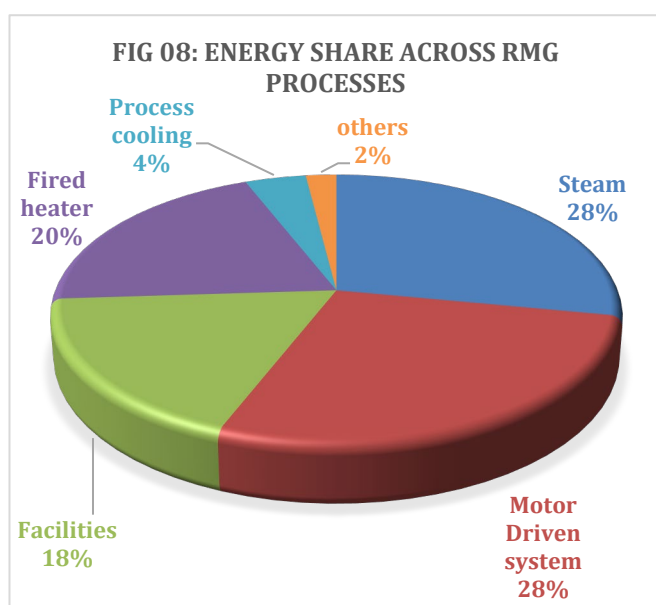
²⁰ Source: [Statement by BGMEA President \(BGMEA website, 2019\)](#)

²¹ Source: As per presentation provided by BGMEA representative



RMG Sector Energy Baseline

The garments sector combined accounts for about 15% of total energy consumption in Industry sector of Bangladesh i.e. 2107kTOE/yr²² which translates to 7.5% of the country's total emissions i.e. 6.7 MtCO₂ approximately. The energy is consumed in a set of production processes which include fabric production (only in case of weaving, knitting), design, pattern production, grading, fabric relaxing, marking, spreading, laying, cutting, sewing, wet processing, spot cleaning/laundry, fusing, pressing and packaging. Based on the end products, the combination of processes may vary for specific units. In order to understand the energy related features of the sector, a survey of 20 units was conducted along with energy audit of 4 units. Based on the information generated from the survey, the share of energy consumed in RMG across type of process is given below:



The energy consumption in the sector is skewed towards use of natural gas, with grid electricity having a minor share. Diesel is mostly used to generate electricity as a back-up and its use depends on non-availability of grid electricity. As per discussions with stakeholders, small RMG units use diesel generators to fulfil power needs. In large units, there are captive natural-gas power plants that supply electricity. While natural gas is used in its primary form as boiler fuel, diesel is almost exclusively used for backup (in most cases) or primary power generation.

In order to enhance the understanding of on-ground realities of RMG sector in Bangladesh and assess energy efficiency opportunities in the context of Bangladesh RMG sector, a series of walkthrough audit and

²² Source: [Information derived from EE & Conservation Master Plan upto 2030 \(DANIDA, 2017\)](#)

detailed energy audit had been undertaken. Energy survey was conducted in a total of 20 units, while detailed energy audit conducted in 4 units. The findings from the studies as well as audits have been presented below.

In order to provide the idea of the manufacturing of RMG products, a brief description of production processes of a standard RMG unit has been explained.

Figure 10: Flow-chart of the manufacturing processes in RMG sector

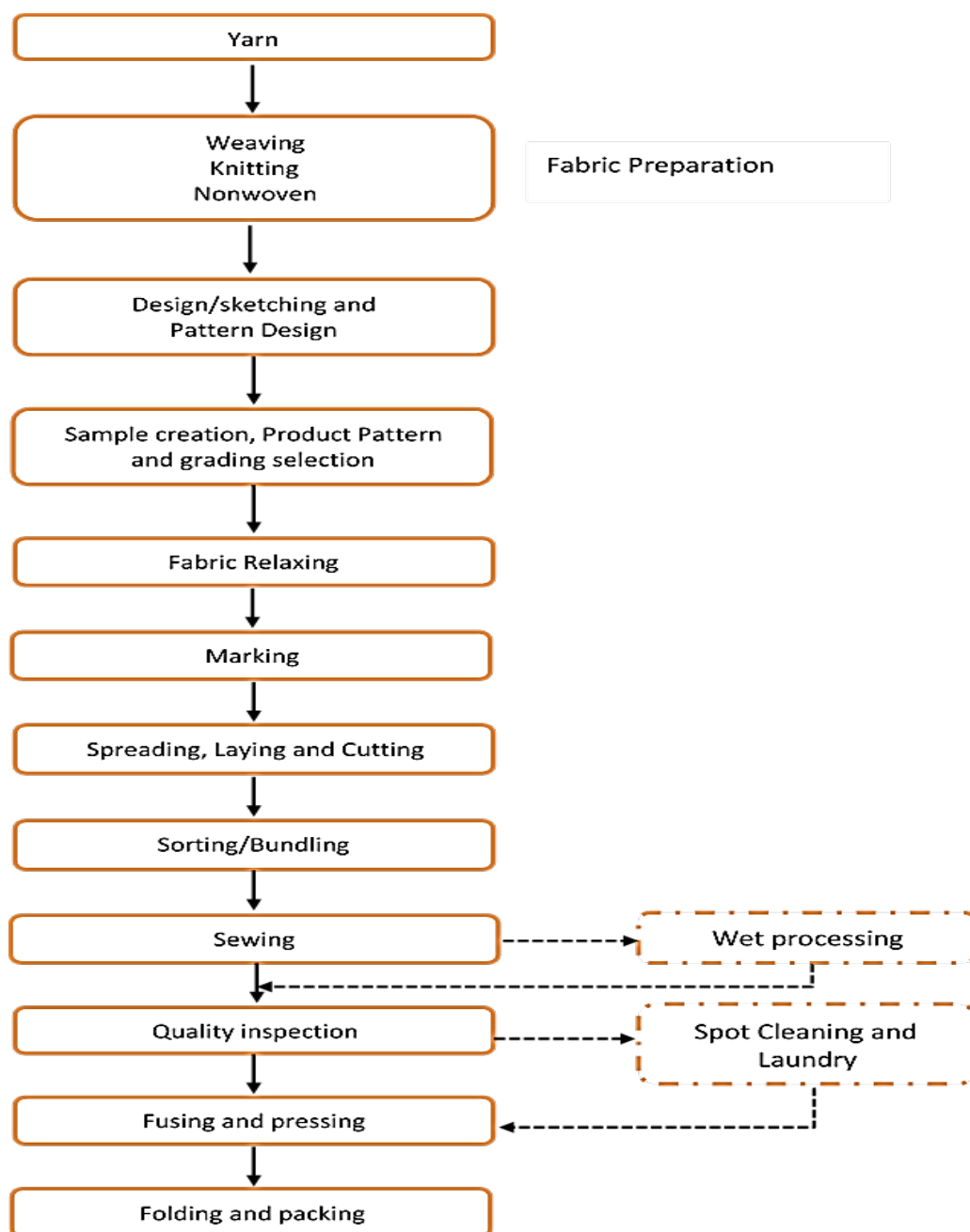


Table 03: Brief Description of RMG unit Processes

Operation	Process description	Method
Fabric production	• Weaving: Involves interlacing two or more yarn systems crosswise and vertically. Woven textiles are used in all textile sectors (apparel, home textiles, and technical textiles). On the weaving machine (loom) the weft yarn is inserted into the lengthwise-oriented warp yarns (shed). • Knitting: In knitting, fabric is formed by interlocking or intermeshing loops of one or more sets of yarns. Knitting is performed using one of two processes: weft or warp knitting. Each of these processes is done on several different types of machines. • Nonwovens: Fabrics made of fibers without intermediate yarn processing are called nonwovens. Nonwovens are essentially sheet or web structures made by bonding and/or interlocking fibers, yarns, or filaments using mechanical, thermal, chemical bond, or solvent means.	Automatic/ manual
Design/sketch	• First step to manufacturing involves designing the product containing measurement defining a particular style	Automatic/ manual
Pattern design	• A sample pattern is prepared based out of the style concept developed from design/sketch. • Pattern provides the design template on the basis of which multiple iterations of a particular style will be produced.	Automatic/ manual
Sample	• A sample garment is produced from the first pattern to analyze pattern fit and design by a panel of designers, pattern makers and sewing specialists. • Any modifications, if required, are incorporated in the pattern	Manual
Production Pattern	• Pattern design is used for creating production patterns which will be used for mass production.	Automatic/ manual
Grading	• Patterns are created based on all (S M L XL XXL or size 10, 12, 14, 16) or else there are standard patterns of and so on for different figure and sizes.	Automatic/ manual

Fabric Relaxing	• “Relaxing” refers to the process that allows the material to relax and contract prior to being manufactured. • Relaxing process allows fabrics to shrink so that further shrinkage during customer use is minimized.	Automatic/ manual
Marking	• Marker making is the process to sketch on fabric before cutting it. • Markers are laid in such a way so that minimum possible fabric gets wasted during cutting operation	Automatic/ manual
Spreading, Laying, and Cutting	• Spreading is the process of superimposing lengths of fabric on spreading table for cutting purpose. • After successfully spreading the fabric and putting all the marks and sketch by the marker; fabric is cut according to design	Automatic/ manual
Sorting/Bundling	• Sorting of pattern shaped cloth according to size and design • Requires much precision since having bundles of mismatched patterns can create severe problems.	Manual
Sewing	• Stitching/sewing undertaken after cut pieces are bundled according to size, color and quantities determined by the sewing room.	Manual
Wet processing	• Includes de-sizing, scouring, bleaching, dyeing, printing and finishing etc. that involve adding colors to fiber, yarn, or fabric • Performed along with a variety of finishing steps that provide certain desired characteristics to the end product.	Automatic/ manual
Quality inspection	• Complete quality checking of product to assess whether it is compliant with quality requirements	Manual
Spot Cleaning and Laundry	• Spots and blemishes are often marked with a sticker and taken to a spot-cleaning area where the garment is cleaned using steam, hot water, or chemical stain removers. • Based on client requirements, some products require to be fully laundered in on-site laundry or with subcontract agreements with off-site laundry operations. • Commercial laundry facilities are equipped with at least three types of machines: washers, spinners, and dryers. Some facilities also have the capability to perform special treatments, such as stone- or acid-washing.	Automatic/ manual

Fusing and pressing	- Finished product released by quality control goes for final pressing. - The ironing station consists of an iron and an ironing platform with steam supplied by an on-site boiler. - In most facilities, the ironing platforms are equipped with a ventilation system that draws steam through the ironing table and exhausts it outside the factory.	Automatic/ manual
Folding and packing	- The finished garments are then folded in a specific dimension, sometimes using a template. - The price tags, hang hags and any other kind of tags are attached to garment after folding. - Different types of packing accessories are used to keep the garment in a desired shape.	Manual

The figure 11 below provided the breakdown of the final energy use in the garments industry as per the walkthrough survey:

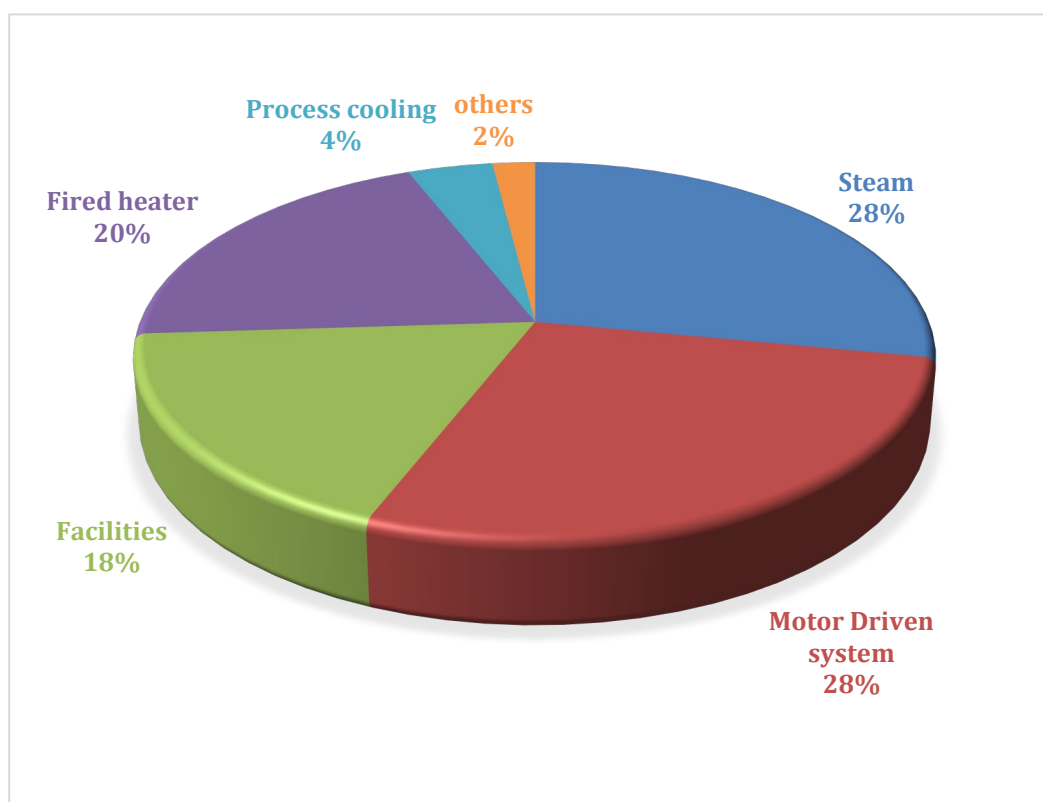


Figure 11: Process wise share of energy consumption in RMG industry

Growth of RMG sector and GHG emission:

Tracking historic growth of the sector for the past 5 years reveals a CAGR of 7.5%. Even though the market is experiencing pressure in generating profits due to high input costs (rising energy, material and labour costs), increased compliance cost, high financing rate, the demand for fast fashion items, which can help in

sustained growth in the sector. While onset of COVID-19 will definitely impede growth in near future, demand in the medium and long term is expected to increase and the sector shall continue its growth trajectory.

Given such a situation, GHG emission will also increase with increased production levels. Given current level of penetration of energy efficient equipment, which is negligible, it is expected that emissions growth will follow a similar trajectory, witnessing a growth of about 7% CAGR in next five years. Thus, increasing penetration of energy efficient equipment would be a key method of stemming the growth of emissions for the sector.

Financial Market baseline for EE&C:

There are several loan facilities for energy efficiency and conservation promotion, however higher lending interest rate and the time required for appraisal have hampered smooth implementation of loan projects. In this context, both minimizing the time for loan procedure through simplification of loan application and speedy appraisal and development of loan having concessional low interest rate and loan scheme will promote energy efficiency and conservation in Bangladesh further.

In Bangladesh progress with industrial energy efficiency has been slow. This situation is expected to prevail with the current market and technical challenges. Hence, interventions in the proposed program aims to increase the rate of adoption of energy efficient machineries in the textile and RMG sectors, which has 25%-30% of energy saving potential²³.

Considering the high cost of changing machinery process optimization is more preferable to the concerned stakeholders. However, the replacement of new machinery can be financially justifiable via low interest loan so that total payment amount for energy efficient equipment (equipment cost and financial cost) break even with acquiring conventional equipment.

The programme will result in a real and visible paradigm shift in the textile and RMG sector of Bangladesh and advance it towards low-carbon sustainable development by limiting the country's GHG emissions, as specifically recommended in the Nationally Determined Contribution 2015, the Bangladesh Climate Change Strategy and Action Plan (BCCSAP) and Energy Efficiency and Conservation Master Plan (EE&C Master Plan) 2030. The programme is expected to result in direct emission reductions of 0.727 tCO₂e annually by facilitating and scaling-up investment in energy efficiency improvements. However, gradually the annual emission reduction amount will increase as the energy mix will contain greater share of coal and oil, as domestic natural gas production is declining.

Context: In describing the mitigation and/or adaptation needs, briefly describe the target region/area of the proposed interventions including information on the demographics, economy, topography, etc.

The Bangladesh economy continues to perform well with robust and stable growth. The strong growth comes with stable inflation, moderate public debt, and greater resilience to external shocks. The country continues to make steady progress in reducing poverty and improving social indicators.

Real GDP growth in FY18 (ending June 30) further accelerated to 7.86 percent from 7.28 percent in the previous fiscal year, led by strong private consumption and investment. The current account turned into a deficit with slower export growth, higher imports, and decline in remittances, while the balance of payments

²³ Source: [Bangladesh: Industrial Energy Efficiency Program \(ADB website, 2014\)](#)

remained in small surplus. The debt-to-GDP ratio has remained stable at around 30 percent with the fiscal deficit well below the 5 percent of GDP budget target.

The macroeconomic situation is expected to remain robust in FY19. Growth is projected above 7 percent with strong domestic demand. Inflation is expected to remain below 6 percent. The current account deficit is projected to widen to close to 2 percent of GDP with stronger import demand for food, industrial raw materials, and capital machinery, while remittances and exports start to recover.

Table 4: Economic and Demographic data of Bangladesh²⁴

	2017	2018	2019
Market Size and Growth			
Real GDP (Bt.B)	9,479.0	10,224.4	10,992.0
Real GDP Growth (%)	7.28	7.86	7.50
Nominal GDP (\$B)	249.7	274.0	301.7
Population (M)	164.7	166.4	168.1
Population Growth (%)	1.10	1.00	1.00
GDP per Capita (\$)	1,516	1,647	1,795
Unemployment Rate (%)	4.37	4.38	4.40
Macroeconomic Stability			
Current Account Balance (\$B)	(6.37)	(7.84)	(8.82)
Consumer Prices Growth (%)	5.70	5.54	5.40
Budget Balance/ GDP (%)	(3.42)	(4.80)	(4.80)
External Sector			
Goods: Exports (fob) (\$B)	31.39	33.72	35.81
Total Imports (cif) (\$B)	47.57	55.39	58.66

In addition to this, prior to the COVID-19 pandemic, ADB reported in its 'Asian Development Outlook 2019' Bangladesh's economy is in good shape & likely to grow. Bangladesh is confident that it will surpass the 8 per cent growth threshold in the current fiscal year on the back of manufacturing and service sectors.²⁵

With this expected increasing trend of economic growth & positive future outlook, apparel industry played a significant role being the largest contributor in export income for the country. Textile sector contributes more than 13% of the GDP of Bangladesh while also accounting for over 86% of the country's export (from Textiles & Textile related products), which is a testament to the huge importance of the sector in the country's growth. However, due to COVID-19 pandemic a lot of the expected economic trend of Bangladesh will change, a brief assessment of impact of COVID-19 on Textile and RMG sector is given below.

²⁴ S&P Global Market Intelligence

²⁵ <https://www.thedailystar.net/country/gdp-growth-rate-of-bangladesh-2019-will-achieve-8-per-cent-adb-1724293>

COVID -19 impact on the Textile and RMG industry:

As on 15th May 2020, Bangladesh has reported 20,065 confirmed cases and 298 deaths following the identification of the first imported covid-19 case on March 8th.

On March 23rd, the government declared a general holiday from March 26th to April 4th, which was extended subsequently. Government offices, private offices, and courts are closed, commercial banks are allowed to operate shorter hours. Individuals have been requested not to leave their homes except to collect daily necessities and emergency supplies, and to wear masks when outside. On 10 April, The Bangladesh Garment Manufacturers and Exporters Association (BGMEA) and Bangladesh Knitwear Manufacturers and Exporters Association (BKMEA) announced in a joint statement that the readymade garment (RMG) factories would remain closed until 25 April.

Directives by the Bangladesh government stated that textile & garment factories, which supply some of the biggest brands in the world and produce 84% of the country's total exports, would be allowed to resume operations after 25th April, provided factories comply with health guidelines and maintain physical distancing.

On 13 May, it was reported that the government is likely to extend the shutdown of most private and public workplaces until 30 May.

Impact:

Beyond the domestic impact of the health crisis, the two main channels through which the Bangladesh economy will be impacted are remittances and exports of textile & apparels. Remittances represent over 5 percent of GDP (\$16.4 billion in FY 19), and a majority of migrant workers are based in Gulf countries that are affected by the abrupt decline in oil prices. The textile & garment sector accounts for more than eighty percent of the country's exports. As, Bangladesh, the second-largest apparel producer after China, is set to lose roughly US \$ 6 billion in export revenue this financial year amid cancellations from some of the world's largest brands and retailers due to Covid-19 outbreak, shared by Bangladesh Knitwear Manufacturers and Exporters Association (BKMEA) Vice President Mohammad Hatem. The industry has been hit by the cancellation or postponement of several billion US dollars in orders from major retailers in importing countries, although top global apparel brands announced on April 1st that they would receive shipments of goods already produced. The decline in national and global demand for manufactured goods, particularly in the garment sector, risks creating unemployment and deepen poverty. The urban poor will be hardest hit while the number of additional poor will be higher in rural areas. The extent of the impact of the pandemic will depend on the duration of the crisis and the mitigation measures taken.

There have been 3 distinct phases, which charts the impact of COVID in textile & particularly RMG sector of Bangladesh. A sample study of 316 RMG units by Penn State Center for Global Workers' Rights (CGWR) provides vital insight on how the COVID situation has shaped the key events for each of these phases. Details are discussed below:

1. **Phase one – Material supply:** With the lockdown in China, the single largest source of raw materials (Chinese fabric import accounts for 46% of total imports for RMG sector²⁶), RMG units reported delays in raw material shipment or significant increase in raw material prices. This set in to motion the first phase of impacts. Delay in material receipt resulted in delay in product shipment, as a result of which multiple units received imposition of penalty by buyers. Almost all RMG units did not receive any extra payments from buyers on account of increased raw material prices.

²⁶ Source: [Article \(LightCaste Partners, 2020\)](#)

2. **Phase two – Delayed payments:** In the second phase, global incidence of lockdowns started affecting the bottom line of buyers, resulting in delay in payments to RMG units. RMG units were experiencing varying levels of delay, relative to contractually stipulated terms. In most cases, the delay was extended beyond a month.
3. **Phase three – Order cancellation:** Further aggravation of market demand due to continued lockdown, led to cancellation of orders by buyers. While there have been reported variations in cases, mostly buyers cancelling orders have refused to pay for raw materials or production costs incurred by RMG units.

Supportive measures:

As is evident from the discussions above, the COVID pandemic has adversely impacted both demand and supply side of the textile and RMG sector in Bangladesh thereby resulting in cash crunch. National Governments like USA, India are announcing relief package as well as multi-laterals like World Bank. The latter expects to deploy USD 160 billion over the next 15 months to help countries protect the poor and vulnerable, support businesses, and bolster economic recovery²⁷.

While traditional markets are failing, demand for protective personal equipment has surged and prices of PPE products have risen dramatically since the beginning of the COVID-19 outbreak: a six-fold increase for surgical masks; three-fold for respirators; and a doubling in the price of gowns²⁸. However, there is a supply bottleneck in the market with backlog of 4–6 months for supply orders²⁹. This provides an opportunity for textile and RMG units to reposition and focus on medical protective kits. The increased demand for personal protective equipment (PPE) is expected to continue in the short to medium term i.e. 1-3 years, given the state of pandemic. Such product diversification can help RMG units to tide over the pandemic in short term.

However, product diversification will also require capital investment, production of specialized equipment requires compliance with established medical standards as well as establishing new supply chains. Given the current scenario, making such adjustments in scale would require support. While Bangladesh government has announced support for the sector, it may not be possible to revive the sectors only through Government help. Exploring financing from other sources can help provide the requisite support to the sector to tide over the short-term demand glut.

Concluding remarks:

IDCOL is seeking concessional finance from Green Climate Fund (GCF) to promote energy efficiency in the textile and RMG sector of Bangladesh. Traditionally, funding from GCF has targeted such projects that can demonstrate positive climate impact, like climate mitigation from energy efficiency. The pandemic offers a unique opportunity to GCF to extend this mandate and bundle an additional supporting financing instrument, like grant, with their traditional debt to help beneficiaries to tide over the immediate aftermath of COVID. The RMG and textile sector in Bangladesh can benefit from such a grant funding – apart from investments in energy efficiency technology, GCF can help promote business recovery in RMG factories and help them emerge stronger and more resilient from the COVID crisis.

Though the impact of covid-19 is being felt in the short run, the growth is expected to recover over the medium term. Textile and RMG industry, the main backbone of the country, is expected to overcome this crisis as a result of various measures taken by the government and support from international developmental organizations.

²⁷ Source: [Article \(World Bank, 2020\)](#)

²⁸ Source: [Report \(ADB, 2020\)](#)

²⁹ Source: [Report \(ADB, 2020\)](#)

The availability of low cost financing through the proposed GCF funded program for energy saving technology upgrades, will be considered positive and beneficial for the sectors and will add to its recovery.

Related projects/interventions: Also describe any recent or ongoing projects/interventions that are related to the proposal from other domestic or international sources of funding, such as the Global Environment Facility, Adaptation Fund, Climate Investment Funds, etc., and how they will be complemented by this project/programme (e.g. scaling up, replication, etc.). Please identify current gaps and barriers regarding recent or ongoing projects and elaborate further how this project/programme complements or addresses these.

Please identify current gaps and barriers regarding recent or ongoing projects and elaborate further how this project/programme complements or addresses these.

Energy Efficiency and Conservation Promotion Financing Schemes in Bangladesh ³⁰

- **Energy Efficiency Finance Program (IEEFP)**

Among the loan facility for EE&C promotion focused on industry sector in Bangladesh, Global Climate Partnership Fund (GCPF) was one of the earliest funds. GCPF which was initiated in 2011 is the loan facility aiming to extend loans to projects owners for buildings and plants improving energy efficiency by more than 20% and renewable energy project such as small-scale solar power generation, small-scale hydropower, wind power generation and biomass power generation. However, provision of GCPF loan has been limited with just one (1) project which was extended by co-financing of 2 two financial institutions. Reasons for poor achievement could be considered as follows: Firstly, higher lending rate for end-user was raised. As is shown in Table-5, lending rate in BDT denomination was set as about 12%, which is higher than term-loan rate for industry, after considering credit risk cost. Secondly, difficulty in raising fund timely is hampering factor. Financial institutions require longer time to conduct technical appraisal which takes more than 3 months. Thirdly, condition of the loan which needs to meet more than 20% conservation of energy is a big hurdle.

- **Energy Efficiency Finance Program (IEEFP)**

Asian Development Bank (ADB) initiated Industry Energy Efficiency Finance Program (IEEFP) in 2011. IEEFP covered only energy efficiency and conservation equipment while GCPF extended loan for not only energy efficiency and conservation equipment but renewable energy equipment. The team found out that IEEFP fund have comprised of technical assistance and loan facility of USD 30 million. However, it is indicated that no loan has been extended and ADB cancelled most of the fund.

- **Partnership for Clean Textile (PaCT)³¹**

IFC has been implementing "Partnership for Clean Textile (PaCT)". PaCT is playing a leading role in driving the long-term competitiveness and environmental sustainability of the textile wet processing sector, by addressing high water, energy, and chemical use through the adoption of best practices in the textile sector. These best practices lead to declining resource consumption, soaring profits, and an enhanced image in the global apparel market.

Outline of PaCT project is as follows: Over the years, IFC has drawn on its global and local experience and has provided a combination of investments and advisory services in Bangladesh. As of June 2018, IFC's committed portfolio in Bangladesh is around USD 1.52 billion (IFC invested USD 800 million in 2017 and USD 1 billion in 2018 for Bangladesh Textile Sector.).

IFC is promoting sustainable growth and private sector development in Bangladesh through investing in critical infrastructure; boosting financial inclusion; enhancing textiles competitiveness and supporting reforms to make doing business easier for private sector.

³⁰ JICA Report & KII report

³¹ Bangladesh PaCT website.

<https://www.thedailystar.net/country/gdp-growth-rate-of-bangladesh-2019-will-achieve-8-per-cent-ADB-1724293>

However, some of the key points to be noted for the PaCT programme, are as follows:

- PaCT is Resource Efficiency Financing and not a dedicated energy efficiency financing.
- Through PaCT, IFC is playing the role of investment facilitation. It doesn't have a dedicated credit line like other funds which are there in Bangladesh market, to promote energy efficiency in Industries. Banks and financial intermediaries (FIs) are involved in the process, they have partner bank who review the investment proposals and evaluate the costs and benefits of implementing the proposed resource efficiency projects. After the review, the bank lend customized finance to factories to invest in the resource efficiency projects.
- PaCT advocacy helped create a USD 200 million Green Transformation Fund that the Bangladeshi government uses to support low-cost financing for resource efficiency in the textile industry which is not successful primarily because of the cumbersome disbursement process.

- **Bangladesh Bank (Central Bank) Funds:**

Green Transformation Fund (GTF)

The Green Transformation Fund was established by Bangladesh Bank in 2016. The size of fund is USD 200 million and is intended to facilitate access to financing to export-oriented textile & garment products and leather goods manufacturing industries when they import capital machinery relevant to environment-friendly /green attributes such as energy efficiency, renewable and so on.

The initial lending rate to IFI was set at 3 Month Libor + 2.25% which was lowered to 3 Month Libor + 1% from August of 2017. The team estimates lending rate to end-users from IFI will reach 6-8%. So far only an estimated 10%-15% of the fund has been disbursed.

Following issues are pointed out from financial institution on GTF:

- (i) Since the loan facility is USD denominated and Libor-based floating rate loan, borrowers will be affected by change of exchange rate and interest rate;
- (ii) Lot of documents are requested from BB;
- (iii) About half year to 1 year is spent for loan decision, and;
- (iv) Completion of plant and equipment is the condition of loan disbursement.

Bangladesh Bank Re-financing Scheme³²

The central bank launched the BDT 2 billion green banking refinance scheme in 2009 that provides loans for renewable energy and energy efficiency projects such as solar, bio-gas plants, industrial ETP (effluent treatment plant), Pet bottle recycling plants, vermicomposting, hydro power, Solar Battery recycling plants, LED bulb manufacturing plants, setting up of hybrid Hoffman kiln in brick manufacturing units under the scheme – to help reduce industrial pollution and increase power supply. The fund has been named the “solar energy, biogas and effluent treatment plant sector refinance scheme”. More sectors have been brought under the green banking refinance scheme to build environment-friendly economy. Under the scheme, the central bank provides funds to financial institutions at 5% interest, which then provide loans for renewable schemes at 8 - 9 % interest. The cumulative amount refinanced under the scheme up to June, 2018 stood at BDT 3797.25 million. In FY18, total disbursement under the BB's refinance increased by 91 percent to BDT 665.47 million which was BDT 348.80 million in FY17.

³² Bangladesh Bank website

Long Term Financing Facility (LTFF) under Financial Sector Support Project (FSSP) 33

BB intends to provide long term financing for private sector firms, mainly midsize manufacturing firms under the Financial Sector Support Project (FSSP) financed by the International Development Association (IDA). The main objective of this facility is to provide long term financing for private sector firms. The financing facility would be provided with a view to facilitate the development of economies and to promote socially and environmentally sustainable development outcomes. Estimated total fund of LTFF is USD 292.50 million (BDT 229,612.50), of which BB's contribution is USD 38.50 million (BDT 30,222.50) and IDA's contribution is USD 254.00 million (BDT 199,390.00) in FY18. This Financing would be offered for 3 to 10 years term in US dollar and provided to the PFIs authorized by BB to deal in foreign exchange on lending/refinancing to the private sector firms.

- **JICA:**

The feasibility report revealed the demand for JICA EE&CPF Project loan remains high due to its attractiveness in lower lending rate and fixed interest scheme. This high demand is reflected in the fact that the private banks approach to IDCOL, asking to participate in large plant and equipment as a loan syndication partner by financing eligible energy efficient equipment with JICA loan. The only challenge with JICA EE&CPF is that it focusses on varied sectors and hence, cannot cater the complete demand of any one of the sectors. The sector approach will help to cater the huge demand and potential of energy efficiency in the sector and will percolate deep inside the sector. The second issue with JICA EE&CPF is that the project fund size is very small as compared to the demand and potential of the entire country. The second issue with JICA EE&CPF involves the limitations in ticket size of one project (maximum BDT 100 Crore/USD 11.62 million³⁴) and only one company from one group may apply for the loan under this project. This further minimizes the reach of this project across industrial sector. The last point is that, this project has a limited list of EE interventions that may be financed under this project. These limitations do not help the project to percolate deep inside any sector with an approach to bring the entire sector to world standards norms and energy intensity standards.

- **Joint Crediting Mechanism (JCM)³⁵:**

The mechanism was introduced, through signing of bilateral document between Bangladesh and Japan in March 2013. The JCM facilitates diffusion of leading low carbon technologies, products, systems, services, and infrastructure as well as implementation of mitigation actions, and contributing to sustainable development of Bangladesh. It appropriately evaluates contributions to GHG emission reductions or removals from Japan in a quantitative manner, by applying measurement, reporting and verification (MRV) methodologies, and uses them to achieve Japan's emission reduction target.

Under this scheme, Japanese and Bangladeshi participants work in close co-operation to facilitate financial, technological and capacity building support for implementation of programs which will lead to reduction of GHG emission. Under this mechanism, 3 projects have been approved out of which 2 projects are related to energy efficiency i.e. installation of high efficiency loom at weaving factory and high efficiency centrifugal chiller for AC system in clothing tag factory.

³³ Bangladesh Bank website

³⁴ Considering 1USD=86 BDT

³⁵ Source: [Basic concept of Joint Crediting Mechanism \(JCM, 2019\)](#)

- **Support of Safety Retrofits and Environmental Upgrades in RMG Sector of Bangladesh (SSREU/SREUP)³⁶:**

A loan fund has been constituted in March 2019 of EUR 50 million to pre-finance technical up-gradation of eligible RMG factories to improve worker safety and reduce environmental footprint. The loan fund amount is provided by AFD in the form of a 20-year subsidised sovereign loan, with a grace period of seven years. The fund is supplemented by a technical assistance component with total cost of EUR 14.29 million sponsored by EU, KfW, GIZ and Bangladesh Bank.

Under this scheme, a maximum loan amount of EUR 1 million (EUR 3 million in case of exceptional cases) with repayment period of 3-5 years (7 years under exceptional case) will be provided to the RMG owners by participating financing institutions (PFI). The maximum interest rate to the end users has been restricted to 7%, while Bangladesh Bank will on-lend to PFIs at 3.5%.

The project has a performance-based grant mechanism under which a portion of principal amount of loan is waived off (10% and 20% in case of safety remediation and environmental/social investments respectively) in case of verified achievement of targets. The breakdown of investment grants among Eligible RMG Companies and PFIs would be 90:10 i.e. 90% and 10% of the grant will be provided to RMG companies and PFIs respectively.

In case of energy efficiency projects, only those initiatives which reduces energy consumption by at least 20% over baseline scenario and exhibit an internal rate of return (IRR) on investment of 10% are eligible for funding under this scheme.

Table 05: Summary of the key EE&C financing scheme in Bangladesh

Name of Loan Facility	IFIs	Amount	Year of Establishment	Currency	Lending rate to end-user	Tenor	Eligible Areas/Sectors/Products
GCPF	City Bank & Southeast Bank	USD 30 m	2011	BDT	About 12%	5-7 years	Energy efficient equipment & Clean energy
IEEFC (ADB)	Industrial and Infrastructure Development Finance Company (IIDFC)	USD 20 m	2011	-	N.A.	N.A.	Energy efficient equipment
EECPFP (JICA)	IDCOL and BIFFL	USD 100m (JPY 11.9 billion)	2016	BDT	4-6 % (Currency fluctuation risk borne by GOB)	Short to long-term duration	Energy Efficient equipment's.

³⁶ Source: [Circular on loan fund of SSREU \(Bangladesh Bank, 2019\)](#)

ADB- Financing Brick Kiln Efficiency Improvement Project	Commercial Banks	USD 50m	2012	BDT	-	-	Applicable for Brick Manufacturers
JCM	Govt. of Japan	Project Specific	2013	-	N/A	N/A	Implementation of low carbon technologies including energy efficient technologies
SSREU/SR EUP	Participating FIs	EURO 01 million	2019	-	upto 7%	3-7 years	Technical up- gradation of eligible RMG factories
Bangladesh Bank Fund							
GTF	Commercial Banks	USD 200m	2016	USD	6-8 %	5 to 10 years with 1- year grace period	Available only for capital machinery import for three selected sectors; export-oriented textiles & textiles products, leather manufacturing and jute sectors for implementing green/ environment-friendly initiatives
Refinan cing Scheme for Green Product s/Initiati ves	Commercial Banks	USD 23.97m (BDT 2 billion)	2009	BDT	8-9 %	4 to 10 years with 3 to 12 months grace period based on product s/initiati ves	Available for specific 51 green products in 8 sectors. The sectors; RE, EE Technology, alternative energy/fuel, waste management, Recycling or Recyclable Goods Manufacture, Environment- friendly brick manufacturing, environment-
LTTF	Commercial Banks	USD 300 m	2015	BDT	6-9 %	3 to 10 years	Available for: a) Purchase of capital machineries and equipment for up- gradation (including improving Health and Safety compliance) b) Relocation of factories to

							designated industrial zones
IFC PaCT	Partner Banks	PACT Phase 1: USD 11 million in development assistance. IFC helped catalyze USD 39 million in additional investments in the programme's member factories. PACT Phase 2: USD 7 million in development assistance.	Phase 1: Launched in 2013 Phase 2: Launched in 2017	BDT/ USD	6-9%	3 to 10 years	Resource Efficiency Financing. The partner banks review the investment proposals and evaluate the costs and benefits of implementing the proposed resource efficiency projects. After the review, the bank lend customized finance to factories to invest in the resource efficiency projects and also provides linkage with relevant technology service providers.
	<i>Note: IFC PACT advocacy helped create a USD 200 million Green Transformation Fund that the Bangladeshi government uses to support low-cost financing for resource efficiency in the textile industry.</i> <i>IFC's Global Trade Supplier Finance (GTSF) programme also offers suppliers financial incentives to improve factories' environmental and social standards. This USD 500 million investment and advisory programme helps suppliers manage working capital, convert sales receivables to immediate cash, and access lower-cost financing.</i>						

As mentioned above, although there are several loan facilities for energy efficiency and conservation promotion, higher lending interest rate and the time required for appraisal have hampered smooth implementation of loan projects.

Demand for JICA EE&CPF Project loan remains high due to its attractiveness in lower lending rate and fixed interest scheme (4-6 %) and simplified loan application procedure. However, JICA is not sector focused and only covers a small part of spinning in textiles and during JICA outreach it was observed that there are many instances of textile units willing to adopt energy efficiency if similar funding is available, hence there is need for a dedicated line for textile sector.

GCF Fund is targeted for overall Textile and RMG Sectors in Bangladesh which will include spinning, weaving, knitting, dyeing, spreading, laying & cutting, sewing, wet processing, cleaning & laundry, fusing & pressing and folding & packaging, which have larger energy saving potential. In this context, fund from GCF, both minimizing the time for loan procedure through streamlining of loan application procedure and speedy appraisal as well as development of loan proceeds having concessional low interest rate and loan scheme will have catalytic effect and promote energy efficiency and conservation in Bangladesh textile and RMG sectors further.

GCF fund will have catalytic impact on energy efficiency market for Textile & RMG sectors as no dedicated fund is currently catering to energy efficiency of Textile & RMG sectors as a whole. Further, most of the

current loans are catering to larger client base whereas in this credit line a mix of smaller and larger textile clients is proposed which would open a new market for financing availability for this clientele.

Thus, GCF fund will not have crowding impact but will have market making effect on energy efficiency for textile & RMG sectors, where other fund can come later on and contribute towards energy efficiency in Textile & RMG sectors of Bangladesh which will lead to multiple benefits such as:

Production floor		
Enhanced productivity	Superior product quality and value	Competitive edge
Lower production costs (energy, labor raw material etc.)	Enhanced asset value	Increased disposable income
Operation and maintenance		
Extended life-time for plant & machinery	Reduced process / plant downtime	Deferred capital costs towards replacing plant and machinery
Working environment		
Improved thermal comfort and lighting	Increased worker safety and well-being	Retain and attract skilled staff
Reduced health insurance costs and medical expenses	Improved labor productivity	Reduced noise levels
Environmental		
Reduced compliance costs, fines etc.	Reduced hazardous waste, dust etc.	Reduced CO, CO ₂ , NO _x , SO _x emissions
Other economic		
Decreased liability/risks	Re-investment of savings/disposable income	New income streams; increased production capacity

Additionally for GCF Fund

1. The engagement is relevant because it addresses the key challenge for Bangladesh - climate change mitigation.
The scope of the programme is in alignment with the SDG targets– SDG 7, SDG 9 and SDG 13. Sustainable energy use remains of critical importance to the Bangladesh economy. Availability and the rising price of gas is a significant and growing constraint on the growth of textile and RMG industries.
2. Presently Bangladesh market is not fully explored or catered from a financing perspective. IDCOL with the help of the Fund and LFI's will solely focus on mainstreaming EE in the various size and type

of textile & RMG factories (by relaxing the major market constraints), to establish the concept as a business as usual practice so that it can be replicated in other industrial sector.

3. The loan product will be directed to increase energy efficiency across complete value chain in Textile and RMG industries of Bangladesh, which will be a unique product catering to only energy efficiency in complete value chain of both the industries (Different from Partnership for cleaner textile (PaCT) led by IFC whose focus area is investment facilitation for financing resource efficiency projects in textile factories)
4. The loan facility will have following features which will make it highly attractive in the market.
 - Simplified loan application procedure and speedy appraisal minimizing the time for loan processing.
 - Availability of Grace period & Long Tenure and low interest.
5. In addition, the broad experience of the Executing Entities (IDCOL and LFI) in financing industrial energy efficiency will have following advantages:
 - Executing Entities have built their capacity by successfully implementing EE&C funds
 - Strong pipeline built by IDCOL and the LFI for Textile and RMG sector while implementing EE&C fund which will ensure fast disbursement of GCF fund.

Major Gaps and Barriers identified in the previous & existing EE&C credit lines and how it will be addressed in the proposed programme

As mentioned above, although there are several loan facilities for energy efficiency and conservation promotion, higher lending interest rate and the time required for appraisal have hampered smooth implementation of loan projects. In this context, both minimizing the time for loan procedure through simplification of loan application and speedy appraisal and development of loan having concessional low interest rate and loan scheme will promote energy efficiency and conservation in Bangladesh further.

The analysis of the alternative financing indicates a number of issues required to be addressed for being successful:

1. Concessional interest rate in comparison to commercial rate

Due to higher initial investment requirement for energy efficient equipment, users may be reluctant to purchase them compared to conventional equipment. As such, incentivizing the purchase of energy efficient equipment through concessional lending would be required.

2. Flexible tenure & grace period in line with project viability

Some available with significant grace period (1-2 years) and longer tenures which ensures that repayment can be made without cash-flow problems. Easy access of fund through experienced financial intermediary of the energy efficient projects have longer payback period which can only be viable if financing is

3. Due diligence compatible with prevailing investment environment

While it is important that credit-worthy borrowers are selected under any programme, it is also important to ensure that the approval process is streamlined and doesn't lead to significant delays. Some of the alternative financing programme involve multiple agencies in the due diligence and approval process which causes delays.

4. Proper management of currency fluctuation if repaid in foreign currency

Foreign currency loans need to be carefully managed to minimize forex risk. This is especially true for USD loans as BDT has been depreciating at an average rate of 2.94% per year in the past and this trend is likely to continue in the future as well. As such, currency fluctuation exposure needs to be managed through adequate hedging mechanisms, if available or by having a currency risk fluctuation premium which will get added to the cost of the lending.

- **Complex technical requirements for processing loans:** In order to access loans, detailed technical documents need to be prepared which provide the financial justification for undertaking the project and helps banks understand the risks in recovery of loans. Development of these documents involve hiring of energy auditors by the RMG units, which further adds to the cost component and also increase the technical complexity of applying for loans for the unit owners.

In some cases, eligibility for loans include presence of company policy documents like Environment and Social (E&S) policy - a practice which is largely absent in the sector. This reduces the financing options available to an RMG unit to fund its energy efficiency measure.

- **Limited number of banks providing energy efficiency financing:** Under existing market scenario, there is limited number of banks in Bangladesh providing energy efficiency financing. Also, large scale units have greater access to market information in this regard as compared to medium and small scale units. This further limits the option available to unit owners to finance their energy efficiency measures.
- **Limited capacity in bank in appraising energy efficiency projects:** Like RMG unit owners, LFIs have limited technical knowledge in assessing an energy efficiency project and are completely dependent on energy auditor for assessment. Also, there is limited market information available for the banking staff to effectively understand energy efficiency as a concept and specifically energy efficiency in RMG sector. This limited understanding is also contributing to lack of active interest in financing energy efficiency.

Hence, it is observed that limited penetration of energy efficiency measures can be effectively overcome by providing access to concessional financing sources at the same time providing access to technical and market related information to both demand side i.e. RMG units and supply side i.e. banks of the financing requirement for the sector. The TA component sought under the programme will support in reviewing all existing and planned public and private EE financing mechanisms in Bangladesh with a brief review of energy consumption in textile industries and their access to financing and identify financial gaps for realization of the planned EE programmes /activities. This will help in understanding the present requirements/needs of the credit line; including but not limited to institutional setting (management of the credit line), human resources need, indicative timeline for establishing the credit line, structure for raising capital or allocation of state budget to establish and maintain the credit line, structure for the utilization of fund and even diversification of financing facilities. Once the financing strategy is finalized, a financing road-map along with time-based action plan will be developed, which will be subject to annual review. The effectiveness of the new financing mechanism in terms of filling the financing gap for reaching the EE targets and physical impact (energy and GHG savings) will be assessed annually and based on the assessment (includes stakeholder consultation) the financing strategy will be revised.

Initiatives by other development partner:

KfW and AFD: In Bangladesh, National Determined Contribution (INDC) by 2030 under Paris Agreement are defined as the following two types. One is unconditional contribution as 5 % below BAU. The other is conditional contribution assuming international support as 15% below BAU. With reference to this

background, KfW and AFD are each planning to formulate investment pipeline to reduce not only energy consumption but also GHG emission.

GIZ :Not in financial support but in technical cooperation fields, GIZ has been continuously supporting SREDA in capacity development in promoting EE&C. Major focused issues are supporting establishment of legal frameworks to promote EE&C and strengthen energy audit skill etc.

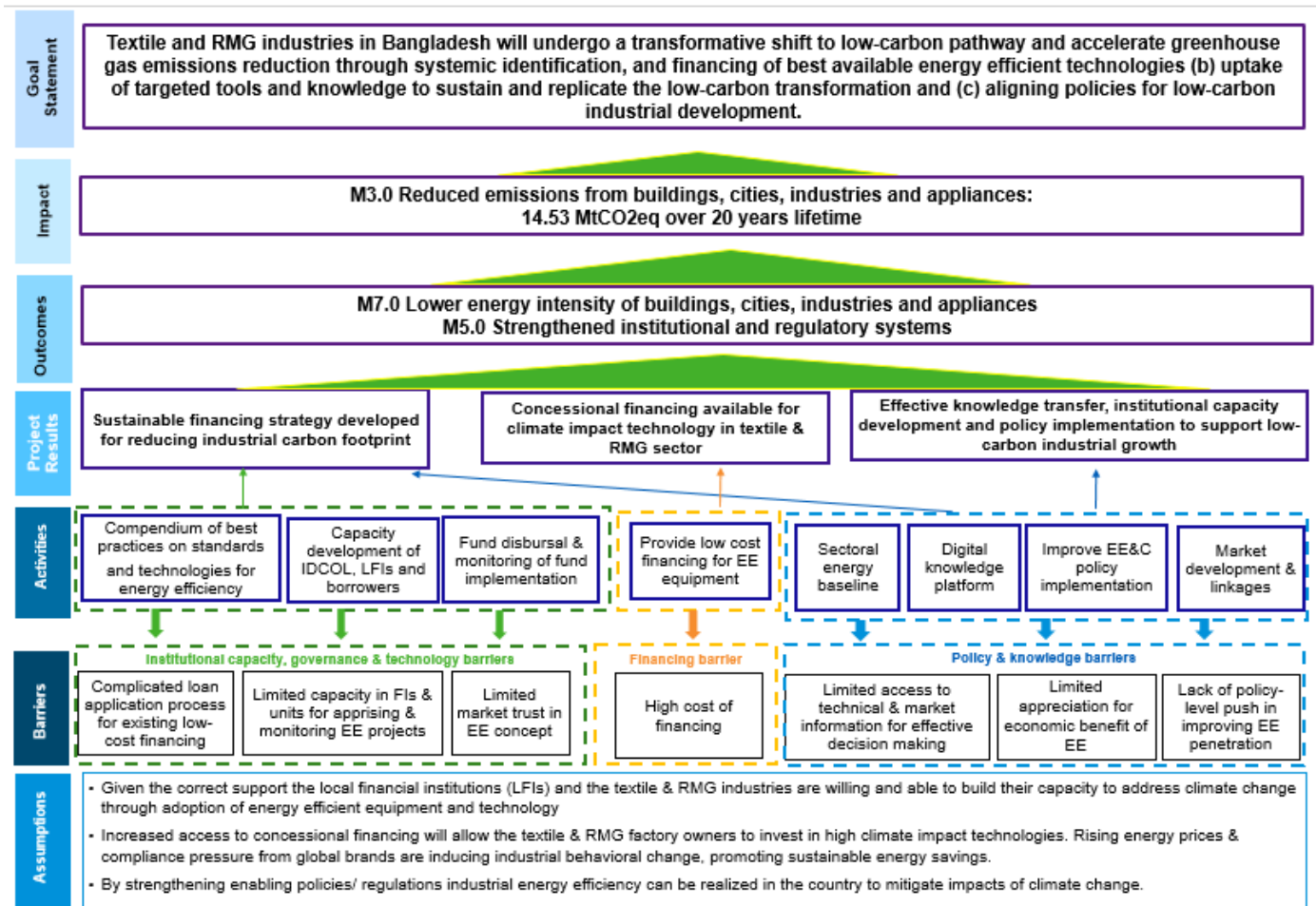
B.2. Theory of change (max. 1000 words, approximately 2 pages plus diagram)

Describe the theory of change and provide information on how it serves to shift the development pathway towards a low-emission and/or climate resilient direction. Provide the diagram of the theory of change (approximately 1 page).

The theory of change should include any barriers (social, gender, fiscal, regulatory, technological, financial, ecological, institutional, etc., as relevant) that need to be addressed. Use a results chain of inputs, activities, outputs, outcomes, and impact statements, and identify the how and why of causal relations to deliver the project's expected results.

Theory of Change of the programme (EE&C in the Textile and RMG Sector):

Figure12:



Bangladesh has been consistently witnessing high economic growth (over 6% GDP growth for the last decade³⁷). This has led to nearly 49% increase in its final energy consumption between 2008 and 2018³⁸. With the country expected to continue in its high growth trajectory, the corresponding energy consumption is also expected to increase significantly. In order to ensure energy access Government of Bangladesh provides generous subsidies in energy supply. The volume has been rising steadily over the past years – national subsidy bill for power generation and energy increased from USD 510 million in 2016-17 to USD 1.62 billion in 2019-20³⁹. Allocation for gas subsidy in annual budget of Bangladesh witnessed a 2x increase in 2019-20. The increasing energy subsidy bill can have a crippling effect of Bangladesh' economy. Efficient use of energy can provide a solution to help reduce the subsidy bill and unlock the investment for overall economic development of Bangladesh and effectively reducing GHG emissions.

Within Bangladesh, Garments and Textiles sector are the highest energy consuming sector within industries – which has the 2nd highest energy consumption after residential sectors^{40,41}. Multiple studies have shown energy savings potential of nearly 25% is present for the sectors, as compared to international best practices⁴². However, there are multiple barriers to ensure benefits are accrued from implementing energy savings measure are realized.

As gathered from stakeholder interviews, the unit owners are generally skeptical of the energy savings that can be achieved from implementing energy efficient equipment which priced significantly higher over normal inefficient variant. The lack of adequate number of demonstration projects for the sector is considered as the reason for the skepticism. Also, under the current financing landscape unit owners have to pay an interest in the range for 11%-14% - under such high interest rates, financing equipment for which lower priced variants are available is unviable for unit owners. The appreciation of cost savings arising of lower energy consumption is absent in the market. The investment decisions are further influenced by the lack of technical knowledge among the unit owners on how to assess the energy savings and the financial benefits are derived. There is also limited information available to the unit owners on the OEMs supplying energy efficient equipment and other information like pricing, which can help assist decision making.

Apart from unit owners, the financing institutions also have limited capacity to assess and apprise an energy efficiency project, leading to lower to no preference of banks to fund energy efficiency projects as compared to other green projects. While, there is currently financing line for energy efficiency through Green Transformation Fund (GTF) of Bangladesh Bank, the complicated loan application process leads to disbursement of loans, delayed by months at a time.

There is also a clear lack of policy push through measures like standards and labelling, minimum energy performance standards etc. that will create a demand side push for energy efficient equipment.

Given this landscape, the introduction of low-cost financing coupled with technical assistance activities can help bring a transformative change aligned with GCF key mitigation area of reducing emissions from industries. Thus in the above shown diagram of Theory of Change we can assume that -

- given the correct support the local financial institutions (LFIs) and the textile & RMG industries are willing and able to build their capacity to address climate change through adoption of energy efficient equipment and technology

³⁷ Source: [Bangladesh Statistical Yearbook 2018 \(Bangladesh Bureau of Statistics, 2019\)](#)

³⁸ Source: [Bangladesh Energy Sankey Diagram \(IEA, 2018\)](#)

³⁹ Source: [Bangladesh national budget \(Daily star, 2020\)](#)

⁴⁰ Source: [Energy Efficiency and Conservation Master Plan up to 2030 \(JICA website, 2015\)](#)

⁴¹ Source: [Information derived from EE & Conservation Master Plan upto 2030 \(DANIDA, 2017\)](#)

⁴² Source: [Bangladesh: Industrial Energy Efficiency Program \(ADB website, 2014\)](#)

- increased access to concessional financing will allow the textile & RMG factory owners to invest in high climate impact technologies. Rising energy prices & compliance pressure from global brands are inducing industrial behavioral change, promoting sustainable energy savings.
- by strengthening enabling policies/ regulations industrial energy efficiency can be realized in the country to mitigate impacts of climate change.

The proposed programme can accelerate investment in a portfolio of high impact climate technologies and supportive changes in business practices especially in the Textile & RMG sectors.

The major barriers that are identified can be classified into three segments (a) institutional capacity, governance & technology barriers: this includes complicated and lengthy loan approval process related to concessional green financing, lack of capacity of the financial institutions to appraise and monitor concessional loans related to industrial energy efficiency and finally the lack awareness in the market about the benefits of energy efficient equipment. (b) Financing barrier: this is lack of availability of concessional financing in the market that can widely cover the premium/additional costs of energy efficient equipment (c) Policy & knowledge barriers: this includes the limited access to technical & market information for effective decision making regarding investments in energy efficiency, limited appreciation for economic benefit of energy efficiency and lastly the lack of policy-level push in improving EE penetration.

In order to overcome these barriers the programme will ensure access to low-cost financing for the program along with effective monitoring of fund implementation will help drive uptake of energy efficient equipment and create demonstrative effect to develop market trust in energy efficiency. In order to facilitate effective appraisal of energy efficiency project application and monitoring capacity building of banks disbursing low cost financing is required, as well as capacity development in units to undertake monitoring and handholding in the loan application process. Handholding activity will include assessment of the key issues in existing loan application process of IDCOL and LFI and accordingly provide recommendations for the modification of disbursement process of the low-cost loan instrument under this programme.

In order to improve access to market and technical information related to energy efficiency for unit owners, a digital platform is envisaged which will contain case studies, guides, manuals of successful energy efficient equipment implementation. The digital platform will help also in registering various stakeholder types (mainly unit owners and energy efficient OEM) and stimulate conversations between the stakeholder' group. The platform will also provide a sectorial baseline energy consumption database which coupled with the information on energy consumption and savings of individual projects within the programme, will help understand sector level impact of energy consumption in Bangladesh.

Apart from virtual conversations, real-life interaction between unit owners and OEMs will be stimulated through technology fairs wherein OEMs will exhibit their energy efficient technologies and will clarify any queries of unit owners. Specific workshops aimed at increasing awareness of units (particularly unit owners) on the benefit of energy efficiency and the various policy levers utilized globally for appraisal of policy makers of Bangladesh is planned to improve awareness of benefits of energy efficiency.

In order to stimulate policy level push, manuals and other supporting documents will be prepared to access to relevant information by policy makers in developing appropriately designed policies/schemes aimed at increasing penetration of energy efficient equipment in Bangladesh. Overall, these activities are expected to bring in the paradigm shift in Bangladesh energy efficiency market and create a replicable model that can be emulated in other sectors of the country to ensure increased penetration of energy efficiency.

Paradigm shift potential: The concessional loan has the potential to trigger behavioural change at industrial sector energy management level to incorporate climate change targets and corporate climate governance principles into strategic decision making. The transformative shift within textile & RMG sectors can be made possible through: the uptake of climate impact technologies; the additional shift triggered by behavioural change at institutional governance and management levels due to capacity development and sensitization on energy efficiency; and all supported by an enabling policy framework, in particular gender-responsive sectorial low-carbon roadmaps. The resulting paradigm shift will be (1) a shift and acceleration of industrial decarbonisation trajectories; (2) transformation of the supply and demand for energy efficiency finance; and (3) acceleration of the market for climate technologies.

B.3. Project/programme description (max. 2000 words, approximately 4 pages)

Define the project/programme. Describe the proposed set of components, outputs and activities that lead to the expected Fund-level impact and outcome results. Components should reflect the project/programme level outcomes.

This should be consistent with the financing by component in section C.2, the results and performance indicators provided in section E.5, and the implementation timetable in annex 5.

Referring to the feasibility study, describe why this set of interventions was selected instead of alternative solutions and how the project/programme can help unlock the needed support in a sustainable manner. Also identify trade-offs of the selected interventions, if applicable.

For Enhanced Direct Access (EDA) proposals and projects/programmes with financial intermediation (loans or on-granting), describe the selection criteria of the sub-project and types.

The proposed programme aims to enable market transformation of Textile and RMG sectors in Bangladesh to increase penetration of energy efficiency measures, leading to a shift to low-emission sustainable development pathways. The programme is broadly divided into 05 components, where Component 1 & 2 addresses the barriers and needs of the textile sector for swift adoption of energy efficiency technologies, whereas Component 3 & 4 does the same for the RMG sector and component 05 addresses regulatory issues. The implementation plan designed for both the sectors are different to improve efficiency in the financial access and also to influence market participation to enable replication in other energy intensive sectors of the country.

DESCRIPTION OF COMPONENT 01 & 02:

Even though the textile sector of Bangladesh has significant energy saving potential, but due to higher cost of energy efficient equipment compared to conventional equipment, end users are often deterred to install them. While there are a few products that have been launched by various financial institutions, however, except a few they have not been able to significantly meet their objectives due to either procedural issues or high cost of funding. The feasibility study indicates the need for ~ 5.7%⁴³ (11%-5.28%) concessionality in the interest rate for energy efficient equipment, compared to commonly available commercial loans (i.e around 11%⁴⁴) in order for EE equipment to be widely adopted in Bangladesh. Using GCF and IDCOL funding, this can be achieved. Similarly minimizing the time for loan procedure through simplification of loan application and speedy appraisal will promote energy efficiency and conservation in Bangladesh further. The key measures proposed to simplify lending are as follows:

⁴³ Please refer to table 06-09 of this document

⁴⁴ <https://www.bb.org.bd/fnansys/interestlending.php>

- Since the financing will be list based, it will be easier to apprise the project. The eligibility would be supported by certificate/catalogue/ brochures/ report from energy auditor or declaration by the supplier. The transactional cost of the parties involved would decrease significantly.
- IDCOL has significantly built its capacity in apprising the loan application of energy efficiency projects. The loan appraisal process for GCF credit line will take significantly less time compared to any other fund. The project will have to meet pre-set criteria approved by GCF and will have to only go through internal approval process of IDCOL. In the case of credit lines where third parties are involved (like central regulatory body SREDA, in case of JICA Fund) or where intermediary financing institutes are involved (e.g. Partner bank in GTF and IFC PaCT project), a significant time (3 month to 1 years) gets wasted in loan approval process which will be saved in this credit line.

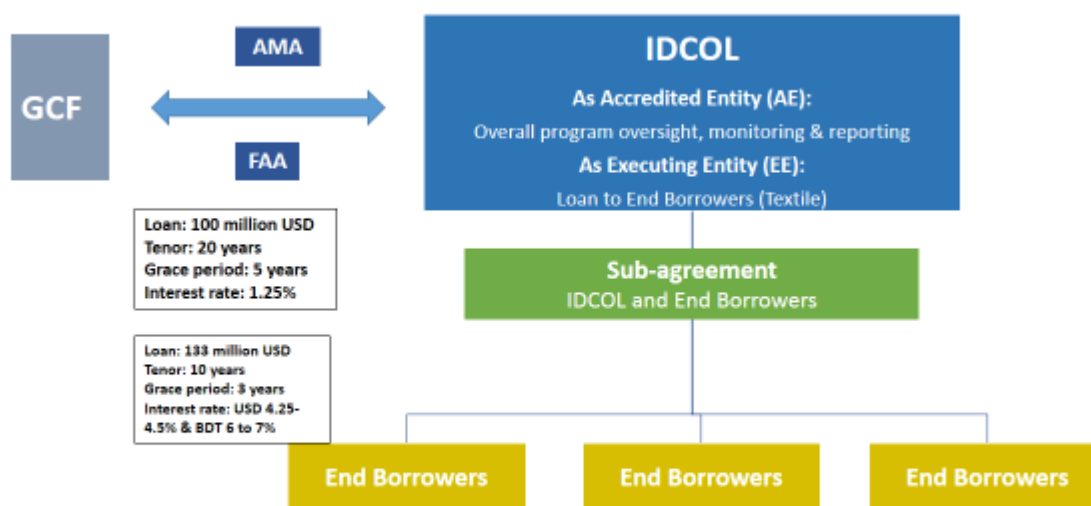


Figure 13: Fund flow from GCF to the end borrowers (reflecting the interest rate and tenor)

This would ensure a catalytic impact on energy efficiency measures in the textile sector.

Component 01 & 02 envisages the creation of concessional fund with USD 133 million, including USD 100.00 million senior loan from GCF and USD 33 million from IDCOL. End borrower's equity investment for USD 33.00 million will be the leveraged financing. IDCOL, an Accredited Entity with GCF, will make available the GCF funds in the form of loans to the end borrowers, who are in this case the private textile manufacturing companies. Using this fund, the projected IRR estimated to be achieved is 7.0 %. The implementation of these energy efficiency measures equivalent envisages annual GHG reductions estimated at 0.20 million tCO₂ and 4.00 million tCO₂ equivalent over the lifetime of the project (20 years) which if included in the economic benefit would lead to an EIRR of 23%. The payback period for various equipment varies between 4-13 years. In addition to these economic and financial benefits, the project will also lead to other intangible social and gender impacts such as better community living standards, providing safe and better working environment leading to higher gender balance in workforce and skill up gradation for workers.

As such, the envisaged program is closely aligned to the GCF objective of catalyzing financing for this initiative and helping scale up the initiative across the country in future through increased showcasing of economic and societal benefits leading to a market transformation in energy efficiency. The investments enabled through this program will help improve the productivity & competitiveness of the local textile industry in Bangladesh thereby leading to sustainable development pathways. The investments would also create a

climate resilient built environment through deep penetration of energy saving technologies, equipment and appliances in the textile industries.

The proposed programme will address the energy efficiency adoption needs of textile sector with following components-

Component 1: USD 133.00 million financing for Energy Saving Equipment & Technology for Textile sector

Component 2: USD 3.05 million GCF Technical Assistance (TA) to develop enabling environment for EE investment in textile sector

Component 1: USD 133.00 million financing for Energy Saving Equipment & Technology for Textile sector

To encourage the industry and businesses to choose energy-efficient equipment which, in many of the cases, is more expensive compared to conventional ones, there is a need to reduce the cost difference by providing concessional lending for energy efficient equipment.

Although energy-efficient equipment may become economical on lifecycle cost basis, equipment buyers are most likely to decide solely based on investment cost, resulting in choosing cheaper, conventional non-energy-efficient equipment. Against this background, the Programme aims to provide a low interest loan to the buyers so that total payment amount for energy efficient equipment (equipment cost and financial cost) will break even with the cost of acquiring conventional equipment.

During the feasibility study, based on the payment simulations for a few types of equipment it was identified that there is a need of for 4.3% to 5.7% concessional interest rate in order to breakeven the cost of EE equipment with that of cheaper traditional equipment. For example, market price of energy efficient and traditional/conventional spinning and boiler machine were compared. The price of energy efficient spinning and boiler machine were found to be 25% and 26% respectively higher than the traditional/conventional equipment. Subsequently, the cost impact of purchase of conventional equipment and energy efficient equipment by changing tenure period and grace period have been analyzed.

The purchase of an energy efficient spinning machine which is ~25% more expensive compared with a conventional type, which will require a 5.53% interest rate loan (considering 8 year loan tenure and 1 year grace period) for the total payment to become par with that of a conventional one. Similarly, a 27% more expensive once-through boiler requires 5.28% interest rate (considering 8 year loan tenure and 1 year grace period) to be even with the conventional type in terms of the total payment amount.

Table 6: Spinning machine with permanent magnet motor (In comparison with induction motor-driven spinning machine)

		Commercial Finance	GCF Loan Finance
		Conventional equipment	Energy Efficient equipment
Price	USD	119,000	149,000
Interest Rate	%	11%	5.53%
Tenure Period	yrs.	8	8
Grace Period	yr.	1	1

Total Payment	USD	189,983	189,983
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Table 7: Small sized once-through boiler (in comparison with fire-tube boiler)

		Commercial Finance	GCF Loan Finance
		Conventional equipment	Energy Efficient equipment
<i>Price</i>	USD	48,000	60,750
<i>Interest Rate</i>	%	11%	5.28%
<i>Tenure Period</i>	yrs.	8	8
<i>Grace Period</i>	yr.	1	1
Total Payment	USD	76,632	76,632

The purchase of an energy efficient spinning machine will require a 6.67% interest rate loan (considering 10-year loan tenure and 2 year grace period) for the total payment to become par with that of a conventional one. Similarly, once-through boiler will require 6.47% interest rate (considering 10 year loan tenure and 2 year grace period) to be even with the conventional type in terms of the total payment amount.

Table 8: Payment Simulation changing loan tenure and grace period. Spinning machine with permanent magnet motor (In comparison with induction motor-driven spinning machine)

		Commercial Finance	GCF Loan Finance
		Conventional equipment	Energy Efficient equipment
<i>Price</i>	USD	119,000	149,000
<i>Interest Rate</i>	%	11%	6.67%
<i>Tenure Period</i>	yrs.	10	10
<i>Grace Period</i>	yr.	2	2
Total Payment	USD	222,860	222,860

Table 9: Small sized once-through boiler (in comparison with fire-tube boiler)

		Commercial Finance	GCF Loan Finance
		Conventional equipment	Energy Efficient equipment
<i>Price</i>	USD	48,000	60,750
<i>Interest Rate</i>	%	11%	6.47%
<i>Tenure Period</i>	yrs.	10	10
<i>Grace Period</i>	yr.	2	2
Total Payment	USD	87,678	89,894

The above analysis clearly demonstrates that to ensure the feasibility of the project, concessional lending is required from GCF to IDCOL to enable IDCOL to extend loan at the low interest to the end borrower. Further analysis on how the proposed financial package is seemed to be appropriate and reasonable is described in section D.6.

However, one thing to be noted is that the credit provided by GCF to IDCOL will be in the form of USD while IDCOL will be lending 60% of the amount to the end users in BDT and the rest 40% in USD. As such, IDCOL will have to manage the currency risk as well in this case which is not the case for other similar credit lines.

In time series analysis done for exchange rate movement of Bangladeshi Taka against USD from 1998 to 2018 it is observed that Bangladeshi Taka is depreciating on an average rate of 2.94%⁴⁵ every year. Hence, IDCOL will be managing the currency risks by charging risk premium to the end borrowers. The feasibility study also indicates that some market exists for USD loans amongst the textile industries, so IDCOL plans to have a portion of the lending in USD as well. Given that such loans will not be exposed to currency fluctuation risk, they can be priced lower than BDT loans.

Eligibility criteria for Textile Borrowers/Sub proponent:

Each of the following criteria must be met by a local project to be considered as an eligible to be financed by GCF credit facility provided through IDCOL. Innovative projects could be considered on a case-by-case basis after having conducted a full-fledged feasibility study to assess whether it complies with the eligibility criteria set under the programme. GCF will be notified in advance in such cases.

Nota bene: if deemed appropriate for the management of the Project, the Lender may agree to modify or further detail the general objectives and technical and financial eligibility criteria in the Operational Manual (to be developed under TA activity). With approval from GCF, the Operational Manual may be updated if needed during the Credit Facility's implementation upon agreement between the Lender and the Borrower.

A. General criteria

- The project developer will be a business entity of limited company only
- For both Green field and Existing project related to up-gradation to energy efficiency technologies/machines. For a Project loan application already being processed at Accredited Entity, a letter of credit (L/C) for purchasing the EE&C equipment, anticipating the use of the Project Loan, may be opened up to 12 months prior to either GCF financing or IDCOL approval of the loan.
- The borrower will be a Textile company
- For procurement of Brand-New machines
- Having Energy Efficient component of proposed equipment to be financed
- Supported by suppliers' declaration/ catalogue/ brochures or report/certificate from independent/certified energy auditor acceptable to IDCOL or the baseline developed under TA activity

⁴⁵ <https://data.worldbank.org/indicator/PA.NUS.FCRF?end=2018&locations=BD&start=2018&view=bar>

- Fulfillment of other required criteria as mentioned in IDCOL's CRM guideline and lending policy
- 100% of the eligible equipment price can be financed

B. Technical Eligibility criteria

Eligibility should be confirmed for each loan application. The Eligible Technology & Equipment List is a predefined list of technologies and equipment as prescribed during the programme formulation. This list will be updated every year, if required, to reflect the accurate market scenario and will be subject to approval by GCF.

Table 10: List of identified EE equipment under feasibility study

Category	Major EE&C Equipment	Condition
Spinning	• Roving Frame • Automatic Winder • Ring Spinning Frame • Air Jet Spinning • Air Compressor • Electric generator • Energy Efficient Boiler • Waste Heat Recovery System • Absorption Chiller • Chiller • LED Light	• Any Equipment/technology eligible for financing , not limited to this list, must have minimum 15% energy saving potential compared to the prevailing inefficient/ less efficient equipment used in the sector • The eligibility would be supported by catalogue/ brochures/declaration form supplier's or report/certificates from independent/certified energy auditor or the baseline developed under TA activity • Common equipment will be interchangeable within the category.
Weaving	• Air Jet Loom • Warper & Sizer • Sewing Machine • Stenters • VFD for Fan, Blower, pump etc. • Air Compressor • Energy Efficient Boiler • Waste Heat Recovery system of Boilers • Waste Heat Recovery System of Stenters • Absorption Chiller • Chiller • LED Light	

Knitting	• Efficient Knitting Loom • Circular Knitting Machine • Flat-bed Knitting Machine • VFD for Fan, Blower, pump etc. • Air Compressor • Energy Efficient Boiler • Waste Heat Recovery System • Absorption Chiller • Chiller • LED Light	
Dyeing	• Energy Efficient Boiler • Jet Flow Dyeing Machine • Cold Pad Batch (CPB) Dyeing for Knit Fabrics • Waste Heat Recovery System • Absorption Chiller	

C. Eligible Investment Cost

The cost of purchasing the energy saving technology and equipment will be the main subject of the loan, which will be utilized by the borrowers to retrofit and/or expand or establish their new factories, processes and building infrastructure.

Financing modality to the Sub-projects / end borrowers can be any of the followings:

A: Financing to procure only energy efficiency Equipment as replacement

B: Financing to procure energy efficiency Equipment for expansion of existing project

C: Financing to setup a new facility comprises of different energy efficiency technologies. In this case, the GCF Loan will only cover the portion of the financing for the energy efficiency technologies rather than the full cost of the facility.

	Investment areas/cost heads	Component 01 -Textile		Component 03-RMG	
		Amount-USD million	%	Amount-USD million	%
GCF contribution	<u>Cost item I :</u> Purchase price with freight, customs duty, related insurance <u>Cost item II :</u> • Transport, assembly, installation • Training and technical transfer; • Customs duty and other administrative costs and	100.00	60%	150.00	60%

	commissions; • Auxiliary / ancillary equipment, connection and accessories, • Other incidental costs.					
IDCOL/LFIs contribution	<u>Cost item I :</u> Purchase price with freight, customs duty, related insurance <u>Cost item II :</u> • Transport, assembly, installation • Training and technical transfer; • Customs duty and other administrative costs and commissions; • Auxiliary / ancillary equipment, connection and accessories, • Other incidental costs. <u>Cost item III :</u> Transport, assembly, installation and related civil construction;	33.00	20%	50.00	20%	
End borrowers contribution	<u>Cost item I :</u> Purchase price with freight, customs duty, related insurance <u>Cost item II :</u> • Transport, assembly, installation • Training and technical transfer; • Customs duty and other administrative costs and commissions; • Auxiliary / ancillary equipment, connection and accessories, • Other incidental costs. <u>Cost item III :</u> Transport, assembly, installation and related civil construction; <u>Cost Item IV:</u> Ancillary infrastructure setup such as civil construction and other engineering and procurement related costs etc.,	33.00	20%	50.00	20%	

	required for the successful installation and subsequent operation of the energy efficient equipment.				
	Total	166.00	100%	250.00	100%

D.1 Financial Criteria

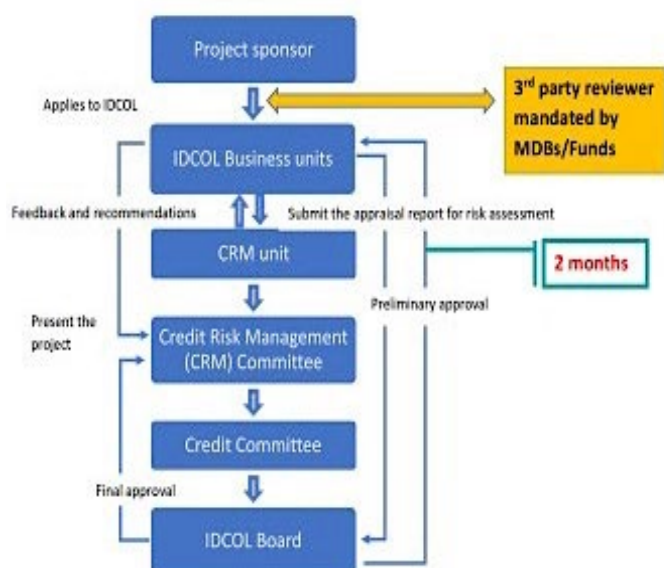
The Eligible Investment should be profitable, to enable the loan repayment with enough cash flow.

- The pay-back period, including financial costs, shall be minimum two (2) years;
- The loan granted by IDCOL to the end borrowers will have a minimum maturity of three (3) years for energy efficiency investments;
- Minimum DSCR of 1.00x
- Any other criteria set by IDCOL approving authority in excess of criteria set out in FP. For deviation of any criteria mentioned in this FP will require GCF approval.

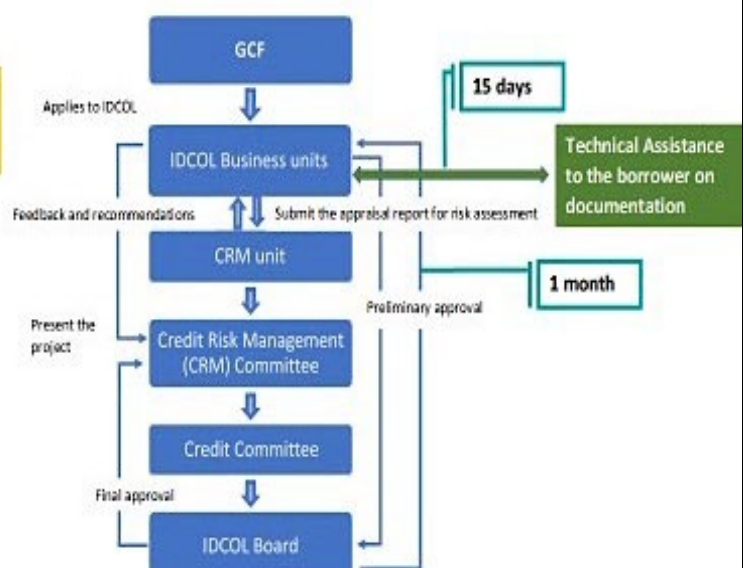
D.2 Lending Conditions- Details in Term Sheet

E. Timing of Loan Application and Appraisal

- Usually IDCOL will consider for loan application from the beneficiary prior to procurement of the equipment. However, for retirement of any machinery L/Cs into GCF project term loan, the L/Cs can be opened maximum 12 months prior to application.



Typical MDB Fund Credit Line



GCF Credit Line

Figure 14: Simplified loan approval process for GCF credit line

A typical loan proposal under other credit lines has a three-month long project approval process where in the third-party reviewers are involved, as mandated by the respective MDBs/funds. In the GCF credit line has the same timeline is envisaged to slash to half of the existing i.e. 1.5 months. This will be possible owing to improvements in the application process leading to streamlined applications which will be implemented as part of the TA component where in end borrowers will be provided with technical assistance for documentation and data sharing which will expedite the entire process of eligibility assessment. Further, since external reviewers are not involved in this process, this again will lead to faster approvals which will be a significant improvement on other credit lines

F. Collateral & security arrangements:

As per “Lending policy” of IDCOL and prevailing industry practice.

Component 2: USD 3.05 million GCF Technical Assistance (TA) to develop enabling environment for EE investment in textile sector

GCF non-reimbursable resources (technical assistance) will support to address real or perceived risks and barriers that are currently limiting EE investments in the textile sector. IDCOL requires the service of a consultant/consultancy firm to perform technical assistance activities. Key output areas under TA will be to (a) develop a sustainable financing strategy to achieve programme milestones and maximize impact potential of the sub-project (GHGE reduction) (b) sensitize and create awareness among the stakeholders about EE equipment to develop a strong project pipeline, which may increase participation of LFI's in future (c) develop capacity and knowledge at the institutional levels required for faster adoption of EE technology (d) develop standards and mechanisms for robust monitoring, reporting and verification of energy savings and GHG emissions reductions and (e) advisory support to the textile borrowers/end users on loan application process to expedite the due diligence process of IDCOL.

The detail activities are described below:

Output 2.1: Sustainable financing strategy developed for large scale adoption of EE equipment in textile sector.

Activity 2.1.1: Devise a financing roadmap and time-based action plan

Activity 2.1.2: Develop operational manual for the program.

Activity 2.1.3: Develop whistle blower policy for the program.

Activity 2.1.4: Classification of Textile units based on scale and type of products and data collection through energy audit of sample plants

Activity 2.1.5: Consolidation of data and developing sector-level baseline

Activity 2.1.6: Interact with stakeholders/market players to prepare a list of equipment already implemented in Bangladesh and having proven energy savings results.

Activity 2.1.7: Revision of baseline by conducting energy audit and data consolidation

Activity 2.1.8: Develop technically-robust, bankable EE project pipeline

Output 2.2: Market awareness on importance of EE equipment in textile sector strengthened

Activity 2.2.1: Organizing programme launching event, to introduce the new concessional credit line and its unique features dedicated to promote EE in Textile Sector.

Activity 2.2.2: Organizing workshops to disseminate to stakeholders on broad policy framework and incentive mechanisms for energy efficiency in textiles, business case and economic benefits, women involvement in energy efficient textile operations (Covering entire key stakeholders of the country related to textile sector. Trade bodies like BTMA, BGMEA .etc. are expected to be on board and be sensitized of EE issues for encouraging their members during decision making. Stakeholders will be selected by IDCOL as per consultation with the expert consultants hired under this TA component. IDCOL's previous stakeholder lists developed during scoping & validation workshops under the PPF, will also be considered).

Activity 2.2.3: Preparing various information materials in the form of brochures/flyers/pamphlets and other related information documents for the dissemination activities

Activity 2.2.4: Develop and launch advocacy campaign on energy conservation and efficiency measures, such as workshops, animations, video clips, brochures, etc.

Output 2.3: Capacity developed at the institutional levels for faster adoption of EE technology

Activity 2.3.1: Perform capacity needs assessment to map the current and desired level of capacity for IDCOL staff and textile end borrowers.

Activity 2.3.2: Conduct capacity building workshops and training sessions for IDCOL and textiles sector end borrowers (participants not limited to end borrowers being eligible for component 01. It will include both eligible & other prospective textile sector end borrowers) who are the applicants for the loan on the tools and requirements to be fulfilled for successfully securing the loan

Output 2.4: Standards and mechanisms for adequate monitoring, reporting and verification of energy savings and GHG emissions developed

Activity 2.4.1: Updating the parameters and requirements for establishing MRV framework, including energy audit and post retrofit performance evaluation for the textile factories (establishing as-is baseline and EE investment-based evaluation KPIs on best effort basis)

Activity 2.4.2: Updating parameters and updating the monitoring framework for environment and gender evaluation of projects for the end borrowers & for IDCOL.

Activity 2.4.3: Carry out yearly energy, financial and environmental audit to establish baseline for monitoring and revisit GHG and other programme targets if required

Activity 2.4.4: Revising and applying guidelines and monitoring methodologies for textile factories

Activity 2.4.5: Requirements for "Management Information Systems (MIS)" in different types of textile factories to be provided to end borrowers for ensuring monitoring and reporting and dashboard preparation covering energy data, environment & social reporting (Common Dashboard for both Component 01-Textile & Component 04-RMG)

Activity 2.4.6: Revision of applicable environmental and social policies of IDCOL and gender specific actions, if required

Activity 2.4.7: Review/audit of environmental, gender and social safeguard compliances

Output 2.5: Advisory support to the Textile Borrowers to shorten appraisal timeline

Activity 2.5.1: Support the end borrowers, eligible under Component 01, in fulfilling the documentation process, completing due diligence during appraisal, checking eligibility and making disbursement process faster.

This will be done through requirement standardization, capacity building of borrowers, facilitating certification from equipment suppliers and empanelment of the vendors all of which will significantly cut down on documentation and disbursement time.

- Create standardized document requirements for loan application under the credit line. The standardization of documents and simplified approval process (internal to IDCOL) will help borrowers cut down on application time
- The TA component will hold workshops to appraise borrowers on the application process and the documentation requirement, so the awareness levels is increased
- Work with the equipment suppliers to facilitate certification and other documentation requirements, so the equipment impact can be clearly mapped
- Vendor's due diligence etc. can be carried out so time can be saved on their selection/finalization as equipment suppliers. They will also work with borrowers to help them speed up application process
- Application process status would be available to borrowers through an MIS tool which they can access, to bring in speed and transparency in the approval process.

Output 2.6: Engagement with global clothing brands to increase EE adoption in textile sectors

The brands play a catalytic role in driving change in the textile and related sectors. The global brands in presence of international competition influence other stakeholders as seen through the feasibility analysis. As part of the TA programme, the connection of the brands to the textile manufacturers can be leveraged by working closely with the brands and developing customized modules to increase the reach and impact of the programme. This will be carried out by:

Activity 2.6.1 Encouraging textile units to be more aligned to the brands' sustainability targets and invest in energy efficient equipment and measures to reduce emissions and encouraging leading global brands by arranging round table discussions

Activity 2.6.2 Encouraging textile units to improve on the social and gender inclusion in workforce ratio to increase the co-benefits impact through sensitization and by stakeholder consultation

- Carrying out capacity building trainings and workshop for the textile units to disseminate the information on the programme and the benefits of the EE adoption (Under activity 2.3.2 above)

The TA component will be executed by IDCOL within the disbursement period of 06 years within project time line (Year 01-06). The main beneficiary of the component will be the textile sector borrowers for sustainable implementation of the program.

DESCRIPTION OF COMPONENT 03 & 04:

The RMG industry is the most energy intensive industrial sub-sector of Bangladesh. As per situation analysis conducted in feasibility studies, energy consuming processes in the sector employ old equipment and exhibit potential for replacement with equipment which lead to efficient use of energy through incorporation of new

technologies. Apart from capital intensive measures, there are system inefficiencies caused due to leakages which can be reduced by undertaking low-cost or no-cost measures. In the context of the RMG sector, it has been established in the previous sections that there is a considerable scope for adopting energy efficiency measures along with the benefits of adopting such measures. However, there are also multiple barriers which are preventing large scale adoption of energy efficiency measures, thereby deferring realization of benefits arising out of it. A key barrier is the availability of concessional financing to support high upfront costs of the energy efficiency equipment and longer payback periods as well as limited market information regarding available options for energy efficiency.

Financing for RMG units is provided from participants in Formal Sector like *Banks, Non-Bank Financial Institutions (FIs), Insurance Companies, Capital Market Intermediaries* like Brokerage Houses, Merchant Banks etc.; *Micro Finance Institutions (MFIs)*. The inherent high level of risk of lending in the country explains the high lending rates, as compared to nearest competitors in RMG export market. A comparison of the prevalent lending interest rates in the competing markets is given below for reference:

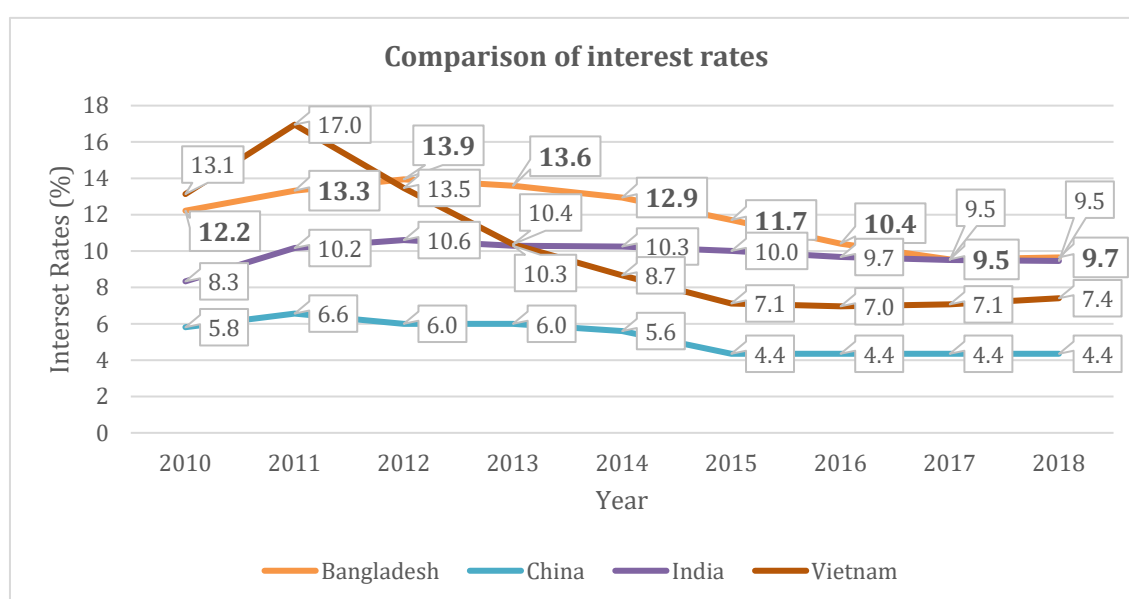


Figure 15: Comparison of interest rates among major RMG exporters

Even though the overall market level interest rates have exhibited a sharp fall in recent times, RMG units in the country⁴⁶ are reported to be receiving funding at 10-12% interest rates from banks and at 11-13% from FIs, depending on financial institution's internal loan process and the creditworthiness of RMG units. In case higher risk is assessed, the interest rates may exceed these limits. Also, the lowering of interest rates has been offset to a degree by increase in labour wages and increased operational costs due to compliance with Accord/Alliance requirements. As exemplified in the figure above, even with the best-case scenario, interest rates in Bangladesh are still considerably higher than its immediate competitors. In the near future, the interest rates are expected to remain in the current state or even increase given the performance of the banking sector in the nation based on key indicators⁴⁷: The feasibility study conducted for the RMG sector indicates the need for ~ 5.7%⁴⁸ (11%-5.28%) concessional in the interest rate for energy efficient

⁴⁶ Source: Stakeholders discussions with bankers to RMG units and individual RMG unit owners

⁴⁷ Source: [Banking Sector in Bangladesh - Moving from diagnosis to action \(CPD, December 2018\)](#)

⁴⁸ Please refer to table 06-09 of this document

equipment, compared to commonly available commercial loans (i.e around 11%⁴⁹) in order for EE equipment to be widely adopted in Bangladesh. Using GCF and LFI funding, this can be achieved.

Also, there are a number of areas related to implementation where the RMG unit requires expert guidance and external consultancy support. Banks, being a key stakeholder, also have limited capacities in assessing an energy efficiency project and estimating the cost savings potential. Given this situation where, on the one hand, both RMG units as well as the banks have limited technical capacity and on the other hand the prevailing market interest rates are considerably high, energy efficiency measures are largely avoided. This Program therefore attempts to address both these barriers through a structured technical assistance program and a concessional line of financing. In order to access financing from the Program, the end borrowers will always be required to take on risk via minimal capital requirements, which is considered as equity participation and considered as part of the project amount.

Component 03 & 04 of the proposed program aims to address the key barriers of RMG sector to increase their energy efficiency by supporting entrepreneurs and business establishments in the garments sector with financial and market resources through the LFI/IDCOL, to avail investment opportunities for energy saving technology upgrades. In order to facilitate the implementation of the program, the Accredited Entity (AE) Infrastructure Development Company Limited (IDCOL) has proposed LFI⁵⁰ to be the Executing Entities for Component 03 and requested financial support by means of concessional financing from Green Climate Fund (GCF).

The proposed program will comprise following two components addressing the energy efficiency needs of the RMG sector:

Component 3: USD 200.00 million financing for Energy Saving Equipment & Technology for RMG sector

Under this component, IDCOL will sign subsidiary agreements with the EEs to create a term loan facility which provides term loans with preferential rates directly to RMG sector end borrowers.

G. General criteria for lending to RMG unit:

- End borrowers i.e. project developer will be a business entity of limited company
- End borrowers should have operations in the production of RMG sector in Bangladesh
- For both Green field and Existing project related to up-gradation to energy efficiency technologies/machines. For a Project loan application already being processed at LFIs, a letter of credit (L/C) for purchasing the EE&C equipment, anticipating the use of the Project Loan, may be opened up to 12 months prior to either GCF financing or LFIs approval of the loan. In case of project having multiple expense heads, only Energy Efficient component of the proposed equipment to be financed
- All financing requests need to be supported by suppliers' declaration/ catalogue/ brochures or report/certificates from independent/certified energy auditor acceptable to the LFIs or the baseline developed under TA activity
- Fulfillment of other required criteria as mentioned in LFIs CRM guideline, lending policy or other applicable policies/requirement
- 100% of the eligible equipment price can be financed

⁴⁹ <https://www.bb.org.bd/fnansys/interestlending.php>

⁵⁰ Initial target LFIs are BRAC Bank Limited, Southeast Bank Limited, City Bank Limited and IDLC Finance Limited

H. Technical Eligibility criteria:

Eligibility should be confirmed for each loan application. The **Eligible Technology & Equipment List** is a predefined list of technologies and equipment as prescribed during the programme formulation. This list will be updated every year, if required, to reflect the accurate market scenario and will be subject to approval by GCF.

Category	Major EE&C Equipment	Condition
Cutting	- Optimized Air Compressor system - Ducting for hot air exhaust in compressors - LED Light - Absorption Chiller - Dehumidifier	- Any Equipment/technology eligible for financing, not limited to this list, must have minimum 15% energy saving potential compared to the prevailing inefficient/less efficient equipment used in the sector - The eligibility would be supported by catalogue/brochures/declaration from supplier's or report/certificates from independent/certified energy auditor or the baseline developed under TA activity - Common equipment will be interchangeable within the category
Sewing	- Servo motors for sewing machine - Optimized air Compressor system - Ducting for hot air exhaust in compressors - Exhaust Fans - LED Light - Gas Generators - Absorption Chiller	
Dyeing	- Steam Boilers with waste heat recovery system - FD fans for boilers - Oxygen control system for flue gas in boilers - Jet Dyeing - Soft Flow - Stenter - Steam Condensate Recovery system - LED Light - Absorption Chiller - Hydro - Insulation of Steam Distribution Lines	
Ironing & Finishing	- Steam Boiler with waste heat recovery systems - FD fans for boilers - Oxygen control system for flue gas in boilers - Insulation of Steam Distribution Lines - Steam Condensate Recovery system - Optimized air Compressor system	

- | | | |
|--|---|--|
| | <ul style="list-style-type: none"> • Ducting for hot air exhaust in compressors • LED Light | |
|--|---|--|

I. Eligible investment cost: Same as “C. Eligible Investment Cost” for Textile.

J. Financial criteria: The eligible investment should be profitable, to enable the loan repayment with enough cash flow.

- The pay-back period, including financial costs, shall be minimum 02 (two) years;
- The loan granted by LFI's to end borrowers will have a minimum maturity of three (3) years for energy efficiency investments;
- Minimum DSCR of 1.00x
- Any other criteria set by LFI's approving authority in excess of criteria set out in FP. For deviation of any criteria mentioned in this FP will require GCF approval.

K. Timing of Loan Application and Appraisal: LFIs will consider loan application from end borrowers for procurement of new energy efficiency equipment. However, for retirement of any machinery L/Cs into GCF project term loan, the L/Cs can be opened maximum 12 months prior to application. As mentioned in Component 1 above, loan processing time will be minimized due to non-existence of external reviewers and pre-set terms & conditions.

L. Lending Conditions- Details in the Term Sheet

M. Collateral & security arrangements: As per “Lending policy” of LFIs and prevailing industry practice

During implementation of the project, LFIs will undertake due diligence activities to determine eligibility of the RMG units for concessional financing. Based on the information provided by the LFIs, IDCOL will provide final approval on the eligibility of the sub-projects for access to concessional line of financing. LFIs will receive funds from GCF as per the disbursement schedule shared under the term sheet. The repayment of the loan from the EEs will also be supervised by IDCOL.

As per the preliminary analysis in the program concept note, it has been estimated that the program would require a financial support of USD 250 million – USD 150 million coming from GCF loan and USD 100 million from co-financing sources. The disbursement period has been considered to be five years and the repayment period has been considered to be of maximum 20 years at this stage.

Component 4: USD 2.30 million GCF Technical Assistance (TA) to develop enabling environment for EE investment in RMG sector

Based on the discussion with key stakeholders it has observed that there is limited market awareness and overall scepticism amongst stakeholders regarding implementation of energy efficiency measures, monitoring and verification of benefits accruing from such measures apart from need for concessional

financing. Banks feel that the institutional capacity in appraising an energy efficiency project and determine bankability of such projects is low as well as the capacity to ensure effective monitoring and verification of these projects to ensure benefits accrual is taking place and hence, repayment is also on track. RMG units, despite appreciating the benefits of energy efficiency, are strained due to reduced profitability and have limited awareness regarding implementation, maintenance and operational requirement of such projects. Also, the lack of demonstration projects within the sector and country, in general, is preventing otherwise interested RMG units in undertaking energy efficiency measures.

Given this context, along with concessional line of financing, there is a definitive need to improve market perception about energy efficiency and provide support to the relevant stakeholders at all stages of implementation energy efficient measures. Also, creating an interface between energy efficiency equipment suppliers or manufacturers and the decision makers behind the investment for energy efficiency i.e. RMG unit owners, can help disseminate the information related to energy efficiency and resolve any queries that the unit owners may have. Policy makers are also required to be apprised the barriers to implementing energy efficiency measures and provide guidance in developing enabling policies, regulations to help overcome these barriers.

It is envisaged that without such enabling activities, a concessional line of financing would not help in generating interest among the stakeholders in undertaking energy efficiency measures. Also, given that the energy efficiency market in Bangladesh is in its infancy, such support activities would be critical to enable market transformation and galvanize the equipment procurement decisions of RMG units. Hence, the TA is viewed to be a critical component to ensure greater uptake of energy efficiency measures during the programme duration as well as ensure sustainability of energy efficiency market for RMG sector beyond the programme duration.

The detail activities are described below:

Output 4.1: Preparatory activities & Base lining energy consumption of RMG sector

Activity 4.1.1: Devise operational manual for the program & selection of prospective LFIs for the program (if required)

Activity 4.1.2 Develop whistle blower policy for the program

Activity 4.1.3 Organizing programme launching event, to introduce the new concessional credit line and its unique features dedicated to promote EE in RMG Sector.

Activity 4.1.4: Classification of RMG units based on scale and type of products and data collection through energy audit of sample plants

Activity 4.1.5: Consolidation of data and developing sector-level baseline

Activity 4.1.6: Interact with stakeholders/market players to prepare a list of equipment already implemented in Bangladesh and having proven energy savings results. List will be integrated with knowledge platform and disseminated through other means to ensure adequate public outreach

Activity 4.1.7: Revision of baseline by conducting energy audit and data consolidation

Output 4.2: Capacity building of Local Financial Institutions (LFIs)/IDCOL

Capacity building of LFIs/IDCOL to assess funding proposal and monitor implementation of energy efficient technologies in factories of RMG borrowers.

Activity 4.2.1: Conduct capacity building trainings/workshops to develop capacity of corporate/head office level officials to effectively conduct banking due diligence of loan applications for implementation of energy efficient technologies.

Activity 4.2.2: Conduct capacity building trainings/workshops to develop technical capacity of banking officials at branch/field level to efficiently monitor implementation of energy efficient technologies over the duration of repayment of loan amount.

Output 4.3: Capacity building of RMG units to enable implementation of energy efficient technologies

Capacity building of RMG units to enable implementation of energy efficient technologies

Activity 4.3.1: Conduct workshop to establish and disseminate business case of energy efficiency among RMG unit owners/top officials to help generate interest in owners in energy efficiency concept as well as provide select real-life case studies to instil confidence among them in pursuing energy efficiency projects

Activity 4.3.2: Impart training to facility/utility managers on the technical requirements for implementing energy efficiency projects as well as necessary steps (i.e. equipment working condition, energy supply stability etc.) to be undertaken during operations to ensure theoretical energy savings are realized.

Activity 4.3.3: Develop awareness of workers on importance of energy efficiency and impart knowledge key skills required to implement energy efficient equipment. This will help them adapt to any possible change in the operations due to introduction of new machinery and ensure seamless transition to new machinery without a loss in productivity.

Output 4.4: Provide on-demand technical support to participating banks on monitoring, reporting and verification (MRV) of energy efficiency projects

Provide on-demand technical support to LFIs/IDCOL on monitoring, reporting and verification (MRV) of energy efficiency projects within TA period of 05 years. On-demand technical support means LFIs/IDCOL will be able to send queries to TA consultant related to any issues faced during MRV of live projects. TA consultant will provide response to the issue or get back to LFIs/IDCOL for further information within 5 days of request being raised.

- *Activity 4.4.1:* MRV Support provided to LFIs/IDCOL (in form of technical trainings, resolve queries on request, handholding support). Impart trainings on technical capacity (i.e. concepts related to energy and emission savings, methods for calculating energy, emission and cost savings etc.) required for appraisal of energy efficiency projects. Resolve queries and provide technical support to LFIs/IDCOL on MRV related issues for projects to develop capacity in the banks to conduct MRV of projects even after TA consultant expires its tenure of 5 years.

Resolve queries and provide technical support to LFI/IDCOL on MRV related issues for projects to develop capacity in the banks to conduct MRV of projects even after TA consultant expires its tenure of 5 years.

Activity 4.4.2: Develop standard solutions document and case study for gender based on experience of commonly issues encountered in previous activity. Identify most commonly faced issues of banks in MRV and develop standard solution document addressing those issues to provide LFI/IDCOL with troubleshooting guide and help them conduct MRV after tenure of TA consultant expires

Output 4.5: Provide assistance to RMG units in loan application process

Output 4.5: Provide assistance to RMG units in loan application process

Rapid Diagnostic study report, Guidance documents, Troubleshooting guide and periodic query resolution response to RMG units

- **Activity 4.5.1** Conducting rapid diagnostic study of key areas related to loan application process for which RMG units require handholding as well as nature of support required.
- **Activity 4.5.2** Prepare guidance documents and filled up templates of loan application forms along with step-by-step explanation of process of loan application along with development of check list on list of supporting documents and guides on where and how they can be available.
- **Activity 4.5.3** Respond to queries related to loan application process received from through Query Management module present in proposed digital platform (discussed under Output 4.8).

Output 4.6: Provide on-demand technical support for implementation to RMG units which have secured GCF funding under this programme

Periodic query resolution and handholding support provided to RMG units; troubleshooting guide

- **Activity 4.6.1:** Implementation support provided to RMG units on request: Resolve queries and provide technical support to borrowers to ensure effective implementation of energy efficiency intervention
- **Activity 4.6.2:** Develop standard solutions document i.e. Troubleshooting manual based on experience of common issues encountered in previous activity: Identify most commonly faced issues of borrowers in implementation and develop standard solution document addressing those issues. The responses and queries will be consolidated to a troubleshooting guide to provide ready reference to any borrower after tenure of TA consultant ends.

Output 4.7 : Technology fair to create market awareness and develop access to energy efficient technology

02 days long Technology fairs organized (in Year 3) with participation of minimum 30 different technology suppliers and representation from 60 different Textile/RMG units

Activity 4.7.1: Organization of 01 technology fair: In order to create market awareness and develop access to energy efficient technology, IDCOL along with the LFIs will organize technology fair to create interface between international manufacturers/suppliers of energy efficient equipment and Textile/RMG unit owners to stimulate market development for energy efficiency in the sector.

This fair will also be covering both Textile sector under component 01 & 02 and RMG sector under Component 03 & 04

The TA component will be executed by IDCOL within the disbursement period of 06 years. The main beneficiary of the component will be the IDCOL, LFIs and Textile-RMG sector borrowers for sustainable implementation of the programme. The TA consultant will be selected from a list of international consultants based on competitive bidding procedures and the scope of work developed from activities given above.

- **Component 5: USD 1.15 million GCF Technical Assistance (TA) to strengthen regulatory & institutional framework at the national level to overcome the operational constraints related to implementing EE&C in the country**

The component has been incorporated to address the policy/regulatory support issue that is preventing faster adoption of EE equipment in the textile & RMG sector of Bangladesh. Sustainable Renewable Energy Development Authority (SREDA) under Ministry of Power, Energy & Mineral Resources, will be the nodal Executing Entity for the proposed activities under this component.

The component includes several activities under two broad outputs (5.1) Development of relevant policies, strategies & plans addressing energy efficiency and (5.2) Strengthening capacity of relevant national institutions to overcome their existing operational barriers related to implementing EE&C projects.

The activities under Output 5.1 are as follows:

- 5.1.1: Review of existing national policies, strategies & plans regarding Energy Efficiency
- 5.1.2 Identify any policy gap & develop mechanism to address such gaps & stock taking of baseline
- 5.1.3 Development of relevant policies, strategies & plans
- 5.1.4 Implementation of the policies, strategies & plans developed under Activity 5.1.3

The activities under Output 5.2 are as follows:

- Activity 5.2.1 Assessment of detail capacity development requirement at policy level
- Activity 5.2.2: Develop training manuals based on the world best practices focusing on performance measurement, EE standards & labelling and financial support mechanism for implementation of energy efficiency projects; (prospective business model with feasible payback and IRR)
- Activity 5.2.3: Organize 03 training of the trainers (ToT) for Government officials for institutionalization of the training manual
- Activity 5.2.4: Organize 10 training session for the policy makers and relevant government official by the trained trainers. National relevant officials mean officials from different government offices related to environment, power, energy, energy efficiency .etc.
- Activity 5.2.5: Organizing 02 workshops to disseminate outcome of the programme to policy makers and EE practitioners and also the learning from implementing energy efficiency in textile and garments sector (financing scenario, business case and economic benefits, women involvement and role of government for long-term sustainability) to ensure institutional capitalization of knowledge.
- Activity 5.2.6: Mass awareness for energy efficiency through other disseminating tools like media coverage, bill board. etc.

The proposed activities under component 05 will facilitate institutionalization of the EE&C measures in the country and will be the key exit strategy to ensure additional concessional funding continues after programme concludes.

Sustainable and Renewable Energy Development Authority (SREDA) has been formed under the provision of “Sustainable and Renewable Energy Development Authority Act, 2012” on 22nd May, 2014 as a nodal agency to ensure energy security and promote, facilitate and disseminate sustainable energy (SE) in both areas of Renewable Energy (RE) and Energy Efficiency (EE).

Earlier, different ministries and departments, including the Bangladesh Power Development Board (BPDB), the Rural Electrification Board (REB), the Local Government Engineering Department (LGED), autonomous bodies like IDCOL and also the private sector was working on renewable energy projects. But the necessity of an authoritative organization in renewable energy sector to maintain coordination among them was being felt. So, an institution like SREDA was inevitable. The institution is now working under Power Division of the Ministry of Power, Energy and Mineral Resources (MPEMR) of the Government of the People's Republic of Bangladesh as a coordination body for the development of the renewable energy in the country.⁵¹

The below diagram represents the implementation arrangement proposed for Component 05:

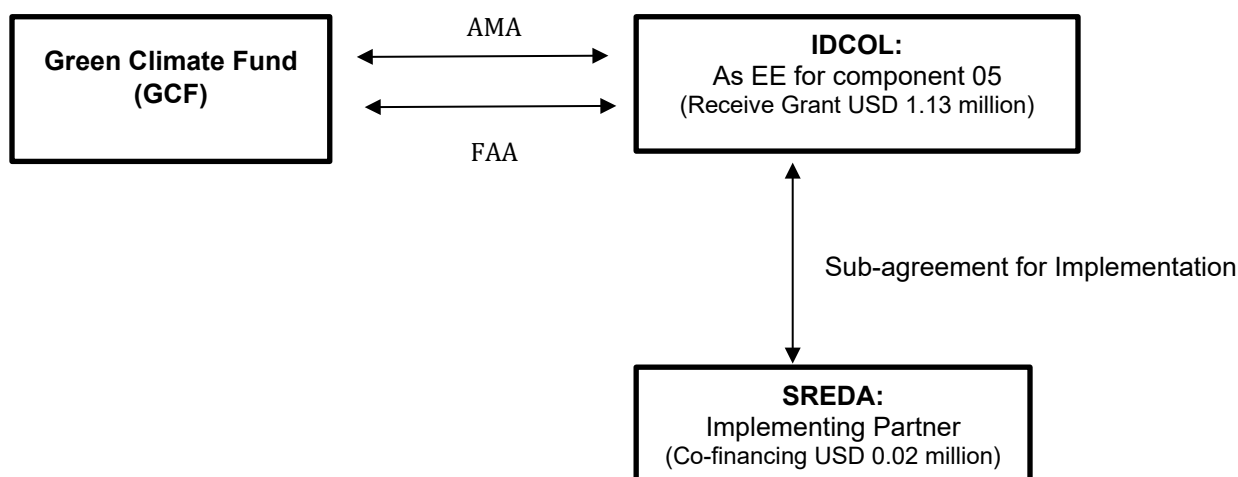


Figure 15.1: Component 05 for SREDA

IDCOL as the accredited entity to GCF will have to sign the Funded Activity Agreement (FAA) with GCF to ensure proposer implementation of Component 05. IDCOL as the EE will be liable to supervise & monitor proper implementation during the implementation timeline. Since, SREDA will be the Implementing Partner for delivering this component, IDCOL will sign a separate sub-agreement with SREDA. IDCOL will engage the required consultant/consultant firm under its own procurement policy in consultation with and engagement of SREDA. SREDA will co-finance for this program in-kind which has been detailed under the budget calculation. SREDA as the Implementing Partner will carry out specific activities mentioned above and IDCOL will facilitate SREDA for the management and operation for component 05.

In excess of activities under component 05, IDCOL will regularly report the energy savings achieved under this entire program to SREDA, in line with reporting to GCF (with the same frequency). This information will be incorporated in national data base for energy savings.

B.4. Implementation arrangements (max. 1500 words, approximately 3 pages plus diagrams)

⁵¹ <https://mpemr.gov.bd/power/details/26>

Provide a description of the project/programme implementation structure, outlining legal, contractual, institutional and financial arrangements from and between the GCF, the Accredited Entity (AE) and/or the Executing Entity(ies) (EE) or any third parties (if applicable) and beneficiaries.

- *Provide information on governance arrangements (supervisory boards, consultative groups among others) set to oversee and guide project implementation. Provide a composition of the decision-making body and oversight function, particularly for Enhanced Direct Access (EDA) proposals.*
- *Provide information on the financial flows and implementation arrangements (legal and contractual) between the AE and the EE, between the EE or any third party and beneficiaries. For EEs that will administer GCF funds, indicate if a Capacity Assessment has been carried out. Where applicable, summarize the results of the assessment.*
- *Describe the experience and track record of the AE and EEs with respect to the activities (sector and country/region) that they are expected to undertake in the proposed project/programme.*

Provide a diagram(s) or organogram(s) that maps such arrangements including the governance structure, legal arrangements, and the flow and reflow of funds between entities.

FOR COMPONENT 01 AND 02:

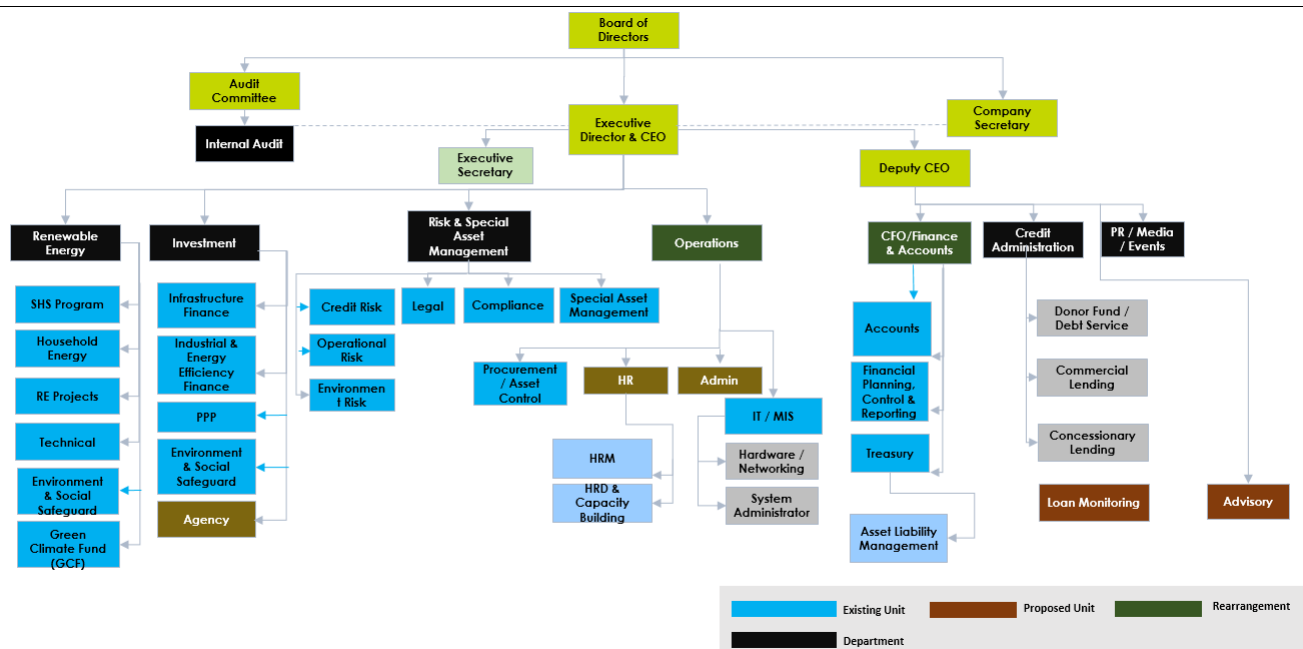
Governance arrangements under the programme for *Component 01 and 02::*

IDCOL will play the significant role as Executing Entity to assess the eligibility of the sub-projects of the Textile sector for financing under the program as per the set criteria and accordingly will place disbursement request to GCF. IDCOL will be the borrower & Executing Entity (EE) where the Textile owners will be the end borrower. IDCOL will channelize the loans to the ultimate beneficiaries, textile borrowers. This business unit(s) of IDCOL will be in primary contact with the end borrowers. IDCOL will have separate role as Accredited Entity (AE) & Executing Entity (EE).

IDCOL's Organogram (Figure 16) shows all the functional departments and detail reporting structure under the respective functional departments including Finance, Internal Audit, Procurement, and other key functions. IDCOL Board has the following three special committees comprising board members: Audit Committee, Credit Committee and Organization Committee (Recruitment and Promotion Committee). Composition and roles and responsibilities of Board Committees is annexed.

Among the list of internal oversight bodies/committees IDCOL has the following committees for maintaining internal oversight in different aspects of company operations: Management Committee, Credit Risk Management Committee, Asset Liability Management Committee, Internal Control and Compliance Unit, Risk Management Forum, Risk Analysis Unit and Integrity Committee. The terms of reference of the aforesaid committees are enclosed in Annex. All these committees are assigned to ensure dedicated internal oversight on vital corporate governance issues. The committees comprise senior management officials of IDCOL with appropriate functional designation and responsibilities that best fit to serve the purpose of the committees.

Figure 16: Organogram of IDCOL

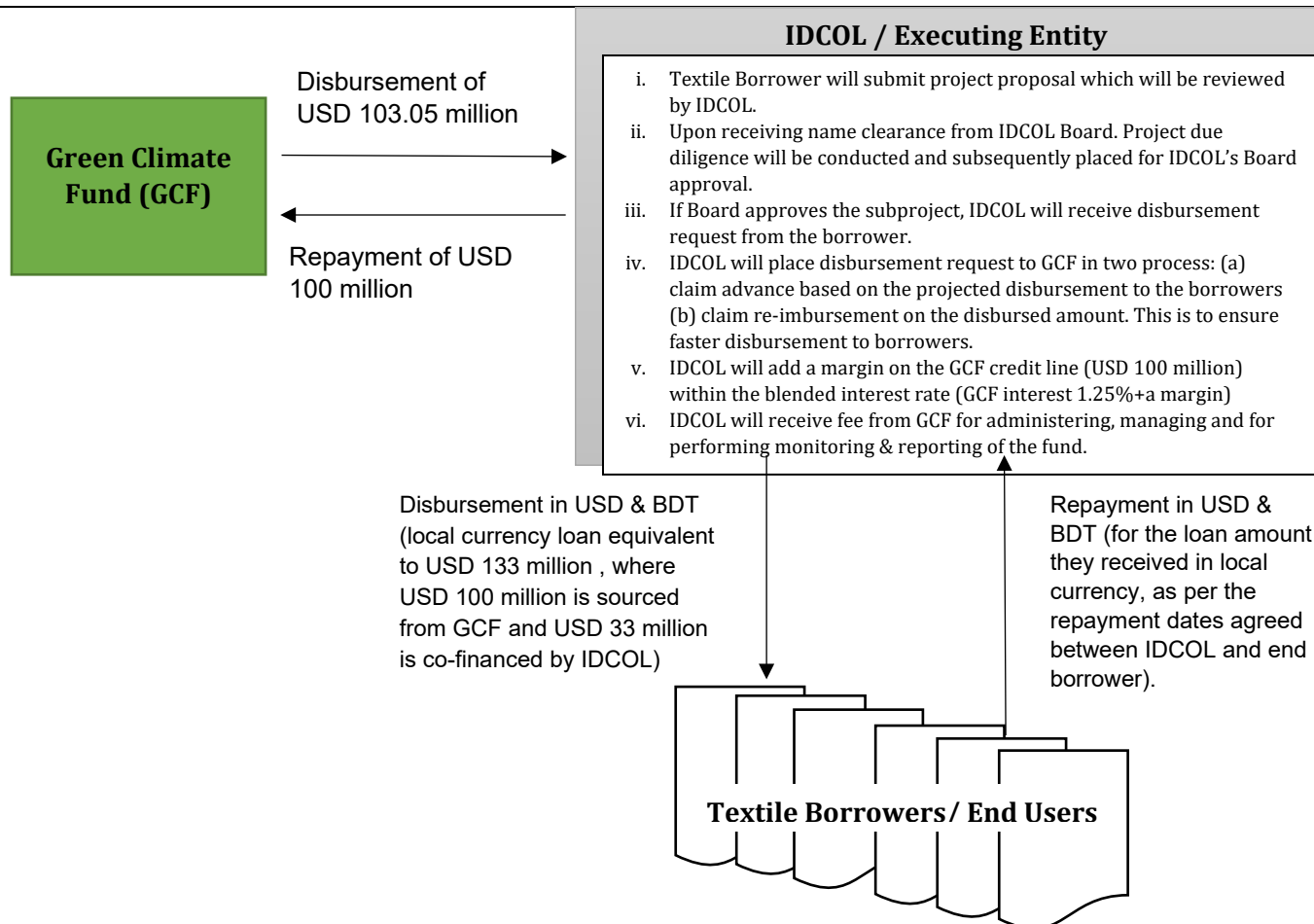


Financing Flow under Component 01: (please refer to figure 17)

1. IDCOL will open a designated GCF "Project Account Textile" for receiving, disbursement & repayment of the GCF fund under this project.
2. After receiving the GCF fund in "Project Account Textile", IDCOL will disburse the fund to the respective end borrower's account.

- **GCF financing to IDCOL-** under component 01 of the programme GCF will provide IDCOL USD 100.00 million senior loan. The senior loan will be passed down to the end borrowers as concessional loan to make EE equipment affordable in comparison to cheaper traditional equipment.
- **IDCOL financing to end borrowers:** IDCOL will on-lend to the end borrower/ textile borrowers in both USD & local currency loan equivalent to USD 133.00 million , where USD 100 million is sourced from GCF and USD 33.00 million is co-financed by IDCOL.

Figure 17: Fund Flow under Component 01&02

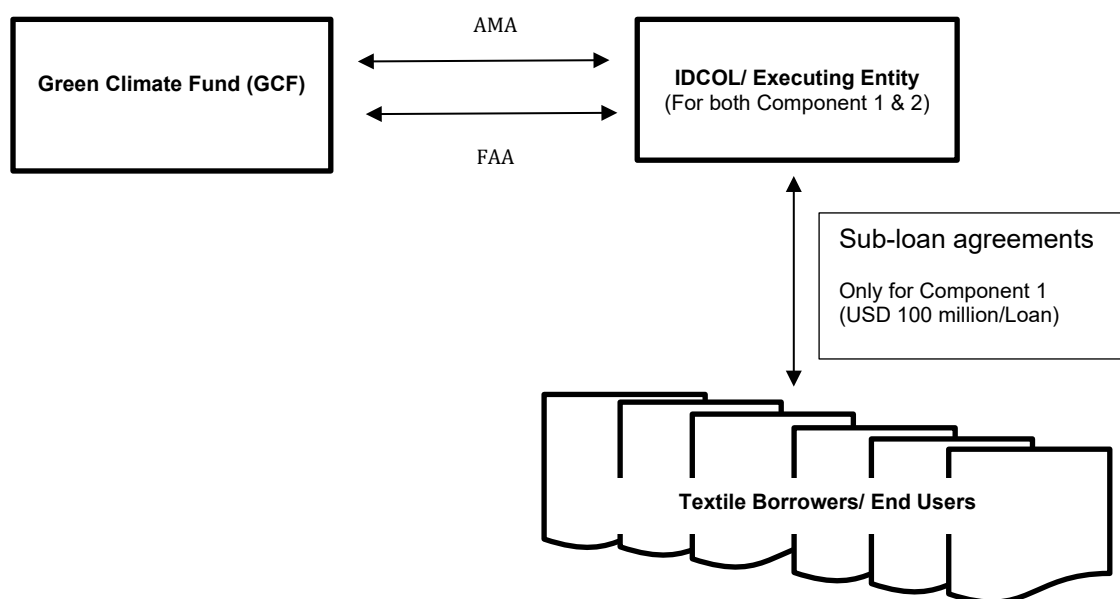


- **Repayment from end user/ textile borrower to IDCOL:** Repayment will be made by the end borrower to IDCOL in USD/local currency for the loan amount they received in USD/local currency, as per the repayment dates agreed between IDCOL and end borrower.
- **Repayment from IDCOL to GCF:** IDCOL will repay USD 100 million to GCF as per the agreed amortisation schedule mentioned in the Term Sheet.
- **Re-investment management:**
 - IDCOL will re-invest a portion of reflows till Year 12.
 - The details management of reflows will be described in Term Sheet.

Financing Flow under Component 02:

- **GCF to IDCOL** – GCF will provide IDCOL USD 3.05 million as technical assistance (TA) for market development in the form of grant.
- **IDCOL to consultant-** IDCOL will execute the TA amount via consultant or consulting firm to carry out activities mentioned in section B.3.

Figure 18 : The Implementation arrangement under the programme for Component 01& 02



IDCOL as an AE and EE of the proposed programme will sign Funded Activity Agreement (FAA) with GCF and subsequently sign sub-loan agreements with textile borrowers/ end borrowers. IDCOL will take the FX risk for Component 01 in arrangement with the end borrowers.

Programme implementation capacity of IDCOL:

IDCOL' experience in EE&C

The Energy Efficiency and Conservation Promotion Financing Project (EECPF) is a 100 Mn USD JICA credit line administered by SREDA, and IDCOL is one of the participating financial institutions for further on-lending this fund at preferential rates to eligible projects based on a predefined criteria. Under the JICA funded Energy Efficiency and Conservation Promotion Financing Project (EECPFP), IDCOL has approved USD 42 million in the in the RMG and Cement sectors, out of which USD 25.15 million has been already disbursed. The financed project would help the end consumer in saving 25 percent and 34 percent of primary energy consumption in the RMG and Cement facility respectively. This means that an electricity of savings of approximately 17 MWh over the life of the equipment. This would also result in a cost saving of approximately BDT 14 Crore per year for these installations for the end consumers

IDCOL has already started approving projects under the second tranche of JICA's EECPF, which aggregates to an amount of USD 84.07 million. IDCOL targets to fully disburse the first and the second tranche of EECPF by the end of 2021.

IDCOL's financial performance

Looking at long-term loan outstanding of IDCOL by sector, renewable energy projects account for 33.15% (December 2018) while that of Infrastructure projects has 66.85% (December 2018). The share of

renewable energy projects has been decreasing from 70% in December 2015. The importance of financing to infrastructure project including energy efficient project is increasing. It implies that due to expected weaker demand for renewable energy project especially for solar home system, loan for infrastructure would become more significant sector of IDCOL in the future.

From IDCOL's financial data, the total assets increased by 22.86% between December 2015 and December 2018 while loans and advances grew by 64%. Observing profit and loss statements, total operating income have increased by 6.94% within 4 years, while profit before tax and provision have increase 3.4% as well during the same period.

Looking at financial indicators ROE and ROA of IDCOL shows better than or almost same level as the Non-Bank average. Regarding Capital Adequacy Ratio (CAR), BB require at least 10% of total Risk Weighted Assets according to the guidelines. That of IDCOL was 11.53% in 2018 and more than 10% from 2015 to 2018.

Financial indicators of IDCOL for 4 years. ⁵²				
	2015	2016	2017	2018
ROE (%)	24.1	6.7	8.4	10.34
ROA (%)	2.0	0.5	0.7	0.85
Capital Adequacy Ratio (%)	11.87	12.44	11.18	11.53

Credit rating obtained from credit risk assessment institutions is as follows:

IDCOL obtained AAA from Credit Rating Agency of Bangladesh⁵³; this rating grade is ranked as first grade in the Bangladesh Bank's rating rate. It is observed under the current corporate finance environment in Bangladesh that loan outstanding has continued to increase by 18% from a year earlier backed by strong credit demand, while the growth rate of deposit which is the source of lending has lowered to 10% or 11%. As a result, advance deposit ratio (ADR: advance/deposit) stays at 77% which is higher than before. This requires tightening in growth of lending which resulted in introduction of private banks' clients to IDCOL or in requesting those NBFIs to participate in loan syndication by private banks.

Private commercial banks and NBFIs were the dominant players in financing most of the categories, such as renewable energy (4.7%), **Energy Efficiency (5.9 %)**, Green Brick Manufacturing (15.4%), recycling and recyclable products (5.4%), setting up green industries (15.8%), and waste management (49.6%) However, IDCOL continues to be the main non-bank financial institution (NBFI) contributor and promoter of renewable energy technologies.

IDCOL's experience with DFIs

Since inception IDCOL has implemented loan and grant amount of more than USD 2.1 billion from DFI's, that has been channeled through the GoB. IDCOL's major development partners are World Bank, ADB, JICA, SNV, KfW, GIZ, IDB, USAID, GPOBA and DFI.

FOR COMPONENT 03 AND 04:

Implementation arrangement for *Component 04: GCF Technical Assistance (TA) to develop enabling environment for EE investment in RMG sector* will be the same as implementation arrangement of

⁵² Annual report of IDCOL

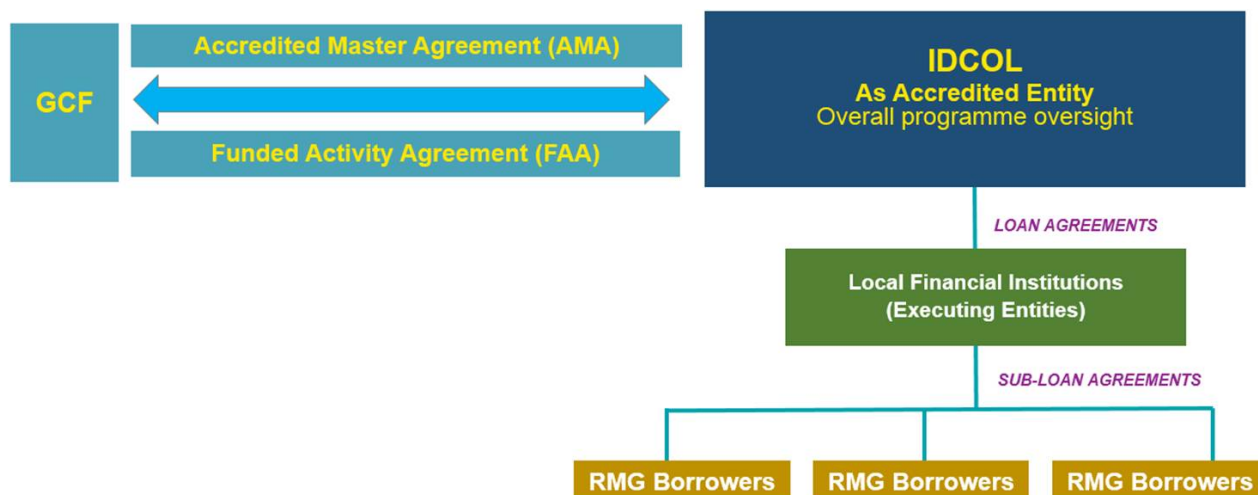
⁵³ As per disclosure on IDCOL's website

Component 2: GCF Technical Assistance (TA) to develop enabling environment for EE investment in textile sector. IDCOL will be the Executing Entity for both component 02 & 04 and will receive fund from GCF for execution.

The implementation modality for RMG loan component 03 is described below:

A. Basic Funding modality:

- Initially, four LFIs namely Southeast Bank Limited, BRAC Bank Limited, City Bank and IDLC Finance Limited have been selected for the RMG component 03. If any of these LFIs does not agree on GCF approved terms & conditions, then IDCOL may select few more LFIs (details in C. Selection process of LFIs). IDCOL can also utilize a portion of RMG fund to lend directly to RMG borrowers (in this case the implementation arrangements would be equivalent to component 1).
- In addition of the roles & responsibilities as the Accredited Entity (AE), IDCOL will play the role of “Borrower” under “Limited Recourse Loan” modality where the repayment obligation of IDCOL will be limited only to the repayment amounts received by IDCOL from time to time from the local FIs; In case IDCOL utilizes any portion of GCF RMG fund, then IDCOL will be liable for repayment of this respective amount.
- The contractual arrangement for the RMG Loan component 03 will be as follows:



B. Funding allocation: Details in Term Sheet

approval.

C. Selection process of LFIs: Details in Term Sheet

D. Approval of eligible sub-project for GCF financing:

- If IDCOL utilizes a portion of RMG fund to lend directly to RMG borrowers, then the approval process will be the same as described for textile loan component (Component 1).
- In case of financing through LFIs, LFIs will submit required information of the eligible sub-projects to IDCOL, as per the specific format, for checking the eligibility of the sub-project.

Information required from LFIs:

- Short description of the project & the borrower

- List of energy efficiency or investment eligible equipment & technology
- Environment Assessment (Category B)
- Co-financing planning with project cost break up
- Board Approval (Tenure, amount, currency, rate)
- After FP approval, specific format for eligibility checking will be developed during Operation Manual preparation under TA support
- Credit Risk Management (CRM) unit of IDCOL will receive the information/document from LFIs (as per specific format along required documents mentioned above). They will check/verify submitted information/document as per eligible criteria of the GCF. After satisfactory review, IDCOL CRMC will provide No-objection to LFI for specific sub-projects.
- Once the sub-project becomes eligible for GCF financing, the LFIs can place disbursement request to IDCOL for that sub-project in future.

E. Disbursement & Repayment of Fund under Component 03:

- IDCOL will open a designated GCF “Project Account RMG” for receiving, disbursement & repayment of the GCF fund.
- “Proceed Account LFI” will be opened & operated by LFI. Subject to BB permission, IDLC will open an FC account with any schedule commercial Bank.
- The LFIs will place disbursement request to IDCOL, for their eligible sub-projects in advanced against their upcoming disbursement or reimbursement for already disbursed fund for their sub-projects. Any regulatory requirement for USD financing to be obtained by LFIs at its own arrangement and inform to IDCOL.
- Credit Administration (CAD) of IDCOL will receive the disbursement requests from LFIs and will verify information/document against disbursement criteria of the GCF Operation Manual. After FP approval, specific format will be developed during Operation Manual preparation under TA support

Required information:

- Sanction advice (commercial terms, acceptance to ESMF/Gender issues. etc.)
- Relevant evidence for compliance with ESMF/Gender policy
- Relevant evidences of disbursement

Upon satisfactory review, IDCOL will approve for disbursement.

- After approval, Accounts Department of IDCOL will submit disbursement request to GCF. IDCOL will place disbursement requests to GCF in any month and several times in a year. After receiving the fund in “Project account RMG”, Accounts Department will disburse the fund to “Proceed Account LFIs” or to “Machinery L/C beneficiary A/C” as per LFI’s discretion.

F. Pricing for co-financing by LFIs: Details in Term Sheet

G. Repayment of Loan & Reflows Management- Details in Term Sheet

Brief Profile of Initial Target Local Financial Institutions (LFIs):

1. **BRAC Bank**, established in 2001, is well known for its SME Banking in Bangladesh. It has also grown rapidly in retail Banking, Corporate Banking, Treasury and other financial services. BRAC Bank Limited’s authorized share is 2,000 million ordinary shares of BDT 10 each. Issued, subscribed and paid-up is

approximately 1,072 million ordinary shares of BDT 10 each (2018). Earnings per share is BDT 5.17 in 2018. Revenue of BRAC Bank in the year 2018 is BDT 19,164 million with a CAGR of 8.30% from year 2015 and net profit after tax is BDT 5,547 million in the year 2018 with a CAGR of 31.57% from year 2015. BRAC Bank is implementing green banking practices through 'Green Reporting' to the Bangladesh Bank. Emphasizing on the conservation of natural resources that include paper, water and energy. At the end of the year 2018, the financing amount in green projects is BDT 12,000 million.

Total asset of the BRAC Bank is BDT 315.42 billion showing CAGR of 12% over the last 4 years while compounded annual growth rate of loans and advances is 17.4% per annum flourishing the business of the bank. Net interest Income as a percentage of total earnings asset stands at 4.49% in 2018 which has slightly declined from 2017. BRAC Bank's interest on deposit has increased resulting in increase in cost of fund and reducing spread over time. Still, spread around 6% in 2018.

The Bank improved their common equity Tier-1 ratio taking it to 12.31% from 11.24%. On the other hand, Capital to Risk (Weighted) Assets Ratio (CRAR) increased to 13.67% in 2018 and this is well above the requirement as per Basel III. On the back of marked improvement in asset quality, that is manifested due to Net NPL ratio being restricted at 0.97%. The Bank is improving its financial health in all criteria. However, it is being strategically aggressive in case of asset liability mismatch but they are maintaining statutory requirement in capital adequacy to withstand any shock. BRAC Bank Limited also maintaining improvement in their profit after tax in the year 2018.

2. **IDLC Finance Limited**, established in 1985, is one of the pioneers in the non-banking sector in Bangladesh. Started as a single product lease finance company, it has a strong and diversified position in Corporate, SME, Retail and Capital Market segments now. In the year 2018, IDLC Finance Limited has Authorised Share Capital of 1,000 million ordinary shares of BDT 10 each. The paid-up capital is 377 million ordinary shares of BDT 10 each. Earnings per share of IDLC Finance Limited in the year 2018 is BDT 4.22. Revenue and net profit after tax in the year 2018 is BDT 4,705 million and BDT 1,591 million, exhibiting a CAGR of 5.9% and 8.54% from year 2015 respectively. IDLC offers 52 green products under 8 categories as defined by Bangladesh Bank to cater the specialized needs of different industries and segments. At the end of the year 2018, the Green Banking portfolio stands at BDT 648 million and cumulative disbursement in green segment is BDT 1,255 million.

Throughout last four year, IDLC is able to maintain AAA (Triple A) long term credit rating and ST-1 short term credit rating to the IDLC Finance Ltd. In 2017, following issuance of Rights Share, their total capital to risk weighted asset ratio taking it to 15.47% from 15.30% and common equity Tier-1 (CET1) capital ratio taking it to 14.62% from 14.57% in the previous year. IDLC has made a marked improvement in asset quality manifested in the improved NPL (Non- Performing Loans) ratio, which was restricted at 2.20%. Total asset of IDLC is BDT 105 billion registering CAGR of 13.6% over the last 4 years while the compounded growth of loan and advance is 15.7% per annum accelerating the growth of the business. Loan productivity has increased steadily over the last 4 years.

The holdings in the portfolio of IDLC remain fundamentally strong and are expected to generate value in coming periods and aid in improving the ROE and ROA, which took a fall in 2018. IDLC has been paying cash dividend consistently to its shareholders over the years.

3. **Southeast Bank Limited** is a private commercial Bank, which launched its activities in Dhaka in 1995, providing conventional banking products and services to retail and business clients, along with specialized products/services like investment fund, automatic fraud monitoring. A dedicated unit named as 'RMG, Textile and Export Financing Unit' at Head Office has been handling the credit portfolio of export-oriented clients, providing both project finance and working capital finance. Total lending amount

in RMG sector in the year 2018 is BDT 50,850 million and total lending amount in green projects is in year 2018 is BDT 12,671 million.

As of 2018, Southeast Bank Limited has Authorized Share Capital of 1,500 million ordinary shares of BDT 10 each. The paid-up capital is 1,045 million ordinary shares of BDT 10 each and Earnings per share of in the year 2018 is BDT 2.33. Total asset is BDT 381.095 million showing CAGR of 13.6% over the last 4 years while compounded annual growth rate of loans and advances is 16.1% per annum flourishing the business of the bank. Revenue in the year 2018 is BDT 14,724 million and net profit after tax is BDT 2,456 million in the year 2018. Capital to Risk Weighted Assets Ratio (CRAR) increased in 2018 despite the growth in loan assets, on the back of marked improvement in asset quality, which is manifested in the improved NPL ratio. SEBL has maintained strong equity base, as on December 31, 2018, and it reflects the bank's long-term viability. Improved CRAR of 12.47% in 2018, which indicates that the bank has kept adequate capital providing cushion against the risk bearing asset.

The bank is improving its financial health in all criteria. However, it is being aggressive in case of asset liability mismatch but they are maintaining statutory requirement in adequacy in capital to withstand any shock. Liquidity management is also satisfactory while maintaining profitability, leading to improvement of earning quality in 2018. The Asset quality has improved too in the most recent years and their effective NPL management has made it one of the few commercial banks in Bangladesh with low NPL.

4. **The City Bank Limited** is one of the prominent private banks in Bangladesh. Established in 1983, the Bank provides a wide suite of deposit and loan products and solutions, catering to requirements of the widest socio-economic population cross-section. The authorized Share Capital of the City Bank Limited is 1,500 million ordinary shares with BDT 10 each. Paid-up capital of the bank is approximately 968 million ordinary shares of BDT 10 each. Earnings per share is BDT 2.08. At the end of the year 2018, the lending amount in RMG sector is BDT 39, 251 Million and cumulative disbursement in green projects is BDT 1,359 million.

The total asset of the City Bank is BDT 324.78 billion in the year 2018 registering CAGR of 14.8% per annum over the last 4 years along with consistent growth of loan and advances. The business of the bank has grown rapidly over the 4 years. Cash flow base accrual ratio was observed to be negative in all years except for 2017 demonstrating that earning of bank was of high quality. The spread of the bank is quite high enough to provide high interest revenue margin. Gross NPL ratio is moving downwards over the years along with net NPL ratio which indicates high quality of the loan asset.

The bank is improving its financial health in all criteria. However, it is being aggressive in terms of ADR. But they are maintaining statutory requirement in capital adequacy to withstand any shock. Liquidity management is also satisfactory. Overall, it can be observed that City Bank's core operational income margin has experienced growth; which indicates, the bank is able to run its main operations efficiently. Moreover, they have had sufficient surplus capital adequacy for risk weighted assets; resulting in both having enough cushion to absorb any unpredicted liquidity shock and complying with Bangladesh Bank's regulations as well.

For component 05:

The implementation arrangement has already described in project description in section B

Explain why the project/programme requires GCF funding, i.e. Why is the project/programme not currently being financed by public and/or private sector? Which market failure is being addressed with GCF funding? Are there any other domestic or international sources of financing?

Explain why the proposed financial instruments were selected in light of the proposed activities and the overall financing package. i.e. What is the coherence between activities financed by grants and those financed by reimbursable funds? How were co-financing amounts and prices determined? How does the concessionality of the GCF financing compare to that of the co-financing? If applicable, provide a short market read on the prevailing of the pricing and/or financial markets for similar projects/programmes.

Justify why the level of concessionality of the GCF financial instrument(s) is the minimum required to make the investment viable. Additionally, how does the financial structure and the proposed pricing fit with the concept of minimum concessionality? Who benefits from concessionality?

In your answer, please consider the risk sharing structure between the public and private sectors, the barriers to investment and the indebtedness of the recipient. Please reference relevant annexes, such as the feasibility study, economic analysis or financial analysis when appropriate.

Need for GCF funding for proposed mitigation intervention for Textile Sector (Component 01 & 02):

- **Addressing highest energy savings potential:** Textile is one of the most energy intensive industry in Bangladesh and a transition into an energy efficient, climate resilient, internationally competitive industry will require a sustainable blend of private and public sector funding. In Bangladesh's context, private commercial banks and NBFIs are the dominant players in financing energy efficiency, however, IDCOL continues to be the main non-bank financial institution (NBFI) contributor and promoter of energy efficiency and renewable energy technologies because of its experience in executing successful EE programs with blended finance.
- **Overcoming existing barriers:** There exist some funding and technical cooperation for energy efficiency and conservation in Bangladesh's industry, such as programmes supported by the GoB and multi/bilateral agencies. These include Global Climate Partnership Fund (GCPF), Asian Development Bank's Industry Energy Efficiency Finance Program (IEEFP), IFC's Partnership for Clean Textile (PaCT), GIZ's support for technical cooperation, Bangladesh Bank's Green Transformation Fund, Bangladesh Bank Re-financing Scheme, Bangladesh Bank's Long-Term Financing Facility (LTFF) under Financial Sector Support Project (FSSP) and JICA's Energy Efficiency and Conservation on promotion financing project (see Feasibility study Annex 2) These initiatives were introduced based on the industrial demand and GoB's commitment towards tackling climate change. However, most of such instruments have not been able to meet their full potential due to several challenges like higher lending rate for end-user; difficulties in approval and disbursement due to delay in operations and loan decision; overly ambitious targets for EE; selective assistance either technical or financial; delayed realization of the investment pipeline etc. Further, due to constraints in availability of traditional financing, energy efficiency is not getting adequate funding from the commercial banks to the textile sector.

The key reasons why this funding from GCF will be very important for the sector as follows:

- The scope of the programme is in alignment with the SDG targets— SDG 7, SDG 9 and SDG 13. Sustainable energy use remains of critical importance to the Bangladesh economy. Availability and the rising price of gas is a significant and growing constraint on the growth of industrial Activity.
- IDCOL will target to offer 50 % of the loan amount to medium and small textile factories in Bangladesh, a market which is not fully explored or catered at present form a financing perspective. This will increase their access to debt-based instruments at affordable terms, and ultimately build

their creditworthiness. Without GCF funding, these smaller segment factories wouldn't be able to access available loans from mainstreamed FIs.

- The loan product will be directed to increase energy efficiency across complete value chain in Textile industries in Bangladesh, which will be a unique product catering to energy efficiency in complete value chain of Textile industry (Different from Partnership for cleaner textile (PaCT) led by IFC whose focus area is investment facilitation for financing resource efficiency projects in textile factories)
- The loan facility will have following features which will make it highly attractive in the market.
 - Simplified of loan application procedure and speedy appraisal minimizing the time for loan processing.
 - Availability of Grace period & Long Tenure and low interest.

In addition, since the programme will be implemented by IDCOL it will have following advantage

- Experienced implementing partner. IDCOL has built its capacity by successfully implementing EE&C for JICA Fund.
- Strong pipeline built by IDCOL for Textile and RMG sector while implementing EE&C under JICA will ensure fast disbursement of fund.
- **Ensuring suitable financing options:** As a result of all of the above-mentioned factors, there is a high need for a programme that can support the transition to energy efficiency in Bangladesh. The market failure due to inefficiency and restricted scope of some of these instruments present a learning experience for any new financial instrument. The proposed programme has been designed after a detailed study of the existing and the past market experiences and have incorporated the lessons learned for long term sustainability of the proposed programme. Since the feasibility studies suggest that the requirement for concessional, low interest rate, loan scheme is quite high, the proposed programme will have a lower lending rate as compared to the other lending instruments in market to attract more beneficiaries especially the small and medium textile units for a greater impact. Moreover, it will minimize the time for loan procedure through simplification of loan application and speedy appraisal.

Relevance of the proposed financial instruments and Concessionality

Existing or past EE financing products available in Bangladesh, provided capital at a very high interest rate to the end users. This restricted the clientele base to only large borrowers who can afford high interest rates. Table 6-9, indicated that due to significant price difference of EE equipment a concessional rate is required with a tenor of 8-10 years and grace period of 1-2 years. Analysis of the viability of few⁵⁴ energy efficient equipment for textile industry indicate that 4% to 5% interest rate for local currency loan will be appropriate to compensate the extra capital cost of EE equipment. However, it has been learned from JICA- Energy Efficiency and Conservation programme (Phase 1) that end users are willing to pay up to 6%.

In addition to GCF funds, the programme is proposed to be co-financed equally from IDCOL's own funds as well as the end borrower's equity, thus ensuring all the stakeholders have a significant stake in the programme's success. The co-financing amounts and prices were determined based on the institutional and financial capacities of the beneficiaries and the trends in market. The associated financial assessment is discussed in detail in the feasibility report in annex 2.

Risk sharing

⁵⁴ Two types of equipment were analysed, comparing the cost impact of purchase of conventional equipment and energy efficient equipment by changing tenure period and grace period.

The programme has been designed to address the anticipated risks and ensure that risks are shared and mitigated by the stakeholders best suited for the same. A detailed risk assessment has been carried out & mitigation measures have been identified during the program design (Annex: Risk Report). While IDCOL will be selecting the borrowers to ensure credit risk is minimized, equipment and technology risk will best be mitigated at the end user level. One of the key risks to IDCOL would be on account of currency fluctuation risk since GCF will be lending in USD while IDCOL will be lending to the end users in BDT.

In time series analysis done for exchange rate movement of Bangladeshi Taka against USD from 1998 to 2018 it is observed that Bangladeshi Taka is depreciating on an average rate of 2.94%⁵⁵ every year,

Need for GCF funding for proposed mitigation intervention for RMG Sector (Component 03 & 04):

While adoption of energy efficiency measures can help in reduction of GHG emissions from the RMG sector and increasing in operating margins, the units must have finances available for their implementation. A comparison of the interest rates across major RMG producing economies is given in Figure 15 above for reference.

Even though the interest rates have exhibited a sharp fall in recent years, RMG units in the country are reported to be receiving funding at 10-12% interest rates from the banks and at 11-13% from other financial institutions namely Non-Banking Financial Institutions (NBFIs). The interest rates depend on the lending agency's risk appetite, loan appraisal process and the creditworthiness of the particular RMG unit. In case higher risk is assessed, the interest rates may go beyond this range as well. As per discussions with stakeholders it was established the actual cost of capital of RMG units may vary between 10-17%. The higher risk perception about energy efficiency loan is also contributed by a combination of insufficient quality of collateral from the units and, low collateral asset value of energy-efficient equipment. The target markets of RMG units in Europe, USA have very low inflation rates, leading to near stagnant selling prices. On the other hand, the input markets of Bangladesh experience higher inflation rates, leading to increasing input costs. The labour wages have also been increasing as well as the overall cost of compliance due to the presence of 'Alliance' in Bangladesh's RMG sector. The increasing input cost and stagnated sales prices are reducing the profitability of RMG units. This tightened economic environment negatively affects RMG units' willingness to invest in energy-efficiency projects.

Even future outlook for financing in the country does not provide any indication of lowered interest rates. It is observed that State-owned Commercial Banks (SCBs), which have a 48% share of loans in the country, have been historically performing poorly in the major banking performance indicators. In order to prevent the SCBs from collapsing, Government of Bangladesh has taken the recourse of recurrent recapitalization under which, it has spent USD 1.8 billion during the period FY2009-FY2017⁵⁶. In order to stimulate growth in the country, it is expected that GoB may continue bank recapitalization to ensure capital infusion in the market. Continued bank recapitalization by the Government is expected to constrain the government exchequer to invest in social and physical infrastructure.

There are a number of financing schemes related to energy efficiency and green financing in general. While all of the schemes include energy efficiency as one area of financing and/or RMG sector as one of the participatory sectors, none of them are focused exclusively on energy efficiency improvement in the RMG sector. Discussions with sector stakeholders were undertaken to identify the extent to which the schemes have been able to achieve their intended targets and understand the gaps, if any, which limits the success of these schemes. Based on the discussion the following key gaps have been identified:

⁵⁵ <https://data.worldbank.org/indicator/PA.NUS.FCRF?end=2018&locations=BD&start=2018&view=bar>

⁵⁶ Source: [Monthly Fiscal Frameworks & Budget Briefs, Finance Division \(CPD, December 2018\)](#)

1. **Lower interest rates required:** In order to ensure competitive advantage, the unit owners are operating with continually decreasing margins. In such a scenario, interest rates much lower than the existing market rate (in the range of 5-6%) would be required to encourage unit owners in taking loans exclusively for energy efficiency projects.
2. **Complicated loan application process:** In order to get access to energy efficiency financing, unit owners are required to conduct energy audits. After the audit, the usual loan appraisal process is undertaken by the banks, making the entire process a lengthy and complicated one. Also, unit owners have to pay upfront the cost of energy audit, further adding to the cost of undertaking energy efficiency.
3. **Limited capacity of banks in appraisal process:** Officials from the banks have limited understanding in the technicalities of energy efficiency project though they understand the RMG business. The bank officials have limited access to the standard tools, techniques and framework for appraisal of energy efficiency projects. This leads to lack of focus in the banks to neither develop financial products targeting energy efficiency investment nor conducts processing of loans for such project in a faster and efficient manner.
4. **Non-availability of loans with banks having existing relationship with RMG units:** RMG owners prefer to continue business relationships with their existing bankers rather than starting new relationships and in case of prolonged relationships, RMG owners secure loans at lower interest rates than prevailing market rates from their existing bank. In case the existing bankers do not provide energy efficiency loans, the RMG owners are wary of reaching out to new banks for energy efficiency loan – a deterrent which adds to the issues of energy efficiency financing.

Given this context of Bangladesh and the key concerns highlighted above, concessional financing can help break this vicious cycle by stimulating the willingness to lend by the banks as well as the willingness to invest by the RMG owners in energy efficiency projects.

Risk sharing: The programme has been designed to address the anticipated risks and ensure that risks are shared and mitigated by the stakeholders best suited for the same. A detailed risk assessment has been carried out & mitigation measures have been identified during the program design (Annex: Risk Report). The credit risk, in this case, will be borne by the Executing Entities.

B.6. Exit strategy and sustainability (max. 500 words, approximately 1 page)

Explain how the project/programme sustainability (financial, institutional, social, gender equality, environmental) will be ensured in the long run after project closure, including how the project's results and benefits will be sustained.

Include information pertaining to the longer-term ownership, project/programme exit strategy, operations and maintenance of investments (e.g. key infrastructure, assets, contractual arrangements). In case of private sector, please describe the GCF's financial exit strategy through IPOs, trade sales, etc.

Provide information on additional actions to be undertaken by public and private sector or civil society as a consequence of the project/programme implementation for scaling up and continuing best practices.

Programme sustainability

The programme activities are designed to impact the sector in a way that the growth is inevitable. A catalytic change in financial, institutional, social, gender equality and environment aspect which will have longer influences far beyond the scope and timeline of this programme. (refer performance indicators annex 11)

Based on the interactions with RMG sector stakeholders, it has been established that the environmental, economic and resource conservation benefits of energy efficiency is recognized by over 60% of the top management of the garment manufacturing firms. However, lower prevalence of energy efficiency is due to combination of factors like

1. **Excessive focus on top-line over bottom-line by the factory owners:** In a competitive market scenario, the factory owners are more inclined to invest in adding production capacity to increase revenue i.e. top-line rather than invest in efficiency upgrade to increase profitability i.e. bottom-line. Hence, investment in energy efficiency is not always a high-priority item for the owners.
2. **Capital investment requirement:** The investment payback in energy efficiency is not immediate and depends on the specific measure being undertaken. In most cases, units do not have adequate capital availability to undertake additional capital investment in increasing production efficiency and hence, are sceptical to undertake such an investment. Also, in order to identify energy efficiency initiatives, owners have to expend on hiring energy audit firms. This adds to the cost component for undertaking energy efficiency measures.
3. **Insufficient demonstration examples:** Examples of actual savings achieved from energy efficient equipment is few and far between in Bangladesh. Owners are normally not hesitant on making additional investment when they have case studies which make help them be fully aware of the on-ground requirements for getting benefits from implementing energy efficiency measures.
4. **Lack of standardization of energy efficient technologies and processes:** There is limited market information available to owners to understand the technical specifications of the energy efficient technologies and are completely dependent on the opinion of energy auditors to implement a specific measure. A list of standardized technologies along with normally used technical specifications that can be used in the RMG sector can prove to be a useful tool to apprise owners as well as the on-ground staff about the energy efficient equipment.
5. **Limited market information:** Also, there is limited information available in the market regarding the processes and procedures to be undertaken to identify energy savings opportunities and for assessing the benefits received from the opportunities. Standardized templates and SOPs can help bridge this knowledge gap and instil a sense of confidence in the owners regarding the entire energy savings assessment process.

The proposed programme aims to address these issues by providing concessional loans and technical assistance to stimulate the energy efficiency market in Bangladesh. Since the programme has been designed as loan to the beneficiaries – only the projects which are bankable in nature i.e. projects which, on an average, can generate returns over and above the cost of capital, are considered under the targeted solution set for the programme. Thus, the programme will help kick starting energy efficiency projects in the RMG sector at an accelerated pace. This will generate a strong sector-wide demonstration effect. Hence, it is expected that after the programme implementation period ends (a) unit owners will be far more willing to invest in energy efficiency projects and, (b) banks and FIs will also be ready to offer a greater number of financial products and instruments designed to cater to the appropriate need of the factory owners for investing in energy efficiency projects.

As mentioned above, the programme is expected to improve the risk perception of investment in energy efficiency in the sector by creating strong demonstration effects and increasing the availability of

standardized energy efficient equipment in the market. This in the long run will also drive down the capital costs of the equipment making energy efficiency investments more attractive and viable. Also, the project will help create new asset classes i.e. industrial energy efficiency, putting the right mechanisms in place and building confidence among investors.

Component 02 & 04, in particular, will help to ensure long-term sustainability of the programme, by providing capacity building and technical support to relevant stakeholders such as the textile and RMG factory owners, operators and the select bank officials. This in turn will augment the local capacity to prepare bankable, high-quality energy efficiency projects as well as to appraise those projects for accessing bank credits. Hence, even after the end of the GCF programme post recovery of loans from IE through, the market for energy efficiency in Bangladesh is anticipated to be more mature and self-sustainable.

Financial	Financial support to the industries to procure EE equipment which would otherwise be filtered out of their options due to capital constraints. The EE equipment procured under the programme, if costly, will not burden the textile and RMG industry owners since the cost saving on energy will, in sometime, reach the break-even and the loan is bearable due to the concessionality.
Institutional	Institutional capacities will be built financially, technologically and in terms of the work force. Internationally competitive EE equipment will facilitate growth in terms of production and reduce cost of production from energy savings. Moreover, the textile and RMG industries will timely be reporting the social impact on their workforce especially gender empowerment and the initiatives towards environmental cleaning of their processes.
Social	Institutional and financial growth of the industries will bring about growth of the communities associated with them. The workforce will experience growth in wages as an indirect benefit of industrial growth/profit-making; with expansion in production, older employees will undergo training and new jobs will be created; as a mandate, the industries will have to provide for basic needs like medical facilities, savings account, provision of safer working environments, etc.
Gender equality	A specialized gender action plan is devised to ensure that neither of men or women experience discrimination. There has to be equity in opportunities for financial opportunities in loan disbursement and job opportunities; skill training; provision of special needs like separate washrooms and maternity leaves, improving work environment, etc.
Environmental	The programme scope is designed to continually prevent a great amount of emissions from textile and RMG sectors of Bangladesh. The endorsement and installation of EE equipment will prevent emissions during the programme life cycle and beyond.

Sustainability ex-post

The programme aims to achieve catalytic impact in the textile and RMG sectors on various parameters. These, if successful, would be replicated across other sectors and ensure significant momentum for the objectives. Post 20 years, the impacts of the programme will continue to benefit the environment, society

and the beneficiaries since, the transformational change in societal mindset and technology transfer would have happened which the beneficiaries of the loan can't revert to the two-decade old practices (ex-ante).

GCF'S EXIT STRATEGY

For Component 01 & 02:

The long-term sustainability of the programme and responsible stakeholder behavior will ensure smooth exit for GCF. IDCOL will execute pre-screening of the potential textile end borrowers to disburse fund and or their progress through the respective deliverables as mention in MRE. Evaluation of progress reports and timely repayment of loan will be done to ensure continued sustainability of the sub-projects. These instruments will reduce financial risks for IDCOL. Repayment of GCF's reimbursable funds is IDCOL's responsibility to facilitate GCF to exit the programme once the tenure has been completed.

For Component 03 & 04:

The programme activities under Component 03 & 04, have been designed with a view of developing market for energy efficiency equipment in RMG sector through capacity building and awareness generation activities i.e. market and technical knowledge related barriers as given under the scope of Component 4. In order to overcome the immediate financial barrier that prevents RMG units considering energy efficient equipment, a concessional loan is envisaged under Component 3.

As a result of the activities in the two components, there will be real-life examples of implementation which will demonstrate the actual benefits that can be achieved and hence establish business case behind adopting energy efficient equipment. This will improve risk perception of energy efficiency measures and create a demand push. The participation of banks in the program will ensure that banking sector will become aware of the demand, thereby increasing availability of financing options in response. The market development activities will help increase operations of equipment supplier, thereby increasing market access of energy efficient equipment. The increased demand is also expect to bring down the prices of equipment, making it more affordable. Thus, the GCF funding will kick-start positive market transformation, which will be self-sustaining and continue after GCF closes the program.

The program design also includes screening of loan application based on established due diligence processes of the 4 EEs. This will ensure only those entities which are capable of repayment are selected for loan disbursement, thereby, insulating GCF from any default risk and facilitate safe exit from programme.

For Component 05:

Under this component certain policy will be developed upon proper assessment of policy gap, if any. Sustainable capacity building of key regulatory stakeholders will be ensured through capacity building initiatives. Energy savings data from this program will be reported to SREDA & will be incorporated into national database.

C. FINANCING INFORMATION

C.1. Total financing

(a) Requested GCF funding (i + ii + iii + iv + v + vi + vii)		Total amount		Currency			
		256.48		million USD (\$)			
GCF financial instrument		Amount	Tenor	Grace period	Pricing		
(i)	Senior loans	250.00	20 years	As per term sheet	As per term sheet		
(ii)	Subordinated loans		Enter years	Enter years			
(iii)	Equity						
(iv)	Guarantees		Enter years				
(v)	Reimbursable grants						
(vi)	Grants (TA)	6.48					
(vii)	Result-based payments						
(b) Co-financing information		Total amount		Currency			
		84.02 million		million USD (\$)			
Name of institution		Financial instrument	Amount	Currency	Tenor & grace	Pricing	Seniority
IDCOL (EE)		Senior Loans	33.00	million USD (\$)	As per term sheet	As per term sheet	<u>pari passu</u>
		In kind (PMC)	1.00	million USD (\$)	-	-	=
LFIs (EEs)		Senior Loans	50.00	million USD (\$)	As per term sheet	As per term sheet	<u>pari passu</u>
SREDA		In kind	0.02	million USD (\$)			
(c) Total financing (c) = (a)+(b)		Amount			Currency		
		340.50			million USD (\$)		
(d) Other financing arrangements and contributions (max. 250 words, approximately 0.5 page)		Not applicable					

C.2. Financing by component

Component	Output	Indicative cost million USD (\$) million USD (\$)	GCF financing		Co-financing		
			Amount million USD (\$) million USD (\$)	Financial Instrument	Amount million USD (\$) million USD (\$)	Financial Instrument	Name of Institutions
Component 01: Energy Saving Technology financing for Textile sector	EE equipment & technology installed	133.00	100.00	Senior Loan	33.00	Senior Loan	IDCOL

Component 02 : USD 3.05 million GCF Technical Assistance (TA) to develop enabling environment for EE investment in textile sector	Creation of enabling situation for EE financing	3.05	3.05	Grants			
Project Management Cost (PMC) for Component 01	-	1.00	-	-	1.00	In-kind	IDCOL
Component 03: USD 200 million financing for Energy Saving Equipment & Technology for garments sector	EE equipment & technology installed	200.00	150.00	Senior Loan	50.00	Senior Loan	LFIs/IDCOL
Component 04 : USD 2.30 million GCF Technical Assistance (TA) to develop enabling environment for EE investment in garments sector	Creation of enabling situation for EE financing	2.30	2.30	Grant	...	...	...
Component 05: USD 1.15 million for strengthening regulatory & institutional framework at the national level to overcome the operational constraints related to implementing EE&C in the country	Strengthenin g regulatory & institutional framework	1.15	1.13	Grant	0.02	In-kind	SREDA
Indicative total cost (USD)		430.50	256.48		84.02		

Special Note:

- For Component 1: Energy Saving Technology financing for Textile sector, IDCOL will co-finance (from own source or will arrange co-financing from other sources at own responsibility) the GCF credit line to the tune of USD 33.00 Million. This co-financing will be blended with GCF funds to develop a lending facility for the borrowers or loan component for GCF funding & co-financing will be separate.

- Similarly, for Component 3 the LFI/IDCOL will co-finance (from own source or will arrange co-financing from other sources at own responsibility) the GCF credit line to the tune of USD 50.00 Million. This co-financing will be blended with GCF funds to develop a lending facility for the borrowers or loan component for GCF funding & co-financing will be separate.
- GCF loan will be pari passu with the IDCOL and LFI loans, which means that it will have the same seniority (both senior loans as opposed to senior from IDCOL and subordinated from GCF).

C.3 Capacity building and technology development/transfer (max. 250 words, approximately 0.5 page)

C.3.1 Does GCF funding finance capacity building activities?

Yes ☒ No ☐

C.3.2. Does GCF funding finance technology development/transfer?

Yes ☐ No ☒

The activities under Component 02, including the package of standardized tools which the programme is expected to develop has a strong replication potential in other market segments and elsewhere. The implementation of the proposed intervention includes a number of consultations, technical assistance activities and capacity building process of key implementing partners of the program (i.e., IDCOL and the textile industries) and a strong dissemination and communications strategy around these standard tools and business model. In addition, the monitoring activities during implementation of the Program will provide information and early lessons from good practices to be further replicated in later implementation stages of the Program and disseminated to market players. This would allow to exchange experiences and replicate results in other regions with similar challenges for investment in EE projects.

The activities under Component 04 of the programme intends to develop capacity across a diverse set of stakeholders operating in the RMG sector in relevant topics which will help address any technical barriers impeding growth of energy efficiency equipment in the sector. In case of EEs, capacity for appraising, monitoring and evaluating energy efficient projects is envisaged – this will help the EEs tap into the energy efficiency market thereby increasing fund flows to the segment. The trainings of unit owners, their facility managers and workers on operational aspects and commercial benefits of energy efficient equipment, will help reduce risk perception towards the concept. The programme also intends to develop capacity of policy makers, which will ensure that policies supporting use of energy efficient equipment adopted by Bangladesh, thereby stimulating the market. Market interface events are also planned which will interested RMG units connect with equipment suppliers and get their queries resolved by them - thereby giving the RMG units a first-hand understanding of the benefits.

D. EXPECTED PERFORMANCE AGAINST INVESTMENT CRITERIA

This section refers to the performance of the project/programme against the investment criteria as set out in the GCF's [Initial Investment Framework](#).

D.1. Impact potential (max. 500 words, approximately 1 page)

Describe the potential of the project/programme to contribute to the achievement of the Fund's objectives and result areas. As applicable, describe the envisaged project/programme impact for mitigation and/or adaptation. Provide the impact for mitigation by elaborating on how the project/programme contributes to low-emission sustainable development pathways. Provide the impact for adaptation by elaborating on how the project/programme contributes to increased climate-resilient sustainable development. Calculations should be provided as an annex. This should be consistent with section E.2 reporting GCF's core indicators.

Designed as per GCF's guideline, the programme aims to achieve the objectives of the fund which is to catalyze a flow of climate finance to invest in low-emission and climate-resilient development, driving a paradigm shift in the global response to climate change. It envisages national implementation of the said targets and so, it's a large-scale project with an integrated approach to meet national climate commitments⁵⁷, energy targets⁵⁸ and streamline industrial development with human development. It will support entrepreneurs and business establishments in the textile and RMG sectors, two of the largest industries of the country, with financial and market resources to avail investment opportunities for energy saving technology upgrades. Consequently, the programme has a substantial impact potential in both the textile and RMG sector of Bangladesh.

IMPACT POTENTIAL IN THE TEXTILE SECTOR-

Energy Saving Potential:

The textile industry is one of the major energy intensive industries in Bangladesh with total industrial demand of 1,701 TOE/year which amount to 17% of total energy demand. As per the feasibility analysis, the manufacturing and operational processes in this sector are energy inefficient due to the use of second-hand machineries to save costs and thus has lot of potential for implementing new energy efficient equipment or retrofitting the existing machineries with energy efficient technologies. Because of energy efficiency implementations under the programme in the textile sector, the total energy saving in the sector is estimated to be 6,547 GWh.

Emission Reduction Potential:

The programme is expected to result in direct emission reductions of 4.00 million tCO₂, by facilitating and scaling-up investment in energy efficiency improvements for a number of textile facilities. Energy efficiency improvement projects include both equipment and technology measures in the selected textile factories. The estimated potential for GHG emission reduction in textile factories depending on baseline fuel, is 0.2 million tCO₂e/year or 4.00 million tCO₂ cumulatively over the 20-year investment life-cycle. Emission reductions are calculated based on avoided quantity of fuel consumption by multiplying baseline energy use by relevant GHG emission factor and lifetime of the investment (which is 20 years in this case).

Emission reduction potential per million USD of investment (Cost per tCO₂eq):

The expected programme cost for Textile sector is USD 133 million of which, USD 100 million will be funded through GCF, USD 33.00 million from IDCOL in the form of loan. There will be leverage financing of \$33.00 million in the form of equity investment from end borrowers. Besides, there is a GCF grant for Technical Assistance (TA) to support associated activities. A multidimensional programme as this, needs to be robust enough to resist national market reflexes. The financial feasibilities show that the estimated cost per tCO₂eq

⁵⁷ Bangladesh's INDC and BCCSAP

⁵⁸ EE&C Master Plan 2030

(a/c) is USD 34.33/ tCO₂e and the estimated GCF cost per tCO₂eq removed (b/c) is approx. USD 25.81/ tCO₂e.

IMPACT POTENTIAL IN THE RMG SECTOR-

The programme aims to achieve the fund's objective and result areas by contributing shift to low-emission sustainable development pathways. The contribution is achieved by providing target market i.e. RMG sector of Bangladesh, access to concessional line of financing for undertaking energy efficiency measures which will result emissions savings as well as provide market participants (RMG units, banks, policy makers) with appropriate technical support required for stimulating the energy efficiency market in Bangladesh, as well as to support implementation of energy efficiency initiatives. Implementation of the programme will also help Bangladesh meet national climate commitments⁵⁹, energy targets⁶⁰, contributing to nation's overall efforts related to climate change. Since the programme aims the RMG sector, a major contributor to Bangladesh economy (sector contributes to 11% of nation's GDP and employs nearly 4 million people) it will help streamline industrial development with human development.

Energy Saving Potential:

The garments industry is the most energy intensive industrial sub-sector of Bangladesh with total demand of 2,107 kTOE/year i.e. 15% of total demand of industrial sector. As per situation analysis conducted in feasibility studies, energy consuming processes in the sector employ old equipment and exhibit potential for replacement with equipment which lead to efficient use of energy through incorporation of new technologies. Apart from capital intensive measures, there are system inefficiencies caused due to leakages which can be reduced by undertaking low-cost or no-cost measures. Based on the assessment, an average inefficient unit exhibits an energy savings potential of 873 GWh/year and considering the potential number of units that can be targeted by the project cost (i.e. USD 250 million), the total energy savings is estimated to be 17,468 GWh throughout programme lifetime (21 years).

Emission Reduction Potential:

The programme is expected to result in direct emission reductions of 0.527 million TCO₂e/year by facilitating implementation of energy efficiency measures in RMG units. Energy efficiency improvement projects include both equipment and technology measures in the selected textile factories. Considering the program lifetime i.e. 21 years, total savings of 10.54 million TCO₂e can be achieved. Emission reductions are calculated based on avoided quantity of fuel consumption by multiplying baseline energy use by relevant GHG emission factor and duration in consideration (21 years in case of program lifetime).

Emission reduction potential per million USD of investment (Cost per tCO₂eq):

The expected project cost is USD 200 million, off which USD 150 million will be funded by GCF through concessional loans and USD 50 million through co-financing. With the co-financing component, USD 50 million will be contributed by LFI by way of loans and leverage financing of USD 50 million will be contributed by RMG units through equity participation. In addition, there is a GCF grant of USD 2.30 million for Technical Assistance (TA) for implementation support and development of energy efficiency market in Bangladesh. The investment from the programme perspective as well as GCF perspective is evaluated by term cost per TCO₂e, calculated based on the emission savings estimated. If the programme lifetime is considered, cost per TCO₂e for programme value and GCF's contribution are USD 19.20/TCO₂e and USD 14.45/TCO₂e respectively.

⁵⁹ Bangladesh's INDC and BCCSAP

⁶⁰ EE&C Master Plan 2030

Leakage emissions from resale of equipment

Detail leakage calculation has been described in Annexure 11.

D.2. Paradigm shift potential (max. 500 words, approximately 1 page)

Describe the degree to which the proposed activity can catalyze impact beyond a one-off project or programme investment. Describe the following, if applicable:

- *Potential for scaling up and replication*
- *Potential for knowledge sharing and learning*
- *Contribution to the creation of an enabling environment*
- *Contribution to the regulatory framework and policies*
- *Overall contribution to climate-resilient development pathways consistent with relevant national climate change adaptation strategies and plans*

The programme enables larger flows of finance for energy efficiency investment and shifting the established paradigm and pathways the investments must be made. The paradigm shift potential for the proposed programme lies in the programme's focus on the private sector as the driving force for investment and implementation of EE retrofits, as opposed to current models which only focus on large industries.

PARADIGM SHIFT POTENTIAL FOR TEXTILE SECTOR

Potential for scaling up and replication

The potential to scale-up the project is great and has been well-incorporated into the project design through the establishment of an efficient financial mechanism and monitoring framework to encourage investments and ease access to finance for textile factories besides enabling further investment decisions.

Its replicability within Bangladesh and abroad is high. A transformational programme as this will invite more textile factories to participate in EE adoption and up gradation, and energy efficiency itself and enable climate mitigation through scaling up of an established market base developed as a project outcome. Neighboring countries like Indonesia, Vietnam, and Cambodia with strong textile industrial sector and similar barriers and risks to EE investments and may benefit by learning from successful projects in Bangladesh which is a high-volume supply market.

Potential for knowledge and learning

The programme will contribute to knowledge creation and sharing by all market players. The two workshops have sensitized the textile fraternity on availability of fund and knowledge sharing.

The programme has a strong component of communication and knowledge sharing through consultations,, capacity building process of key implementing partners of the program (i.e., IDCOL and the textile industries) and a strong dissemination strategy around these standard tools and business model. In addition, the monitoring activities during implementation of the programme will provide information and early lessons from good practices to be further replicated in later implementation stages of the programme allowing exchange of knowledge and learnings.

Contribution to the creation of an enabling environment

The programme will enable market-based environment through policy, finance, technical / capacity de-risking and barrier removal. There will be deep penetration of energy saving technologies, equipment and appliances in the textile industries. The program enables to create a self-reliant ecosystem for both end users and suppliers of energy saving technologies. Helping to build scale and volumes for energy saving technology suppliers, the program will bring down the costs of these technologies and make them attractive even at commercial financing rates.

Contribution to the regulatory framework and policies

The program will contribute to the country-wide policies/programs to bring down the sectorial emissions and the cost of energy saving technologies and systemically promote investment in low-emission or climate-resilient development.

Overall contribution to climate-resilient development pathways consistent with relevant national climate change mitigation/adaptation strategies and plans

As per the commitments of GoB⁶¹, programme will mainstream climate-resilient development pathways into policies and regulatory frameworks and decision-making processes at national, regional and local levels, including private-sector decision-making through this programme will contribute in emission reduction by promoting energy efficiency and establishing self-sufficient markets. The investments enabled through this program will lead climate resilient development.

PARADIGM SHIFT POTENTIAL FOR RMG SECTOR

The programme exhibits substantial potential to affect a paradigm shift in the application of energy efficiency measure for RMG sector in Bangladesh and catalyse impact beyond a one-off project or programme investment.

- **Potential for scaling up and replication:** The model adopted by the programme will help establish the bankability of energy efficiency initiatives and help create a number of demonstration projects which will help overcome the market apprehensions regarding energy efficiency which is leading lower uptake of such technologies. The two-pronged approach of providing concessional financing and providing technical support for implementation, will help address the barriers preventing growth of energy efficiency markets for RMG sector. With industries sector in Bangladesh having substantial savings potential, this approach exhibits potential for replication within the country to achieve further emission savings in future.
- **Potential for knowledge sharing and learning:** As per the programme design, TA consultants will provide implementation support to RMG units and MRV support to Executing Entities (EEs)/ LFI for the project duration. In conclusion of the term, TA consultants will prepare documents which summarize the key issues faced by the two stakeholder groups along with the implemented solutions. This will provide future project proponents with access to learnings from previous projects and accordingly, these learnings can be incorporated to the project design to ensure better implementation and benefits realization.
- **Contribution to the creation of an enabling environment:** The TA component will help develop the energy efficiency market in Bangladesh for RMG sectors. Overall, the programme, through its TA and concessional loan components, aims to overcome the financial and technical barriers which are preventing uptake of energy efficiency measures. Technology fairs are planned in the program design to create an interface with the end users i.e. RMG units and the energy efficient equipment manufacturers/suppliers. The increased uptake through the programme is expected to create scale required for improved access to energy efficient equipment, in terms of availability and lower costs.
- **Contribution to the regulatory framework and policies:** Under the TA component, capacity building of policy makers of Bangladesh in fields related to energy efficiency in industrial sphere is planned. This is expected the policy makers with insights on the potential contribution energy efficiency can have on

⁶¹ Bangladesh's INDC and BCCSAP

Bangladesh's economy along with the steps taken globally to develop enabling policies and regulations to promote energy efficiency. Such capacity building activities along with the demonstration effect of the programme is expected to increase market consciousness regarding importance of energy efficiency and accordingly, contribute to future steps in developing enabling policies and regulations.

- **Overall contribution to climate-resilient development pathways consistent with relevant national climate change mitigation strategies and plans:** The programme aims to contribute to the NDC targets of Bangladesh to reduce GHG emissions in the power, industry and transport sectors by 15% below 'business-as-usual' GHG emissions by 2030 through support is received from developed countries. The programme will also contribute to the targets of reducing energy consumption for industrial sector as given in the Energy Efficiency and Conservation Master Plan up to 2030. Hence, the programme is making positive contribution to the national climate change mitigation strategies and plans.

D.3. Sustainable development (max. 500 words, approximately 1 page)

Describe the wider benefits and priorities of the project/programme in relation to the Sustainable Development Goals and provide an estimation of the impact potential in terms of:

- *Environmental co-benefits*
- *Social co-benefits including health impacts*
- *Economic co-benefits*
- *Gender-sensitive development impact*

The objective is to utilize an integrated suite of interventions to systematically de-carbonize the textile and RMG sector factories to realize both energy and financial savings, without compromising sustainability and sectorial growth/development. Delivering a large-scale retrofit initiative through this programme will have large and important development benefits whose impacts will increase over time as energy prices rise. Further, the availability of domestically produced gas is also severely constrained which will necessitate more textile units to depend on grid for electricity needs. The changing energy scenario will stress the government to revise fuel prices due to import costs and manage the existing supplies. The proposed programme will:

- Share the load of the government by reducing the energy consumption on the demand side
- Support energy conservation targets of the Government.

Further, the proposed programme will address the gender related concerns in the textile and RMG sector and promote gender equality. Based on gender assessment it is evident that gender-based discriminations in terms of wages, overtime, promotion, skill development, healthcare facilities and occupational health and safety are some of the burning issues in the textile and RMG sector in Bangladesh. The programme will try to address these issues through its Gender Action Plan and ensure that women feel safe, equal and encouraged to hold their position in the sector for longer period. Moreover, the programme also intends to promote leadership capacity among the female workers for enhanced sustainability of the sector and overall development.

SUSTAINABLE DEVELOPMENT IN THE TEXTILE SECTOR

Environmental co-benefits

There are direct and indirect benefits to the environment through the proposed programme.

- Most importantly, the installation and operation of EE equipment will lead to GHG emission reduction contributing significantly to climate change mitigation.

- Contribution to improved air quality due to reduced emissions as a macro-impact
- Use of EE effluent treatment plants (ETPs), preventing discharge of untreated water into streams and saving energy while on 24x7 running status
- Reduction in wastes of product, water, hazardous materials, raw materials use and effective reutilization of waste heat

Social co-benefits including health impacts

EE technologies and equipment will result in direct benefits on the society i.e. worker community and the communities residing near the textile factories. Industrial growth and development will lead to better skills and wages thereby up scaling the social status of the textile dependent households.

- Streamlining of local and sectorial economies through increased hiring of local labor, creation of green jobs
- Capacity building and skill training through training modules proposed in the programme to encourage social inclusion
- Improved worker safety and health (resulting in reduced lost work) from user friendly technology upgrades, upgraded operational practices, decreased noise and improved air quality

Economic co-benefits

The economic benefits from the programme will influence almost all the relevant stakeholders. There's a great potential to economically benefit from the concessions provided under this programme, and the consequent energy saving.

- Major economic savings due to reduced spending on energy and, as a result, reduction of energy poverty among textile factories.
- Improved production and capacity utilization and reduced production costs (operations and maintenance, raw materials), which leads to enhanced productivity and competitiveness
- Potential to enable increased investments for energy saving technologies enabling positive externalities in the form of expanded and enhanced job markets, job creation and poverty alleviation increased energy security etc.
- Potential for utilities to improve their supply-side efficiency due to reduced energy demand from textile sector and increased demand side energy affordability of textile factories.

Gender-sensitive development impact

As per the Fund's standards, it's mandatory to encourage and ensure women's participation all through the programme lifetime and beyond.

- Improved access of women to investments and to information about textile sector energy efficiency
- Broader participation of women in opportunities for skill training, business and financial independence.
- Reducing manually operated machinery through deployment of easily operable by women
- Raised awareness of gender equity through implementation of the programme's Gender Action Plan
- Increased efficiency would also facilitate flexible working shift for women

Deployment of EE equipment will increase production efficiency and call for job creation and suitably designed work shifts for women

SUSTAINABLE DEVELOPMENT IN THE RMG SECTOR

- **Environmental co-benefits:**
 - Installation and operation of energy efficient equipment will lead to GHG emission reduction contributing significantly to climate change mitigation.
 - Due to reduction in emissions, improved air quality is envisaged as a macro-impact
 - Resource efficiency resulting from effective reutilization of waste heat, reduction in wastes of product, water, hazardous materials, raw materials use will lead to lower environmental degradation caused due to resource extraction.
- **Social co-benefits including health impacts**
 - Streamlining of local and sectorial economies through increased hiring of local labor, creation of green jobs
 - Capacity building and skill training through training modules proposed in the programme to encourage social inclusion
 - Improved worker safety and health (resulting in reduced lost work) from user friendly technology upgrades, upgraded operational practices, decreased noise and improved air quality
- **Economic co-benefits**
 - Increased profitability of RMG units by reduction of fuel costs leading to positive economic contribution
 - Creation of new jobs among equipment suppliers, operations and maintenance technicians due to market development of energy efficiency equipment
 - Positive impact on financial conditions of RMG units, potentially leading to reinvestment in capacity expansion that can help create new jobs within the sector.
- **Gender-sensitive development impact**
 - Increased entrepreneurship and leadership in RMG sector.
 - Reduced gender gap in terms of wages and thus encourage women to continue their job for longer periods.
 - Creating enabling environment for women by providing them medical facilities, childcare facilities, WASH and special provisions for pregnant women so that women do not have to leave their job due to health or family related issues.
 - Broader participation of women in opportunities for skill training, business and financial independence
 - Increased awareness of gender equity through implementation of the programme's Gender Action Plan
 - Develop a gender sensitive grievance redressal mechanism for better functioning and monitoring of the proposed interventions.
 - Thus, the gender action plan will address the issues mentioned in the section 5 in the gender assessment and enhance gender equality in the RMG units, enhanced probability of capacity expansion, leading to a gender-sensitive sustainable development.

D.4. Needs of recipient (max. 500 words, approximately 1 page)

Describe the scale and intensity of vulnerability of the country and beneficiary groups and elaborate how the project/programme addresses the issue (e.g. the level of exposure to climate risks for beneficiary country and groups, overall income level, etc.). Describe how the project/programme addresses the following needs:

- *Vulnerability of the country and/or specific vulnerable groups, including gender aspects (for adaptation only)*
- *Economic and social development level of the country and the affected population*

- *Absence of alternative sources of financing (e.g. fiscal or balance of payments gap that prevents government from addressing the needs of the country; and lack of depth and history in the local capital market)*
- *Need for strengthening institutions and implementation capacity*

Bangladesh is one of the most adversely affected countries in terms of climate change. The topography is flat, and majority of the landmass lies within 10 meters above mean sea level. The geographic location at the convergence of the three great rivers – the Ganges, Brahmaputra and Meghna – and near the Eastern Himalayas. All land types except highlands are exposed to monsoon flooding for part or whole of the year⁶². Floodplains located in the north-western, central, south-central and north-eastern regions are subject to regular flooding at different frequency and intensity while the coastal plain is subject to cyclones and storm surges, salinity intrusion and coastal inundation. Pleistocene terrace land is characterized by moisture stress while flash flood is common in the hilly areas and the piedmont plains in the northeast and north-western parts of the country. Such Climate events are disrupting the socio-economic growth and stability of the country. Moreover, Bangladesh is becoming increasingly vulnerable for air pollution, sea level rise, high tide and river flood by cyclone potentially caused by GHG emissions from rising fossil fuel consumption.

This is despite the total GHG emissions in the country being less than 0.35% of the global emissions. The country's fossil emissions since 2000 have grown by 6.8%⁶³ Without a collective ambition to limit GHG emissions internationally, the impact of climate change to Bangladesh could be catastrophic resulting in annual loss of 2% of GDP by 2050 and 9.4% of GDP by 2100. The country is listed on the top 10 most affected countries of the world in the Climate Risk Index for 2017⁶⁴.

The proposed programme directly contributes to the collective global action under UN Paris Climate framework to help limit GHG emissions and mitigate climate change impacts in the country. Being a developing country, Bangladesh is also in the list of LDCs by GCF and hence a focus area for the fund.

Also in the current energy supply scenario, the public sector investments in Bangladesh is focused more on supply side interventions in order to ramp up the generation capacity and bridge supply gaps. This limits the budgetary resources available with the government to drive public sector investments for demand side interventions, especially towards energy saving technologies. In this scenario, boosting private sector investments is critical to keep investment cycle going in important markets such as industry, buildings and household appliances. However, one of the major barriers for driving private sector investments for energy efficient technology upgrades is the limited availability of attractive financing availability. Financing barriers such as weak capital market, high cost of lending, high investment costs as well as off-taker credit quality and complex approval process all limit the availability of financing for energy efficiency initiatives. In this scenario, supporting businesses with attractive financing products and mechanisms for energy saving technology upgrades could go a long way in getting the attention of corporate top management and overcoming this important market barrier.

For Textile and RMG manufacturers, available means of financing is bank loans or equity contribution. Under the current market scenario Textile and RMG units have to lend at a rate of 9-15% based on the credit worthiness and credit history of unit. While the bank-lending has grown over 15% since 2016⁶⁵, credit risk in the sector continues to be high. It has been observed that credit risk in banking sector will continue to be elevated⁶⁶ owing to loan-restructuring practices, historically weak credit-risk-management practices, concentrated loan books, exposure to single large borrowers, and government-directed pressure on state-owned banks' lending decisions. The inherent high level of risk of lending explains the high lending rates, as

⁶² Bangladesh NAPA (<https://unfccc.int/resource/docs/napa/ban01.pdf>) 2005; updated 2009

⁶³ <https://www.worldometers.info/co2-emissions/bangladesh-co2-emissions/>

⁶⁴ German watch

(https://www.germanwatch.org/sites/germanwatch.org/files/Global%20Climate%20Risk%20Index%202019_2.pdf)

⁶⁵ Source: [Bangladesh Systemic Risk Dashboard \(Bangladesh Bank, December 2018\)](#)

⁶⁶ Source: [Banking sector developments in Bangladesh \(IHS Markit, 2018\)](#)

compared to nearest competitors in RMG export market. Current efforts to stabilize banks by recapitalization by Government of Bangladesh is expected to continue leading to expectations that high interest rates will continue to prevail. Such high interest rates negatively affect credit influx into the sector at a time when profitability is strained due to high input costs caused by inflation and increased labour wages and due stagnated market prices in customer markets in USA and EU.

Besides the financing barriers, energy efficiency uptake also technical challenges such as lack of awareness and understanding of the long term benefits by consumers, perceived underperformance risks in energy savings, focus of short term benefits and equipment availability all of which also hamper the growth of energy efficiency products. These can be mitigated to some extent by demand side awareness and capacity building. Based on discussion with stakeholders, it was established that there is limited capacity in banks to assess bankability of energy efficiency projects as well as continued MRV of such projects to assess whether project objectives are realized. There are also technical knowledge gaps in RMG units regarding implementation of energy efficient measures. Also, policies and regulations supporting proliferation of energy efficiency in industries sector should be enhanced to ensure market barriers are removed. Accordingly, capacity building of policy makers on global best practices on energy efficiency policy development is essential to provide guidance on developing similar policies/regulations for Bangladesh.

D.5. Country ownership (max. 500 words, approximately 1 page)

Please describe how the beneficiary country takes ownership of and implements the funded project/programme. Describe the following:

- Existing national climate strategy
- Existing GCF country programme
- Alignment with existing policies such as NDCs, NAMAs, and NAPs
- Capacity of Accredited Entities or Executing Entities to deliver
- Role of National Designated Authority
- Engagement with civil society organizations and other relevant stakeholders, including indigenous peoples, women and other vulnerable groups

The programme directly contributes to the country's ambitions and goals under the UN Paris Climate framework. Moreover, the program is consistent with the country's national strategy (Energy Efficiency Master Plan) to provide 'Energy Efficiency Finance Program' to boost the investments in energy efficiency products with low interest loans as one of the key financial incentives. Furthermore, the program also contributes to the GoB target to achieve 10% of energy consumption reduction in the industry sector by 2030 and 20% improved energy intensity by 2030.

Bangladesh's NDC identified three key emitting sectors, amongst which power sector accounts for 33% of total emission, transportation 26% and industrial sector 41%. Taking this into consideration, improving industrial energy efficiency, expanding renewable energy and modernising transport infrastructure have been identified as the three key mitigation interventions. This has been embedded in our Five-Year Planning process that is central to the development of Bangladesh. NDC has set the target to reduce 15% GHG emission below Business As Usual (BAU) emission level by 2030.

To arrange necessary funding for low emission development, the Economic Relations Division of the Ministry of Finance, also the NDA of GCF in Bangladesh developed a Country Programme in 2018 with support from GCF. The screening process of NDA filtered the most strategically important climate change projects/programme that should be funded under GCF. The Country Program includes a robust pipeline of projects and programs which will be submitted to GCF for its financial support. IDCOL as a Direct Access

Entity of Bangladesh is now working on developing funding proposals for a couple of these projects. The proposed programme is also listed in the Country Programme.

For IDCOL's and LFI's capacity to execute the proposed programme please refer to section B.4. The NDA advisory committee conducted a due diligence of the proposed programme and IDCOL as an executing entity prior to issuing NOL.

This proposal has been developed in consultation with civil society groups and other relevant stakeholders and would continue to seek their future engagement in accordance with the GCF's environmental and social safeguards and stakeholder consultation guidelines. The proposal also places decision-making responsibility with in-country financial institutions and uses domestic systems to ensure accountability.

D.6. Efficiency and effectiveness (max. 500 words, approximately 1 page)

Describe how the financial structure is adequate and reasonable in order to achieve the proposal's objectives, including addressing existing bottlenecks and/or barriers, and providing the minimum concessionality to ensure the project is viable without crowding out private and other public investments. Refer to section B.5 on the justification of GCF funding requested as necessary.

FOR TEXTILE SECTOR

The programme's results chain is an approach to market transformation for energy efficiency. This approach is based on the fact that, due to the high upfront capital intensity of energy efficient investments in the textile sector, access to large quantities of low-cost financing is critical to cost-effectively drive energy efficiency in the textile sector.

By conducting payment simulations for a few types of equipment in B.3, the proposed financial structure was deemed to be adequate and reasonable to achieve programme's objective:

- The purchase of an energy efficient spinning machine which is ~25% more expensive compared with a conventional type will require a 5.53% interest rate loan (considering 8 year loan tenure and 1 year grace period) for the total payment to become par with that of a conventional one. Similarly, a 27% more expensive once-through boiler requires 5.28% interest rate (considering 8 year loan tenure and 1 year grace period) to be even with the conventional type in terms of the total payment amount.
- On the other hand, in order to overcome the barrier of longer payback period, the loan tenor was increased to 10 years, with 2 years grace period. With such, the purchase of an energy efficient spinning machine will require a 6.67% interest rate loan (considering 10-year loan tenure and 2 year grace period) for the total payment to become par with that of a conventional one. Similarly, once-through boiler will require 6.4 % interest rate (considering 10-year loan tenure and 2 year grace period) to be even with the conventional type in terms of the total payment amount.

The above analysis clearly demonstrates that to ensure the feasibility of the subproject, concessional lending is required from GCF to IDCOL to enable IDCOL to extend loan at the low interest to the end borrower. IDCOL is currently implementing JICA's Energy Efficiency and Conservation Promotion Financing Project (EECPF) which has been so far the most successful EE credit line in Bangladesh. Thus, for the proposed programme IDCOL is using EECPF project as the benchmark for the end user financing package.

Besides just comparing the market value of the traditional and EE equipment, further analysis in terms of assessing the NPV of operational cost of both types of machineries were done to determine the right interest rate that will provide appropriate IRR for the end users. Please refer to Table 12.

The table 12 below compares the NPV of additional investments required for installing the energy efficient spinning and boiler machines the proposed GCF credit line vis-a-vis regular equipment under the standard commercial financing:

Table 12: NPV & IRR comparison of commercial finance, JICA EECPPF project and proposed programme

Machine Type	Δ investment NPV if GCF loan is availed (i=5%,tenor-10yrs, moratorium-2 yr) compared to commercial loan is (i=11.5%,tenor 8 yrs, moratorium-1 yr)	Δ investment NPV if GCF loan is availed (i=6 %,tenor-10yrs, moratorium-2 yr) compared to commercial loan (i=11.5%,tenor 8 yrs, moratorium-1 yr)
Spinning machine	BDT 4,81,998	(BDT 29,986)
Boiler	BDT 1,87,760	(BDT 17,823)

The above table clearly indicates that only an interest rate in the range of 5%-6% for the GCF credit line would make the additional investment in EE equipment feasible for the borrowers. Thus, crowding out of private investors will not occur at this rate and at the same time the financial structure will be viable to the end users. The Δ investment in the above table signifies the overall savings that a borrower will accrue when an energy efficient equipment is installed viz-a-viz a conventional equipment. Even though the costs of EE equipment are on higher side than that of a conventional machine, the savings in terms of operating cost over a lifetime and a tenure of 8-10 years, outweigh the higher cost of EE equipment. The life of machine is considered to be 15 years in both the cases. It is assumed that the spinning machine runs on electricity and an incremental electricity tariff of 2% is considered for calculating the operating. The EE equipment for each machine type shall have a minimum efficiency of 10% over conventional machine,

With the current JICA EE&CPF credit line, there are some challenges faced such as (a) it focusses on varied sectors and hence, cannot cater the complete demand of any one of the sectors. Thus, sector approach proposed in this programme will help to cater the huge demand and potential of energy efficiency in the textile sector and will percolate deep inside the sector. (b) The second issue with JICA EE&CPF involves the limitations in ticket size of one project (maximum BDT 1 billion) which further minimized the reach of this project across industrial sector. Thus, in the proposed programme the project ceiling has been extended to BDT 2 billion. (c) The last point is that, this project has a limited list of EE interventions that may be financed under this project. These limitations do not help the project to percolate deep inside any sector with an approach to bring the entire sector to world standards norms and energy intensity standards. Hence keeping this in mind the proposed programme is designed to accommodate all kind of textile equipment having a minimum efficiency level of 10%.

FOR RMG SECTOR

High upfront capital investment and high interest rates of borrowing has been informed by stakeholders as a key barrier from implementing energy efficient technologies, even if the technologies exhibit a high

expected rate of return. The program intends to address this barrier through concessional financing and supplement it with market development activities which would help stimulate market demand, increase availability of energy equipment and develop capacities of market players.

The level of concessionality which will stimulate market for energy efficiency has been determined by simulating loan payment scenarios (considering 8-year loan tenure and 2-year grace period) for 2 sample equipment:

- **Conventional vs Energy Efficient boiler:** The energy efficient gas boiler is around 25% more expensive than a conventional gas boiler and will require a 5.15% interest rate to have the same repayment amount as that of a conventional gas boiler being financed by loans at market rate i.e. 11%. Considering a 10% premium on repayment, the interest rate becomes 7.55%.
- **Conventional vs energy efficient stenter:** Similarly, a 23% more expensive energy efficient stenter requires 5.35% interest rate to have the same repayment amount as a conventional stenter with financing at market rates. 10% premium makes the interest rate 7.85%.

Accordingly, a loan payment simulation has been conducted to compare the rate of interest at which an energy efficient equipment will have the same repayment as a loan taken at market rate for a comparable conventional equipment, all other terms of the loan remaining the same (viz. loan tenor and moratorium/grace period). The simulation has been conducted for two sample equipment viz. gas fired boiler and stenter machines as given below.

It is assumed that since energy efficient equipment gives an additional benefit of monetary savings through energy conservation, users i.e. unit owners can be inclined to pay a small premium to acquire the energy efficient variant. This would also help financial institutions give some margin to accommodate any operational costs. The results from the simulation is given below:

Table 13: Price simulation for Energy Efficient Gas Boiler

Parameter	Unit	Commercial Finance for conventional equipment	Commercial finance for energy efficient equipment	Concessional finance for energy efficient equipment	
				At par	10% Premium
Price	USD	57,600	72,000	72,000	72,000
Interest Rate	%	11%	11%	5.15%	7.55%
Tenure Period	yrs.	10	10	10	10
Grace Period	yr.	2	2	2	2
Total Payment	USD	89,543	111,929	89,661	98,527

Table 14: Price simulation for energy efficient stenter

Parameter	Unit	Commercial Finance for conventional equipment	Commercial finance for energy efficient equipment	Concessional finance for energy efficient equipment	
				At par	10% Premium
Price	USD	64,800	80,000	80,000	80,000
Interest Rate	%	11%	11%	5.35%	7.85%
Tenure Period	yrs.	10	10	10	10
Grace Period	yr.	2	2	2	2
Total Payment	USD	100,736	124,365	100,428	110,736

Considering a wider range of equipment, a range of interest between 5.0%-7.5% should be ideal market prices of loan instruments to generate interest in unit owners for energy efficient equipment.

Hence, a minimum concessional financing at ~5.0-7.5% would be required to make energy efficient equipment attractive to unit owners. Given the additional benefit of using energy efficient equipment is monetary savings (in the form of energy savings), it is expected that unit owners will be more inclined to buy an energy efficient variant over a non-energy efficient one, even with a 10% premium on repayment, thereby creating a demand push for efficient variants. This would, however, require extensive awareness building among unit owners on the business case for energy efficiency as well as increasing access of energy efficient equipment.

The financial structure for the program includes 04 local financial institutions, which will act as EE for the program. However, during programme implementation new LFIs may be selected based on the selection criteria described in B4, notably IDCOL may also be an EE if required. For now the shortlisted 04 LFIs. These banks have proven experience of operating in RMG financing market as well as offering green financing instruments. They have established project due diligence procedures aligned with the statutory requirement of Bangladesh. The presence of these institutions in the financial structure will help in taking the concessional loan to a greater number of RMG units, given existing customer base of the banks as well as increase efficiency in disbursement and monitoring given the alignment of the bank in operating green financing instruments.

The co-financing provided by the banks will help in infusing private sector financing in the sector and with the success of the program, it is expected more financial institutions (both public and private sector) will increase participation in the sector.

Apart from the issues discussed above, a number of stakeholders were consulted during program design, which has highlighted need for capacity building in appraising, implementing, monitoring and evaluating energy efficiency projects as well as improve market interface between equipment suppliers and RMG units. These non-financial barriers, which are also critical to the success of the program, have also been addressed through activities proposed in Component 2.

Please describe the efficiency and effectiveness of the proposed project/programme, taking into account the total financing and mitigation/ adaptation impact the project/programme aims to achieve, and explain how this compares to an appropriate benchmark.

Please specify the expected economic rate of return based on a comparison of the scenarios with and without the project/programme.

Please specify the expected financial rate of return with and without the Fund's support to illustrate the need for GCF funding to illustrate overall cost effectiveness.

FOR TEXTILE SECTOR

As mentioned above, this programme envisages the creation of concessional fund with USD 133 million, including USD 100 million senior loan from GCF and USD 33 million from IDCOL. IDCOL, an Accredited Entity with GCF, will make available the GCF funds in the form of loans to the end borrowers, who are in this case the private textile manufacturing companies. Using this fund, the projected IRR estimated to be achieved is 7.0 %. The implementation of these energy efficiency measures equivalent envisages annual GHG reductions estimated at 0.2 million tCO₂ and 4.00 million tCO₂ equivalent over the lifetime of the project (20 years) which if included in the economic benefit would lead to an EIRR of 23.00%. The payback period for various equipment varies between 4-13 years. In addition to these economic and financial benefits, the project will also lead to other intangible social and gender impacts such as better community living standards, providing safe and better working environment leading to higher gender balance in workforce and skill up gradation for workers.

FOR RMG SECTOR

The proposed programme envisages to infuse USD 200 million in financing energy equipment in RMG units. This will include USD 150 million of concessional loans from GCF and USD 50 million from EEs i.e. LFI/IDCOL in the form of loans. The loans will be provided by LFI/IDCOL to the end borrowers at a rate substantially lower than prevailing market rates (~11%). Using this fund, the projected IRR is 5.00%. The program will lead to annual GHG emissions of 0.527 MtCO₂ and savings of 10.54 MtCO₂ across the program lifetime. Considering the economic benefit of the emission savings, the program returns an EIRR of 44.5%. In addition to these economic and financial benefits, the project will also lead to other intangible social and gender impacts such as better community living standards, providing safe and better working environment leading to higher gender balance in workforce and skill up gradation for workers. It will also lead to higher operating profit for units, a part of which is expected to get reinvested in capacity expansion, leading to job creation. The increased demand for energy efficient equipment will also lead to creation of green jobs.

Please explain how best available technologies and practices have been considered and applied. If applicable, specify the innovations/modifications/adjustments that are made based on industry best practices.

FOR TEXTILE SECTOR

Sector wide best practices can be broken down into the process improvement, metering water conservation, fuel efficiency and reduction in electricity consumption. These measures represent the best practices for resource efficiency and a cost effectiveness perspective in the textile processes.

Table 15: Current best practices for energy efficiency in Textile sector

Type	Detail
Energy Management and	- Policies and standard operating procedures for energy management

process improvement	- Development of skills and knowledge management for awareness - Information systems for monitoring and reporting - Improve planning and investment - Marketing and communications for encouraging other units/firms - Process automation to control dyeing and printing processes - Improve machine utilization, particularly for the most energy-intensive machines - Benchmark energy and water use and set concrete reduction targets
Metering	- To enable factory management to closely track resource and energy consumption for specific processes and to identify those resource intensive processes where efforts to improve efficiency should be focused
Water conservation is energy conservation	- Water leak detection, preventive maintenance, improved cleaning - Improving boiler efficiency - Improve efficiency of steam systems and maintain steam traps - Insulation of equipment and tanks - Heat recovery and reuse: Reuse cooling water, Reuse condensate, Reuse process water
Fuel savings from above mentioned water conservation and reuse by:	
Efficient use of Fuels	a) Improving the operation and efficiency of the boiler b) Increasing efficiency in the use of steam in the production process to reduce the quantity of steam the boiler must generate.
Reduction in Electricity consumption	- Optimization of compressed air systems: fixing leaks in the air system and checking and optimizing pressure - Electricity savings from improved boiler efficiency - Electricity savings from heat recovery

The above-mentioned practices are easy-to-implement and low-cost that come with multiple co-benefits. Besides energy efficiency, these would bring in resource efficiency, environmental benefits, cost effectiveness and increase in productivity of textile units. These can be taken up as capacity building part of the TA programme.

FOR RMG SECTOR

Garment production is an organized activity consisting of sequential processes such as laying, marking, cutting, stitching, checking, finishing, pressing and packaging. Based on best practices in each process interventions were suggested to the stakeholders. These measures represent the best practices for resource efficiency and a cost effectiveness perspective in the garment manufacturing processes.

Table 16: Current best practices for energy efficiency in RMG sector

Type	Detail
	- Installation of VFD on major motors

Energy Management and Conservation	- Optimization of compressed air system - Installation of auto oxygen control in boilers - Installation of energy conservation turbine
Fuel Reduction	- Improving boiler efficiency - Improve efficiency of steam systems and maintain steam traps - Insulation of equipment and tanks - Heat recovery and reuse: Reuse cooling water, Reuse condensate, Reuse process water - Increasing efficiency in the use of steam in the production process to reduce the quantity of steam the boiler must generate.
Reduction in Electricity consumption	- Optimization of compressed air systems: fixing leaks in the air system and checking and optimizing pressure - Electricity savings from improved Lighting system - Electricity savings from use of clutch motors - Installation of IE3 motors - Installation of energy efficient screw compressors
The above-mentioned energy conservation measures are easy-to-implement and low-cost that come with multiple co-benefits. Besides energy efficiency, these would bring in resource efficiency, environmental benefits, cost effectiveness and increase in productivity of RMG units. These can be taken up as capacity building part of the TA programme.	

E. LOGICAL FRAMEWORK

This section refers to the project/programme's logical framework in accordance with the GCF's [Performance Measurement Frameworks](#) under the [Results Management Framework](#) to which the project/programme contributes as a whole, including in respect of any co-financing.

E.1. Paradigm shift objectives

Please select the appropriated expected result. For cross-cutting proposals, tick both.

- ☒ Shift to low-emission sustainable development pathways
- ☐ Increased climate resilient sustainable development

E.2. Core indicator targets (RMG & Textile combined)

E.2.1. Expected tonnes of carbon dioxide equivalent (t CO ₂ eq) to be reduced or avoided (mitigation and cross-cutting only)	Annual	0.727 million t CO ₂ eq
	Lifetime	14.53 million t CO ₂ eq
E.2.2. Estimated cost per t CO ₂ eq, defined as total investment cost / expected lifetime emission reductions (mitigation and cross-cutting only)	(a) Total programme financing	340.50 million USD
	(b) Requested GCF amount	256.48 million USD
	(c) Expected lifetime emission reductions	14.53 million t CO ₂ eq
	(d) Estimated cost per t CO₂eq (d = a / c)	23.43 USD / t CO₂eq
	(e) Estimated GCF cost per t CO₂eq removed (e = b / c)	17.65 USD / t CO₂eq
E.2.3. Expected volume of finance to be leveraged by the proposed project/programme as a result of the Fund's financing, disaggregated by public and private sources (mitigation and cross-cutting only)	(f) Total finance leveraged	167.02 million USD
	(g) Public source co-financed	0.02 million USD
	(h) Private source finance leveraged	167.00 million USD
	(i) Total Leverage ratio (i = f / b)	
	(j) Public source co-financing ratio (j = g / b)	0.0000780
	(k) Private source leverage ratio (k = h / b)	0.65
E.2.4. Expected total number of direct and indirect beneficiaries, (disaggregated by sex)	Direct	Click here to enter text. Click here to enter text.% of female
	Indirect	Click here to enter text. Click here to enter text.% of female
	<i>For a multi-country proposal, indicate the aggregate amount here and provide the data per country in annex 17.</i>	
E.2.5. Number of beneficiaries relative to total population (disaggregated by sex)	Direct	Click here to enter text. (Expressed as %) of country(ies)
	Indirect	Click here to enter text. (Expressed as %) of country(ies)
	<i>For a multi-country proposal, leave blank and provide the data per country in annex 17.</i>	

E.3. Fund-level impacts

Expected Results	Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions
				Mid-term ⁶⁷ (After Year 06)	Final ⁶⁸ (After Year 12)	
M3.0 Reduced emissions from buildings, cities, industries and appliances	M3.1 Tonnes of carbon dioxide equivalent (t CO ₂ eq) reduced or avoided - buildings, cities, industries, and appliances	- Annual report on energy savings from end borrowers & reflected in dashboard and its respective GHG emission savings. - Yearly Energy Audits and technical support to the LFIs on MRV during TA implementation period (please refer to Activity 2.4.3, 4.4.1 & 4.4.2). - After the end of TA implementation period, IDCOL and LFIs will continue to monitor and collect data from the borrowers. As per the loan agreement the end borrower for EE loan sub-project (proponent) should submit energy efficiency data formats provided in Annex 11 (to IDCOL and LFIs) on yearly basis.	0 tCO ₂ e	1.07 million tCO ₂ e	6.66 million tCO ₂ e	- Improved results in energy savings and avoidance of GHG emissions - Key assumptions used in calculating the ex-ante GHG impact is described in section D.1 - Programme lifetime: 20 years.⁶⁹ - Machine lifetime: 15 years - Mid-term target is defined as cumulative GHG savings expected to be achieved by mid-term point; by this time GCF loan amount and IDCOL's & LFI's co-financing amount will be disbursed to borrowers. Since disbursements will start from Year 02 (from

⁶⁷ Midterm point defined as 6 (six) years after Programme start (FAA Effective Date)

⁶⁸ Final point defined as 12 (twelve) years after Programme start (FAA Effective Date)

⁶⁹ The subprojects under the programme has a lifetime of 10 years. IDCOL and LFIs will re-invest the reflows from the programme till Year 10, which will be fully recovered by Year 20, hence the programme lifetime is 20 years.

		4. Furthermore, the end borrower for EE loan sub-project (Proponent) should submit data and information required during inspection and energy audit to IDCOL and GCF.				FAA effective date) the maximum operational year of sub-project/s will be no more than 05 years. 6. Actual values of cumulative GHG savings achieved is will be reported annually. 7. As per GCF approved methodology the average annual GHG emission reduced is 0.73 MtCO₂. Subsequently with a machine lifetime of 15 years and programme lifetime of 20 years the lifetime GHG emission reduction is 14.53 MtCO₂.
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E.4 Fund-level outcomes

Expected Outcomes	Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions
				Mid-term (After Year 06) ⁷⁰	Final (After Year 12) ⁷¹	
M7.0 Lower energy intensity of buildings, cities, industries and appliances	<i>M7.1 Energy intensity/improved efficiency of buildings, cities, industries and appliances as a result of Fund support</i>	1. Annual report on energy savings from end borrowers & reflected in dashboard	Baseline for textile industry-0 GWh	Textile industry-2618 GWh	Textile industry-5,237 GWh	1.Rebound effect due to lower energy intensity is limited 2. Energy saving data reported to GCF will be disaggregated

⁷⁰ Midterm point defined as 6 (six) years after Programme start (FAA Effective Date)

⁷¹ Final point defined as 12 (twelve) years after Programme start (FAA Effective Date)

		2. Yearly Energy Audits during TA implementation period. 3. After the end of TA implementation period, IDCOL and LFIs will continue to monitor and collect data from the borrowers. As per the loan agreement the end borrower for EE loan sub-project (proponent) should submit energy efficiency data formats provided in Annex 11 (to IDCOL and LFIs) on yearly basis. 4. Furthermore, the end borrower for EE loan sub-project (Proponent) should submit data and information required during inspection and energy audit to IDCOL and GCF.	Baseline for RMG industry- 0 GWh Cumulative baseline - 0 GWh	RMG industry- 2988 GWh Cumulative- 5,608 GWh	RMG industry- 9,773 GWh Cumulative- 15,011 GWh	industry wise. Annual energy saving per industry (i.e. textile & RMG) will be presented along with programme cumulative number. 3. Key assumptions in calculating the ex-ante energy savings impact is described in section D.1 4. Mid-term target is defined as cumulative energy savings expected to be achieved by mid-term point; by this time GCF loan amount and IDCOL's & LFI's co-financing amount will be disbursed to borrowers. Since disbursements will start from Year 02 (from FAA effective date) the maximum operational year of sub-project/s will be no more than 05 years. 5. As per GCF approved methodology the average annual energy saving is 1,310 GWh Subsequently with a machine lifetime of 15 years and programme lifetime of 21
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						years the lifetime energy saving is 24.015 GWh
M5.0 Strengthened institutional and regulatory systems	<i>M5.1 Institutional and regulatory systems that improve incentives for low-emission planning and development and their effective implementation</i>	-Report on gap identification -Policy, strategy or plan developed - Number of Government officials trained under Output 5.2 verified using attendance list. - Effectiveness of the GoB capacity development activities verified using training feedback forms, outputs of facilitation techniques used and interacting activities performed.	0 systems	Development & operationalization of new effective national policy, strategist plan (for example. Energy labeling for efficiency)	Policy implementation to increase private investment in EE	1. Presently there is an opportunity create enabling policies/ regulations to prompt uptake of energy efficient equipment in the industrial sector. By identifying the policy gaps a concrete action plan will be developed. However, top of the mind recommendation of SREDA is to strengthen the EE labelling program. 2. The EE&C Master Plan, mentions about promoting EE Labeling program which will be targeted at large energy consuming entities and equipment in the industrial, residential and commercial sectors. 3. Expected plans to be included in the policy: (a) Accreditation of laboratories for energy efficiency measurement tests (c) evaluate energy manager/auditor licensing system

						(d) increase participation of manufacturers in the EE labeling program. 4. An indicator-based assessment framework having a transparent underlying theory on how policies would impact on the selected indicator and what other factors may also have an influence. This will be developed under Activity 5.1.4.
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E.5. Project/programme performance indicators

Expected Results	Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions
				Mid-term (After Year 06)	Final (After Year 12)	
Financing for Energy Saving Equipment & Technology for Textile & RMG sector	Amount of loan fund approved during the targeted project timeline	Approval letters form IDCOL/LFIs & acceptance by the end borrowers	0	Textile: USD 133 million RMG: USD 200 million	Textile: USD 164.93 million RMG: USD 251.56 million	1. Amount considering expected market demand in the Textile & RMG EE sector. 2. Given access to appropriate finance, textile & RMG factories will invest in high climate impact technologies. 3. To maximize the impact of the programme fund the principal payment received

						from GCF loan, on-lent to the end borrowers will be re-invested till Year 10 (starting from FAA effectiveness).
	EE equipment & technology has been successfully installed in the factory	Factory Visit report by IDCOL/LFIs	0	43 factories	55 factories	1. Considering that all the regulatory & other external requirements of the projects have been completed smoothly. 2. Disbursements under the programme will start from Year 02 (since FAA effectiveness), by Year 06 GCF loan amount and EE's co-financing amount will be fully disbursed to the end borrowers so the number of factories installing EE equipment will be higher for the first 6-7 years of the programme.
	Sector-level baseline, operational manual, list of renowned EE equipment and financing roadmap & time	Acceptance letters from IDCOL/LFIs for adopting operation manual, sector-level baseline, list	0	Adoption and functioning of operation manual, whistle blower policy &	Adoption and functioning of operation manual, whistle blower	1. GCF will approve the procurement process and the prepared documents by 2nd quarter of Year 02 2. A baseline by 2nd quarter of Year 02

<i>Sustainable financing strategy developed for large scale adoption of EE equipment in textile & RMG sector.</i>	based action plan both textile & RMG programs are developed	of renowned EE equipment, financing road map & time based action plan under the programme		baseline information	policy & baseline information	to be developed by consolidating data based on classification of Textile & RMG units (scale and type of products) and data collection through energy audit of sample plants 3. All the required factory visits for baseline are completed within stipulated time
	Regulatory & institutional framework at the national level strengthened to overcome the operational constraints related to implementing EE&C in the country	1.Report on policy gap identification 2.Policy, strategy or plan developed 3. An indicator-based assessment framework having a transparent underlying theory on how policies would impact on the selected indicator and what other factors may also have an influence. This will be developed under Activity 5.1.4. 4. Number of Government officials trained under Output 5.2 verified using attendance list.	0	1.Developm ent & operationali zation of new effective national policy & strategic plan 2. Organize 03 ToTs of the government officials 3. Organize 10 training session for the policy makers and relevant government official by the trained trainers. 4. Organizing 02 workshops to disseminate outcome of the programme to policy makers and	1.Develop ment & operational ization of new effective national policy & strategic plan 2. Organize 03 ToTs of the governmen t officials 3. Organize 10 training session for the policy makers and relevant governmen t official by the trained trainers. 4. Organizing 02 workshops to disseminat e outcome of the	1. The EE&C Master Plan, mentions about promoting EE Labeling program which will be targeted at large energy consuming entities and equipment in the industrial, residential and commercial sectors. 2. Expected plans to be included in the policy: (a) Accreditation of laboratories for energy efficiency measurement tests (c) evaluate energy manager/auditor licensing system (d) increase participation of manufacturers in the EE labeling program.

		5. Effectiveness of the GoB capacity development activities verified using training feedback forms, outputs of facilitation techniques used and interacting activities performed.		EE practitioners and also the learning from implementing energy efficiency in textile and garments sector (financing scenario, business case and economic benefits, women involvement and role of government for long-term sustainability) to ensure institutional capitalization of knowledge.	programme to policy makers and EE practitioners and also the learning from implementing energy efficiency in textile and garments sector (financing scenario, business case and economic benefits, women involvement and role of government for long-term sustainability) to ensure institutional capitalization of knowledge.	
	Programme Launching event arranged for both Textile & RMG program	Attendance list ensuring key stakeholders attended	0	Covering entire key stakeholders of the country related to textile & RMG sector.	Covering entire key stakeholders of the country related to textile & RMG sector.	1. Covering entire key stakeholders of the country related to textile & RMG sector. Trade bodies like BTMA, BGMEA, BKMEA etc. are expected to be on board. 2. Programme Launching event for the for potential borrowers, local & foreign suppliers, O&M service providers, Trade

<i>Market awareness on importance of EE equipment in textile & RMG sector strengthened</i>						Bodies, Fashion Brands and other key private and public sector stakeholders to create awareness on the new GCF credit line
	Develop effective communication materials for information dissemination (brochure, flyer, pamphlet, etc) and advocacy campaign for Textile	-Number of communication materials developed -Quick survey to solicit information recipient's feedback and check their reactions and comprehension.	0	Minimum 70% of the target audience reached as per the action plan	Minimum 70% of the target audience reached as per the action plan	Communication materials will be developed and will focus the information need of various stakeholders.
	Technology fair with participation of technology suppliers and representation of Textile/RMG units organized	Attendance list ensuring key stakeholders attended Feedback survey of key technology suppliers and representatives of Textile/RMG	0	-Minimum 70% of the target participants should be satisfied	- Minimum 70% of the target participants should be satisfied	1. One international technology fair will be arranged for both textile & RMG industries 2. Technology fair with participation of 30 different technology suppliers and representation of 60 different Textile/RMG units organized.
<i>Institutional capacity building at IDCOL/LFIs and at end borrowers level</i>	Improved capacity/skills of stakeholders, end borrowers, IDCOL & LFIs ensured to better cater to EE financing	Training feedback forms, outputs of facilitation techniques used and interacting activities performed	0	-Minimum 80% of the target audience should be satisfied - Minimum 80% of the target audience should be able to	-Minimum 80% of the target audience should be satisfied - Minimum 80% of the target audience should be able to	The training sessions will be arranged by the consultant and the participants will be selected by LFIs/IDCOL.

				articulate the message delivered.	articulate the message delivered.	
E.6. Activities (Component 01&2, EE&C in Textile Sector)						
Activity	Description	Sub-activities	Deliverables			
Financing for Energy Saving Equipment & Technology for Textile sector						
Activity 1.1.1 Disbursement of USD 133.00 million credit line for EE equipment in textile sector	IDCOL will extend USD 133 million amount of loans to textile borrowers for purchasing EE equipment over a tenor of 10 years with maximum 03 years of grace period. From the USD 133 million, GCF loan contribution will be USD 100 million while the rest USD 33 million will be co-financed by IDCOL as the Executing Entity. IDCOL will revolve the loan over 10 years.	1.1.1.1 Prepare financing strategy 1.1.1.2 Develop strong project pipeline & mapping 1.1.1.3 Approach large number of prospective textile borrowers 1.1.1.4 Selection of end borrowers 1.1.1.5 Approval and disbursement of USD 133 million against USD 33 million equity	Loans of USD 133.00 million extended to implementing entities			
Sustainable financing strategy developed for large scale adoption of EE equipment in textile sector.						
Activity 2.1.1: Devise a financing roadmap and time-based action plan	The financing roadmap will be a comprehensive document that will list the tasks for achieving programme's objectives.	2.1.1.1 Prepare financing strategy via primary and	1. IDCOL and GCF approved financing roadmap 2. Time-based action plan			

	Each action will be clearly described so as to avoid confusion later on. Every action will have set timeline, designated resources and a detail implementation plan. The action plan will a strategic document for IDCOL in order to ensure concrete steps needed to attain programme goals are undertaken (usually covering the coming year), a timetable for each task, a description of who will do what and a follow-up process.	secondary data collection. 2.1.1.2 Prepare a detail action plan keeping in mind the targets set in the programme	
Activity 2.1.2: Develop operations manual for the program.	- An operations manual will be developed to ensure stakeholders can reliably and efficiently carry out their tasks with consistent results as per GCF policies & guidelines. It will reduce human error and inform everyone precisely what they need to do, who they are responsible to and who they are responsible for. - A documented operations manual will aid in making EE projects more scalable. When there will be documented procedures for reference, it will be much easier to implement the same procedures on a larger scale or at other locations/ sector.	2.1.2.1 Study GCF policies and guidelines that are applicable under the programme. 2.1.2.2 Study market best practices for similar programme. 2.1.2.3 Develop operations manual for the program	IDCOL and GCF approved Operational Manual for the programme.
Activity 2.1.3: Develop whistle blower policy for the program.	A whistle blower policy will be prepared for the programme as per the guidelines of GCF and Bangladesh Bank to maintain transparency. This policy will be	2.1.3.1 Analyze GCF policies and guidelines that are applicable under the programme.	IDCOL and GCF approved Operation Manual for the programme.

	applicable only for Textile sector EE financing as a separate whistle blower policy will be prepared under RMG TA component.	2.1.3.2 Study market best practices for similar programme. 2.1.3.3 Develop whistleblower policy based on best practices.	
Activity 2.1.4: Classification of Textile units based on scale and type of products and data collection through energy audit of sample plants	There are various sizes of textile factories in the country. Generally they can be widely categorized into three segments i.e large, medium and small. Each categories have different energy consuming pattern. Therefore segregation of factories based on their production capacity will be done primarily. And then the baseline scenario of each category will be identified separately in the sector-level baseline. This is important to understand appropriate energy saving scope of the industry. Rather than an individual factory baseline prior to every loan application is to ensure (i) Loan application process does not get elongated; (ii) Baseline numbers reflect accurate conditions and are not influenced by owners to depict a conducive scenario for loan application; (iii) Cost burden does not fall on borrower completely.	- 2.1.4.1 Sample textile units for the baseline study to be divided into large, medium and small-scale plant, based on its production data and predetermined criteria that gives proper representation of the textile industry.	1 nos. report on Textile unit classification
Activity 2.1.5: Consolidation of data	Baseline for the programme for the selected indicators in the logframe will be	2.1.5.1 In Year 01 (since FAA effectiveness) energy audits to be conducted in	1 nos. document containing data sheets (which includes baseline information)

and developing sector-level baseline	developed under this activity. A sector wide study, undertaking several variables with a definitive methodology (described in the operations manual) will be conducted at the beginning of the programme. This data will be used for 5 years and later a second baseline study will be conducted. (please refer to activity 2.1.8) The main reason for proposing a consolidated baseline rather than an individual factory baseline prior to every loan application is to ensure (i) Loan application process does not get elongated; (ii) Baseline numbers reflect accurate conditions and are not influenced by owners to depict a conducive scenario for loan application; (iii) Cost burden does not fall on borrower completely.	multiple plants ensuring adequate representation from each category (large, medium & small) and each type of product (viz. knitwear, woven, non-woven). For each category and type of product, audits will help establish production processes, equipment involved in each process and specific energy consumption (SEC) for each process. Based on the representative numbers and total population figures, the baseline energy consumption will be prepared for Textile sector 2.1.5.2 Consolidate data from energy audit 2.1.5.3 Identify appropriate assumptions to ensure realistic extrapolation of audit results 2.1.5.4 Undertake extrapolation of audit results	
Activity 2.1.6: Interact with stakeholders/market players to prepare a list of equipment already implemented in Bangladesh and having proven energy savings results. List will be integrated with knowledge platform and	Establishing a real-time database of energy efficient machineries related to both the sectors that has strong track record. By track record it is meant those EE equipment that are widely accepted and used in Bangladesh markets as energy efficient	2.1.6.1 Consultant to conduct textile sector EE equipment's market study based on primary and secondary data 2.1.6.2 Consultant to develop list of renowned & reliable EE equipment that can be financed to the textile sector borrowers.	1 nos. document containing equipment list

disseminated through other means to ensure adequate public outreach	machineries and also those which has been financed under other DFIs/ donor organization in the country.	2.1. 6.3 The EE equipment list developed is to be made widely accessible through online knowledge platform developed under Activity 2.4.6.	
Activity 2.1.7: Revision of baseline by conducting energy audit and data consolidation	Subsequent to activity 2.1.6 the proposed activity is the second baseline study that is to be conducted by the end of 5th year. The main reason for proposing a consolidated baseline rather than an individual factory baseline prior to every loan application is to ensure (i) Loan application process does not get elongated; (ii) Baseline numbers reflect accurate conditions and are not influenced by owners to depict a conducive scenario for loan application; (iii) Cost burden does not fall on borrower completely.	2.1.7.1 In year 06 (since FAA effectiveness) energy audits to be conducted in multiple plants ensuring adequate representation from each category (large, medium & small) and each type of product (viz. knitwear, woven, non-woven). For each category and type of product, audits will help establish production processes, equipment involved in each process and specific energy consumption (SEC) for each process. Based on the representative numbers and total population figures, the baseline energy consumption will be revised for the Textile sector 2.1.7.2 Consolidate data from energy audit 2.1.7.3 Identify appropriate assumptions to ensure realistic extrapolation of audit results 2.1.7.4 Undertake extrapolation of audit results	1 nos. report on revised energy baseline for sector
Activity 2.1.8: Develop technically-robust, bankable EE project pipeline	This will be an ongoing activity for 6 years. Consultant/consulting firm will be responsible to promote the	2.1.8.1 Approach large number of prospective textile borrowers. 2.1.8.2 Develop strong project pipeline & mapping	Technically-robust, bankable EE project pipeline for textile sector.

	programme in the textile sector and develop a strong relationship with prospective clients/borrowers. Activities 2.2.1 to 2.2.4 will be some of the key platforms to connect with the borrowers and sensitize them on EE. This pipeline will be very important to ensure smooth implementation of the programme within the specified time.		
Market awareness on importance of EE equipment in textile sector strengthened			
Activity 2.2.1: Organizing programme launching event	A grand opening of the programme through a social event that will be intended to introduce the new concessional credit line and its unique features dedicated to promote EE in Textile Sector. And also to introduce the relevant public and private sector stakeholders to a new business modality using EE equipment.	2.2.1.1 An event organizing firm will be hired through a competitive bidding process. The procurement process will be facilitated by the consultant/consulting firm. Or IDCOL can also arrange it at own arrangement 2.2.1.2 The participants (local & foreign) will be selected by IDCOL 2.2.1.3 Photographs, attendance list, post-event report, media publication of the programme launching event will be shared with GCF.	1 nos. program launch event
Activity 2.2.2: Organizing awareness raising workshops	This activity focuses on developing the market for the programme by reaching out to the relevant private sector stakeholders through awareness raising workshops. The workshops will	2.2.2.1 Case studies and workshop presentation materials to be developed by consultant/consulting firms with support from IDCOL. 2.2.2.2 The participants (local &	Awareness raising workshops

	disseminate to stakeholders on broad policy framework and incentive mechanisms for energy efficiency in textiles, business case and economic benefits, women involvement in energy efficient textile operations (Covering entire key stakeholders of the country related to textile sector. Trade bodies like BTMA, BGMEA .etc. are expected to be on board). With interactive approaches, participants will be able to acquire new knowledge, share their experiences and develop new practices together related to adopting EE equipment in the textile sector.	foreign) will be selected by IDCOL 2.2.2.3 On board NDA for country ownership. 2.2.2.4 Interactive sessions with the participants to understand the effectiveness of the workshop. (group work, individual/group presentation, open floor discussion, Q&A sessions with experts, feedback forms etc.) 2.2.2.5 Photographs, attendance list, post-event report, media publication of the programme launching event will be shared with GCF.	
Activity 2.2.3: Preparing various information/communication materials	Written and visual communication materials to be developed in the form of brochures/flyers/pamphlets and other related information documents for dissemination activities and for effective marketing of the programme.	2.2.3.1 Consultant/consulting firm will develop written and visual communication materials for marketing of the programme amongst both the public and private sector stakeholders. 2.2.3.2 Consultant/consulting firm as well as IDCOL to ensure access to communication materials for the key stakeholders. 2.2.3.3 IDCOL to measure the effectiveness of the communication materials through survey.	Brochures, flyers, pamphlets, press release, website, and other communication materials.
Activity 2.2.4: Develop and launch advocacy campaign on energy	Advocacy campaign via workshops, animations, video clips, brochures, etc will be the starting point	2.2.4.1 Develop campaign materials to raise awareness in the	Campaign materials (brochure, pamphlet, video clips, case studies on best practices, round table

efficiency & conservation.	for implementation of Component 05. This campaign will be focused on improving/reforming the current policy design and institutional framework regarding adoption of EE equipment in the textile sector and ensure its effective implementation.	public and private sector. 2.2.4.2 Learning dissemination document and share success stories	discussion, press release, etc) Quick survey reports to solicit information recipient's feedback and check their reactions and comprehension regarding the campaign materials.
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Capacity developed at the institutional levels for faster adoption of EE technology

Activity 2.3.1: Capacity needs assessment of IDCOL and potential borrowers.	Perform capacity needs assessment to map the current and desired level of capacity for IDCOL staff and potential borrowers.	2.3.1.1 Consultants will develop need assessment report mapping IDCOL's strengths and weaknesses to implement the programme	A report stating the institutional gaps of IDCOL to implement the programme, ways of overcoming its and the required resources.
Activity 2.3.2: Conduct capacity building workshops and training sessions for IDCOL, stakeholders and end users who are the applicants for the loan on the tools and requirements to be fulfilled for successfully securing the loan	Capacity building and training, including support for the development of necessary financial instruments, risk management tools, procedures and the creation of an adequate knowledge base to evaluate and extend EE loans	2.3.2.1 Consultant will develop training manuals in close consultation with IDCOL and sectorial experts. 2.3.2.2 Consultant along with IDCOL will identify potential participants for the trainings. 2.3.2.3 Consultant will conduct the training via experts. 2.3.2.4 Training reports will be developed identifying the effectiveness of the training and the satisfaction of the attendees	- Training modules - Attendee list - Post-training/workshop reports - Attendee feedback forms - Photographs of the training

Standards and mechanisms for adequate monitoring, reporting and verification of energy savings and GHG emissions developed

Activity 2.4.1: Updating the parameters and requirements for	As per the Bali Action Plan to ensure the programme is implemented in "measurable,	2.4.1.1 All relevant quantitative and qualitative information	MRV framework, including energy audit and post retrofit performance evaluation for the textile factories
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establishing MRV framework	reportable and verifiable” manner a framework will be developed. - The MRV framework will enhance transparency of the tracking of GHG emission levels, climate finance flows received or the impact of mitigation actions proposed under the programme. - MRV framework will also facilitate sharing information and lessons learnt and allow assessing whether set targets have been achieved under the programme with relevant stakeholders, especially the GoB. - MRV framework will also strengthen trust of climate finance donors and other investors for replication of EE&C projects.	to be identified and verified. 2.4.1.2 Detail methodology on how to measure qualitative and quantitative indicators should be developed. 2.4.1.3 Perform energy audit and post retrofit performance evaluation for the textile factories (establishing as-is baseline and EE investment-based evaluation KPIs) 2.4.1.4 Plan the monitoring process 2.4.1.5 Ensure institutions, entities, arrangements and systems involved in domestic MRV 2.4.1.6 Update information in the framework.	
Activity 2.4.2: Updating parameters and develop the monitoring framework for environment and gender evaluation	- Gender-sensitive monitoring and evaluation is important to reveal whether the programme addresses the different priorities and needs of women and men, to assess if it has an impact on gender relations, and to determine the gender aspects that need to be integrated into monitoring and evaluation systems. - Monitoring is fundamental to environmental impact assessment (EIA) both to assess	2.4.2.1 All relevant quantitative and qualitative information to be identified and verified. 2.4.2.2 Determine indicators 2.4.2.3 Detail methodology on how to measure qualitative and quantitative indicators should be developed. 2.4.2.4 Plan the monitoring process 2.4.2.5 Ensure institutions, entities, arrangements and	Monitoring framework for environment and gender evaluation under the programme

	adherence to standards of the programme and to support management options. The use of indicators will assure that the key variables associated with significant environmental impacts are addressed as per proposed ESMF and improves communication and reporting processes.	systems involved in monitoring framework. 2.4.2.6 Update information in the framework.	
Activity 2.4.3: Yearly energy, financial and environmental audits	To report achievements against the set targets in the annual progress report to GCF, yearly energy, financial and environmental audits will be conducted to monitor and track the progress against the indicators.	2.4.3.1 Subcontracting of audit firms for independent evaluation if required. 2.4.3.2 Conduct audits in selected locations and prepare audit reports.	Audit reports
Activity 2.4.4: Revising and applying guidelines and monitoring methodologies for textile factories	- Revise guidelines and monitoring frameworks after scheduled review. - A pragmatic model for assessing whether guidelines and monitoring framework need to be updated will be identified in the operational manual.	2.4.4.1 Schedule review of the guidelines and monitoring frameworks 2.4.4.2 Perform literature searches 2.4.4.3 Model for assessing whether a guideline or monitoring framework needs updating must be determined in the operations manual 2.4.4.4 Drafted guidelines and monitoring frameworks to be shared with GCF for review and comments.	Draft revised guidelines and monitoring frameworks to be shared with GCF. The revised documents will be adopted under the programme after GCF and IDCOL Board approval

Activity 2.4.5: Develop an interactive MIS dashboard/database	To ensure organized collection of data that will be stored and easily accessed electronically from a computer system an interactive MIS dashboard/database will be developed. This activity will be conducted by the consultants under TA. The consultant/consulting firm will be responsible for development, operation, maintenance & handover of a MIS system for knowledge management, business networking, query management, loan application process and automating M&E process. This MIS will be accessible to key public and private sector stakeholders.	2.4.5.1 Gather business requirements 2.4.5.2 Create IT architecture 2.4.5.3 Platform development, testing & deployment 2.4.5.4 Maintenance till 5th year 2.4.5.5 Knowledge transfer to AE through trainings, guidebooks etc 2.4.5.6 Handover to AE	1 operational MIS/digital platform with functioning modules
Activity 2.4.6: Revision of applicable environmental and social policies of IDCOL and gender specific actions, if required	• Revise guidelines, policies and action plans after scheduled review. • A pragmatic model for assessing whether guidelines and monitoring framework need to be updated will be identified in the operational manual.	2.4.6.1 Schedule review of the documents 2.4.6.2 Perform literature searches 2.4.6.3 Model for assessing whether documents needs updating must be determined in the operations manual 2.4.6.4 Drafted documents to be shared with GCF for review and comments.	Draft revised documents to be shared with GCF. The revised documents will be adopted under the programme after GCF and IDCOL Board's approval

Activity 2.4.7: Review/audit of environmental and social safeguard compliances	Monitor borrowers' compliance to environmental and social safeguard compliances as per the lending agreement.	2.4.7.1 Subcontracting of audit firms for independent evaluation if required. 2.4.7.2 Design audit as per guidelines of programme's ESMF 2.4.7.3 Conduct audit	Audit report of borrowers' environmental and social safeguard compliances
Advisory support to the Textile Borrowers to shorten appraisal timeline			
Activity 2.5.1: Guiding and supporting borrower's for faster disbursement	Prepare guidance documents with step-by- step explanation of process for (a) IDCOL's project appraisal/due diligence (b) Development of check list on list of supporting documents and guides on where and how they can be available. (c) This will be accompanied with a vendor list to provide a ready reference for sourcing equipment and seeking quotation. This activity will be performed by the consultants/consulting firms procured under the TA. The consultants/consulting firms will interface with the customers to make the credit application process user-friendly.	2.5.1.1 Guidance documents 2.5.1.2 Face to face guidance services 2.5.1.3 Compile probable list of vendors and integrate with document	Guidance document for borrowers containing all relevant information
Engagement with global clothing brands to increase EE adoption in textile sectors			
Activity 2.6.1 Round table discussions with global brands to align textile units to the brands' sustainability targets	Through round table discussion this activity aims to strengthen participation of global fashion brands in the programme to ensure wider penetration of the	2.6.1.1 Select participants in discussion with fashion brands, trade bodies and SREDA. 2.6.1.2 Organize 01 round table discussion	• Post event report • Attendance list • Photographs/Video • Press brief • Presentation materials

	programme activities in the textile sector. By partnering with the global fashion brands the programme will strengthen and promote EE adoption in the textile sector by establishing a healthy business environment, training the workers, improving social compliance status and building relationships among the manufacturers, exporters and importers, with the goal of increasing Bangladesh's export market.	between fashion brands, textile factories & policy makers	
Activity 2.6.2 Stakeholder consultation in collaboration with global brands	IDCOL in collaboration with the global brands will organize stakeholder consultation meetings to identify ways in which textile units can be encouraged to improve on the social and gender inclusion in workforce ratio to increase the co-benefits impacts under the programme.	2.6.2.1 Identify the participants 2.6.2.2 Organize stakeholder consultation meetings	- Meeting minutes - Attendance list - Photograph - Presentation materials
Activities (Component 03&04, EE&C in RMG Sector)			
Activity	Description	Sub-activities	Deliverables
Financing for Energy Saving Equipment & Technology for RMG sector			
Activity 3.1.1 Disbursement of USD 200.00 million credit line for EE equipment in textile sector	IDCOL through the EEs (LFIs) will extend USD 200 million amount of loans to RMG borrowers for purchasing EE equipment over a tenor of 10 years with maximum 03 years of grace period. From the USD 200 million, GCF loan contribution will be USD 150 million while the rest	3.1.1.1 Prepare financing strategy 3.1.1.2 Develop strong project pipeline & mapping 3.1.1.3 Approach large number of prospective RMG borrowers 3.1.1.4 Selection of end borrowers 3.1.1.5 Approval and disbursement of USD	Loans of USD 200 million extended to implementing entities

	USD 50 million will be co-financed by Executing Entities. IDCOL will revolve the loan over 10 years. EEs/ LFI will re-invest the principal amount received on the GCF contribution which will to additional to GCF proceeds	200 million against USD 50 million equity	
Preparatory activities & Baseline energy consumption of garments sector			
Activity 4.1.1: Devise operation manual for the program	Develop manual elucidating stakeholders involved in the program, activities to be undertaken, roles and responsibilities etc.	4.1.1.1 Understand program design 4.1.1.2 Identify stakeholders and role in program 4.1.1.3 Create program manual aligned with specified timelines	1 no. operational manual
Activity 4.1.2 Develop whistle blower policy for the program	Develop policy to ensure stakeholders can flag fraudulent activities or activities which deviate from stipulations set in program manual	4.1.2.1 Understand requirements for whistle-blowing 4.1.2.2 Develop policy laying out procedure for recording and assessing any complaint and safeguarding the whistle-blower	1 nos. whistle blower policy document
Activity 4.1.3 Organizing programme launching event,	Programme launch event, to introduce the new concessional credit line and its unique features dedicated to promote energy efficiency in RMG Sector	4.1.3.1 Identifying invitees and sending out invites 4.1.3.2 Other logistics to organize event 4.1.3.3 Conduct event	1 nos. program launch event
Activity 4.1.4: Classification of RMG units	Classify RMG units in Bangladesh based on scale and type of products to aid baseline process	4.1.4.1 Conduct energy audit of sample plants 4.1.4.2 Assess audit results to develop segments of RMG units based on scale	1 nos. report on RMG unit classification

		of operations and type of products	
Activity 4.1.5: Consolidation of data and developing sector-level baseline	Data consolidation and develop baseline of energy consumption for RMG units in Bangladesh in the Year 1 to aid future assessment of sector-level impact of program	4.1.5.1 Consolidate data from energy audit 4.1.5.2 Identify appropriate assumptions to ensure realistic extrapolation of audit results 4.1.5.3 Undertake extrapolation of audit results	1 nos. document containing data sheets (which includes baseline information)
Activity 4.1.6: Prepare a list of renowned EE equipment with proven track records	Interact with stakeholders/market players to prepare a list of equipment already implemented in Bangladesh and having proven energy savings results in the RMG sector. List will be integrated with knowledge platform and disseminated through other means to ensure adequate public outreach	4.1.6.1 Identify relevant stakeholders 4.1.6.2 Undertake discussion to identify energy equipment in use in Bangladesh and has evidence of energy savings 4.1.6.3 Gather evidences of energy savings for each equipment	1 nos. document containing equipment list
Activity 4.1.7: Revision of baseline	Revise baseline derived in previously based on energy audits of plants and estimate actual change in energy consumption due to program	4.1.7.1 Conduct fresh set of energy audits 4.1.7.2 Analyse data as per methodology established for previous baseline study 4.1.7.3 Derive revised results of baseline	1 nos. report on revised energy baseline for sector
Capacity building of Local Financial Institutions (LFIs)			
Activity 4.2.1 Capacity building of LFIs (corporate office)	Conduct capacity building trainings/workshops to develop capacity of corporate/head office level officials to effectively conduct banking due diligence of loan applications for implementation of energy efficient technologies.	4.2.1.1 Identify training needs by interacting with sample within target group 4.2.1.2 Develop capacity building material to address needs 4.2.1.3 Identify participants for workshop	02 combined capacity building trainings/workshops with participants of 40-60 for each training/workshop from LFIs. Participants will include 10-15 corporate/head office members selected from each LFI

	Participants will be corporate/head office level officials (02 combined capacity building trainings/workshops with participants of 40-60 for each training/workshop from LFIs. Participants will include 10-15 corporate/head office members selected from each LFI)	4.2.1.4 Organize workshop for dissemination of capacity building material 4.2.1.5 Feedback forms taken from participants and interactive discussion stimulated between trainer and trainee	
Activity 4.2.2: Capacity building of LFIs (branch level executives)	Conduct capacity building trainings/workshops to develop technical capacity of banking officials at branch/field level to efficiently monitor implementation of energy efficient technologies over the duration of repayment of loan amount. (02 combined capacity building trainings/workshops with participants of 60-80 for each training/workshop. Participants will include 15-20 branch/field level officials selected by each LFI)	4.2.2.1 Identify training needs by interacting with sample within target group 4.2.2.2 Develop capacity building material to address needs 4.2.2.3 Identify participants for workshop 4.2.2.4 Organize training for dissemination of capacity building material 4.2.2.5 Feedback forms taken from participants and interactive discussion stimulated between trainer and trainee	02 combined capacity building trainings/workshops with participants of 60-80 for each training/workshop. Participants will include 15-20 branch/field level officials selected by each LFI.
Capacity building of RMG units to enable implementation of energy efficient technologies			
Activity 4.3.1 Awareness building workshops on EE&C	Conduct workshop to establish and disseminate business case of energy efficiency among RMG unit owners/top officials to help generate interest in owners in energy efficiency concept. (02 workshops at national level with participants of 60-80. These relevant participants will be	4.3.1.1 Identify training needs by interacting with sample within target group 4.3.1.2 Develop capacity building material to address needs 4.3.1.3 Identify participants for workshop 4.3.1.4 Organize training for dissemination of capacity building material	02 workshops at national level with participants of 60-80. These relevant participants will be selected by IDCOL upon consultation with LFIs.

	selected by IDCOL upon consultation with LFIs).	4.3.1.5 Feedback forms taken from participants and interactive discussion stimulated between trainer and trainee	
Activity 4.3.2: Building capacity for technical skills of facility/utility managers	Impart training to facility/utility managers on the technical requirements for implementing energy efficiency projects as well as necessary steps (i.e. equipment working condition, energy supply stability etc.) to be undertaken during operations to ensure theoretical energy savings are realized.(02 combined trainings at with participants of 40-60 within TA implementation period of 05 years. These relevant participants will be selected by IDCOL upon consultation with LFIs)	4.3.2.1 Identify training needs by interacting with sample within target group 4.3.2.2 Develop capacity building material to address needs 4.3.2.3 Identify participants for workshop 4.3.2.4 Organize training for dissemination of capacity building material 4.3.2.5 Feedback forms taken from participants and interactive discussion stimulated between trainer and trainee	02 combined trainings with participants of 40-60 within TA implementation period of 05 years.
Activity 4.3.3: Increase awareness and develop capacity of factory workers	Develop awareness of workers on importance of energy efficiency and impart knowledge key skills required to implement energy efficient equipment in the RMG factory. (This will be done through inclusion of energy efficiency & other gender-social relate issues in regular trainings of the factory workers. Each LFI will ensure such inclusion in 05 sub-projects during TA period of 05 years. Apart from it, each LFI will also arrange at least 01 safety training for respective end borrowers within TA period of 05 years.	4.3.3.1 Identify training needs by interacting with sample within target group 4.3.3.2 Develop capacity building material to address needs 4.3.3.3 Identify participants for workshop 4.3.3.4 Organize training for dissemination of capacity building material 4.3.3.5 Feedback forms taken from participants and interactive discussion stimulated between trainer and trainee	Each LFI will ensure such inclusion in 05 sub-projects during TA period of 05 years. Apart from it, each LFI will also arrange at least 01 safety training for respective end borrowers within TA period of 05 years

Provide on-demand technical support to participating banks on monitoring, reporting and verification (MRV) of energy efficiency projects

Activity 4.4.1: MRV support provided to 4 LFIs	Enhance capacity of LFIs in undertaking MRV to assess the benefits realized from the project. This will be achieved in the form of technical trainings, resolve queries on request, handholding support	4.4.1.1 Receipt of request for MRV support 4.4.1.2 Resolution of queries by consultation with sector energy efficiency expert	4 LFIs to be supported in MRV through trainings and handholding in real-life scenarios
Activity 4.4.2: Develop standard solutions document based on experience of commonly issues encountered in previous activity	Identify most commonly faced issues of banks in MRV and develop standard solution document addressing those issues to provide banks with troubleshooting guide and help them conduct MRV after tenure of TA consultant expires	4.4.2.1 Consolidate all queries resolved in previous activity 4.4.2.2 Prepare document listing query and resolutions	1 document containing standard solutions to MRV issues commonly faced by banks

Provide assistance to RMG units in loan application process

Activity 4.5.1 Conduct rapid diagnostic study to understand key requirements for loan application support	Conducting rapid diagnostic study of key areas related to loan application process for which RMG units require handholding as well as nature of support required. This would involve conducting focused interviews with RMG units to understand the key issues they face in the loan application process for any conventional corporate loan instrument. Subsequently, identify kind of assistance RMG units require as well as provide recommendations LFIs on possible ways to simplify loan application process for RMG units.	4.5.1.1 Discussion with sample RMG units to understand key issues in loan application 4.5.1.2 Discussion with banks to understand constraints leading to issues for applicant 4.5.1.3 Identify methods of resolving issues and provide recommendations	1 nos. report which gives areas which require improvement and recommendations on the change in the application process required to increase participation by RMG units.
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Activity 4.5.2 Develop FAQs, manuals, check-lists, filled-up templates to elucidate loan application process	Prepare guidance documents and filled up templates of loan application forms along with step-by-step explanation of process of loan application along with development of check list on list of supporting documents and guides on where and how they can be available. This will be accompanied with a vendor list to provide a ready reference for sourcing equipment and seeking quotation.	4.5.2.1 Based on recommendations prepare loan application templates and step-by-step guide 4.5.2.2 Compile probable list of vendors and integrate with document	1 consolidated document containing all relevant information
Activity 4.5.3 Respond to queries related to loan application process	Respond to queries of RMG units related to loan application process. The responses and queries will be consolidated to a troubleshooting guide to provide ready reference to any borrower after tenure of TA consultant ends.	4.5.3.1 Compile all queries of RMG units 4.5.3.2 Develop troubleshooting guide based on resolution provided for queries	1 consolidated document containing all relevant information
Provide on-demand technical support for implementation to RMG units which have secured GCF funding under this programme			
Activity 4.6.1: Implementation support provided to RMG units on request	Resolve queries and provide technical support to borrowers to ensure effective implementation of energy efficiency intervention	4.6.1.1 Compile requests from RMG units 4.6.1.2 Identify activities required for resolution 4.6.1.3 Handhold RMG units in implementing recommendations	No. of RMG units with queries resolved
Activity 4.6.2: Develop standard solution documents	Identify most commonly faced issues of borrowers in implementation and develop standard solution document i.e. troubleshooting manual based on experience of common issues encountered in previous activity addressing those issues. The responses and queries will be consolidated to a troubleshooting guide to	4.6.2.1 Compile all queries of RMG units 4.6.2.2 Develop troubleshooting guide based on resolution provided for queries	1 consolidated document containing all relevant information

	provide ready reference to any borrower after tenure of TA consultant ends		
Technology fair to create market awareness and develop access to energy efficient technology			
Activity 4.7.1: Organize technology fair	In order to create market awareness and develop access to energy efficient technology, IDCOL along with the LFIs will organize technology fair to create interface between international manufacturers/suppliers of energy efficient equipment and Textile/RMG unit owners to stimulate market development for energy efficiency in the sector. The fair will be organized in Year 03 (since FAA effectiveness) with participation of minimum 30 different technology suppliers and representation from 60 different Textile/RMG units	4.7.1.1 Identify invitee list for event and send invite 4.7.1.2 Ensure interaction between unit owners, technology OEMs etc.	1 nos. combined technology fair for RMG and Textiles sector
Development of relevant policies, strategies & plans addressing energy efficiency			
5.1.1: Review of existing national policies, strategies & plans regarding Energy Efficiency	- The Sustainable & Renewable Energy Development Authority (SREDA) was established by Bangladesh Parliament in May 2012 as a national nodal organization for promoting demand-side energy efficiency and conservation in the country. - IDCOL will be working with SREDA on this component to identify the areas of national policies, strategies and plan	5.1.1.1 Develop ToR for the consultant/ consulting firm jointly with SREDA 5.1.1.2 IDCOL will conduct the procurement of the consultant/ consulting firm in discussion with SREDA 5.1.1.3 The consultant at this stage will be engaged in data collection, both from primary and secondary sources and will be analysing the findings against set criteria, which will be preliminary	Stock taking report on existing national policies, strategies & plans regarding Energy Efficiency

	that can be further strengthened. Initially there will be a stock taking of the information from both primary and secondary sources, which will be analyzed on the basis of the GoB's target, available resources and international best practices.	determined in discussion with IDCOL and SREDA	
5.1.2 Identify any policy gap & develop mechanism to address such gaps	Subsequent to Activity 5.1.1 the consultant/consulting firm will – - Identify the gaps in policy implementation. - Identify the barriers that is restricting the performance of the existing policies to reach the country set EE&C targets. - Identify the areas of improvements that can be achieved under the programme. (at par with allocated budget and activity timeline)	5.1.2.1 Consultant/consulting firm will develop a methodology for conducting the policy implementation gap study. 5.1.2.2 Consultant/consulting firm will develop a draft report based on primary & secondary data 5.1.2.3 Consultant/consulting firm will finalise the report in consultation with IDCOL & SREDA	A comprehensive report on policy gaps (design or implementation) and institutional gaps, along with possible mitigation strategies. Discussion papers
5.1.3 Development of relevant policies, strategies & plans	Based on the finding from the report produced in activity 5.1.1 and 5.1.2, relevant policies, strategies and plans to strengthen the country's EE&C implementation capacity	5.1.3.1 IDCOL and SREDA will conduct at least two stakeholder consultation meeting/workshops where the findings of the study conducted in activity 5.1.1. & 5.1.2 will be presented and through stakeholder discussion possible mitigation action plan will be finalised.	- Workshop report where the discussion paper will be presented. - Draft policies strategies and plans developed - Further consultation with the stakeholders (At this stage it is necessary to seek help from stakeholders to fine tune the wording, clarify meaning and make adjustments to the policy before it is finalized.) - Pictures/ Videos of the event - Attendees list

5.1.4 Implementation of the policies, strategies & plans developed under Activity 5.1.3	When the stakeholders of the policy development process are satisfied that all issues and concerns about the policy have been aired and dealt with, it is time to finalize the policy. The final policy document needs to be formally adopted by the GoB with an appropriate record entered in to the minutes. Following formal adoption of the policy it should be communicated far and wide throughout the organizations and stakeholders.	5.1.4.1 Consultant will prepare the upgraded policy document as per the outcomes of activity 5.1.3. 5.1.4.2 Relevant institutions of GoB will adopt the policy document. 5.1.4.3 Policy implementation to be monitored by SREDA. 5.1.4.4 Wide media coverage is to be ensured. Required communication documents will be developed for creating awareness among public & private stakeholders.	• Policy documents with strategies and action plans. • Minutes of the event where GOB adopts the policy. • Media coverage.
Capacity development & Market awareness on importance of EE equipment strengthened			
Activity 5.2.1 Assessment of detail capacity development requirement at policy level	As per the policy document prepared, an institutional capacity assessment will be carried out to understand the policy adoption & implementation capacity of the involved institutions.	5.2.1.1 Analysis report developed on capacity need assessment of SREDA for short, medium and long term implementation of EE&C projects. 5.2.1.2 Consultant/consulting firm will collect both primary and secondary data.	Institutional capacity assessment report
Activity 5.2.2: Develop training manuals	Following formal adoption of the policy, training sessions may need to be conducted to ensure that organization personnel are fully informed and able to implement the policy. If the policy is not well communicated it may fail. Hence a training manual will be developed	5.2.2.1 Audience will be identified as per the assessment conducted under activity 5.2.1 5.2.2.2 Training tools will be finalised 5.2.2.3 Training materials will be developed	Training manual/s developed

	based on the world best practices focusing on performance measurement, EE standards & labelling and financial support mechanism for implementation of energy efficiency projects (prospective business model with feasible payback and IRR). The training manual will ensure consistency in presentation of content. It may also ensure that all training information on skills, processes, and other information necessary to perform tasks is together in one place. This will be important for institutionalizing the expected outcome of component 5.	5.2.2.4 Assess the training materials through demo training and collect feedback	
Activity 5.2.3: Organize 03 training of the trainers (ToT) for Government officials	Due to rotation, GoB officials often get transferred to the other departments or ministries hence it is important to develop pool of trainers who can continue the training or knowledge dissemination activities beyond programme's implementation timeline, so that new batch of officials can continue the implementation of EE&C projects.	5.2.3.1 Select participants for ToT (they are often senior government officials) 5.2.3.2 Conduct at least 03 training sessions for the trainers	Post event report on ToT Attendees list Photographs /videos
Activity 5.2.4: Organize 10 training session for the policy makers and relevant government official by the trained trainers	The trainers will now conduct training for the officer who will be responsible for monitoring and implementing EE&C policies.	5.2.4.1 Select participants 5.2.4.2 Conduct training 5.2.4.3 Collect feedback and have an interactive session	• Post training report • Attendees list • Photographs /videos
Activity 5.2.5: Organizing 02 dissemination	Programme results will be presented in national conferences where participants from	5.2.5.1 Dissemination workshops will be	• Report on dissemination workshop. • Attendees list

workshops on programme outcomes	government, professional associations, trade bodies, local community groups and other key stakeholders will be present for effective replication in other sectors. Through wide dissemination of programme outcomes to policy makers and EE practitioners and also the learning from implementing energy efficiency in textile and garments sector (financing scenario, business case and economic benefits, women involvement and role of government for long-term sustainability) the activity will ensure institutional capitalization of knowledge.	arranged in collaboration with SREDA	Photographs /videos
Activity 5.2.6: Develop tools for mass awareness on EE&C	The main objective of this activity is to achieve mass awareness for energy efficiency through disseminating tools like media coverage, bill board. etc. Required communication materials on the importance of EE&C in the country will also be further developed. Information on the programme success will also be widely shared for awareness and replication through increased demand.	5.2.6.1 ToR for a consultant/consulting firm to be prepared. 5.2.6.2 Procurement to be conducted by IDCOL with support from SREDA. 5.2.6.3 Communication materials to be developed targeting both public and sector stakeholders	Communication materials

E.7. Monitoring, reporting and evaluation arrangements (max. 500 words, approximately 1 page)

Arrangements for monitoring, reporting and evaluation for component 01 & 02 (EE&C in Textile sector):

Monitoring:

- IDCOL has a proven track record of highly efficient, effective monitoring and supervision of the previously led and current internationally funded country-wide program. This has been made possible because of the programme specific MRE plans devised and implemented by IDCOL suited to the needs of each programme.
- As IDCOL will act both as AE & EE for the program, separate units of IDCOL will play different roles. Business Unit of IDCOL will be responsible for both the execution of TA support & concessional funding to the end borrower. As the AE, "GCF unit" of IDCOL will be responsible for administrating & managing the fund as well as supervision, evaluation and reporting. Monitoring will be ensured for each sub-project, to be financed under this program, in different project phases through multiple engagements:

Stage	Responsibility	Monitoring action
Sub-project approval stage	Business unit of IDCOL	Select eligible borrowers suitable for GCF financing & assess project viability under regular process
	Credit Risk Management (CRM)	Ensure due diligence of borrowers and compliance with existing lending policy of IDCOL & other GCF facility criteria
	Management/ Board of IDCOL	Approve funding proposals as per IDCOL's vision & mission ensuring viability, integrity, governance, sustainability and other related aspects.
Sub-project disbursement phase	Business unit, Legal unit, Credit Administration Department (CAD)	Ensure disbursement of GCF fund in line with FAA
Sub-project implementation stage	Business unit	Regularly monitor the project through timely progress report from end borrowers and visiting the project premise to ensure smooth operation for anticipated mitigation impact at the project end

Reporting & evaluation mechanism:

- As per GCF's mandate for a participatory MRE, there will be frequent reporting from the EE to the AE and the AE to the Fund as pre-determined intervals. As the EE, while the Business unit of IDCOL will conduct their investment activity as usual as the EE, the GCF unit of IDCOL which is a dedicated wing other than the business unit will supervise & monitor generation of supporting/documents for reporting & verification of different components under the log frame.

- GCF Unit of IDCOL will periodically report to GCF on the energy usage & tCO₂ reduction for each of its sub-project by compiling the data received from different sub-projects (by IDCOL business unit/third party M&E consultant through a common dashboard/online portal). Periodic energy savings reporting will contain actual savings from ongoing sub-projects. In case sub-project loan is adjured, an estimate for energy savings will be calculated for the remaining life of the equipment. In order to set the baseline & energy saving potential through energy efficient equipment & technology, IDCOL will engage consultant/ energy auditor under TA component which will be used as reference at the time of each proposal approval. The textile borrowers will install energy measurement equipment at the factory premise to identify energy usage/savings for the equipment/technology financed under this specific GCF program & will report periodically to IDCOL/ third party M&E consultant.. The expected mitigation outcome will be verified through 1) Energy savings and 2) Increased production output by using EE equipment & technology.
- The reporting and evaluation tools of the programme will be designed and facilitated by a technical assistance consultant under the TA activity as per the followings:

Reporting tool	Reporting line	From date of consultant engagement
All relevant reporting formats for: - Energy savings & tCO₂ reduction - Gender balance and empowerment report (as per Gender Action Plan) - All other relevant reporting formats Detail methodology and format for interim & final evaluation/verification	From end borrower to IDCOL	As per "Implementation Timetable"
	From IDCOL to GCF	
	From IDCOL to GCF	

- The mid-term & final evaluation of the program will be carried out by Independent Evaluator for reporting to GCF.
- Proper execution of the TA component, for which grant of USD 3.05 million has been requested, will be assessed under the Interim evaluation after the end of year 06.

Arrangements for monitoring, reporting and evaluation for component 03 & 04 (EE&C in RMG sector):

Monitoring:

- IDCOL will play the role as an Accredited Entity (AE) & Local Financial Institutions (LFIs)/IDCOL will be the Executing Entity (EE).
- LFIs will have direct relationship with their respective end borrowers i.e. the RMG unit. The monitoring action at LFIs level will be as follows:

Stage	Responsibility	Monitoring action
Sub-project approval stage	Business unit of respective LFI	Select eligible borrowers suitable for GCF financing & assess project viability under regular process
	Credit Risk Management (CRM) of respective LFI	Ensure due diligence of borrowers and compliance with existing lending policy of LFI

	Approving Authority of respective LFI (Board/Credit Committee/Management)	Approve funding proposals as per LFI's vision & mission ensuring viability, integrity, governance, sustainability and other related aspects.
Sub-project disbursement phase	Credit Administration Department (CAD) of respective LFI	Ensure disbursement of GCF fund in line with regulatory requirement, Banks own policy and FAA
Sub-project implementation stage	Business unit of respective LFI	Regularly monitor the project through timely progress report from end borrowers and visiting the project premise to ensure smooth operation for anticipated mitigation impact at the project end

- IDCOL has a proven track record of highly efficient, effective monitoring and supervision of the previously led and current internationally funded country-wide program. This has been made possible because of the programme specific MRE plans devised and implemented by IDCOL suited to the needs of each programme.
- As the AE, IDCOL will be responsible for administrating & managing the fund as well as supervision, evaluation and reporting. LFIs will have to approach IDCOL for approving their each sub-project. IDCOL will assess the eligibility of each sub-project (i.e. project of one RMG Units) as per the terms & conditions of the approved GCF program. The pre-specific format will be used to check the eligibility (format will be developed during Operation manual development stage).

Reporting & evaluation mechanism:

- As per GCF's mandate for a participatory MRE, there will be frequent reporting from the EE to the AE and the AE to the Fund as pre-determined intervals. End borrowers will report to Executing Entity i.e. LFIs and LFIs will then report to Accredited Entity i.e. IDCOL. IDCOL will supervise & monitor generation of supporting/documents for reporting & verification of different components under the log frame.
- IDCOL will periodically report to GCF on the energy usage & tCO₂ reduction for each of sub-project by compiling the data received from different sub-projects (by LFIs/third party M&E consultant through a common dashboard/online portal). Periodic energy savings reporting will contain actual savings from ongoing sub-projects. In case sub-project loan is adjured, an estimate for energy savings will be calculated for the remaining life of the equipment. The baseline & energy saving potential of energy efficient equipment & technology will be ascertained by the borrower with the help of consultant/ energy auditor. The appointment of the consultant/ energy auditor may co-financed under the TA. The End Borrowers will install energy measurement equipment at the factory premise to identify energy usage/savings for the equipment/technology financed under this specific GCF program & will report periodically to LFIs/ third party M&E consultant. The expected mitigation outcome will be verified through 1) Energy savings and 2) Increased production output by using EE equipment & technology
- The mid-term & final evaluation of the program will be carried out by Independent Evaluator for reporting to GCF.
- Proper execution of the TA component, for which grant of USD 2.30 million has been requested, will be assessed under the Interim evaluation after the end of 06 years.
- Following milestones/reports will be ensured under this program as per AMA & project design: Same as for component 1& 2 mentioned above

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F. RISK ASSESSMENT AND MANAGEMENT

F.1. Risk factors and mitigations measures (max. 3 pages)

FOR COMPONENT 01 & 02 (EE&C IN TEXTILE SECTOR):

The main type of risk is technical & operation risks associated with project & less social & environment risks. Based on the above assessment, the maximum risk category is Low-Moderate and, therefore, the risk category for the overall project is **Moderate**. A summary of the risk is as follows:

Risk Factor	Category	Probability	Impact
1. Demand risk and market risk in the textile sector which may slow down investment	Credit	Low	Medium
2. Political risks associated with changing attitude towards support & priority for EE in Textile		Low	Low
3. Financial risks from Credit and Currency fluctuations	Forex	Medium	High
4. Social unrest affecting BAU scenario	Technical & Operational	Low	Low
5. Delays in project implementation due to time consuming process of review and clearance as well on ground operationalization.		Low	Low
6. Potential technological risks like equipment failure & supply chain failure		Low	Medium
7. Safety risks to equipment and employees.		Low	Medium
8. lack of awareness and knowledge about the benefits of energy efficiency	Other	Low	Medium
9. Social risks like exclusion of social groups and indigenous communities from project benefits		Low	Low
10. Gender risks regarding non-access to project benefits		Low	Medium
11. Environment risks caused due to inadequate disposal of waste generated from textile process		Medium	Low
12. Health Risks occurring from the processes in textile units		Low	Low

Selected Risk Factor 1

Category	Probability	Impact
CreditCredit	LowLow	MediumMedium
Description		
Demand risk and market risk in the textile sector which may slow down manufacturing, marketing and profit making and fail to attract investment		
Mitigation Measure(s)		
While chances of demand and market risks are low given the potent role of Bangladesh's textile sector in National and international clothing sector.		
IDCOL will take mitigation measures such as collateral loans for distributing loans to end consumers and also ensure hypothecation of energy efficient equipment finance if required.		

Selected Risk Factor 2

Category	Probability	Impact
CreditCredit	LowLow	LowLow
Description		

Political risks associated with changes in directives that may modify their support and priority for energy efficiency in the sector.		
Mitigation Measure(s)		
Energy efficient measures makes business sense for end consumers as they achieve great savings over project life time, especially with concessional lending being proposed which mitigated this political risk. Moreover, in an energy deficient country like Bangladesh, significant climate goals to be achieved, government is very keen on promoting EE foreseeable future.		
Selected Risk Factor 3		
Category	Probability	Impact
ForexForex	MediumMedium	Select
Description		
Significant foreign exchange risk at the time of repayment of GCF fund in future as borrowing from GCF will be in foreign currency (USD) and re-lending to the end user in local currency (BDT).		
Mitigation Measure(s)		
GCF will provide USD loan to IDCOL & then IDCOL will re-lend it to end borrowers in local currency i.e. BDT. Considering the trend of 2.94% yearly devaluation rate of BDT against USD for the last few years due to a number of macro-economic factors, the repayment of foreign currency loan (USD) to GCF in the future may wipe out the concessional benefits of this program. If end borrowers have to bear the currency fluctuation cost then sub-projects financed under this program will not be viable anymore. The absence of long-term hedging market for BDT is another barrier for such currency fluctuation management.		
This risk will be addressed by partly lending in USD to the borrowers thereby reducing their BDT exposure and also, having currency risk fluctuation premium as part of BDT interest rate. As Executing Entity for Component 1, IDCOL will bear the FX risk.		
Selected Risk Factor 4		
Category	Probability	Impact
Technical and operationalTechnical and operational	LowLow	LowLow
Description		
Social unrest caused due to external factors leading to disruption in the BAU scenario for textile sector		
Mitigation Measure(s)		
Social instability caused from external influences can be mitigated through involvement and healthy relations with the NGOs, social welfare groups and associations working together for worker's development and welfare.		
Compliance with the programme's ESS will foster a healthy social relationship, ensure social participation and empowerment and reduce the discrimination amongst the associated community. The end borrowers of the fund will have the sole responsibility of mitigating this risk.		
Selected Risk Factor 5		
Category	Probability	Impact
Technical and operationalTechnical and operational	LowLow	LowLow
Description		

Delays in project implementation due to time consuming process of project review and clearance as well on ground operationalization.

Mitigation Measure(s)

IDCOL has a streamlined project approval process to reduce any delays in approval. To reduce the time consumed in installation, the supplier of the equipment will be considered responsible for installation. Furthermore, clear specifications of funding terms including documentation required for the project being eligible for this project will be communicated in a timely manner.

Delay in project implementation by the end borrowers will be monitored by IDCOL as per this program monitoring framework.

Selected Risk Factor 6

Category	Probability	Impact
Technical and operational Technical and operational	Low Low	Medium Medium

Description

Potential technological risks like equipment failure; supply chain failure such as delays or collapse of supply chain for technology or associated equipment; lack of qualified service technicians, etc.

Mitigation Measure(s)

Companies who have past experience of successful operations in Bangladesh as service provider through setting service centers and experienced maintenance staff will be preferred. Also, latest EE equipment can be procured from any of these reputed manufacturer / suppliers at discretion of end user/ customer.

Skill development of the manpower for the operation & maintenance of the EE technology will be ensured through trainings and other initiatives.

Selected Risk Factor 7

Category	Probability	Impact
Technical and operational Technical and operational	Low Low	Medium Medium

Description

Safety risks like accidents, injuries and incidents of fire and other occupational damages to equipment and employees.

Mitigation Measure(s)

It's mandated to ensure compliance with project ESIA which provides for requisite avoidance, mitigation and compensation measures; National Laws and Policies ⁷² and register with Accord on Fire and Building Safety in Bangladesh (Accord)/ the Alliance for Bangladesh Worker Safety (Alliance)/ the National Action Plan on Fire Safety and Structural Integrity (NAP) or any other regulatory requirement set by the Government as and when required.

Synchronization with these will significantly reduce risks of accidents, injuries and incidents of fire and other occupational damages.

Compliance with Regulatory Standards of the Ministry of Labour and Employment & IDCOL EHS policies and guidelines & clear contractual rules covering the need for occupational health and safety technical activities will be monitored.

⁷² Environment Conservation Act (ECA) 1995; Environment Conservation Rules (ECR) 1997; Environment Court Act (ECA) 2010

Selected Risk Factor 8		
Category	Probability	Impact
OtherOther	LowLow	MediumMedium
Description		
Risks from lack of awareness and knowledge about the benefits of energy efficiency in textile sector and availability of the financial instrument for the same.		
Mitigation Measure(s)		
Under the technical assistance component of the program, IDCOL will arrange a number of national stakeholder engagement events to raise awareness about EE in textile and availability of a funding in this regard. The knowledge gap will be addressed through development of knowledge products, marketing material, workshops, and outreach programs to reduce and mitigate risks associated with lack of awareness about the existence and function of the financial facility.		
Selected Risk Factor 9		
Category	Probability	Impact
OtherOther	MediumMedium	LowLow
Description		
Social risks like exclusion of social groups and indigenous communities from skill up gradation, trainings, workshops and job opportunities from new equipment and machinery.		
Mitigation Measure(s)		
Compliance with the Social regulatory frameworks in Bangladesh, IDCOL's ESSF, GCF's guidelines and the ESIA would ensure none of the listed social risk to occur at any given time.		
Selected Risk Factor 10		
Category	Probability	Impact
OtherOther	LowLow	MediumMedium
Description		
Gender risks from lack of gender equity in trainings and skill up gradation required for integration of new equipment and machinery.		
Mitigation Measure(s)		
Gender Inclusiveness and equity will be promoted throughout project implementation. The project participants to comply with the National policies like Labour Act and Labour Rules, Women development policy 2011 and GCF's Gender policy to ensure reduction and mitigation of gender risks. Compliance with Gender Action Plan of this program will ensure the participation of women in such capacity building initiatives.		
Selected Risk Factor 11		
Category	Probability	Impact
OtherOther	MediumMedium	LowLow
Description		
Environment risks caused due to inadequate disposal of solid and liquid generated industrial plant modernization processes.		

Mitigation Measure(s)

The ESIA identifies and analyses upstream environmental issues that may affect the project and the sector and suggests mitigation measures for the same. It also consists of an appropriate management plan for implementing, monitoring and reporting of the suggested environmental mitigation and enhancement measures. Pre-identification of environmental risks through ESIA would prepare to reduce the occurrence and intensity of these risks.

It essential to comply with the ESIA as it conforms to the national regulations and IDCOL guidelines.

Selected Risk Factor 12

Category	Probability	Impact
OtherOther	LowLow	LowLow

Description

Health Risks from Respiratory & Dermal contact with hazards causing risk of lung cancer and injury to the bronchial tubes, dust particles, VOCs, heat and chemicals occurring from the processes in textile units

Mitigation Measure(s)

Health risks are low level and have low probability to occur. Separate mitigation will not be required as this risk is already reduced due to the implementation of Energy Efficient equipment that are certified, user and environment friendly as compared to the existing equipment. The end-users will have to ensure that the equipment and its installation is compliant with programme's ESIA and conforms to the national regulations and IDCOL guidelines.

FOR COMPONENT 03 & 04 (EE&C IN RMG SECTOR):

The program is exposed to majorly technical and operational risks, which includes environmental and social risks and the combined rating of impact and probability of the various risks is observed to be mostly Medium. A summary of the risks is given below:

Risk Factor	Category	Probability	Impact
1. Lack of capacity in equipment operation and maintenance (O&M)	Technical & Operational	Medium	Medium
2. Limited market awareness		Medium	High
3. Disposal of inefficient machinery		Medium	Medium
4. Reduced job opportunities for women & other social groups		Medium	Medium
5. Health and safety related issues		Medium	High
6. Credibility of executing entities	Credit	Medium	High
7. Default in re-payment by end borrowers (RMG units):		Medium	High
8. Foreign exchange fluctuation	Forex	Low	High
9. Limited capacity in banks for due diligence of energy efficiency project	Governance	Medium	High
10. Long lead time in loan disbursement		Medium	Medium
11. Reduced demand from client markets	Other	Medium	High
12. Decline in sectoral growth		Low	High
13. Money laundering/Terrorism Financing risk:	ML/FT	Low	High

Accordingly, the overall program has a **Moderate** Risk Level due to greater incidence of Medium-level risk elements.

Selected Risk Factor 1		
Category	Probability	Impact
Technical and operational Technical and operational	MediumMedium	MediumMedium
Description		
Lack of capacity in equipment operation and maintenance (O&M): If penetration of advanced EE equipment low, adequately trained personnel for O&M of machine would be low. Also, availability of spares may be low and cost of spares high. Hence, overall cost of O&M would be high increasing skepticism of potential borrower and reducing uptake of EE equipment.		
Mitigation Measure(s)		
Technical Assistance (TA) activity designed to develop capacity within set of RMG units in O&M of energy efficiency equipment. Uptake and continued use of energy efficient equipment in targeted units under the programme will help create real-life demonstration of the benefits as well as prompt other parties to take up similar measures.		
Selected Risk Factor 2		
Category	Probability	Impact
Technical and operational Technical and operational	MediumMedium	HighHigh
Description		
Market Awareness: If penetration of advanced EE equipment low, adequately trained personnel for O&M of machine would be low. Also, availability of spares may be low and cost of spares high. Hence, overall cost of O&M would be high increasing skepticism of potential borrower and reducing uptake of EE equipment.		
Mitigation Measure(s)		
As part of TA activity, technology fairs proposed involving RMG unit owners and energy efficient equipment manufacturers, to generate awareness on technical and commercial queries. Also, capacity building sessions are planned as part of TA activities, to provide RMG unit owners, facility managers and workers with information on facets of energy efficiency relevant to their particular category. Additionally, LFIs are expected to pursue independent efforts to push the loan products in the market by improving overall awareness regarding the benefits of energy efficiency		
Selected Risk Factor 3		
Category	Probability	Impact
Technical and operational Technical and operational	MediumMedium	MediumMedium
Description		
Reduced job opportunities for women & other social groups: Women currently constitute 80% of work force of RMG units, most of them area employed in sewing area. Depending on the specific type of energy efficient equipment, there might be a need for re-skilling of women workforce. Re-skilling increases risk of reduced job opportunities. This can have an adverse effect on the prospect of women empowerment. Similar risks are applicable for other social groups, most of whom are unskilled to low-skilled.		
Mitigation Measure(s)		

Under TA activity, capacity building activities are planned specifically for women workers which will disseminate relevant information for operating new energy efficient equipment. Under Gender Action Plan, targets for capacity building of women workers is also provided to ensure there is no risk of reduced job opportunities for women. Similar capacity building activities for workers are planned as part of the TA activity

Selected Risk Factor 4

Category	Probability	Impact
Technical and operational Technical and operational	MediumMedium	MediumMedium

Description

Issues related to health and safety: Installation of new equipment can lead to increased health related issues due to generation of material harmful to the health of human beings. The deviation in regular operations for the workers due to installation of new equipment might lead to safety related issues.

Mitigation Measure(s)

Under TA activity, capacity building activities are planned for workers group to ensure they are better equipped to handle new energy efficient equipment and there is limited loss of productivity. These activities will help workers to get acquainted to the new equipment and impart knowledge on implication of improper handling on safety

Selected Risk Factor 5

Category	Probability	Impact
Technical and operational Technical and operational	MediumMedium	HighHigh

Description

Disposal of inefficient machinery: The program aims to replace existing inefficient equipment with efficient equipment variant. The inefficient variants if discarded, can pose serious environmental issues. With no existing laws related to disposal of discarded equipment in place, proceeding with the replacement process may lead to detrimental environmental

Mitigation Measure(s)

The ESIA identifies and analyses environmental issues arising out of disposal and pollution that may affect the project and the sector. It provides the guidelines under which potentially harmful activities can be undertaken, taking into account the possible mitigation measures. The guidelines are integrated into an appropriate management plan for implementing, monitoring and reporting of the suggested environmental mitigation and enhancement measures. It essential to comply with the ESIA as it conforms to the national regulations and IDCOL guidelines

Selected Risk Factor 6

Category	Probability	Impact
CreditCredit	MediumMedium	HighHigh

Description

Credibility of borrowing entity: The key financial indicators for banking sector in Bangladesh indicates in spite of substantial credit growth, it is plagued with high default rates and banks are not able maintain the minimum capital adequacy norms. In the current scenario, the banking system can be considered be stable with medium level of risk of instability in the short to medium term.

Mitigation Measure(s)

As part of the feasibility study, due diligence of the LFIs has been undertaken. This has helped establish the current and future financial strength of the selected banks, ensuring assured repayment to GCF and flow of funds in subsequent tranches.

Selected Risk Factor 7

Category	Probability	Impact
CreditCredit	MediumMedium	HighHigh

Description

Default in re-payment: Even though RMG sector has been witnessing three-year average credit growth of over 15% since 2016⁷³, credit risk in the sector continues to be high. Furthermore, sector is plagued by long and difficult legal processes to recover loans and continued case of defaults may lead to reduced interest of banks in lending to RMG sector units.

Mitigation Measure(s)

As part of TA activity, templates for verifying energy and cost savings of project in the loan application will be prepared. This coupled with established process to determine financial health of a borrowing entity, will help in mitigation. Also, project design ensures periodic assessment by executing entities and accredited entities to monitor project progress, templates for which have been prepared as part of project preparatory activities.

Selected Risk Factor 8

Category	Probability	Impact
ForexForex	LowLow	HighHigh

Description

Foreign exchange fluctuation: LFIs will be receiving funds from GCF in USD and in case it chooses to extend loans to RMG units in BDT, it would be exposed to Forex rate fluctuation risk. Since repayment from the borrowing units to the LFI would be in BDT and bank would be repaying the amount to GCF in USD, in case of fluctuation in forex rate, banks would have to repay more or less than original amount received.

Mitigation Measure(s)

Since program design leads to lower risk rating, it is recommended to ensure that structure of loan disbursement be continued during program implementation

Selected Risk Factor 9

Category	Probability	Impact
GovernanceGovernance	MediumMedium	HighHigh

Description

Limited capacity in LFIs for due diligence of energy efficiency project: Under current scenario, bank officials have limited technical knowledge about energy efficiency, the requirements for energy efficient equipment, assessment of cost savings realized by installing such equipment and how the savings affects loan repayment. The lack of technical capacity can lead to incorrect assessment of borrowers and hence, increase risk of loan repayment.

Mitigation Measure(s)

TA activity can help identify information gap among the personnel of LFIs and accordingly develop training material to bridge the knowledge gap. The knowledge material can be disseminated through workshops with branch level officials as well as mid management officials. Non-financial standardized documents will be prepared which will help establish the rules for engagement of the borrower and the LFIs. This will help create the necessary framework which can be adopted by the LFIs.

Selected Risk Factor 10

⁷³ Source: [Bangladesh Systemic Risk Dashboard \(Bangladesh Bank, December 2018\)](#)

Category	Probability	Impact
GovernanceGovernance	MediumMedium	MediumMedium
Description		
Long lead time in loan disbursement: To design an energy efficiency project, an independent energy audit required where entire process may take considerable time, spanning into 7-8 months in worst case scenario. Under such circumstances, the premise under which the loan proposal was prepared might change leading to invalidity of the project and accordingly, potential borrowing entities might be discouraged from pursuing such projects.		
Mitigation Measure(s)		
In order to ensure faster turn-around of the loan proposal, evaluation of the proposal would be conducted by the EEs. As part of the project preparatory activity, the necessary monitoring framework and templates by which accredited entity i.e. IDCOL can comprehensively monitor and evaluate the fund disbursement by LFIs will be developed. Accordingly, development of adequate capacity in LFIs would be undertaken as part of TA activity		
Selected Risk Factor 11		
Category	Probability	Impact
OtherOther	MediumMedium	HighHigh
Description		
Reduced demand from client markets: RMG is an export-oriented industry and change in demand pattern in the target market can affect the sector as a whole. In case of reduced demand, profitability of units would decrease leading to reduced finance availability to fund energy efficiency measures		
Mitigation Measure(s)		
Since the risk arises from external factors which are beyond the scope of action of the stakeholders, tolerating the risk level is the only risk treatment option available		
Selected Risk Factor 12		
Category	Probability	Impact
OtherOther	LowLow	HighHigh
Description		
Decline in sectorial growth: RMG sector is witnessing a period of tepid growth after decades of high growth along with effect of rising costs and stagnant output prices. Maintaining profitability under such scenario will be strained and accordingly, RMG owners can be disincentivized to undertake EE measures		
Mitigation Measure(s)		
Since the risk arises from external factors which are beyond the scope of action of the stakeholders, tolerating the risk level is the only risk treatment option available		
Selected Risk Factor 13		
Category	Probability	Impact
ML/FT	LowLow	HighHigh
Description		
Money laundering/Terrorism Financing risk: Finance disbursed under the program can also be utilized to finance terrorist activities, money laundering or prohibited practices by the banks or the end borrowers. While on one hand the intended objective will not be met, on the other hand the finance would be utilization would be counterproductive. Also, given the fact that it is a concessional line of financing, perceived probability of money laundering can also be high		
Mitigation Measure(s)		

Bangladesh has strong regulatory system to monitor financing issues of the Financial Institutions under the purview of central Bank i.e. Bangladesh Bank. The detail system has been explained in attachment Annex 02_Reply_ "Compliance Note".

G. GCF POLICIES AND STANDARDS

G.1. Environmental and social risk assessment (max. 750 words, approximately 1.5 pages)

Provide the environmental and social risk category assigned to the proposal as a result of screening and the rationale for assigning such category. Present also the environmental and social assessment and management instruments developed for the proposal (for example, ESIA, ESMP, ESMF, ESMS, environmental and social audits, etc.). Provide a summary of the main outcomes of these instruments. Present the key environmental and social risks and impacts and the measures on how the project/programme will avoid, minimize and mitigate negative impacts at each stage (e.g. preparation, implementation and operation), in accordance with GCF's ESS standards. If the proposed project or programme involves investments through financial intermediations, describe the due diligence and management plans by the Executing Entities (EEs) and the oversight and supervision arrangements. Describe the capacity of the EEs to implement the ESMP and ESMF and arrangements for compliance monitoring, supervision and reporting. Include a description of the project/programme-level grievance redress mechanism, a summary of the extent of multi-stakeholder consultations undertaken for the project/programme, the plan of the Accredited Entity (AE) and EEs to continue to engage the stakeholders throughout project implementation, and the manner and timing of disclosure of the applicable safeguards reports following the requirements of the GCF [Information Disclosure Policy](#) and [Environmental and Social Policy](#).

Describe any potential impacts on indigenous peoples and the measures to address these impacts including the development of an Indigenous Peoples Plan and the process for meaningful consultation leading to free, prior and informed consent, pursuant to the GCF [Indigenous Peoples Policy](#).

Attach the appropriate assessment and management instruments or other applicable studies, depending on the environmental and social risk category as annex 6.

FOR COMPONENT 01 & 02 (EE&C IN TEXTILE SECTOR):

IDCOL has conducted risk screening for the project using the IFC's Performance Standards and World Bank's EHS Guidelines, besides national legislation of GoB and evaluate their significance level (i.e. level of risk and probability). The envisaged environmental and social safeguards (ESS) category for this project under consideration is Category B (medium risk category)⁷⁴. The potential risks (refer to Section F) are few; low to medium in magnitude and occurrence and largely reversible using the prescribed mitigation measures and good international industry practices.

Environmental and Social Management Framework

During the PPF activity, an ESMF has been developed by IDCOL to identify potential risks to the environment and social matters from the projects and outline strategies for managing those risks and minimising undesirable environmental and social impacts. The programme's ESMF will:

Provide energy efficient solutions to the textile sector of Bangladesh to reduce energy consumption in the sector and in turn the emissions of greenhouse gases

Encourage the deployment of energy efficient equipment and technology through planning, commitment and continuous improvement

Comply with all applicable laws, regulations and standards for the protection of the environment and society; and

Adopt the best practicable means available to prevent or minimise environmental impact.

Describe all monitoring procedures required to identify impacts on the environment; and

Provide an overview of the obligations of Project staff and contractors regarding environmental obligations

⁷⁴ IFC's approach to risk categorization as adopted by GCF

Environmental Audit

An environmental audit was conducted on one of the textile factories to establish the past performance, environmental issues or non-compliances if any. It assessed the entire processing area and the area under which the utility section of unit operates. Environmental and safety issues with respect to liquid effluent, solid waste and emissions, as well as light, sound and ventilation were covered under this audit. Furthermore, the audit included assessment of energy conservation and energy efficiency opportunities at the factory.

Documents to be developed during programme implementation

Site-specific screening check-lists and Environmental and Social Management/ Mitigation Plans (ESMPs), based on the ESMF, will be prepared for each site where construction works will be implemented, publicly consulted and disclosed locally before procurement commences for the civil works.

Key Environmental and Social Risks have been presented in detail with respective and highly specific mitigation measures in Section F. To summarise, the relevant risks and mitigation measures are:

Discrimination/Exclusion of social groups

Social exclusion may happen due to multiple reasons like structural/economic (iniquitous economic conditions, low wages, dual and segregated labour markets, historical oppression (colonialism), discrimination, absence of legal/political recognition, institutional/civic non-acceptance, self-exclusion, etc. These can be managed by complying with social regulatory frameworks in the country, sectorial mandates, IDCOL's ESSF and GCF's guideline for social inclusion.

Gender based discrimination/exclusion

Gender-disaggregated data will be used for monitoring outputs, outcomes and impacts during project implementation to ensure women's participation in employment opportunities in the textile sector. Hence, the gender issues and mitigation is addressed in almost all the programme design documents namely, ESMF, Gender assessment and action plan, Logical Framework (Performance indicators) and the MRE Report.

Occupational safety

Bangladesh Accord was created to enable working in a fearless environment for concerns like fires, building collapses, or other accidents that could be prevented with reasonable health and safety measures. Besides, IDCOL's ESS, GCF's ESMF and the programme ESIA developed specifically for this case, identifies workplace hazards to minimize their risks with appropriate control measures and thereby avoid the financial costs of accidents & occupational ill health.

Waste Disposal

Under Section 12 of the Bangladesh Environment Conservation Act 1995 (ECA 1995) no industrial unit or project can be established or undertaken without obtaining an Environmental Clearance Certificate (ECC) from the DOE. The programme ESIA should be made to ensure compliance with the legislations/regulations associated to solid waste management and Bangladesh Standards and Guidelines for Sludge Management and other related guidelines.

Health Risks

Health related risks will be minimal as the certified, user friendly energy efficient equipment will reduce exposure to most of the health hazards as they minimise handling of chemicals, heat, dust with automated machines. This will be a low-level risk with least chances of occurrence.

Institutional arrangement

The textile factories (EEs) with the assistance of the Project Management team from IDCOL (AE), will be responsible for ensuring sound environmental performance of the contractor in charge of construction

throughout the project and ensure compliance with the ESMF. IDCOL will supervise and advice on monitoring and reporting under this component.

Multi-stakeholder consultations

Gender balanced scoping and validation consultations were conducted where distinguished guests, key and primary stakeholders from Ministries, associations, trade unions, industries, academicians and development professionals participated to state their insights and discuss several associated issues.

Workshop on 'Importance of energy efficiency in textile sector and its influence on environment, society and gender' on 29th May 2019 in Dhaka.

Dissemination and stakeholder consultation workshop on 'Impact of Energy Efficiency on Environment, Social and Gender in Textile Sector of Bangladesh' on 29th July 2019, Dhaka

The project management team will continue to monitor the programme implementation and timely report the developments as per the MRE schedule, i.e. an annual report on Environmental and Social compliance (as per the ESMF) to monitor the degree of implementation of the ESMF and an annual Gender balance and empowerment report. This will involve stakeholder engagement and knowledge dissemination across various stakeholder groups throughout the implementation.

FOR COMPONENT 03 & 04 (EE&C IN RMG SECTOR):

IDCOL has undertaken risk screening of the project based on the framework provided in IFC's Performance Standards and World bank's EHS Guidelines as well assessed compliance with national legislation of GoB. Based on this assessment, the significance level (i.e. level of risk and probability) of each risk has been determined and subsequently a composite score has been assigned for the risk the project faces. Accordingly, the envisaged environmental and social safeguards (ESS) category for this project is Category B (medium risk category)⁷⁵. As observed, the potential Environmental and Social risks (mentioned in Section F) are limited with low to medium risk rating. Also, these risks can be easily mitigated, by incorporating adequate measures in the program design.

Environmental and Social Management Framework

As part of PPF activity, an ESMF has been developed by IDCOL as the environmental and social assessment and management instrument for the proposed program. The document provides a framework for managing environmental and social risks specific to the projects constituting the proposed program to minimise undesirable environmental and social impacts. The programme's ESMF will aid in project development by executing entities in the following way:

Help identify environmental and social risks specific to their projects

Apprise on E&S regulations of GoB and related guidelines of GCF

Help develop plan to effectively manage identified environmental and social risks

Guidance on ensuring stakeholder participation in developing ESMP

Establish robust grievance redressal mechanism to ensure stakeholder participation during implementation phase

Provide guidance on institutional mechanism required to ensure effective implementation of ESMP

⁷⁵ IFC's approach to risk categorization as adopted by GCF

Environmental Audit

In order to understand environmental and social risks at program level, survey was conducted for 20 RMG units and further supplemented by survey of 150 households located around the selected REMG units. Under the survey, existing working conditions, health and safety hazards, social issues within factory premises were assessed along with interviews with unit workers and facility managers. This formed the basis for developing the ESMF.

Documents to be developed during programme implementation

Site-specific screening check-lists and Environmental and Social Management/ Mitigation Plans (ESMPs), based on the ESMF, will be prepared for each project site, publicly consulted and disclosed locally before project implementation.

Management of key environmental and social risks

Key Environmental and Social Risks have been presented in detail along with their respective mitigation measures in Section F. To summarise, the relevant environmental and social risks and mitigation measures are:

1. Discrimination/Exclusion of social groups

Probability of social exclusion occurring due to the program is limited since implementation of energy efficient equipment is independent of which social group is involved in O&M. Also, impact is medium since introduction of energy efficiency equipment would lead to minimal disruption since semi-skilled labour, required for operating the equipment, is readily available in Bangladesh. Hence, overall risk arising from this factor is low for the program.

2. Gender based discrimination/exclusion

Introduction of energy efficient equipment may require upskilling for female workers in the Garment factories, most of whom are in low-skill functions. Given the risk, capacity building activities are planned specifically for women workers which will disseminate relevant information for operating new energy efficient equipment and risk of reduced job opportunities for women is mitigated.

3. Occupational safety

Introduction of new machinery would lead to deviation from existing operations of workers and inappropriate handling may lead to safety related risks. This can not only lead to injury of concerned worker, but also unrest among worker groups or loss of life in the extreme case, causing high to medium levels of loss of productivity. In order to mitigate this risk, capacity building activities are planned for workers group to ensure they get acquainted with new equipment and gain knowledge on implication of improper handling on safety

4. Waste Disposal

Lack of regulatory bindings on disposal of inefficient equipment would mean unit owners will adopt the least cost method of disposal, which may not be environment friendly. Accordingly, ESMF provides guidance on safe disposal of all wastes arising from project activities, thereby minimizing environmental damage

5. Health Risks

Majority of energy efficient RMG equipment do not contain harmful or hazardous material that can adversely affect workers. Difference in generation of noise between efficient and inefficient equipment is also low. Hence, it is perceived that low health risks will occur as a result of the program

Institutional arrangement

A detailed institutional arrangement is provided where specialist in procurement, social development, financial management, environment etc. will support the project director and the project management unit to ensure compliance with ESMP. In order to ensure Capacity development of End-borrowers in effective monitoring and evaluation, specific capacity building programmes have been proposed as part of the program design. Also, disclosure of ESMP documents as well as periodic compliance monitoring reports is

proposed to be translated to local language and disseminated to all the concerned stakeholders as well as made available to the public, through the website of the End Borrower and any other suitable online public platform.

A Grievance Redressal Mechanism (GRM) has also been developed that allows affected people to voice concerns regarding environmental and social impacts for sub-project activities through a predictable, transparent, and credible process, resulting in outcomes that are seen as fair, effective, and lasting. It provides the institutional structure and grievance resolution process along with insight on GCF's perspective on GRM.

Multi-stakeholder consultations

Gender balanced scoping and validation consultations were conducted where distinguished guests, key and primary stakeholders from Ministries, associations, trade unions, industries, academicians and development professionals participated to state their insights and discuss several associated issues.

Scoping workshop on environmental, social and gender considerations for the program on 4th September 2019 in Dhaka.

Dissemination and stakeholder validation workshop on environmental, social and gender considerations for the program on 23rd October 2019, Dhaka

As part of the program design, stakeholder consultations for a key part of development and ESMP, as well during its implementation. The design also provides scope of stakeholders reaching out to executing entities by establishing a robust grievance redressal mechanism.

G.2. Gender assessment and action plan (max. 500 words, approximately 1 page)

Provide a summary of the gender assessment and project/programme-level gender action plan that is aligned with the objectives of GCF's [Gender Policy](#). Confirm a gender assessment and action plan exists describing the process used to develop both documents. Provide information on the key findings (who is vulnerable and why) and key recommendations (how to address the vulnerability identified) of the gender assessment. Indicate if stakeholder consultations have taken place and describe the key inputs integrated into the action plan, including: how addressing the vulnerability will ensure equal participation and benefits from funds investment; key gender-related results to be expected from the project/programme with targets; implementation arrangements that the AE has put in place to ensure activities are implemented and expected outcomes will be achieved, monitored and evaluated.

Provide the full gender assessment and project-level gender action plan as annex 8.

FOR COMPONENT 01 & 02 (EE&C IN TEXTILE SECTOR):

Gender assessments were done through analysis of various parameters like Socio-economic, Health & Safety, Sanitation and access to basic facilities to understand gender-based vulnerabilities and specific needs. The data from primary (industrial survey, KII, two national level gender balanced stakeholder consultations) and secondary sources (National statistics, progress reports and government published documents) was equated with the international relevant standards and policies for gap analysis.

Key Findings:

The assessments highlighted a number of disparities ranging from gender ratio, wages, job opportunities, education, financial independence, skill and trainings, health, safety, provisioning of basic facilities and most importantly decision making. Women have been observed to be far left behind in terms of job security and

financial independence. Unlike the RMG sector, textile is a male dominated sector with limited/negligible women participation at execution or decision-making level.

Industrial survey under the PPF activity indicate that women workers are paid around 8 to 10% less on average than their male compatriots and have lesser opportunities in the sector. Their participation in trainings for skill upgradation and safety also has been quite low. Surveys show that women are under-privileged due to lack of skills/technical knowledge and experience, attrition due to unplanned pregnancy, selective availability for single day shift, unwillingness to stretch for more hours at work, lack of higher degrees to handle supervisory/decision-making roles.

Programme interventions:

The programme encourages gender equality throughout the development and implementation phases i.e. gender balanced consultations, decision making and embedding gender specific benefits during the implementation of the project. Women participation as key stakeholders for project design and development:

Table 17: Percentage of women participation in stakeholder engagement

Stakeholder Engagement	Stakeholder Type	% of women participants
Key Informants Interview (KII)	Primary Stakeholders: Ministries, Associations, Multilaterals, Research/Academia and Industry	27%
Workshop/Group consultation	Primary and Secondary stakeholders: Ministries, Associations, Multilaterals, Research/Academia and Industry	Workshop 1 23%
		Workshop 2 24%
Industry Survey	Industry: Women led, Workers and Community	Women led industries: 35% factories surveyed have women leaders in capacities of Directors/DGM/CEO
		Executives/workers: 30%

Gender sensitive vision and objectives have led to specific log-frame indicators to embed gender inclusion in project design. The M&E log frame will measure the performance of the gender-based activities.

Specific activities have been listed in the Gender action plan with strict targets to be achieved as per the given timeline. These programme activities for gender empowerment are:

- Encouragement of women entrepreneur to venture in the textile sector
- Prioritization of existing woman headed manufacturing units
- Reducing the gender wage gap through unbiased employment as per the labour rules and encouragement of similar wages for men and women for similar work type
- Facilitate women participation in the sector through provision of medical facilities, child care and sanitation facilities
- Training and Skill Development for machine use and operating on programming software through development of training modules and periodic trainings and evaluation
- Safety trainings while operating on new machinery/equipment through development of training modules and periodic trainings and evaluation
- Grievance Redressed through formulation of grievance committee and establishment of grievance redressed process

- Formulation of a Gender Balanced Project Committee with women participating in decision-making from IDCOL's programme management team and textile factories

The programme's MRE scheme will measure outputs and track progress of gender specific activities through monitoring and reporting the achievement towards mid-term and final targets for each activity with a specific performance indicator and means of verification. Timely reporting will facilitate evaluation of the performance of gender based programme activities.

FOR COMPONENT 03 & 04 (EE&C IN RMG SECTOR):

The gender assessment was conducted by examining the engagement of men and women in the RMG sector in Bangladesh, labor force participation, access and control to technology and financial resources, as well as power relations. Such analysis enabled to explore into a plethora of factors namely Socio-Economic, Health & Safety, sanitation and access to basic facilities, which in turn facilitated to clearly understand the magnitude of adversities based on gender and the precise requirements necessary to mitigate them.

Primary data was collected through industrial surveys, household surveys, Key Informant Interviews (KIIs), two national level stakeholder consultations etc. The data from secondary sources was gathered through relevant reports, about national policies, guidelines and statistics (environmental, social, gender), documents published by the government which focused upon potential risks identified in the RMG sector, historical studies and research on Energy Efficiency (EE) in Bangladesh's RMG sector. This was compared with the international best practices, considering gender equity to identify and scrutinize into the gap prevalent in terms of gender considerations in the sector.

Key Findings:

The findings derived from the gender assessment emphasized upon the discrepancies prevailing within the RMG sector which essentially includes male-female employment ratio, wage discrimination, employment opportunities, financial independence & inclusion, occupational health & safety, lack of access to skill & trainings and basic facilities, welfare provision, social protection and freedom of association, collective bargaining and social dialogue as well as playing a decision making role at the managerial level.

Women in the labor force have been at the forefront of Bangladesh's recent successful economic growth. They make up most of the workers in the ready-made garment (RMG) sector, the country's flagship export-oriented industry. However, even though the RMG sector has played a tremendous role for women empowerment, there has been discrimination in labour force participation in this sector.

From data collected within factories, 4 out of every 5 production line workers are female, whilst just over 1 in 20 supervisors is a woman. This is mainly due to lack of education, trainings, technical knowledge and social responsibilities of women towards their families. Also, women are engaged more in sewing or as helpers compared to men. Less number of women are seen in RMG units working in cutting, quality control or supervising compared to men.

Numerous surveys (both industrial and household) carried out under the PPF reflect that although gender-based wage discrimination does not exist predominantly in the sector, discrimination exists in areas of equal treatment— women are forced to do overtime more than the men and the cut from the overtime allowance is more for women workers than men. Women workers face severe discrimination with regard to the scope of promotion and their career prospect is limited.

Participation of women is increasing in the Bangladeshi labor market, but socio-cultural norms envisage that women and men's position is not equal. Women tend to be concentrated in certain low-paying, low-skilled sectors in the garment industry, and within it, are often stuck in lower paying occupations such as sewing machine operators and helpers.

Due to stereotypes regarding women's and men's aspirations, preferences and capabilities, employers' perceptions of women's and men's skills and attitudes are affected. Thus, majority of women are not privileged enough to avail considerable access to enhancing their skillset and technical knowledge through training & experience and hence they remain 'unskilled' compared to their male counterparts. In addition, social and work place safety, maternity and pregnancy issues, unwillingness to work long hours/overtime, lack of transport facilities and the care burden all contribute to the fact of significantly limited presence of women engaged in more sophisticated employment positions such as managers/supervisors in the RMG sector.

Programme interventions:

During the course of the proposed programme, gender balance, equity and equality has been promoted all through the development and execution stages. This is inclusive of gender balanced participation in stakeholder consultations, incorporating gender induced benefits & enhancements during the design and implementation of the projects. The following illustrates participation of female stakeholders in various engagements undertaken within project planning and development phase:

Table 18: Percentage of women participation in stakeholder engagement

Stakeholder Engagement	Stakeholder Type	% of women participants
Key Informants Interview (KII)	Primary Stakeholders: Ministries, Associations, Multilaterals, Research/Academia and Industry	25%
Workshop/Group consultation	Primary and Secondary stakeholders: Ministries, Associations, Multilaterals, Research/Academia and Industry	Workshop 1 15%
		Workshop 2 11%
Industry Survey	Industry: Women led, Workers and Community	Women led industries: 0% factories surveyed have women leaders in capacities of Directors/DGM/CEO
		Executives/workers: 40%

Coalescing gender responsive perception and goals under the programme will enable to constitute indicators for the logical framework that in turn will establish gender inclusion into the programme design and evaluate their performance based on gender sensitive activities.

The Gender Action Plan (GAP) within this assessment mirrors the logical framework of the project and is an integral part of project/program design. GAPs include clear targets, gender design features and measurable performance indicators to ensure women's participation and benefits. Key aspects of the GAP are incorporated into project/program assurances to encourage buy-in from AEs and other partners. The activities listed within the GAP are as follows:

- Encouragement of women entrepreneur to venture in the garment sector
- Reducing the gender wage gap through unbiased employment as per the "Labour rules (2015)" and Encourage similar wages for men and women for similar work type as per the "National Wage Board"
- Facilitate women participation in the sector through provision of: Medical Facilities, Child care and Sanitation Facilities
- Training and skill development for machine use and operating on programming software through development of training modules and trainings and evaluation
- Safety trainings while operating on new machinery/equipment through development of training modules and periodic trainings and evaluation
- Grievance Redressed through Formulation of a project committee for addressing grievance and Establishing grievance redress process
- Formulation of a project committee with women participating for monitoring of the activities under the program
- Training & skill development of relevant Business Units of LFIs

The programme monitoring, reporting, evaluation & verification procedures will quantify and evaluate outputs and track progress through reporting the results by aligning them with mid-term and final targets for each activity with a specific performance indicator and means of verification. Periodic and prompt reporting in accordance with the designated timeline will enable to actualize evaluation of the performance of gender based project/programme activities.

G.3. Financial management and procurement (max. 500 words, approximately 1 page)

- IDCOL, as a development financial institution under the ownership of Ministry of Finance of Government of Bangladesh, has substantial experience in financing and monitoring industrial development projects and management of funds provided by development organizations like World Bank, ADB, JICA .etc. The projects funded by IDCOL have successfully been accounted and audited as per the international standards, in line with national procedures in financial management.
- Financial management of the program will be guided by IDCOL's own accounting policies & procedures, as well as procedures related to the financial & operational aspects of project management administration. When not acting as an Executing entity, IDCOL, as an AE will comply with the terms & conditions set forth for financial management of GCF proceeds as per the Accreditation Master Agreement (AMA). Again for Component 03, IDCOL in its role as AE will ensure that the executing entities i.e. LFIs/IDCOL will comply with internationally accepted accounting standards for the project related activities at their end.

- The energy efficiency component procured under this program will be the subject matter of the loan financing. Thus the procurement will be carried out by the end borrower for their respective projects as per the business as usual practice. Along with the disbursement request/procurement application, the end borrowers will be submitting the evidences/supporting for implementation of the respective procurement under the programme. These will be screened by IDCOL/LFIs to approve the loan application.
- After the disbursement of GCF fund as per the implementation arrangement in section B4, IDCOL/LFIs will monitor that the proceeds of the funding are efficiently utilized for intended purposes only and in compliance with GCF and GoB. Fund disbursement and repayment will be conducted as per scheduled (annex 14).
- IDCOL has an independent internal audit division, reporting directly to the Board, to add value through an independent appraisal of all IDCOL's financial operations and activities. The result of such appraisal is improved operational efficiency, risk analysis & management and internal control system that will ensure achieving IDCOL's corporate objectives. Moreover, IDCOL's financial statements are audited externally by a qualified, internationally recognized auditing firm, in accordance with international standards of auditing. This firm is competitively selected by IDCOL.
- As per the AMA, the procurement of goods and services for funded activities, whether by the Accredited Entity itself, and Executing Entity or by a third party, shall be done in accordance with the rules, policies and procedures of the Accredited Entity i.e. IDCOL to the extent and scope of its Accreditation. Thus IDCOL will follow its own "Procurement Policy" for the operations of this program. Procurement audit is conducted periodically by the internal audit division to ensure proper compliance with the policy.

G.4. Disclosure of funding proposal

Note: The Information Disclosure Policy (IDP) provides that the GCF will apply a presumption in favour of disclosure for all information and documents relating to the GCF and its funding activities. Under the IDP, project and programme funding proposals will be disclosed on the GCF website, simultaneous with the submission to the Board, subject to the redaction of any information that may not be disclosed pursuant to the IDP. Information provided in confidence is one of the exceptions, but this exception should not be applied broadly to an entire document if the document contains specific, segregable portions that can be disclosed without prejudice or harm.

Indicate below whether or not the funding proposal includes confidential information.

☐ **No confidential information:** The accredited entity confirms that the funding proposal, including its annexes, may be disclosed in full by the GCF, as no information is being provided in confidence.

☒ **With confidential information:** The accredited entity declares that the funding proposal, including its annexes, may not be disclosed in full by the GCF, as certain information is being provided in confidence. Accordingly, the accredited entity is providing to the Secretariat the following two copies of the funding proposal, including all annexes:

- ☐ full copy for internal use of the GCF in which the confidential portions are marked accordingly, together with an explanatory note regarding the said portions and the corresponding reason for confidentiality under the accredited entity's disclosure policy, and
- ☐ redacted copy for disclosure on the GCF website.

The funding proposal can only be processed upon receipt of the two copies above, if containing confidential information.

H. ANNEXES

H.1. Mandatory annexes

- ☒ Annex 1 NDA no-objection letter(s) : **Attached**
- ☒ Annex 2 Feasibility study - and a market study, if applicable- **Attached**
- ☒ Annex 3 Economic and/or financial analyses in spreadsheet format- **Attached**
- ☒ Annex 4 Detailed budget plan – **Attached**
- ☒ Annex 5 Implementation timetable including key project/programme milestones – **Attached**
- ☒ Annex 6 E&S document corresponding to the E&S category (A, B or C; or I1, I2 or I3):
[\(ESS disclosure form provided\)](#)
 - ☐ Environmental and Social Impact Assessment (ESIA) or
 - ☐ Environmental and Social Management Plan (ESMP) or
 - ☐ Environmental and Social Management System (ESMS)
 - ☐ Others (please specify – e.g. Resettlement Action Plan, Resettlement Policy Framework, Indigenous People's Plan, Land Acquisition Plan, etc.)
 - ☒ Environmental and Social Management Framework (ESMF) **Attached**
- ☒ Annex 7 Summary of consultations and stakeholder engagement plan-**Attached**
- ☒ Annex 8 Gender assessment and project/programme-level action plan- **Attached**
- ☒ Annex 9 Legal due diligence (regulation, taxation and insurance)- **Attached**
- ☒ Annex 10 Procurement plan- **Attached**
- ☒ Annex 11 Monitoring and evaluation plan- **Attached as part of “Program Indicator, Reporting & Monitoring Framework”**
- ☒ Annex 12 AE fee request- **Attached**
- ☐ Annex 13 Co-financing commitment letter, if applicable [\(template provided\)](#)
- ☒ Annex 14 Term sheet including a detailed disbursement schedule and, if applicable, repayment schedule - **Attached**

H.2. Other annexes as applicable

- ☐ Annex 15 Evidence of internal approval [\(template provided\)](#)
- ☐ Annex 16 Map(s) indicating the location of proposed interventions
- ☐ Annex 17 Multi-country project/programme information [\(template provided\)](#)
- ☐ Annex 18 Appraisal, due diligence or evaluation report for proposals based on up-scaling or replicating a pilot project
- ☐ Annex 19 Procedures for controlling procurement by third parties or executing entities undertaking projects financed by the entity
- ☐ Annex 20 First level AML/CFT (KYC) assessment
- ☐ Annex 21 Operations manual (Operations and maintenance)
- ☒ Annex 22 GHG Emission Methodology **Attached**
- ☒ Annex 23 Risk Assessment Report **Attached**

Annex 24 Environment Audit Report **Attached**

** Please note that a funding proposal will be considered complete only upon receipt of all the applicable supporting documents.*

A Acronyms

Acronym	Full form
ADB	Asian Development Bank
ADR	Advance Deposit Ratio
AE	Accredited Entity
AFD	Agence Francaise De Development
AMA	Accredited Master Agreement
BAU	Business as usual
BB	Bangladesh Bank
BCCSAP	Bangladesh Climate Change Strategy and Action Plan
BDBL	Bangladesh Development Bank Limited
BDT	Bangladesh Taka
BERC	Bangladesh Energy Regulatory Commission
BGMEA	Bangladesh Garments Manufacturers and Exporters association
BIFCL	Bangladesh Industrial Finance Company Limited
BIFFL	Bangladesh Infrastructure Finance Fund Limited
BISR	Bangladesh Institute of Social Research
BKMEA	Bangladesh Knit Manufacturers and Exporters association
BTMA	Bangladesh Textile Mills Association
BTMC	Bangladesh Textile Mills Corporation
CAD	Credit Administration Department
CAGR	Compound Annual Growth Rate
CAR	Capital Adequacy Ratio
CCPP	Construction of Combined Cycle Power Plant
CEO	Chief Executive Officer
CLASP	Collaborative Labelling and Appliance Standards Program
CN	Concept Note
CO2	Carbon dioxide
CRM	Credit Risk Management
CSR	Corporate Social Responsibility
DAE	Direct Access Entity
DFI	Development Finance Institution
DFID	Department for International Development
DoE	Department of Environment
DSCR	Debt Service Coverage Ratio
E&S	Environmental and social
ECA	Environment Conservation Act
ECC	Environment Clearance Certificate
ECR	Environment Conservation Rules

Acronym	Full form
EDA	Enhanced Direct Access
EE	Energy Efficient
EE&C	Energy Efficiency and Conservation
EE&CPF	Energy Efficiency and Conservation Promotion Financing
EECPF	Energy Efficiency and Conservation Promotion Financing
EIRR	Economic Internal Rate of Return
EPZs	Export Processing Zones
ERD	Economic Relations Division
ESIA	Environmental and Social Impact Assessment
ESMF	Environmental and Social Management Framework
ESMP	Environmental and Social Management/ Mitigation Plans
ESMS	Environmental and Social Management System
ESS	Environment and Social Safety
ESSF	Environmental and Social Safety framework
ETP	Effluent Treatment Plant
EY	Ernst & Young
FAA	Funded Activity Agreement
FBCCI	Federation of Bangladesh Chambers of Commerce & Industries
FCBs	Foreign Commercial Banks
FIRR	Financial Internal Rate of Return
FIs	Financial Institutions
FP	Funding Proposal
FSSP	Financial Sector Support Project
FY	Financial year
GCF	Green Climate Fund
GCPF	Bangladesh, Global Climate Partnership Fund
GDP	Gross Domestic Product
GHG	Green House Gas
GIZ	Deutsche Gesellschaft für Internationale Zusammenarbeit
GoB	Government of Bangladesh
GTSF	Global Trade Supplier Finance
GTF	Green Transformation Fund
GWh	Giga watt hour
ICB	Investment Corporation of Bangladesh
ICCCAD	International Centre for Climate Change and Development
ICS	Improved Cook Stove
IDA	International Development Association
IDCOL	Infrastructure Development Company Limited
IDP	Information Disclosure
IEA	International Energy Agency
IP	Implementation Partner
IEEF	Industrial and Energy Efficiency Finance

Acronym	Full form
IEEFP	Energy Efficiency Finance Programme
IFC	International Finance Corporation
IFI	International financing institution
INDC	Intended Nationally Determined Contribution
IP	Implementing Partner
IRR	Internal Rate of Return
JICA	Japan International Cooperation Agency
KIIs	Key Informant Interviews
KPIs	key Performance Indicators
RFP	Request for Proposal
kWh	Kilowatt Per Hour
L/C	Letter of Credit
LED	Light-emitting Diode
LFI	Local Financial Institutions
LNG	Liquefied Natural Gas
LTFF	Long Term Financing Facility
M&E	Monitoring and evaluation
MIS	Management Information System
MoEF	Ministry of Environment and Forests
MoEFCC	Ministry of Environment, Forest and Climate Change
MoLE	Ministry of Labour and employment
MoPEMR	Ministry of Power Energy and Mineral Resources
MoT	Ministry of Textile
MoU	Memorandum of understanding
MoV	Means of Verification
MoWCA	Ministry of Women and Children Affairs
MRV	Monitoring, Reporting and Verification
MT	Million Tonne
MTOE	Million tonnes of oil equivalent
NAP	National Action Plan
NAPA	National Adaptation Programme of Action
NBFI	Non-Bank Financial Institution
NDA	National Designated Authority
NDC	Nationally Determined Contribution
NIP	New Industrial Policy
NPV	Net Present Value
OE	Original Equipment
PaCT	Partnership for Clean Textile
PC	Power Cell
PCBs	Private Commercial Banks
PFI	Participating Financial Institution
PFI	Power Factor Improvement

Acronym	Full form
PLS	Profit-Loss Sharing
PO	Partner Organizations
PPF	Project Preparation facility
PPP	Public Private Partnership
PSMF	Power System Master Plan
PSMP	Power System Master Plan
PTS	Primary Textile Sector
RE	Renewable Energy
REB	Rural Electricity Board
RMG	Readymade garment
ROA	Return on asset
ROE	Return on equity
SCB	State Commercial Bank
SDBs	Specialized Banks
SDG	Sustainable Development Goals
SHS	Solar Home System
SLL	Shuttle less looms
SMEs	Small and medium enterprises
SOCB	State Owned Commercial Banks
SREDA	Sustainable & Renewable Energy Development Authority
SSL	Shuttle Less Looms
TA	Technical Assistance
TCF	Trillion Cubic Feet
TOE	Tonnes of oil equivalent
MRE	Monitoring, Reporting and Evaluation
UN	United Nations
UNDP	United Nations Development Programme
UNFCCC	United Nations Framework Convention on Climate Change
USD	United States Dollar
VFD	Variable Frequency Drive
VSDs	Variable Speed Drives



D.O. No-09.00.0000.105.05.011.17- 71

14 October 2020

Secretary
Economic Relations Division
Ministry of Finance
Government of Bangladesh

Re: **Funding proposal for the GCF by IDCOL regarding the programme "Promoting private sector investment through large scale adoption of energy saving technologies and equipment for Textile and Readymade Garment (RMG) sectors of Bangladesh"**

Dear Sir,

We refer to the programme "Promoting private sector investment through large scale adoption of energy saving technologies and equipment for Textile and Readymade Garment (RMG) sectors of Bangladesh" in Bangladesh as included in the funding proposal submitted by IDCOL to us on 13 October 2020.

The undersigned Fatima Yasmin is the duly authorized representative of the National Designated Authority of Bangladesh.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the programme "Promoting private sector investment through large scale adoption of energy saving technologies and equipment for Textile and Readymade Garment (RMG) sectors of Bangladesh" as included in the funding proposal.

By communicating our no-objection, it is implied that:

- The government of Bangladesh has no-objection to the programme "Promoting private sector investment through large scale adoption of energy saving technologies and equipment for Textile and Readymade Garment (RMG) sectors of Bangladesh" as included in the funding proposal;
- The programme "Promoting private sector investment through large scale adoption of energy saving technologies and equipment for Textile and Readymade Garment (RMG) sectors of Bangladesh" as included in the funding proposal is in conformity with Bangladesh's national priorities, strategies and plans;
- In accordance with the GCF's environmental and social safeguards, the programme "Promoting private sector investment through large scale adoption of energy saving technologies and equipment for Textile and Readymade Garment (RMG) sectors of Bangladesh" as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the programme "Promoting private sector investment through large scale adoption of energy saving technologies and equipment for Textile and Readymade Garment (RMG) sectors of Bangladesh" as included in the funding proposal has been duly followed.

We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the programme, including the textile sector projects and activities and garment sector projects and activities.

We acknowledge that this letter will be made publicly available on the GCF website.

Kind regards,


14.10.2020
Fatima Yasmin

Executive Director
Green Climate Fund
Songdo Business District, 175 Art Center-daero
Yeonsu-gu, Incheon 22004, Republic of Korea

Environmental and social safeguards report form pursuant to para. 17 of the IDP

Basic project or programme information	
Project or programme title	Promoting private sector investment through large scale adoption of energy saving technologies and equipment for Textile and Readymade Garment (RMG) sectors of Bangladesh
Existence of subproject(s) to be identified after GCF Board approval	Yes
Sector (public or private)	Private
Accredited entity	Infrastructure Development Company Limited (IDCOL)
Environmental and social safeguards (ESS) category	Category I-2
Location – specific location(s) of project or target country or location(s) of programme	Bangladesh
Environmental and Social Impact Assessment (ESIA) (if applicable)	
Date of disclosure on accredited entity's website	N/A
Language(s) of disclosure	N/A
Explanation on language	N/A
Link to disclosure	N/A
Other link(s)	N/A
Remarks	N/A
Environmental and Social Management Plan (ESMP) (if applicable)	
Date of disclosure on accredited entity's website	N/A
Language(s) of disclosure	N/A
Explanation on language	N/A
Link to disclosure	N/A
Other link(s)	N/A
Remarks	N/A
Environmental and Social Management (ESMS) (if applicable)	
Date of disclosure on accredited entity's website	Friday, October 9, 2020
Language(s) of disclosure	English and Bengali
Explanation on language	Bengali is the official and national language of Bangladesh and spoken by the beneficiary communities.
Link to disclosure	Environmental and Social Management Framework for RMG English: https://idcol.org/download/Environment%20and%20Social%20Management%20Framework%20RMG%20(English).pdf Bengali: https://idcol.org/download/Environment%20and%20Social%20Management%20Framework-RMG%20(Bengali).pdf

	Environmental and Social Management Framework for Textile English: https://idcol.org/download/Environment%20and%20Social%20Management%20Framework Textile%20(English).pdf Bengali: https://idcol.org/download/Environment%20and%20Social%20Management%20Framework-%20Textile%20(Bengali).pdf
Other link(s)	N/A
Remarks	An ESMS consistent with the requirements for a Category I-2 project is contained in the “Environmental and Social Management Framework: Promoting private sector investment through large scale adoption of energy saving technologies and equipment for garment sector” and the “Environmental and Social Management Framework: Promoting private sector investment through large scale adoption of energy saving technologies and equipment for textile sector of Bangladesh.”
Any other relevant ESS reports, e.g. Resettlement Action Plan (RAP), Resettlement Policy Framework (RPF), Indigenous Peoples Plan (IPP), IPP Framework (if applicable)	
Description of report/disclosure on accredited entity's website	N/A
Language(s) of disclosure	N/A
Explanation on language	N/A
Link to disclosure	N/A
Other link(s)	N/A
Remarks	N/A
Disclosure in locations convenient to affected peoples (stakeholders)	
Date	Friday, October 9, 2020
Place	Hard copies of the disclosure were made available at the IDCOL headquarters, located at: Infrastructure Development Company Limited (IDCOL) Head office UTC Building, 16th Floor, 8 Panthapath, Kawran Bazar, Dhaka-1215, Bangladesh
Date of Board meeting in which the FP is intended to be considered	
Date of accredited entity's Board meeting	TBD
Date of GCF's Board meeting	Monday, November 9, 2020

Note: This form was prepared by the accredited entity stated above.

Independent Technical Advisory Panel's assessment of FP150

Proposal name:	Promoting private sector investment through large scale adoption of energy saving technologies and equipment for Textile and Readymade Garment (RMG) sectors of Bangladesh
Accredited entity:	Infrastructure Development Company Limited (IDCOL)
Project/programme size:	Medium

I. Assessment of the independent Technical Advisory Panel

1.1 Impact potential Scale: Medium/High

- Bangladesh, in South Asia, forms the larger and eastern part of the Bengal region. It is situated in one of the largest deltas in the world, formed by a dense network of the distributaries of the rivers Ganges, Brahmaputra and Meghna, and more than 230 major rivers and their tributaries. The total land area of 147,570 square kilometres consists mostly of low and flat land. With a population exceeding 166.5 million, Bangladesh is ranked as one of the most densely populated countries in the world.
- Although its status is being reviewed, Bangladesh is currently regarded as a least developed country. The country's economy recorded remarkable growth since 2010 as a result of strong private consumption and investment. The country's real gross domestic product (GDP) grew from 7.3 per cent in 2017 to 8.2 per cent in 2019.
- Bangladesh is frequently cited as one of the countries most vulnerable to climate change because of its geographic location, the dominance of floodplains, and low elevation from the sea, among other things. The country experiences severe climate change impacts such as higher summer temperature, irregular monsoon and erratic rainfall patterns, floods and droughts, coastal and riverbank erosion, and salinity intrusion along the coast.
- Although the total greenhouse gas (GHG) emissions of Bangladesh amount to less than 0.35 per cent of total annual global emissions, the country's fossil fuel-related emissions have grown by 6.8 per cent since 2000. GHG emissions per capita increased from 0.18 tonnes of carbon dioxide equivalent (tCO₂eq) in 1997 to 0.47 tCO₂eq in 2014, with an average annual growth rate of 6 per cent. The increase was driven by rapid industrial expansion, and growth of energy production from fossil fuel sources.
- Bangladesh is the world's second-largest apparel producer after China. Textile industries are one of the fastest growing and among the most important industries of the country. The sector's contribution to GDP has grown from 3 per cent in 1991 to more than 13 per cent in 2018 representing more than 80 per cent of the country's annual exports. The textile industry (along with the garment industry) is one of the largest employers in the country, employing over 4 million people, or 6 per cent of the country's industrial employment.
- The industrial sector in Bangladesh accounts for 47.8 per cent of commercial energy consumption, predominantly in the form of natural gas and electricity. However, the sector is not efficient in its energy use because of continuous usage of old or poorly maintained machines and inadequate energy management. The industries' lack of interest in energy-efficient and conservation technologies is assessed to be due to inadequate financial incentives, and lack of

technical expertise. If the economic growth persists with the current industrial energy intensity, Bangladesh could face difficulties in managing the rising energy demand and consequently in achieving its GHG emission reduction targets under the Paris Agreement.

7. Although energy-saving opportunities are present across the various industrial sectors, the textile sector (spinning, weaving, knitting and dyeing), and ready-made garment (RMG) sector (cutting, sewing, dyeing, ironing, and finishing) in particular are considered to have substantial scope for energy savings. The RMG sector is the most energy intensive of all industry sectors, consuming 15.4 per cent of the total industrial energy demand in Bangladesh. The textile sector follows the RMG subsector, with its demand representing 12.4 per cent of the industry sector.¹

8. Investment in energy efficiency is regarded by textile and RMG factories as an additional financial burden so is rarely undertaken, especially in small to medium-sized factories. The feasibility study conducted for the textile sector suggests that the larger textile facilities are typically 10–15 per cent more energy efficient than those in the small or medium categories, with efficiency differences being as high as 50 per cent in some cases. In the RMG sector, small factories having small-scale investments are more oriented towards increasing short-term profits than increasing productivity and profits in the long run.

9. The proposed programme “Promoting private sector investment through large scale adoption of energy saving technologies and equipment for Textile and Readymade Garment (RMG) sectors of Bangladesh” offers concessional funding to local financial institutions to on-lend to private sector textile and RMG entities to procure energy-saving equipment and technology for replacement or expansion with the aim of reducing GHG emissions from the subsectors, and ultimately from Bangladesh. The programme period is 20 years including an implementation period of 11 years.

10. IDCOL of Bangladesh is the accredited entity (AE). In addition to IDCOL, four local financial institutions (LFIs) will be executing entities (EEs), BRAC Bank Limited, City Bank Limited, IDLC Finance Limited, and South East Bank Limited. The Sustainable and Renewable Energy Development Authority (SREDA) under Ministry of Power, Energy & Mineral Resources will be the implementing entity for the policy component.

11. The programme is comprised of five components: two funding components and three technical assistance (TA) components.

12. Component 1 entails USD 133.0 million of financing for energy-saving equipment and technologies for the textile sector. Under component 1, IDCOL, as the EE, will make GCF funds available to the private textile manufacturing businesses in the form of senior loans on concessional terms together with its own contribution of USD33.0 million, which IDCOL would provide from its own account or mobilize from other sources. Disbursements will be made from IDCOL in USD and local currency for the businesses to procure energy-efficient equipment and technologies. Technical eligibility criteria require a 15 per cent energy-saving potential for the equipment to be funded. Technical eligibility will be confirmed by IDCOL based on manufacturers’ catalogues or certifications, or energy auditors’ reports. Component 1 will finance replacement and expansion of existing equipment and facilities. The maximum amount of a single loan is set as USD 20 million.

13. In component 2, TA amounting to USD 3.05 million is to be provided from GCF to support IDCOL’s funding activities in component 1. This will help IDCOL to establish an implementation strategy and build the capacity of IDCOL staff for efficient adoption of energy-efficient technologies as well as effective monitoring, reporting and verification of energy

¹ Energy Efficiency and Conservation Master Plan up to 2030 (2015) prepared by SREDA and the Power Division, under the Ministry of Power, Energy and Mineral Resources

savings and GHG emission reductions. Market awareness of energy-efficient equipment and technologies will be enhanced to generate a project pipeline and for swift implementation of the programme. Government agencies will be approached so that challenges identified during the programme can be shared to develop enabling policies and regulations for investments in energy efficiency in Bangladesh.

14. Component 3 provides USD 200.0 million of financing for energy-saving equipment and technologies for the RMG sector. As EEs, the four LFIs will on-lend the GCF concessional funding in USD to private sector RMG businesses for investment in energy-saving equipment and technologies. IDCOL, as the AE, will channel the GCF funds to the four LFIs. The LFIs will provide an additional USD 50 million on their own account in both USD and local currency in total. Technical eligibility criteria require a 10 per cent energy-saving potential for the equipment to be funded. Similar to component 1, technical eligibility will be confirmed by the LFIs based on manufacturers' catalogues or certifications, or energy auditors' reports. Component 3 will finance replacement and expansion of existing equipment and facilities. The maximum amount of a single loan is set as USD 20 million.

15. Component 4 is for TA amounting to USD 2.3 million to support the lending activities of the four LFIs in component 3, and to promote energy efficiency investments in the RMG sector. This component includes support for the EEs to undertake preparatory work for implementation, and capacity-building for EE staff and private sector entities in the RMG sector. Technical support for the EEs and private sector entities is also provided for monitoring, reporting and verification of the savings achieved and for programme implementation. In addition, this component envisions a technology fair to be held within three years of the start of implementation, to promote market awareness and access to EE technology. This knowledge dissemination event will cover both the textile and RMG subsectors.

16. Component 5 is for TA amounting to USD 1.15 million to strengthen the regulatory and institutional framework at the national level to overcome the operational constraints related to implementing energy efficiency and conservation in the two subsectors in the country. It offers technical support for assessing the existing regulatory framework and preparing policies and strategies conducive to the promotion of energy efficiency in Bangladesh. It further includes capacity development and training of government officials and other stakeholders on energy efficiency measures and financing. The component will be implemented by IDCOL, as the EE, and SREDA.

17. The programme requires total financing of USD 340.5 million, of which GCF is requested to provide a USD 250.0 million loan and USD 6.48 million TA grant. The balance of USD 84.02 million is to be provided by IDCOL (a USD 33.0 million loan plus USD 1.0 million contribution in kind), the four LFIs (a USD 50.0 million loan) and SREDA (a USD 20,000 in-kind contribution). The GCF loans have a tenor of 20 years including a 5-year grace period.

18. The AE estimates that the programme will reduce or avoid about 18.6 MtCO₂eq of GHG emissions in total over the 20-year lifetime of the investments by offering financial and technical resources to private companies operating in the textile and RMG sectors to access investment opportunities for energy-saving technology upgrades. In that estimation, 13.1 MtCO₂eq, or 70 per cent of the programme's target GHG emission reduction will be achieved in the RMG sector while 5.5 MtCO₂eq or 30 per cent will be achieved in the textile sector.

19. The estimation is made taking into account the emission factors of the major sources of energy (natural gas, diesel, electricity from grid) being consumed by factories, and emission savings that can be achieved by introducing energy efficiency technologies. The AE further estimates that 50 per cent of the total funding would be used by large factories. The rest would be directed to finance medium- and small-scale factories.

20. In response to a question from the independent Technical Advisory Panel (TAP) on treatment of the existing equipment replaced with energy-efficient equipment funded by the programme, the AE clarified that the replaced equipment might be resold in the market if it is still considered to have economic lifetime and monetary value. The AE agreed to take into account the estimated GHG emissions that are likely to continue if the replaced equipment, which is less energy efficient, remains in operation for some time. The AE explained that it assumes that the existing equipment has a machine life of 10–15 years and could be replaced and resold at years 6–10. Subsequently, the equipment would be used for 4–5 years by the new buyer before it is fully retired. According to the AE, these ongoing emissions can be estimated to be 1.5 MtCO₂eq and 2.6 MtCO₂eq in the textile and RMG sectors, respectively.
21. Accordingly, the AE estimates that the total amount of GHG emission savings the programme is expected to deliver amount to 14.5 MtCO₂eq, of which the saving in the textile sector is 4.0 MtCO₂eq (27.6 per cent), and that of the RMG sector is 10.5 MtCO₂eq (72.4 per cent).
22. The TAP assesses the impact potential of the programme to be medium/high.

1.2 Paradigm shift potential

Scale: Medium/High

1.2.1. Potential for scaling up and replication

23. The programme offers USD 416 million for concessional financing and technical support to IDCOL and the four LFIIs to support investments in energy efficiency in the sectors. Long-term concessional financing will be required from GCF so that loans from IDCOL and the four LFIIs to the sectors can be priced low enough so that the present value of the energy-efficient equipment can be at par with that of conventional equipment when financed with ordinary market-priced loans. In the Bangladesh market today, the cost of energy-efficient equipment is 20–25 per cent higher than conventional equipment.
24. The TAP considers that the programme scalability and replicability are largely dependent on the availability of concessional financing. The bankability of energy efficiency investments cannot be established in the absence of concessional financing. The textile and RMG sectors contain more than 1,400 and 4,000 factories respectively, among which only a fraction operates energy efficiently. Therefore, if the programme is to upscale, the funding requirement would be large. Accordingly, the TAP considers the programme's potential for scaling up and replication to be limited unless a large amount of concessional financing is secured.

1.2.2. Potential for knowledge and learning

25. The TA components of the programme will contribute to knowledge creation and dissemination about energy-efficient equipment and technologies and their benefits for the textile and RMG industries among key stakeholders in Bangladesh. Knowledge and lessons learned will be accumulated and shared through consultations and, more importantly, through the monitoring activities to be carried out with the assistance of consultants during implementation.

1.2.3. Contribution to the creation of an enabling environment

26. The programme provides components that will help develop the energy efficiency market for the textile and RMG sectors in Bangladesh. The loan components support the private companies in the sector to overcome the financial and technical hurdles to investing in energy-efficient equipment and technologies. Technology fairs are planned in the programme design to

create an interface with the investors/users, and the manufacturers/suppliers of energy-efficient equipment. Knowledge and lessons learned collected throughout the programme's implementation will be shared among key stakeholders including financial institutions, industrial communities and government agencies. The programme may contribute to bring down the costs of the energy-efficient equipment by creating a new market for energy-efficient equipment and technology in these sectors.

1.2.4. Contribution to the regulatory framework and policies

27. The programme promotes capacity-building of the policymakers in fields related to energy efficiency in industry especially in the textile and RMG sectors in Bangladesh. Component 5 of the programme, to be implemented jointly by the AE and SREDA, offers technical support for assessing the existing regulatory framework and preparing policies and strategies conducive to promoting energy efficiency, as well as training of government officials and other stakeholders.

28. The TAP views the paradigm shift potential of the programme as medium/high.

1.3 Sustainable development potential Scale: Medium/High

1.3.1. Environment co-benefits

29. The programme will promote reduction in the use of fossil fuels (natural gas, diesel and coal) as energy sources in the sectors' operations, thus contributing to improved air quality in Bangladesh. Energy consumption in the textile and RMG sectors is skewed towards use of natural gas with grid electricity having a minor share. Diesel is mostly used to generate electricity as a backup and its use depends on non-availability of grid electricity. Although large factories are equipped with captive natural gas power plants that supply electricity, small RMG factories rely heavily on diesel generation sets to fulfil their power needs. The share of diesel has increased significantly, by 2,440 per cent, while the shares of natural gas and grid electricity have declined by 22 per cent and 75 per cent, respectively.

1.3.2. Economic co-benefits

30. The programme will help the industry to improve production and capacity utilization and reduced production costs, which will lead to improved productivity and competitiveness. The industry contributes more than 13 per cent of the country's GDP and 86 per cent of the country's export. It also generates more than 65 per cent of the country's industrial employment, employing over 4 million people of whom 80 per cent are women. In addition, the industry promotes development of small-scale industries, because 99 per cent of ancillary activities such as dyeing, printing and manufacturing zippers and labels are carried out locally.

31. The programme contributes to environment and economic development of the country by enhancing energy efficiency in the industry which contributes more than 13 per cent to the nation's GDP. The sector is also the largest sector of employment. Although the industry has a workforce comprising more than 80 per cent women, any benefits that the programme generates are likely be indirect. As such, the TAP considers the sustainable development potential of the programme as medium/high.

1.4 Needs of the recipient

Scale: High

1.4.1. Economic and social development level of the country and affected population

32. Bangladesh remains a least developed country although its economy recorded remarkable growth since 2010 as a result of strong private consumption and investment.

33. Bangladesh is one of the most adversely affected countries in terms of climate change. It is listed in the top 10 most affected countries of the world in the Climate Risk Index for 2017. The topography is flat, and the majority of the landmass lies within 10 metres above mean sea level. All land types except highlands are exposed to monsoon flooding for part or all of the year. Floodplains located in the north-western, central, south-central and north-eastern regions are subject to regular flooding at different frequencies and intensity. The coastal plain is subject to cyclones and storm surges, salinity intrusion and coastal inundation. It is estimated that the impact of climate change on Bangladesh could result in an annual loss of 2 per cent of GDP by 2050 and 9.4 per cent of GDP by 2100.

1.4.2. Absence of alternative sources of financing

34. The programme requires total financing of USD 340.5 million, of which a USD 250.0 million loan and a USD 6.48 million TA grant are required from GCF. The GCF loans will be priced at a concessional rate and have a tenor of 20 years including a 5-year grace period.

35. The public sector investments in Bangladesh are presently focused more on supply-side interventions in order to increase generation capacity and bridge supply gaps. Budgetary resources available to drive public sector investments for demand-side interventions, especially towards energy-saving technologies, are therefore constrained.

36. Under the current market scenario, the industry may access commercial loans at an interest rate of 9–15 per cent based on creditworthiness and credit history. Although bank lending to the sector has grown by over 15 per cent since 2016, the sector is still perceived as being a high credit risk. There are several loan facilities for energy efficiency and conservation promotion; however, higher lending interest rates and the time required for appraisal have hampered smooth implementation of loan projects.

37. The TAP notes that alternative sources of financing are not currently available, especially of this size and tenor.

1.4.3. Needs of strengthening institutions and implementing capacity

38. Although the Energy Efficiency and Conservation Master Plan up to 2030 (2015) prepared by SREDA and the Power Division under the Ministry of Power, Energy and Mineral Resources confirms the potential and importance of energy efficiency in the industry and building sector for economic development, the implementation and promotion of energy-efficient equipment and technologies has been slow in Bangladesh due to lack of concessional funding and technical capacity. The programme includes capacity-building activities not only for financial institutions and industry but also for policymakers in fields related to energy efficiency in industry, especially in the textile and RMG sectors. Technical support is included for preparing policies and strategies conducive to promoting energy efficiency, as well as training of government officials.

39. The TAP assesses the needs of the recipient of the programme as high.

1.5 Country ownership

Scale: Medium/High

1.5.1. Existence of a national climate strategy

40. Bangladesh submitted its intended nationally determined contribution to the United Nations Framework Convention on Climate Change (UNFCCC) in September 2015. Bangladesh ratified the Paris Agreement on 21 September 2016 and its nationally determined contribution (NDC) can now be found on the UNFCCC's interim NDC Registry.

41. Bangladesh's NDC describes its plans for reducing GHG emissions through a progressive approach to developing its economy on a low-carbon pathway. In the NDC Bangladesh committed to reducing GHG emissions in the power, industry and transport sectors by 5 per cent below the business as usual scenario by 2030 using only domestic resources, or by 15 per cent below the business as usual scenario by 2030 if sufficient and appropriate international support is received.

42. Bangladesh's NDC identified three key emitting sectors: the industrial sector accounts for 40.6 per cent of total emissions; the power sector accounts for 32.8 per cent; and transportation 26.6 per cent. Accordingly, improving industrial energy efficiency is identified as one of the three key mitigation interventions together with renewable energy expansion, and transport infrastructure modernization. This has been embedded in the country's 5-year planning process.

1.5.2. Coherence with existing policies

43. The Energy Efficiency and Conservation Master Plan up to 2030 (2015) contemplates a programme to extend low-interest loans for energy efficiency and conservation undertakings in industrial and building sectors. The programme will contribute to the government's target to achieve a 10 per cent reduction in energy consumption by the industry sector by 2030, and 20 per cent improvement in energy intensity by 2030.

1.5.3. Capacity of accredited entities or executing entities to deliver

44. IDCOL is the AE of this programme. It is also the EE of all the financial and technical components except component 3 where the four LFIs extend loans to the RMG sector. IDCOL will assess the eligibility of the borrowers in the textile sector for financing under the programme as per the criteria set prior to loan execution.

45. IDCOL is a non-banking institution established in 1997 and is owned by the Government of Bangladesh. It has a mission to catalyse and optimize private sector participation in promotion, development and financing of infrastructure, as well as renewable energy and energy efficiency projects in a sustainable manner through public-private partnership initiatives.

46. The financial performance of IDCOL is compatible with or better than other non-banking institutions in Bangladesh. The total assets of IDCOL increased from 66,980 million Bangladeshi taka (BDT)² in 2015 to BDT 90,876 million in 2019 (i.e. by 35.7 per cent) while its loan book grew from BDT 36,964 million to BDT 68,799 million (86.1 per cent). Its total revenue increased from BDT 4,324 million to BDT 5,378 million (24.4 per cent) during the same period. It is rated AA1 by Credit Rating Agency of Bangladesh. The IDCOL loan portfolio in 2018 included renewable energy projects (33.2 per cent) and infrastructure (including energy efficiency) projects (66.9 per cent).

² As of October 2020, 1 USD = 84.70 BDT

47. IDCOL has implemented loans and grants amounting to more than USD 2,100 million channelled through the government from development finance institutions. Its development partners include World Bank, Asian Development Bank, Japan International Cooperation Agency, KfW, Deutsche Gesellschaft für Internationale Zusammenarbeit, Inter-American Development Bank, United States Agency for International Development and GPOBA (now known as Global Partnership for Results-Based Approaches, GPRBA). Under the USD 100 million Energy Efficiency and Conservation Promotion Financing Project funded by the Japan International Cooperation Agency, IDCOL has approved USD 42 million for the RMG and cement sectors, out of which USD 25 million has been already disbursed.

48. Component 3 will be executed by four LFI: BRAC Bank Limited, City Bank Limited, IDLC Finance Limited and South East Bank Limited.

49. BRAC Bank Limited, established in 2001, is well known for its banking services aimed at small and medium-sized enterprises in Bangladesh, with half of its lending portfolio being small and medium companies. In 2015-2019, revenue and net profit after tax increased from BDT 6,835 million and BDT 2,436 million to BDT 9,948million and BDT 5,646 million (average growth rate of 11.4 per cent and 32.9 per cent), respectively. Total assets of BRAC Bank Limited amount to BDT 367,920 million in 2019 showing an average growth rate of 16.0 per cent during the same period, while its portfolio of loans and advances grew by 19.8 per cent per annum. In 2018 it reported BDT 12,000 million of financing for green projects. The bank is rated as AA1 by the Credit Rating Agency of Bangladesh.

50. City Bank Limited was established as Bangladesh's first private commercial bank in 1983. It provides a wide range of deposit and loan products and solutions catering for the needs of its broad client base. The total assets of City Bank increased at an annual growth rate of 14.8 per cent since 2014to reach BDT 324,780 million in 2018. In 2019 it reported that its loan portfolio and profit after tax had increased to BDT 246,944 million and BDT 2,472 million with an annual growth rate of 6.7 and 22.5 per cent, respectively. The total assets of City Bank were reported to be BDT 354,698 million in 2019. In 2018 its lending amount to the RMG sector was BDT 39,251 million, and cumulative disbursements to green projects amounted to BDT 1,359 million. It is rated as AA2 by Credit Rating Agency of Bangladesh.

51. IDLC Finance Limited, established as a single product lease company in 1985, is regarded as one of the pioneers in the non-banking sector in Bangladesh. It currently has a diversified operation across different segments in banking services. Revenue and net profit after tax in 2019 amounted to BDT 4,815 million and BDT 1,522 million, exhibiting a growth rate of 7.7 per cent and 5.7 per cent since 2014, respectively. During the same period total assets increased from BDT 57,160 million to BDT 114,655 million at a rate of 14.9 per cent. IDLC Finance Limited offers 52 green products under eight categories as defined by Bangladesh Bank, the Central Bank of Bangladesh. In 2018, the green banking portfolio stood at BDT 648 million and cumulative disbursements in the green segment reached BDT 1,255 million. IDLC Finance Limited is rated AAA.

52. South East Bank Limited is a private commercial bank launched in 1995. It provides conventional banking products and services to retail and business clients, along with specialized products/services such as an investment fund and automatic fraud monitoring. Its total assets in 2018 were reported to be BDT 381.095 million, which was the result of annual growth of 13.6 per cent over the previous four years. Revenue was BDT 14,724 million and net profit after tax was BDT 2,456 million in 2018. Its exposure to the RMG sector was BDT 50,850 million in 2018. The total lending to green projects was BDT 12,671 million. Southeast Bank Limited is rated AA by Credit Rating Information and Services Limited of Bangladesh.

1.5.4. Engagement with civil society organizations and other relevant stakeholders

53. As part of the project development, the AE carried out consultation workshops for key stakeholders including government agencies, financial institutions, representatives of textile and RMG companies, academia and civil society organizations. Reports of the consultation workshops are attached to the funding proposal.

54. A no-objection letter issued by the Ministry of Finance, Government of Bangladesh, the GCF national designated authority, is attached to the funding proposal.

55. The TAP considers the programme's country ownership as medium to high. The programme is in line with the national policies and strategies that support energy efficiency in the industrial sector. It was developed in consultation with key stakeholders. The TAP considers that the AE has sufficient experience and capacity to implement the programme as presented. While the four LFIs do have exposure to the RMG sector, their capacity to implement the programme cannot be fully confirmed.

1.6 Efficiency and effectiveness Scale: Medium

1.6.1. Cost-effectiveness and efficiency regarding financial and non- financial aspects

56. As per the AE's projection, the programme intends to deliver total GHG emission savings of 14.5 MtCO₂eq. Accordingly, the cost of GHG emission reductions per tCO₂eq is estimated to be USD 23.4 against the total cost of funding of USD 340.5 million. With respect to GCF funding of USD 256.5 million, it is projected to be USD 23.4 per tCO₂eq.

57. The cost varies between the textile and RMG sectors: USD 34.3 per tCO₂eq and USD 19.2 per tCO₂eq, respectively, against the programme funding. With respect to GCF funding, it is estimated to be USD 25.8 and USD 14.5 per tCO₂eq for the textile and RMG sectors, respectively.

1.6.2. Amount of co-financing

58. The co-financing ratio of GCF financing is estimated to be 0.33. The programme requires a total financing of USD 340.5 million, of which USD 256.48 million is provided by GCF in the form of loan and grant. The balance of USD 84.02 million is to be provided by IDCOL, the four LFIs and SREDA in the form of a loan and in-kind contribution.

59. The total amount of funding to be mobilized for purchasing energy-efficient equipment is expected to be higher than the total amount of the programme. IDCOL and four LFIs expect borrowers to provide equal amounts of funds to their loans. Accordingly, an additional USD 83.0 million can be mobilized in the form of borrower's equity contribution. If that amount is included, then the co-financing ratio goes up to 0.65.

1.6.3. Programme/project financial viability and other financial indicators

60. The programme promotes investments in energy efficiency by providing concessional financing so that the total cost of investment, including equipment, financial and operational costs that a company incurs for its purchase of energy-efficient equipment would not be disadvantageous when compared with that of conventional equipment. On the basis of a survey, the AE estimates that the upfront cost of energy-efficient equipment is approximately 25 per cent higher than that of conventional equipment. Therefore, the interest rate of the programme's loans for the companies will be set lower than the market interest rate so that the present values of investment in energy-efficient equipment and conventional equipment become compatible. The loan tenor will extend up to 10 years to help companies manage yearly cash outflow.

1.6.4. Industry best practices

61. The programme will help the textile and RMG sectors to showcase the financial and operational merits of energy-efficient equipment and technologies and establish energy-efficient practices in Bangladesh. The three technical components include assistance to the EEs and private sector companies in the sectors to enhance monitoring and market awareness. A technology fair will be organized within three years of the start of implementation, to promote market awareness and access to energy-efficient technologies for both sectors. The government agencies will be supported so that challenges identified in the programme can be shared and lessons learned can be reflected in developing enabling policies and regulations for investments in energy efficiency in Bangladesh.

62. The TAP assesses the efficiency and effectiveness of the programme as medium.

II. Overall remarks from the independent Technical Advisory Panel

63. The TAP recommends that the Board consider the proposed programme with the following two conditions:

- (a) The AE shall monitor and report to GCF actual annual energy consumption of each piece of eligible energy-efficient equipment and technology which has been financed by the Programme, for as long as any amount of the related sub-loan is outstanding; and
- (b) The AE shall monitor and report to GCF annual energy consumption of each piece of equipment and technology that has been replaced as a result of the Programme, from the date it was replaced until:
 - (i) The sixth anniversary of the replacement; or
 - (ii) The relevant equipment or technology has been scrapped and retired from operation, whichever is earlier.

64. The programme offers concessional funding to finance energy efficiency investments for replacement and expansion in the textile and RMG sectors in Bangladesh. The AE assumes that replaced equipment would be resold at the secondary market rate for replacement of even less efficient equipment. Accordingly, there would be no leakage with respect to emission reductions. However, the TAP is unable to fully agree to the AE's assumption, because that would not be always the case. Rather, the likely case is that replaced equipment would be purchased on the secondary market for the purpose of expansion to catch up with market growth, rather than replacement, especially if textile and RMG factories have limited interest in energy efficiency.

65. Further, the TAP sees a risk that the replaced equipment would be operated longer in total than its expected lifetime so the estimated impact on emission reductions could be larger than that assumed. In that case, the estimated emission reductions of the programme would turn out to be less. To validate and confirm the leakage of emission reductions, it is imperative that actual energy consumption of all equipment replaced is actually monitored so that GCF can reasonably evaluate the programme's ultimate impact with respect to GHG emission reductions.

66. Without the strong monitoring plan, the TAP would recommend that the Board consider a request to the AE to ensure all of the replaced equipment to be scrapped and retired from operation.

Reply to the Independent Technical Advisory Panel assessment findings (FP150)

Proposal name: Promoting private sector investment through large scale adoption of energy saving technologies and equipment for Textile and Readymade Garment (RMG) sectors of Bangladesh

Accredited entity: Infrastructure Development Company Limited (IDCOL)

Impact potential

As rightfully mentioned in the ITAP assessment that the the textile sector (spinning, weaving, knitting and dyeing), and readymade garment (RMG) sector (cutting, sewing, dyeing, ironing, and finishing) have substantial scope of energy savings. The RMG sector is the most energy intensive in the industry consuming 15.4 per cent of the total industrial energy demand in Bangladesh. The textile sector follows the RMG subsector with its demand representing 12.4 per cent of that of the total industry. The proposed programme “Promoting private sector investment through large scale adoption of energy saving technologies and equipment for Textile and Readymade Garment (RMG) sectors of Bangladesh” aims to achieve the Fund’s objective and result areas by contributing shift to low-emission sustainable development pathways. The contribution is achieved by providing target market i.e. Textile and RMG sector of Bangladesh, access to concessional line of financing for undertaking energy efficiency measures which will result in 14.53 MtCO₂ emissions savings as well as provide market participants with appropriate technical support required for stimulating the energy efficiency market in Bangladesh, as well as to support implementation of energy efficiency initiatives.

Paradigm shift potential

The paradigm shift potential is rightly described in the ITAP assessment, IDCOL as a government owned development financial institution will strongly support government’s target and vision to enable the industrial energy transition to low carbon energy system through major increase in energy efficiency. The success of the proposed programme will enable replication of similar programmes in other sectors of the country.

Sustainable development potential

IDCOL agrees with the remarks made by ITAP regarding the programme’s sustainable development potential. Delivering a large-scale retrofit initiative through this programme will have large and important development benefits whose impacts will increase over time as energy prices rise.

Needs of the recipient

Bangladesh is in dire need of concessional financing, capacity development and marketing activation to realize the potential of industrial energy efficiency. The need of the proposed programme is indeed high, especially considering the severe impact that the manufacturing sector has faced due to impacts of COVID-19.

Country ownership

As mentioned in the ITAP assessment the Country Ownership of the proposed programme is relatively higher. The proposed programme is regarded in the Country Programme (A16 & A17) as of strategic importance and considered to be (a) well aligned with national development and climate priorities and (b) reflective of the national priorities with a focus on appropriate action in the field of mitigation.

Country Program:

http://nda.erd.gov.bd/files/1/Publications/CC%20Policy%20Documents/Bangladesh_GCF-CP_Draft.pdf

Efficiency and effectiveness

As rightfully mentioned in the ITAP assessment the proposed programme is designed to optimize the use of funding available in terms of leverage and sustainability, investing concessional resources optimally in bankable projects to enable adequate financing and creating market instruments to manage risks among the market actors. Furthermore, the programme will help the textile and RMG sectors to showcase financial and operational merits of energy efficiency equipment and technologies, and establish energy efficient practices in Bangladesh.

Overall remarks from the independent Technical Advisory Panel:

The ITAP recommended in its assessment, that IDCOL has to consider following two conditions.

- 1) *The AE shall monitor and report to GCF actual annual energy consumption of each piece of eligible energy-efficient equipment and technology which has been financed by the Programme, for as long as any amount of the related sub-loan is outstanding; and*
- 2) *The AE shall monitor and report to GCF annual energy consumption of each piece of equipment and technology that has been replaced as a result of the Programme, from the date it was replaced until:*
 - a) *The sixth anniversary of the replacement; or*
 - b) *The relevant equipment or technology has been scrapped and retired from operation, whichever is earlier.*

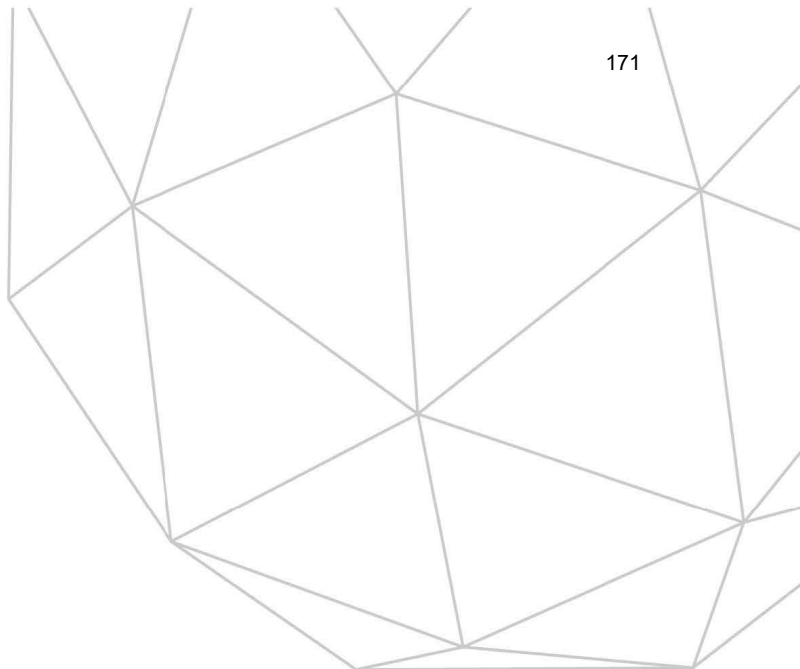
IDCOL's response:

IDCOL agrees to the above conditions and will work with GCF to implement a reporting system.



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GENDER ASSESSMENT REPORT

Promoting private sector investment through large scale
adoption of energy saving technologies and equipment for
garment sector in Bangladesh

Table of Contents

Executive Summary	4
1 Introduction	5
2 Rational of Gender Assessment	6
3 Methodology of the Gender Assessment.....	7
3.1 Literature Review.....	7
3.2 Factory Survey and Household Survey.....	7
3.3 Key Informant Interview.....	7
3.4 Stakeholder Engagement Workshop.....	8
4 Policies and Regulations.....	9
4.1 National Policy, Strategy and Framework.....	9
4.2 International Policy and Framework.....	12
4.2.1 Green Climate Fund (GCF).....	12
4.2.2 World Bank Group.....	13
4.2.3 Asian Development Bank (ADB).....	14
4.2.4 International Finance Corporation (IFC)	15
4.2.5 Japan International Cooperation Agency (JICA).....	16
4.3 Gender Related Policies of IDCOL and Local Financial Institutions (LFIs)...	17
5 Findings from Gender Assessment.....	19
5.1 Introduction.....	20
5.2 Labor Force Participation	21
5.3 Constrains for Female Employment	26
5.4 Educational Status	28
5.5 Gender Perspectives of Energy Efficiency	30
5.6 Decision Making	31
5.7 Wage and Benefits	32
5.8 Workplace Environment	33
5.9 Social beliefs, perceptions, and stereotypes.....	37
5.10 Access to Information	38
5.11 Women and Health	39
5.12 Access to Financial Services by RMG Workers	39
5.13 Leadership Opportunities for Women in RMG Sector	40
6 Gender Inclusion Practices in Bangladesh.....	42
7 Legal and Administrative Framework Protecting and Promoting Gender Equality in Bangladesh.....	44

8 Alignment of the Project with GCF Gender Assessment Guideline	47
9 Grievance Redressal System (GRS)	49
10 Recommendations	51
11 Gender Action Plan	52
11.1 Outline	52
11.2 Monitoring of Gender Action Plan	59

List of Tables

Table 5-1 Trends in labour force and employment	22
Table 5-2 Annual average change in labour force and participation rate ⁷	23
Table 5-3 Revised wage structure by GoB in 2019	32

List of Figures

Figure 5-1 Gender scenario in labour force participation in RMG units	22
Figure 5-2 Gender distribution of workers in selected RMG units	24
Figure 5-3 Gender of facility managers	24
Figure 5-4 Major constraints to female employment perceived by different stakeholders	27
Figure 5-5 Education Attainment of the Female Workers	28
Figure 5-6 Education Attainment of the male Workers	29
Figure 5-7 Educational qualification of facility managers	29
Figure 5-8 Country-wise Garments Monthly Minimum Wage 2019 / Source: NYU Stern Center for Business and Human Rights	33
Figure 5-9: Workers' Responses on Maternity Care	35
Figure 5-10 Respondents receiving training on chemical handling and storage	35
Figure 5-11 Respondents receiving training on fire management	36
Figure 5-12 Respondents receiving environment, health and safety training	36

Executive Summary

Ready-made garment (RMG) sector in Bangladesh plays a key role in the economic development of the country as one of the biggest industrial contributors to the country's gross domestic product (GDP). The RMG industry now contributes over 83% to Bangladesh's export income. Moreover, this sector is one of the major sources of employment for women in Bangladesh, especially for the poor and unprivileged women. Women were the main workforce in the country's apparel sector since the sector began its journey in 1970's. The RMG sector has been a key enabler in overall socio-economic development of Bangladesh. In common perspective, a majority of the 4-5 million people employed in the sector are women - 80% of the workers and more than 60% of the total workforce for the sector are women.

The employment of these women, mostly from rural areas, have provided them with financial freedom and greater social independence. Compared to other industries, women employment rate is higher in RMG sector in Bangladesh. So, for implementing any project in the RMG sector it is very important to ensure equal participation of both male and female in all the project interventions. The GCF also promotes environmental, social, economic and development co-benefits through taking a gender-sensitive approach. In this regard, this project aims to assess the gender dimensions in RMG sector in Bangladesh during project preparation, implementation and monitoring stage. For this purpose, the study team conducted stakeholder consultations and field surveys to collect information and analyzed those data through a gender lens. The results were incorporated throughout the other project documents and mainstreamed into the mitigation and adaptation initiatives to obtain in such a way that gender co-benefits are obtained.

The objective of the Gender Assessment was to identify the gender aspects of the project activities, including selection of employees and their trainings, to collect information from various stakeholders such as workers, supervisors, managers, key informants (women and men) and to learn the impact of the project on empowerment of different categories of women and men. This tool also helped to promote gender – relevant entry points, policies and identify opportunities for enhancing gender equality in this project.

Through this gender action plan the project aims to ensure that the services and technologies provided by the project/program be available and accessible to both women and men, equal access for the women and men from vulnerable communities to information and opportunities necessary to participate and benefit fully from the anticipated outcomes of the project and promote leadership capacity of women.

1 Introduction

Improved energy efficiency in Bangladesh can lead to greater business growth and profitability at the local level; at the national level it supports GDP growth and enhances energy security, and at a global level it helps mitigate climate change and ensures sustainable global growth. For industries, energy efficiency reduces production costs and increases profits. This also reduces their carbon footprint at no additional cost (an important consideration in the face of international buyers' requirements). Among all the sub-sectors, Garment and textile sector (manufacturing sector) consumes maximum energy. As Garment and Textile sector consumes maximum energy, it also holds a significant potential to reduce energy consumption. Valuing these benefits, in the current environment in Bangladesh, if energy intensive industries (Garment and textile) in sectors with high savings potential start utilizing energy efficient technologies they will be poised to capture huge future economic opportunities. In addition to such economic benefits, the environmental impact of decreased energy usage and reduced carbon emissions, social impact by providing well-being and employment will create a more sustainable Bangladesh for future generations.

Based on the existing scenario of RMG sector in Bangladesh, it is evident that adoption of energy efficiency measures can considerably improve socio-economic impact of the sector as well as reduce environmental impact. Infrastructure Development Company Limited (IDCOL) intends to realize these benefits by developing and implementing a program to support entrepreneurs and business establishments with financial and market resources to avail investment opportunities for energy saving technology upgrades. IDCOL has solicited financial support from Green Climate Fund (GCF) to execute this program. This fund will be then utilized as concessional loans to finance garment business via four selected banks i.e. City Bank Ltd, Southeast Bank Ltd, BRAC Bank Ltd and IDLC Finance Ltd.

Ready-made garment (RMG) sector in Bangladesh plays a key role in the economic development of the country as one of the biggest industrial contributors to the country's gross domestic product (GDP). The RMG industry now contributes over 83% to Bangladesh's export income.¹ Moreover, this sector is one of the major sources of employment for women in Bangladesh, especially for the poor and unprivileged women. Women were the main workforce in the country's apparel sector since the sector began its journey in 1970's. The RMG sector has been a key enabler in overall socio-economic development of Bangladesh. In common perspective, a majority of the 4-5 million people employed in the sector are women - 80% of the workers and more than 60% of the total workforce for the sector are women.

The employment of these women, mostly from rural areas, have provided them with financial freedom and greater social independence. Compared to other industries, women employment rate is higher in RMG sector in Bangladesh. However, the percentage of female workers have declined in the recent years. According to a survey of Center for Policy Dialogue (CPD), released on March 2018, among garment worker 60.8% are female & 39.20% are male. The survey was conducted on 2270 workers in 193 factories in Dhaka, Gazipur, Narayanganj & Chottogram. This situation has been created in the sector due to increasing use of technology,

¹ Dey, P. 2019. RMG sector's contribution to GDP downs by 3%. TextileToday. <https://www.textiletoday.com.bd/rmg-sectors-contribution-gdp-downs-3-2/>

lack of skills of women workers compared to males, increasing interest of men for salary rise, social inequalities and due to labor-intensive nature of the sector. Even, the average age of women working in this sector is lower than that of men. In addition, there is discrimination in areas of service book, working hour including overtime, leaves (casual, sick, annual), separation of rest room, social protection like provident fund and group insurance, and representational rights.

After the Rana Plaza incident, more emphasis has been given on the compliance issues in the RMG sector. All the owners are obliged to ensure health and safety issues for the workers and ensure equal facilities for both genders. International buyers also give importance on compliance issues while placing order and usually select those that are more complaint. However, due to lack of financial support, many small and medium enterprises often lag behind to ensure the required facilities for women, especially pregnant women and mothers with dependent children. So, with this programme there will be an opportunity for the RMG units to include better facilities for women and address compliance issues as well. This way they will be able to attract more international buyers and access better opportunities for future development.

As such, gender equity is crucial to the sustainable development of all sectors of the economy. GCF also gives special attention on ensuring gender equity and focuses on gender-responsive climate action programmes and projects that benefit women and men. The GCF Governing Instrument states that: “The Fund will strive to maximize the impact of its funding for adaptation and mitigation ... promoting environmental, social, economic and development co-benefits and taking a gender-sensitive approach.” In this regard, this programme aims to assess gender dimensions in RMG sector in Bangladesh during project preparation, implementation and monitoring stage. For this purpose, the study team conducted stakeholder consultations and field surveys to collect information and analyzed those data. The results were incorporated throughout the other project documents and mainstreamed into the mitigation and adaptation initiatives to obtain in such a way that gender co-benefits are obtained.

2 Rational of Gender Assessment

The Gender Assessment will help to understand relationship between men and women, their access to resources, their activities, and the constraints they face relative to each other. A comprehensive gender assessment entails the examination of the different roles, rights, needs, and opportunities of women and men, boys and girls in this project context. It is a tool that will help to promote gender – relevant entry points, policies and identify opportunities for enhancing gender equality in this project. GCF also puts emphasis on the ensuring gender equity in all project level activities. In this regard, a Gender Assessment was conducted for mainstreaming gender into the project level activities. The objective of the Gender Assessment was-

- to identify the gender aspects of the project activities, including selection of employees and their trainings.
- to collect information from various stakeholders such as workers, supervisors, managers, key informants (women and men);
- to learn the impact of the project on empowerment of different categories of women and men;

3 Methodology of the Gender Assessment

The assessment studied the engagement of men and women in the RMG sector in Bangladesh, labor force participation, access and control to technology and financial resources, as well as power relations. The study team gathered information through literature review, surveys and stakeholder analysis. Also, household surveys were conducted to get the perspective of the community people living in the vicinity of the factory. Detailed methodology is described below:

3.1 Literature Review

At the inception phase of the project the team performed literature review from secondary sources to understand the baseline situation of the RMG sector in Bangladesh to understand the baseline situation of RMG sector and the broad scope for EE applications in the sector considering gender dimensions. In this regard, the project team analyzed relevant reports, about national policies and guidelines (environmental, social, gender equality), potential risks identified in the RMG sector, historical studies and research on Energy Efficiency (EE) in Bangladesh's RMG sector, international best practices, potential EE interventions etc. considering gender equity.

3.2 Factory Survey and Household Survey

The study team conducted open inquiries of facility managers and workers of RMG factories and community people living near the selected RMG units on energy efficiency, environment and social issues related to RMG sector focusing on gender. For this purpose, 80 respondents from 20 RMG units and 150 households were selected in four key garment hubs i.e. in Dhaka, Gazipur, Narayanganj and Chattogram district. In each of the RMG units, key informant interviews (KII) were conducted with facility managers and workers and field group discussions (FGD) were conducted with a group of 10-15 people including workers and facility managers. While choosing the respondents for FGD, the study team tried to maintain gender balance by including both male and female representatives. However, during the survey period, no female facility manager was identified so there is no representation of women in the information shared by facility managers.

3.3 Key Informant Interview with Experts

Key informant interviews (KIIs) have been conducted to supplement the information mined from secondary sources as well as to enhance the understanding of the ground realities of the RMG sector. For KII representatives were selected from government ministries, NGO, INGO, development partners, academicians, financial institution, monitoring agencies for RMG sector etc. A total of 25 KIIs were conducted based on a questionnaire which contain queries related to the following thematic areas:

- Growth prospects of RMG sector
- Compliance with environmental, social and gender (ESG) related requirements
- Energy efficiency adoption in sector
- Financing energy efficiency initiatives.

During the KIs the study team tried to maintain a gender balance by including both male and female representatives from each sector. But in most of the cases it was not possible due to absence of female leadership in the sector. However, at least one woman was interviewed from selected stakeholder categories i.e., from selected government ministries, NGO, INGO, development partners, academicians, financial institution, monitoring agencies (BGMEA, BKMEA) working for RMG sector etc.

3.4 Stakeholder Engagement Workshop

While preparing the proposal, stakeholder engagement was a prerequisite. In this regard, the information gathered during the literature review and surveys were presented during the stakeholder workshops where stakeholders and beneficiaries discussed on existing issues and concerns pertaining to Environment, Society and Gender in the Garment sector. A wide range of key stakeholders i.e. energy efficiency practitioners including regulators, policy makers, experts and garment sector entrepreneurs, as well as environment, social and gender experts actively participated and shared their insights and experiences. The workshop clarified the framework of the proposed GCF proposal along with the key indicators with the stakeholders and participants. The outcome of the workshops was noted and relevant changes in the project documents were made accordingly. The detail findings from the workshops have been presented in a separate document of workshop report.

Both the workshops were well attended. A wide range of energy efficiency practitioners including regulators, policy makers, experts, academicians, researchers, garment sector entrepreneurs, NGO, INGO, development organizations, CSOs as well as environment, social and gender experts were attended the workshop. In the first workshop total 46 participants were present among them 15% were female participants and in the second workshop there were 45 participants where 11% were female participants. Even though RMG sector occupies a high number of women workers, there few women involved in the higher management. Also, very few organizations have women involved in project management or research regarding RMG sector related issues. So, the representation of women was less compared to men in the workshops. However, in both the workshops it was maintained that at least one woman was present from relevant government ministries and departments, monitoring agencies (BGMEA, BKMEA), academicians, researchers, garment sector entrepreneurs, NGO, INGO, development organizations, CSOs and sectoral experts.

4 Policies and Regulations

4.1 National Policy, Strategy and Framework

The government of Bangladesh recognizes the need for coordinated efforts on gender issues and the need to mainstream gender issues in development initiatives to increase resilience. At present Bangladesh is engaged in aligning national legislation and policies with its international commitments to strengthen gender dimensions. Some key policies are with regards to climate change and adaptation and the provision of gender issues in those policies are highlighted below. In addition, there is also a set of supporting 'Rules' and 'Regulations' that include important provisions.

The Constitution of Bangladesh has recognized fundamental rights of women workers at work places including necessary social protection (Articles, 28, 29, 34, 38, 14, 15 and 20(1), (The Constitution of Bangladesh). Bangladesh Constitution provides that the state shall not discriminate against any citizen on the grounds of religion, race, caste or place of birth and women shall have equal rights with men in all spheres of the state or public life (Article 28). Bangladesh as a state is responsible for its citizens to ensure the right to work that is the right to guaranteed employment at a reasonable wage having regard to the quantity and quality of works, and reasonable rest, recreation and leisure (Article 15). Bangladesh Constitution recognizes that everyone shall be paid for work on the basis of the principle from each according to his abilities (Article 20). In addition, the fundamental rights guaranteed in Chapter III of Bangladesh Constitution, especially relevant to workers' rights, prohibits all forms of forced labour and makes it a punishable offence (Article 34) and guarantees the right to freedom of association and to form trade unions (Article 38).

Bangladesh Labour Act, 2006 (BLA) has included a large number of labour rights for protection of women workers. BLA has various positive points relating to labour rights, but due to its various limitations, absence of adequate penalty in statutory law for employers and lack of proper code of conduct by the garment authorities, there is scope of violation of labor rights in some areas.

The BLA 2006, mainly focus on the following major components:

- a. Sanitation and Hygiene:** From section 51-59, BLA 2006, the terms and conditions of cleanliness, ventilation and temperature, dust and fume, disposal of waste and effluents, overcrowding, lighting, drinking water, latrines and urinals, were discussed. In section 59, it is clearly mentioned to provide distinct toilet facility for men and women along with proper cleaning maintenance.
- b. Maternity Benefit:** Every pregnant women worker can get the maternity benefit of eight weeks, preceding the expected day of her delivery and eight weeks immediately following the day of her delivery (Section 45-46). The procedure and amount of regarding payment of maternity benefit are also mentioned in Section 47- 49.
- c. Working conditions and hours:** Employment of women, adolescents or children in specific works (such as the dangerous operations, underground or underwater work) is prohibited or restricted (Section 39, 40, 42 and 79). Furthermore, without the consent of women worker, working between 10.00PM to 06.00AM is not allowed (Section 109).

- d. Safety:** According to Section 89, At least one first-aid box for every one hundred fifty workers must be accessible and maintained. Also, for the establishments with forty or more workers, a child care room with proper facilities should be provided (Section 94).
- e. Trade Union:** If 20% of the total workers or members are women than while establishing trade union, the executive body must comprise minimum 10% women members (Section 176e, BLA, 2013).
- f. Indecent behavior:** Despite of the rank or status of a woman worker, no one of the establishment can treat or behave in indecent or improper manner with her (Section 332, BLA, 2013).
- g. Institutional framework:** If factories occupies 50 or more workers than safety committees must be formed (Section 90A, BLA, 2013), a Safety welfare officers shall be appointed and the cost of treatment of occupational diseases will be covered (Section 150(3), BLA, 2013) if the establishment holds more than 500 workers (Section 89(8), BLA, 2013) and workplaces with more than 5000 members or staffs should establish workplace health centres (Section 89(6), BLA, 2013).

Bangladesh Labor Rules, 2015 emphasizes on gender equity and has special provision for pregnant women. According to the Bangladesh Labor Rules, 2015, the owner and other workers should have the following responsibilities to a pregnant worker, such as:

- (a) Such behavior or comment should not be made so that she gets insulted or humiliated
- (b) Not engage her in hazardous work declared by the government or any work hazardous for her health.
- (c) To transfer or post her to any work where there is no hazard.
- (d) To give priority to use the lift during the work.
- (e) After the delivery, to make opportunity for breastfeeding and ensure the congenial atmosphere for this.

Right to Occupational Safety and Health BLA, 2006 prescribes provisions relating to occupational hygiene, occupational diseases, safety measures, industrial accidents, protection of women and young persons in dangerous occupations and also covers conditions of work, working hours, welfare facilities, holidays, leave etc. (Sections 11, 51, 52, 53, 56, 57, 58, 61, 62, 91, 92, 93, 94 and, 109, BLA, 2006). Most of the provisions lack in standard values and are not specific rather general in nature. In terms of occupational safety and health issues, the status of Bangladesh is very poor and women workers are extremely vulnerable in this regard.

Section 195 of the Bangladesh Labor Act, 2006 (amended 2013) outlaws numerous “unfair labor practices.” For example, no employer shall “dismiss, discharge, remove from employment, or threaten to dismiss, discharge, or remove from employment a worker, or injure or threaten to injure him in respect of his employment by reason that the worker is or proposes to become, or seeks to persuade any other person to become, a member or officer of a trade union.”

The National Women Development Policy (2011) briefly promotes women’s role in environmental management and the importance of ensuring facilities for and the security of pregnant women in the event of natural disasters. It emphasizes on elimination of all forms of violence to women including physical and mental abuse and sexual harassment, rape, dowry, family abuse and acid throwing in family, society and in the workplace. The policy ensures

participation of women in decision making regarding environment management and pollution control and program implementation. It also ensures gender equal rate of wages, increase participation of women in the labor market, equal opportunity at the workplace, ensured security and removal of disparities in employment.

The policy aims to motivate all employment agencies to give women all kinds of equal opportunities under government quota and employment policy and create necessary environment to promote access of women to employment areas at increasing rate, their position as such and continue their advancement. In addition, it promotes to make necessary reforms of all concerned law, rule and policy for wider employment of women. In recognition of the contribution of women in the management of natural resources conservation and a safer environment to give them opportunity of equal participation in environment preservation policy and programs reflecting a women perspective. Also, the policy emphasizes to undertake all out initiative for employment of the educated and illiterate women labor.

The 7th Five Year Plan (2016-2020) has also highlighted the aspects for worker safety. It was mentioned in the 7FYP that higher growth in Bangladesh was benefitted from a rising labour force both due to the demographic transition that produced a larger share of working population and an increasing labour force participation rate of the female labour. But wage rates tend to be lower for female workers relative to male and proper employment regulations to facilitate the participation of female workers (e.g. maternity leave, skill development, child care facilities, etc.) have to be put in place in most areas of employment. The RMG industry has shown the way. However, safety standards in RMG are becoming a major challenge for sustaining the continued expansion of RMG sector. Thanks to recent coordinated efforts of the Government with the Apparel Manufacturers, and the buyer-driven initiatives. Substantial progresses in ensuring worker safety standards and improvement in working conditions have been achieved.

Bangladesh has ratified International Labor Organization (ILO) conventions 87 and 98 on freedom of association and collective bargaining and is required to protect the rights contained in them. At the international level, Bangladesh has also ratified the ***UN Convention on the Elimination of all forms of Discrimination against Women (CEDAW)***, as well as different ILO Convention.

Bangladesh Climate Change Strategy and Action Plan 2009 (BCCSAP) has a programme for strengthening gender consideration for climate change management. Under this programme it was mentioned that gender consideration is needing to be included in all climate responsive activities.

Bangladesh Climate Change and Gender Action Plan (ccGAP: Bangladesh) 2013 has been prepared with an aim to ensure gender equality in climate change-related policies, strategies and interventions. This policy has highlighted the importance of equal participation of women and men in climate change mitigation practices and low carbon development. Some of the major points that are addressed in the action plan are following:

- a. Ensure that gender considerations are addressed in the process of reviewing energy and technology policies.
- b. Develop gender responsive programmes to reduce GHG emission at household (HH) while ensuring women's access to energy and power technologies.

- c. Improve women's knowledge on climate change mitigation issues.
- d. Develop women's entrepreneurship related to waste management.
- e. Increase women's participation in solid waste and drainage management.
- f. Mainstream gender considerations in coastal and social forestry programmes or initiatives.

In addition, the **ccGAP: Bangladesh 2013** has also put emphasis on enhancing knowledge and access to financial instruments for women.

The primary responsibility for protecting the rights of workers rests with the Bangladesh government. However, there are some professional bodies such as Bangladesh Garment Manufacturers and Exporters Association (BGMEA), and the Bangladesh Knitwear Manufacturers and Exporters Association (BKMEA) who can do to ensure compliance with such provisions, and to sanction companies that abuse worker rights. However, factory owners can carry considerable political clout in Bangladesh and this can act as a barrier to holding them to account for violating workplace rights, as well as health and safety provisions.

The United Nations Collaborative Programme on Reducing Emissions from Deforestation and Forest Degradation in Developing Countries + (REDD+) Readiness Roadmap (2012) pays attention to gender by taking initiative to identify gender-specific risks of the REDD+ Readiness development process in Bangladesh. In addition, the roadmap describes activities and institutional development that ensure 'gender-neutral outcomes.'

International companies that purchase clothes and other products from Bangladesh factories also have a responsibility to ensure that worker safety and rights are maintained throughout their supply chains. Many factory conditions described in this report not only violate Bangladesh's labor law, but also breach the standards that Western high street retailers insist that their suppliers follow.

4.2 International Policy and Framework

4.2.1 Green Climate Fund (GCF)

The GCF provides clear definitions and differences between key gender terms; Gender, Gender Equality, Gender Equity and Gender Sensitivity.

In accordance with the GCF Gender Policy and Action Plan, the core elements of gender mainstreaming required in GCF project design include:

- Country ownership in terms of alignment with national policies and priorities;
- A mandatory initial socioeconomic and gender assessment, to proactively build in a gender-sensitive approach to project planning design and implementation arrangements;
- Gender equitable stakeholder consultations based on the gender parameters in the policy;
- Determination of how the project/programme can respond to the needs of women and men in view of the specific climate change issue;
- Identification of drivers of change and gender dynamics required to achieve the project/programme adaptation or mitigation goals;

- Identification and design of specific gender elements to be included in project/programme activities.

Mainstreaming gender in GCF projects can be synthesized into a three-pronged approach on a gendered project cycle:

- Gender analysis: to understand the social, economic and political factors underlying climate change-exacerbated gender inequality, and the potential contributions of women and men to societal changes in order to build resilience to and address climate change;
- Gendered actions: methods and tools to promote gender equality and reduce gender disparities in climate action; and
- Gender-sensitive M&E: measuring the outcomes and impacts of project activities on women and men's resilience to climate change through gender responsive M&E.

4.2.2 World Bank Group

The World Bank recognizes in its Gender and Development Policy Framework that gender issues are important dimensions of its poverty reduction, economic growth, human well-being and development effectiveness agenda. According to the gender mainstreaming strategy, "The term gender refers to culturally based expectations of the roles and behaviors of males and females. The term distinguishes the socially constructed from the biologically determined aspects of being male and female."

The Bank Operational Policy Focused Specifically on Gender and Development (OP/BP 4.20) has been developed with the following objectives:

- Providing a country-level, strategic Approach to mainstreaming gender issues
- Country-level Gender Assessments
- Integration of gender assessment findings in the Country Assistance Strategy (CAS) developed by the bank for each donor country. It draws on and discusses the findings of the Gender Assessment and links policy provisions for gender (OP/BP 4.20) and policy provisions for CAS preparation and consultation (BP 2.11)
- Integration of gender issues into operations of interventions established under the CAS.

Based on the policy guidelines, a Gender Assessment (whether stand alone, as part of a broader ESW, or done outside the Bank) should typically include the following aspects:

- a. A brief description of:
 - i. The different socioeconomic roles of females and males, including their participation in both the market and household economies;
 - ii. male-female disparities in access to, control over, and use of assets and productive resources;
 - iii. male-female disparities in human development indicators;
 - iv. The relative participation of females and males in development decision-making; and
 - v. Laws, institutional frameworks, norms, and other societal practices that lead (implicitly or explicitly) to gender discrimination and/or gender inequality;

- b. A review of the country context, including the country's policies, priorities, legal and regulatory framework, and institutional arrangements for implementing its gender and development goals; and
- c. Based on (a) and (b) above, suggested gender-responsive policies and interventions that are relevant for reducing poverty and enhancing economic growth, human well-being, and development effectiveness.

4.2.3 Asian Development Bank (ADB)

ADB's policy on Gender and Development (GAD) adopts mainstreaming as a key strategy in promoting gender equity. The key elements of ADB's policy include the following.

- a. Gender sensitivity: to observe how ADB operations affect women and men, and to consider women's needs and perspectives in planning its operations.
- b. Gender analysis: to assess systematically the impact of a project on men and women, and on the economic and social relationship between them.
- c. Gender planning: to formulate specific strategies that aim to bring about equal opportunities for men and women.
- d. Mainstreaming: to consider gender issues in all aspects of ADB operations, accompanied by efforts to encourage women's participation in the decision-making process in development activities.
- e. Agenda setting: to assist DMC governments in formulating strategies to reduce gender disparities and in developing plans and targets for women's and girls' education, health, legal rights, employment, and income-earning opportunities.

In 2013, ADB adopted a new Gender Equality and Women's Empowerment Operational Plan 2013-2020 that sets out the strategic directions and guiding framework for moving the gender equity agenda forward, and for better aligning gender operations with Strategy 2020.

The outcome of the gender equality operational plan is improved gender equality results of ADB operations by 2020. While gender mainstreaming will remain the primary and priority approach, on its own it may not be sufficient to narrow persisting gender gaps and inequalities. Targeted public policies and priority investments in women and girls are required to accelerate closing remaining gender gaps and achieving better gender equality outcomes.

Opportunities for direct investment in women and girls to narrow gender disparities will be explored in areas such as:

- a. girl's secondary education and completion;
- b. access to productive assets, labor saving technology, jobs, and income earning opportunities;
- c. school-to-work transition programs and skills training for employment, especially for young women;
- d. business development services for women entrepreneurs; (v) financial services and access to credit;
- e. policy and legal reforms to tackle issues of gender-based violence and anti-trafficking
- f. giving women voice in decision-making in both formal and informal institutions.

These complimentary and more targeted initiatives are more likely to deliver gender equality outcomes faster and to equip women to better capture and benefit from the opportunities provided through gender mainstreaming.

4.2.4 International Finance Corporation (IFC)

Supporting gender equality is a strategic priority for IFC, which provides advice and investments in partner countries and companies that help expand and strengthen the role of women. IFC's strategic priorities for Gender:

1. Closing gaps in health and education
2. Supporting more and better job opportunities
3. Helping increase access to finance, insurance, technology, and land
4. Enhancing women's leadership roles

IFC works to reduce gaps between men and women in the private sector, enabling companies and economies to improve their performance. Too often, women's private-sector participation is under-valued. Integrating women better into business operations can increase companies' profitability and contribute significantly to shared prosperity. However, women have less access to equal employment opportunities and capital to grow their businesses, experience high levels of sexual violence, are often barred from owning or inheriting land or other property, and have less income to invest for the future.

IFC and its partners are addressing these challenges by working with clients to ensure that women and men can equally participate in creating business value and secure influence and income for themselves. When men and women alike are allowed to make decisions, pursue employment, entrepreneurship, and leadership, societies can reach their economic potential.

In the IFC Performance Standards (PS 1-8), gender inclusion is addressed as a cross-cutting issue across multiple Performance Standards especially for assessment and management of environmental and social risks and impacts (PS-1) and labor and working conditions (PS-2).

Further, IFC has developed a toolkit for addressing the specific gender related issues. The toolkit is a modular, customizable guide that includes four tool suites focused on different issues:

a. TOOL SUITE 1: Increasing Gender Diversity from the Workforce to the Boardroom

Tool Suite 1 presents a detailed business case for increasing gender balance on boards, in senior management, and in the workforce. It features 16 tools to help companies:

- i. Assess their gender-balance performance at all levels of their organization
- ii. Increase gender balance in their workforce
- iii. Monitor and evaluate progress towards gender diversity targets

b. TOOL SUITE 2: Women-Owned Businesses and the Supply Chain

Tool Suite 2 presents a detailed business case for increasing the gender diversity of businesses that are part of an OGM company's supply chain. It features six tools to help companies:

- i. Evaluate their current supply chain
- ii. Develop gender-sensitive criteria for qualified vendors to improve outreach to and development of women-owned businesses

- iii. Monitor and sustain progress towards a more gender-diverse supply chain
- c. **TOOL SUITE 3: Women and Community Engagement**
Tool Suite 3 presents the business case for considering the issues, perspectives, and concerns of female community members in a company's community engagement activities. It features 10 tools to help companies:
 - i. Assess the degree of gender sensitivity in their current community engagement approaches
 - ii. Integrate gender into existing community engagement assessments and activities
 - iii. Monitor and sustain progress towards a more gender-equitable community engagement strategy and activity portfolio
- d. **TOOL SUITE 4: Addressing Gender-Based Violence (GBV) in the Workforce**
Tool Suite 4 addresses and defines GBV, including intimate partner violence (IPV), sexual exploitation and abuse (SEA), and sexual harassment. It features seven tools to help companies:
 - i. Recognize, address, and support victims of workplace sexual harassment
 - ii. Support employees who may have experienced IPV and SEA in the community
 - iii. Design activities as part of their community engagement portfolio, to recognize and address forms of GBV occurring in the community
 - iv. Monitor and sustain progress towards reducing GBV

4.2.5 Japan International Cooperation Agency (JICA)

JICA has three Strategic Development Objectives (SDOs) for “gender and development,” with effective approaches, trends, and possible future directions. These three objectives are introduced below. The first and the second objectives correspond to the “principal objectives” in DAC classification of aid contribution, while the third corresponds to the “significant objective” of the same.

JICA envisions “Inclusive and Dynamic Development” in its efforts for promoting human and to provide timely, comprehensive support to those in developing countries. JICA recognizes women as a key agent for change in development and ensures that women have an equal voice and leadership in decision-making in all its interventions. JICA will also enhance women's participation in leadership roles in all stages of natural disasters from DRR, to emergency relief and to reconstruction. It will also facilitate the formation of policies from a gender and diverse group perspective. JICA's work will be responsive to the different needs and priorities of women and diverse groups as well as their security as it actively provides support in building societies full of hope and dignity.

Priority Areas:

- i. Women's Economic Empowerment; Promote women's Livelihood, employment, entrepreneurship and business development
- ii. Women's Rights and Security; Ensure women's rights and security in conflicts, natural disasters and other crisis including Trafficking in persons and Gender based violence
- iii. Women's Education and lifetime Health; Promote women's lifetime health and education to enhance their autonomy

- iv. Gender Responsive Governance; Promote gender equal policies, strategies and institutions
- v. Gender Responsive Infrastructure; Promote infrastructure to enhance women's mobility, economic opportunities and well-being
- vi. Enhancing Women's Capabilities, Eliminating Discrimination and Improving Basic Infrastructure.

4.3 Gender Related Policies of IDCOL and Local Financial Institutions (LFIs)

4.3.1 IDCOL

IDCOL makes the gender equity and equality a priority to maintain a safe, secured, positive, and collaborative culture in its work environment. IDCOL's ESMF provides detail directions on integrating gender issues in development and assessing projects/programs which are following:

- a) IDCOL will assess internal mechanism and capacities to promote greater visibility of gender and social inclusion considerations in the design of its projects and programs.
- b) Ensure that all recipient projects and borrowers under the projects will identify and maximize opportunities for women's participation in all employment opportunities during construction, operation and maintenance of infrastructure, with potential adoption of evidence-based gender related targets.
- c) Promote adherence to national core labor standards and commitments to gender equity and women's empowerment stated in the legislations and regulations of the Government of Bangladesh.
- d) Assess the social and gender impacts of the Solar Home Systems program and the potential use of other identified women friendly technologies for greater social gender related benefits.
- e) Ensure the use of the Gender Action Framework.

This Policy will guide in promoting gender equality, equity and awareness in its internal policies and procedures.

Additionally, IDCOL has developed a Gender Policy for its organizational purpose. IDCOL believes that all staff members share a common interest in achieving the objectives of the organization and collaborative effort of the employees will ensure creating an environment that would promote teamwork and mutual respect. This Policy has been formulated to institute a systematic approach within the organizational practices in promoting gender sensitiveness, awareness, equity and equality in its internal environment and activities. Therefore, the Gender Policy along with ESMF will lead to implement gender sensitive approaches at IDCOL, its activities, and adherence to all other relevant laws, guidelines and policies.

4.3.2 Local Financial Institutions (LFIs)

There will be four local financial institutions (LFIs) involved in the programme who will channel the fund as concessional loans to finance garment business. These financial institutions are- City Bank Ltd, Southeast Bank Ltd, BRAC Bank Ltd and IDLC Finance Ltd. All of these institutions are well aware of the gender related concerns as they have experience of working

with readymade garments (RMG) sector. RMG sector in Bangladesh is one of those sectors that follows rules and regulations regarding gender quality especially in wage distribution, labor rights and occupational health and safety. Also, the current monitoring system in the RMG sector is well maintained and there is pressure from the international buyers as well. As a result, the LFIs are well aware of the gender related issues and its monitoring. A brief description of the existing policy and experience of the LFIs are following:

IDLC Finance Limited follows the ten Principles of the United Nations Global Compact (UNGC) which are derived from Universal Declaration of Human Rights, International Labor Organization's Declaration on Fundamental Principles and Rights at Work, Rio Declaration on Environment and Development, United Nations Convention Against Corruption. These Principles covers Human Rights, Labor, Environment and Anti-Corruption, and IDLC takes consent from respective clients for the same to approve loan applications. Thus, IDLC will consider gender issues to finance Energy Efficient projects under GCF by taking consent from clients considering wage level, overtime, leave policy, workplace health and safety (i.e., protective gears, fire safety, first-aid), freedom of association and/or collective bargaining provisions, and child labor.

BRAC Bank has always been very much aligned with the gender equality, women empowerment and promoting leadership capacity of women when it comes to adhering ESG requirements within its organizational framework. On the policy level, they have their own Environmental & Social Safeguard Policy and Speak-up Policy where gender issues are highlighted with utmost priority.

They have agreed to follow the following process under the guidance of Bank's Sustainable Finance Committee and on the working level, Sustainable Finance Unit will carry forward following activities to adhere GCF's ESG requirements:

- 1) Review the existing Environmental & Social Safeguard Policy and add the new ESG requirements of GCF as part of it and get it approved by the management/board.
- 2) Rollout capacity building sessions for business and risk assessment officials on the new ESG guidance note as prescribed by GCF.
- 3) Follow GCF's prescribed methodology as well as already operating ESG risk assessment framework prior financing any under GCF.
- 4) Yearly/half yearly monitoring of Bank as well as its client's performance in dealing with ESG issues.

Apart from Sustainable Finance Unit, BRAC Bank has their own Credit Inspection Team to monitor client's performance all over the year, hence conducting half-yearly or annual inspection will be a regular activity for us.

Southeast Bank Limited (SEBL) does not have any specific gender related policy but they conduct periodic inspections at the project sites. Periodic and surprise inspections are among the keys risk management tools of SEBL. Most of the RMG clients of SEBL work with renowned international buyers who also have certain conditions precedent on gender related matters. For the proposed sub-projects, SEBL will conduct onsite as well as offsite periodic inspection for ensuring proper practice of gender related issues in the project. Also, SEBL will assess the client's willingness and ability to implement gender issues as required by GCF and other regulatory authorities in the appraisal of loan proposal.

City Bank does not have any separate gender related policy for external project monitoring. However, they have an ESMS that is prepared following IFC Performance Standards and consider gender inclusion into the projects. Their customers have to fill up a declaration form before getting loan above a certain amount. This declaration form also has some criteria regarding gender issues. Also, they have a gender policy within their institutional framework to ensure gender equality in their organization and staff.

City Bank conducts period audits and monitoring at project level and are positive about conducting half-yearly/ yearly monitoring after providing loans to the RMG units. During the monitoring procedure they will follow the gender action plan developed for this programme along with national policies and GCF guidelines for ensuring gender equality and women empowerment.

5 Findings from Gender Assessment

5.1 Introduction

Since its journey in Bangladesh, the RMG sector has created opportunities for people, especially for the poor and disadvantaged women to lead a dignified life. The industry plays a key role in employment generation and in the provision of income to the poor – it employs nearly 4.4 million workers². The sector has also played a significant role in the socio-economic development of the country with women comprising nearly 80% of the total workforce in the sector³. It also plays a pivotal role to promote the development of linkage small scale industries, since 99% of the accessories like dyeing, printing, zippers, labels are manufactured in Bangladesh⁴. This had helped open employment opportunities through indirect economic activities, thereby contributing to the country's overall socio-economic development, woman empowerment and poverty alleviation.

It has seen tremendous growth and dominated the industrial sector of the country. Currently, there are more than 4,000 RMG firms (The Bangladesh Garment Manufacturers and Exporters Association, 2019) in Bangladesh. Majority of those firms are locally owned with the exception of a few foreign firms located in export processing zones. Nearly 98% of RMG firms of Bangladesh are located mainly in 4 districts viz Dhaka (38.0%), Gazipur (28.9%), Chittagong (16.1%) and Narayanganj (14.7%)⁵. Most of the factories are in different industrial clusters in these four districts due to availability of better infrastructural facilities, access to major trade routes for import and export, logistic and banking facilities as well as easier availability of workers. Even within these regions there is a spread of RMG units within and beyond Export Processing Zones (EPZs) – only 1.5% of the total RMG enterprises are located in EPZs. This is due to the change in government legislation in 2010, allowing foreign direct investment (FDI) outside EPZs as well. Another key characteristic of the RMG sector in Bangladesh is that almost half of the units i.e. 48.9% are small enterprises which hire less than 500 workers⁶.

However, women working in RMG sector are plagued with various problems, some of these are insufficient wage, irregularity in timeliness of wage payments and unsafe working conditions. Specifically, women workers have many issues that need to be taken care of especially in areas like workplace discrimination and career prospect, harassment, work and working time including rest, occupation health and safety, welfare provision, social protection and freedom of association, collective bargaining and social dialogue. Although gender-based wage discrimination is not seen that much now a days, discrimination exists in areas of equal treatment— women are forced to do overtime more than the men and the cut from the overtime allowance is more for women workers than men. Women workers face severe discrimination with regard to the scope of promotion and their career prospect is limited. A study shows that only 5-10% of the women workers eventually become supervisors (The Daily Star, March 8,

² Source: Information disseminated by BGMEA

³ Source: [Statement by BGMEA President \(BGMEA website, 2019\)](#)

⁴ Source: As per presentation provided by BGMEA representative

⁵ Source: [New Dynamics in Bangladesh's Apparels Enterprises \(CPD website, 2018\)](#)

⁶ Source: [New Dynamics in Bangladesh's Apparels Enterprises \(CPD website, 2018\)](#)

2015). Women workers' workplaces are not free from sexual harassment which ranges from simple insults and demeaning remark to unwelcome touching and grabbing.

The Gender Assessment mainly focused on the existing gender scenario of the RMG sector Bangladesh and tried to identify the problems, gaps and potential measures to overcome those. For the growth and sustainability of the sector, integrating women's participation and contribution is a major concern. In this project, survey was conducted in 20 RMG units of different size, location & energy intensity to get an overview of the baseline scenario and identify the existing issues for ensuring gender equity. Along with this, KII, FGD and stakeholder consultations were conducted to get an overall picture of the current gender dynamics of the RMG sector in Bangladesh.

5.2 Labor Force Participation

Women in the labor force have been at the forefront of Bangladesh's recent successful economic growth. They make up most of the workers in the ready-made garment (RMG) sector, the country's flagship export-oriented industry. The creation of a large microfinance system focused on women—a subject of much study and emulation around the world—has also changed the economic position of women. Some of the major issues behind the participation of women in the RMG sector in Bangladesh are the following according to a study⁷:

- a) Women from extreme poor households have a significantly higher probability of participation compared to non-poor. The difference between moderate poor and non-poor are insignificant.
- b) Poorer women are more likely to participate in low paid wage employment whereas the non- poor participate in self-employment.
- c) Presence of young children has a positive effect on self-employment and a negative effect on wage employment.
- d) Being married has a negative effect. This result may have been influenced by the fact that paid employment dominated by the RMG sector requires long hours of work. The study finds that there is a growing acceptance of outside employment among young unmarried workers.

In socioeconomic context of Bangladesh, women participate less on formal livelihood activities in rural areas. Even if they do, there is discrimination in wages compared to men and socially it is not seen positively that a woman is working outside home. Also, there is limited livelihood opportunity due to lack of industries in rural areas. So, women from rural areas prefer RMG sector as there is good opportunity for women even with no education or skill, they get equal or almost equal pay as men, it is a formal employment and they get a regular salary each month. Moreover, they move to new places for work and there they can live independently and without judged by the society. On the other hand, as the women coming from the rural area are comparatively less educated or skilled than those of urban areas, the employers can hire them with minimum wages. Also, they are more vulnerable, docile and manageable than women in urban areas. They accept without protest the flexible terms of employment and hardly raise their voice against anything. This is why employers prefer to hire them.

⁷ Bridges, S; Lawson, D; Begum, S. 2011. "Labour Market Outcomes in Bangladesh: The Role of Poverty and Gender Norms", European Journal of Development Research, Vol. 23, www.palgrave-journals.com/ejdr.

Even though the RMG sector has played a tremendous role for women empowerment, there has been discrimination in labour force participation in this sector. From data collected within factories, 4 out of every 5 production line workers are female, whilst just over 1 in 20 supervisors is a woman.⁸ This is mainly due to lack of education, trainings, technical knowledge and social responsibilities of women towards their families. Also, women are engaged more in sewing or as helpers compared to men. Less number of women are seen in RMG units working in cutting, quality control or supervising compared to men. A recent study⁹ showed that Male workers are found to be less in helper and sewing department by about 86% and 78% respectively.

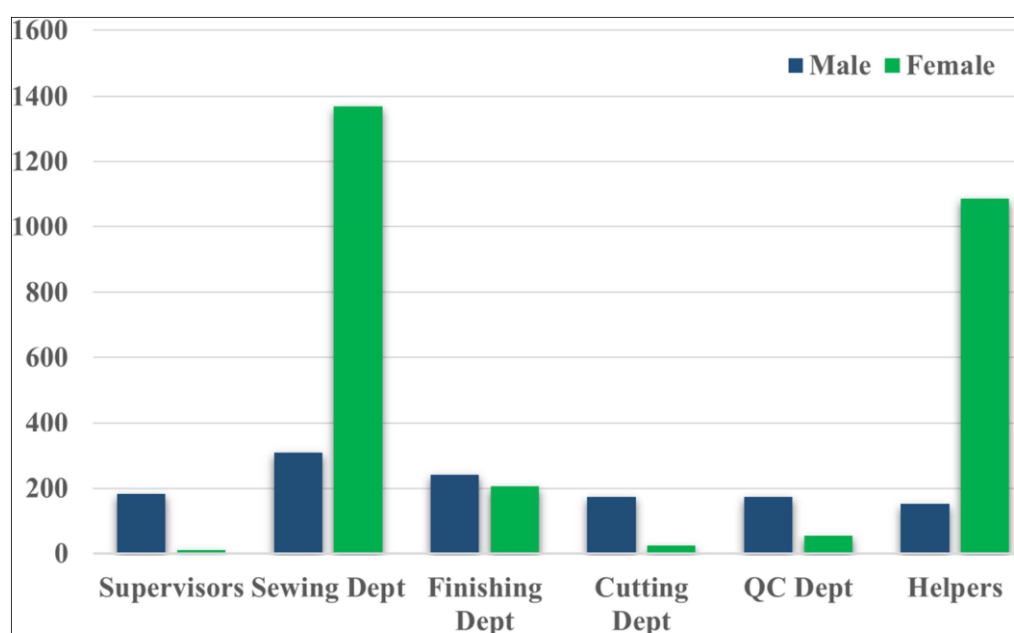


Figure 5-1 Gender scenario in labour force participation in RMG units

The Labour Force Surveys (LFS), conducted by the Bangladesh Bureau of Statistics (BBS), are the major source of data related to labour market and employment in the country. There are a number of challenges in using LFS data, as there are concerns with respect to the comparability of data from different survey years, owing to changes in methodology. Keeping in mind these limitations, we here present the figures on labour force participation and employment for the period between 1999/00 and 2016/17.

Table 5-1 Trends in labour force and employment¹⁰

	1999/00	2005/06	2010	2013	2015/16	2016/17
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⁸ <https://www.theigc.org/blog/female-empowerment-in-the-bangladeshi-garment-industry/>

⁹ Mahmud, M. I. & Afrin, S. 2017. Sustainability of Female Workforce in the Ready-Made Garment Industries of Bangladesh. International Journal of Science and Qualitative Analysis. Vol. 3, No. 5, pp. 49-54. doi:10.11648/j.ijjsqa.20170305.11

¹⁰ Raihan, S. & Bidisha, S. H. (2018). Female employment stagnation in Bangladesh. EDIG Research Paper Five

	All areas	Rural	Urban	All areas	Rural	Urban	All areas	Rural	Urban	All areas	Rural	Urban	All areas	Rural	Urban	All areas	Rural	Urban
Labour force (millions)																		
Both	40.7	31.5	9.2	49.5	37.8	11.7	56.7	43.4	13.3	60.7	43.5	17.1	62.1	44.8	17.3	63.5	45.7	17.8
Male	32.2	25.1	7.1	37.3	28.5	8.9	39.5	30.2	9.3	42.5	30.5	12	43.1	30.6	12.5	43.5	30.7	12.8
Female	8.6	6.4	2.2	12.1	9.3	2.8	17.2	13.2	4.0	18.2	13.1	5.1	19.1	14.3	4.8	19.9	14.9	5.01
Employed population (millions)																		
Both	39.0	30.3	8.7	47.4	36.1	11.3	54.1	41.7	12.4	58.1	41.9	16.2	59.5	43	16.5	60.8	43.9	16.9
Male	31.1	24.4	6.7	36.1	27.5	8.6	37.9	29.1	8.8	41.2	29.6	11.6	41.8	29.7	12.1	42.2	29.8	12.4
Female	7.9	5.9	2	11.3	8.6	2.7	16.2	12.5	3.7	16.8	12.3	4.5	17.8	13.3	4.4	18.6	14.1	4.6
Labour force participation rate (%)																		
Both	54.9	54.6	55.8	58.5	59.4	55.7	59.3	60.0	57.3	57.1	57.3	56.7	58.5	59.6	56	58.2	59.3	55.7
Male	84.0	84.0	83.7	86.8	88.0	83.2	82.5	83.3	80.2	81.7	81.6	81.7	81.9	81.9	81.7	80.5	80.3	81
Female	23.9	23.1	26.5	29.2	29.8	27.4	36.0	36.4	34.5	33.5	33.7	32.9	35.6	37.6	30.8	36.3	38.6	31

Source: LFS, BBS (Data from several years)

Table 5-2 Annual average change in labour force and participation rate⁷

	1999/00– 2005/06	2005/06– 2010	2010– 2013	2013– 2015/16	2015/16– 2016/17
Annual average growth rate in labour force (%)					
Male	2.64	1.47	2.53	0.38	0.96
Female	6.78	10.5	1.94	1.65	4.49
Annual average change in labour force					
Male	0.85	0.55	1.00	0.20	0.40
Female	0.58	1.23	0.33	0.30	0.80
Annual change in participation rate (percentage)					
Male	0.47	-1.08	-0.27	0.07	-1.4
Female	0.88	1.70	-0.83	0.7	0.7

Source: LFS, BBS (Data from several years)

According to the labor force survey (LFS), the female labor force participation rate increased from 2003 to 2006 and then increased dramatically from 2006 to 2010, reaching 36.1%. The LFS in 2013, found women's participation declined to 33.5%, or a 2.6 percentage point fall between the two surveys.¹¹ According to a recent survey of Center for Policy Dialogue (CPD), the ratio of female worker participation in the RMG sector, known as a women-driven sector, has seen a decline in recent years. According to the survey, there are 3,596 active RMG factories in Bangladesh with 3.5 million workers, of which 60.8% are female and 39.2% are male. In the previous survey in 2015, the workers were 36% male and 64% female. The survey was conducted on 2,270 workers in 193 factories in small, medium, and large categories in Dhaka, Gazipur, Narayanganj, and Chittagong.¹²

During this study it was observed that the percentage of female workers is less than male workers. This scenario might have occurred because of the recent increase in wages which

¹¹ <https://www.adb.org/sites/default/files/publication/203906/women-work.pdf>

¹² <https://cpd.org.bd/declining-female-participation-in-rmg-sector-cpd-study/>

attracted more male workers to join the labour force in RMG sector. Also, the employers are now becoming interested to hire male workers over females as they would not have to provide additional facilities (i.e., maternity leave, special support for pregnant or lactating mother, workplace safety during night shifts) to men. Moreover, men are available anytime as they do not have family bindings or security related concerns in the perspective of Bangladesh.

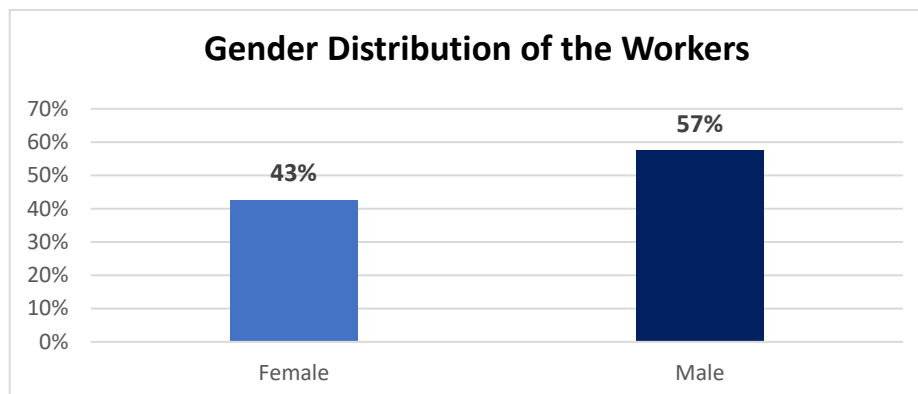


Figure 5-2 Gender distribution of workers in selected RMG units

The RMG units were selected were based on their technical performance mainly and the survey was conducted with the Facility Managers to get an overall idea about the factory. However, in case of facility managers, all the facility managers in the sample are males. There was no representation of women in the higher management of the selected RMG units as well. This is a very common scenario in Bangladesh as the participation of women in managerial posts in garments sector is negligible. According to a study conducted on women workers in Bangladesh RMG sector by CPD in 2018, only about 0.5% of managers in RMG enterprises are female, and only about 9.3% of HR managers in the RMG sector are female. On the production floor, of the 60.8% females, more women are employed in the sewing section (73.9%) and less women work in the cutting section (22.7%).¹³

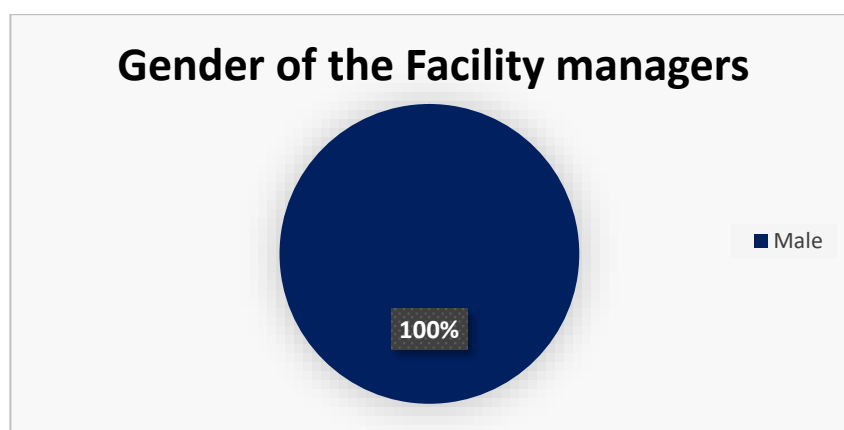


Figure 5-3 Gender of facility managers

Lack of training facilities, skill and proper knowledge of technology are the main reasons for female workers to lag behind in the sector, the surveys reveal. Besides, male workers are taking more interests in the sector due to increasing wages. During FGD, female workers said

¹³ Uddin, M. 2018. Declining female participation in RMG sector: CPD Study. Centre for Policy Dialogue. <https://cpd.org.bd/declining-female-participation-in-rmg-sector-cpd-study/>

that there are trainings for workers but women do not participate because they feel most of the participants are male. Also, women are promoted less in higher positions. So, the female workers do not feel interested to participate in the trainings. However,

It was revealed from KII with experts that in terms of gender equality, the sector has very low wage discrimination at similar work-levels although some works are exclusive terrains of male workers. The sector does, however, has limited participation from women in decision-making positions - only 7% of management professionals are women. This leads to entry barriers in terms of market perceptions. Also, women themselves are also less inclined to get into such job roles.

The tapering off the number of female workers can be stopped only by organizing training and skill development works, both by public and private organizations, the experts said. Over 3.31 million people are, at present, working in this sector. Majority of them are laborer. Earlier in 2013, the BBS survey found the percentage of male workers in the RMG sector 43.14 while the female workers 56.86%. The total workers in this sector were 2.99 million at that time. The reason of the decline is mainly due to changing technology, women have no opportunity to rule the readymade garments sector by utilizing sewing skill only. A comparison between the findings of the two BBS surveys also revealed that the average growth rate of employment of male workers in the RMG sector has increased 11.54% between 2013 and 2016-2017. On the other hand, the rate of employment for women has decreased by 6.56% during the same period.

Specifically, on women workers in Bangladesh RMG sector, the CPD study reveals that only about 0.5% of managers in RMG enterprises are female, and only about 9.3% of HR managers in the RMG sector are female. On the production floor, of the 60.8% females, more women are employed in the sewing section (73.9%) and less women work in the cutting section (22.7%). The study points to the skill level comparison between males and females in terms of knowledge of using machines, and finds that compared to male workers, female workers are proportionately less knowledgeable about operating different machines.

There are numerous estimates on the number of factories and number of labourers present. The list below gives the estimates from reputed establishments:

- a. **Department of Inspection for Factories and Establishments (DIFE):** DIFE has published a full database of 4841 RMG factories in 2017 (DIFE 2017). According to this data base, 2,155,453 workers are currently employed in the RMG industry of them 43% are men and 57% are women.
- b. **Bangladesh Garment Manufacturers and Exporters Association (BGMEA):** As per BGMEA documentation, there are 4,482 RMG factories employing about 4 million workforces of them 3.2 million are women (BGMEA, 2017).
- c. **Bangladesh Bureau of Statistics (BBS):** BBS conducts Labour Force Survey which is counted as having the biggest sample sizes in Bangladesh. According to this survey, there are 3.1 million workforces employed in the RMG factories of them 1.7 million are men and 1.4 million are women (QLFS, 2017) i.e. 54% and 46% men and women respectively.

Government policies can affect female labor force participation rates, positively or otherwise. Some important policies are already in place and appear to have had good effect. The various

programs that have been implemented for increasing female enrollment in primary and secondary schools have been notably effective in eliminating the gender education gap. Bangladesh already has a relatively generous maternity leave policy. Employers are required to provide 16 weeks of paid maternity leave, as specified by the 2006 Labor Law. While in principle this policy is likely to increase female participation rates, it is unclear how much impact it has had in practice. It is unlikely to have any effect at all on anything but formal paid employment, which limits it to a small minority of working women. No systematic evaluations appear to have been made, but there are reports that paid maternity leave is generally not granted at most RMG factories and that most workers are not aware of their rights in this regard.

5.3 Constrains for Female Employment

The KIIs were conducted with policy makers, BGMEA, BKMEA, RMG factories, development partners, noted economists and asked mainly about their views on the present status of women in the RMG sector, the challenges of female employment and the strategies that can be adopted to raise female labour force participation in Bangladesh.¹⁴

Figure 5-4 presents the seven major constraints to employing women as mentioned by the stakeholders. These constraints include lack of education, required skill and knowledge, social and work place safety, social norms, maternity and pregnancy issues, long working hours/overtime, lack of transport facilities and the care burden.

Male-female equal contribution is very important for social coherence and balance. However, due to social constrains, female workers have face problems and sometimes have to sacrifice their career. After the updated increased PayScale, more men are showing their interest to join RMG sector. The owners also prefer to employ a male worker rather than a female worker. This is mainly because of –

- Lack of skill in female workers.
- Female workers need more breaks after marriage or after having children.
- Safety issues need to be assured for female workers and mostly they do not want to stay for night shifts.

Most of the female workers in the RMG sector are very young, 18-29 years and do not tend to work in the sector for a long period. This is mainly due to occupational hazard, health issues, damage to eyesight, having children etc. Also, employers do not prefer aged experienced workers in the factory to reduce cost and prefer to higher fresh workers at lower wages.

However, social and gender issues are better than previous because maternity leave, and other facilities are given to the workers in most of the RMS units.

¹⁴ Annex 1 presents a list of the KII participants



Figure 5-4 Major constraints to female employment perceived by different stakeholders

From social perspective, male ratio is higher in management sector. It will be solved by creating awareness and providing training programs. There are some barriers like mind set up, lack of equity, lack of awareness, etc. It will be solved in future. BKMEA also provide lactating mother training (who have two children under 2 years age) and get 900 take-ups per month (total 36000 take-ups in two years) for extra care of mother (Government is paying the amount) and also female pregnant workers get maternity leave.

Currently, rate of educated women has increased in the RMG sector with the increased education rate in the country overall. Young women joining the sector have mostly completed at least primary education with a good percentage of completion of eighth grade and above as well. However, women usually get less opportunity for technical skill development trainings due to the common perspective of the management that men are technically sounder than women and can learn new skills easily. This situation makes women lag behind from their male competitors and they cannot get promotions. In many cases, women have to be stagnant in a position for years or lose their job. To ensure increased women's engagement, mandatory gender-sensitive trainings for all workers is needed. In this way female workers will not be dropped off for automation only due to lack of trainings.

Jobs in the garments sector are divided among several sections and there are different workers for each section. At initial level the workers do not need much technical skills. But in later stages higher technical knowledge is required for operating modern machineries. No worker, both male and female enter into the sector knowing the technical skills; they gradually learn those after several trainings. The problem lies in gender inequality because male workers are given more opportunities and involved in decision making than female workers.

So, it is essential to ensure that both women and men get access to skill development opportunities. Also, the higher management need to change their concept of sex-segregation of work and provide equal scope for both genders.

The project will increase awareness of the women working in the sector regarding their rights and encourage them to raise their voice whenever they face any injustice. Also, monitoring mechanism will be at place so that it can be assessed whether the management is addressing issues to ensure women empowerment and gender equality.

5.4 Educational Status

In RMG sector, at the worker level recruitments are done through interview and by skill test of the worker. No standard certification system exists to assess the skill level of the worker. RMG employers prefer at least class 5 level education while recruit workers and at least class 8 for the technical occupations and supervisors. According to a survey of Center for Policy Dialogue (CPD), released on Mar'18, among garment worker 60.8% are female & 39.20% are male. The survey was conducted on 2270 workers in 193 factories in Dhaka, Gazipur, Narayanganj & Chottogram. These large portions of women are mostly less or uneducated.

Among the female workers surveyed, most have attained some form of education (SSC or HSC level, primary level or education up to class ten) and 4% have no formal education at all.

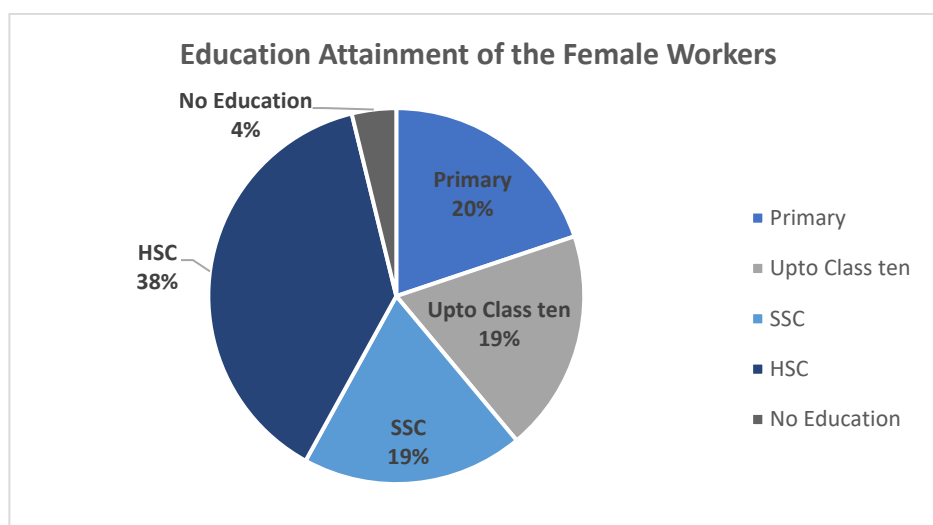


Figure 5-5 Education Attainment of the Female Workers

Among the male workers surveyed, there is a greater incidence of individuals with higher educational qualifications (34% have SSC qualification, 14% have HSC qualification, 9% have primary education attainment, 29% have education attainment up to class ten and 14% have a diploma or a bachelor's degree) and there are no individuals without any formal education.

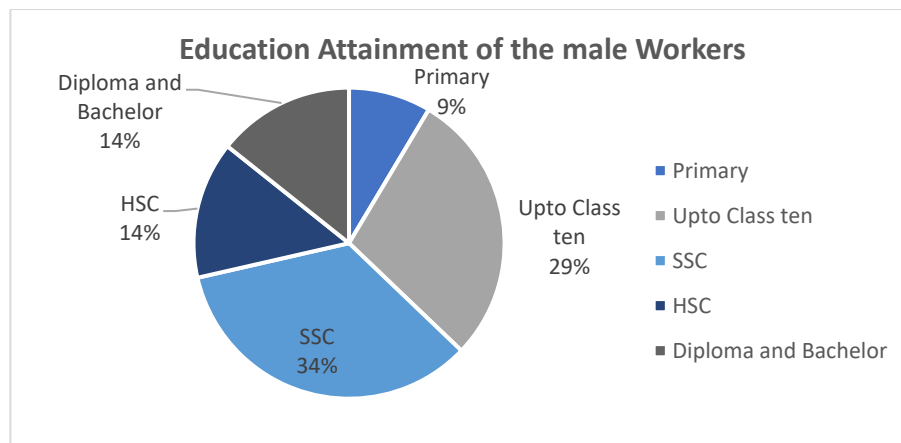


Figure 5-6 Education Attainment of the male Workers

Overall, the male workers perform better in terms of education attainment than the females.

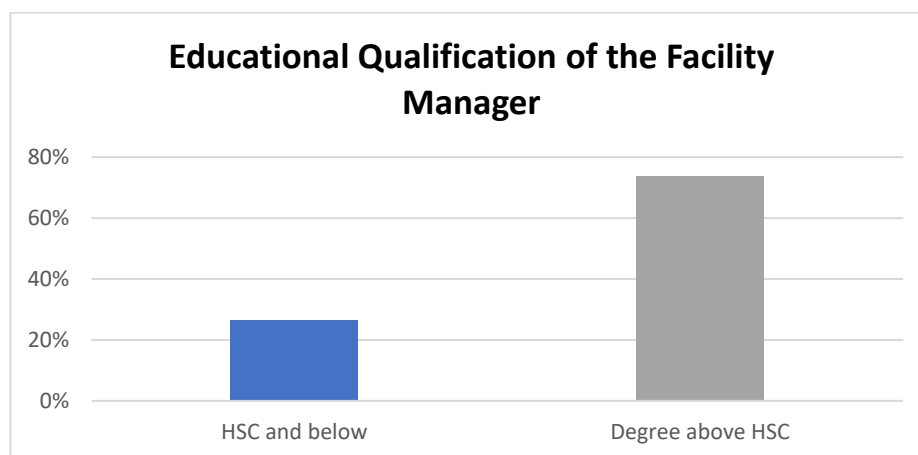


Figure 5-7 Educational qualification of facility managers

Three-fourths of the sampled facility managers have education higher than HSC and one-fourths of the facility managers have education qualification of HSC or below.

In RMG sector the practice of formal training is still not so popular. Only technicians and managerial staff undergo some formal training system but for the workers and supervisors the skills are achieved mostly through informal training which is well known as on-the-job training. Only a small number of entrepreneurs have been investing for developing the skills of their workforce.

The apparel industry in Bangladesh needs to draw the attention of the youth of the country, to show them the huge career opportunities that are to be found in the RMG sector. This could include a stronger emphasis on providing young people with a basic understanding of the sector, including the various levels of employment opportunities at the floor level, to merchandisers, designers, product developers and beyond.

In education, there is a strong need for greater levels of knowledge transfer between education and business. Many western countries have clear and well-developed knowledge transfer partnerships in place which sees ideas developed in colleges and universities being shared

with the industry. Bangladesh also needs to think carefully about how it can develop an education system to provide the RMG sector with the industry leaders of the future who will allow the sector to compete at the higher end of the value chain and address the need for competent middle management. It is unfortunate that the RMG industry in Bangladesh often hires managers from neighboring countries such as Sri Lanka and India when there is an abundance of potential talent domestically available—if only they were nurtured the right way. Around 13 percent of the country's garments factories have employed foreign experts in the top posts who remit over USD 5 billion from Bangladesh every year (CPD 2018). In the absence of a skilled workforce, particularly in merchandising, design and marketing, as well as in operation of sophisticated machines, the factory owners have hired experts from China, Japan, India, and Pakistan to fill the gap. Out of 126 universities (46 public and 80 private) there are two universities in the country—BGMEA University of Fashion and Technology (BUFT) and Bangladesh University of Textiles (BUTEX)—that have been established for the garments and textiles industry specifically.¹⁵ Even though the number is not up to the mark, these two universities are producing skilled graduates for the garments industry. Other engineering and technical universities should also include courses relevant to textile/RMG sector as it is a potential industry in the country.

Moreover, current industry leaders have a vital role to play here, visiting schools and colleges and providing input to government agencies about the long-term needs of the RMG industry. There is an urgent need to develop a cadre of upper level managers. The country's business schools, in cooperation with BGMEA and BKMEA can develop training programmes for lower level managers to become upper level managers. Bangladesh should stop dependence on expatriate managers. Producing high skilled workers for the RMG industry will contribute to the economy by reducing the amount of foreign remittance that leaves the country.¹³

At present the demand for skilled workers is increasing. Besides, it is becoming difficult for new women to get involved in this sector due to its labor-intensive nature of work. Women are quitting their jobs to take care of their children. Without initiatives taken at the public-private level to make women skilled, it will be difficult for them to sustain in this profession. Moreover, Woman worker cannot work more than seven years on average. A number of experts and people involved with the sector said a large section of women workers quit jobs after having a child.

5.5 Gender Perspectives of Energy Efficiency

Different opinions were gathered on gender perspectives of energy efficiency from the stakeholder engagement workshop where a wide range of energy efficiency practitioners including regulators, policy makers, experts, academicians, researchers and garment sector entrepreneurs, as well as environment, social and gender experts participated. During the discussion it was identified that the women who are joining mainly are either unskilled, uneducated, or has joined without any training. As a result, no expertise is being created. With increased automation and use of advanced technology in machineries, already a decreasing trend of female workers is seen in the past 4 years. Experts are concerned that with the introduction of energy efficient machineries and equipment the percentage of female workers

¹⁵ Rahman, S. 2019. Are educational institutions failing the RMG industry? The Daily Star. November 20, 2019. <https://www.thedailystar.net/opinion/news/are-educational-institutions-failing-the-rmg-industry-1829314>

might reduce even more if they cannot compete with the male workers in terms of technical knowledge and skills. Some recent studies also showed similar types of findings.

According to a study, it was underlined that women workers are more vulnerable than their male counterparts to losing their jobs due to automation in 5 key sectors in Bangladesh including the readymade garment industry. It said women in low-wage occupations with less education are likely to be impacted more due to high risk of automation in readymade garment and textile, furniture, agro-processing, leather and footwear and tourism and hospitality sectors at the beginning of Fourth Industrial Revolution (4IR).¹⁶ The research also mentioned that the workers with primary education are going to be more vulnerable than those having secondary level of education. To overcome this situation, more trainings and knowledge sharing activities need to be promoted especially for the female workers as they are generally the ones with lower level of education and technical skills compared to the male workers in the context of Bangladesh.

Another issue identified during stakeholder consultation was considering special needs of men and women while installing any energy efficient equipment. For instance, replacing regular lights with energy efficient lights is a very common measure towards attaining energy efficiency. However, some studies by BKMEA showed that workers face difficulties with color shading when LED lights are used and it ultimately caused eyesight damage especially to women as more women are engaged in sophisticated tasks like sewing in RMG sector. So, there should be some standard for choosing energy efficient machinery and equipment has no negative impact on health.

5.6 Decision Making

Bangladesh RMG sector plays a significant role to reduce poverty and to increase women empowerment. RMG industry has been swaggering in the country since long time as a catalyst for sustainable development and growth. Apart from job opportunities, the RMG sector gave underprivileged women to step out of their homes and live independently. Through working in the sector, women became more aware of their rights and started to raise voice for equality. Also, higher education can give increased wage which encouraged a lot of poor and underprivileged women to complete secondary (SSC) and higher secondary (HSC) level of education. With the job women can support their families financially and have secured a place in decision making at family level. Moreover, this sector plays a significant role in changing mindset of women as these working women realize the importance of education and tend to spend more on their children's better education.

However, there is still a gap in terms of decision making at the workplace. Workers' organization/committees of different forms are available at workplaces but most of the women workers generally are not members of the organization/committees available at workplaces. Moreover, majority are not members of any organization/association at workplace.

¹⁶ <https://bd.apparelresources.com/business-news/sustainability/automation-weighs-heavy-women-garment-workers/>

While women workers are increasingly becoming representatives at the factory level, they are still not being equally represented. Also, leadership positions in union confederations and federations are still largely dominated by men.

This is partly due to widely-recognized gender ideologies, as well as to the reality that women are still expected to assume the majority of childcare responsibilities after they return home, discouraging them from serving as trade union representatives.

5.7 Wage and Benefits

In every five year, the government forms a new Minimum Wage Board for the RMG sector to formulate a new wage structure for apparel workers. The board is usually led by a senior district judge as chairman. In addition to representative from the owners and workers in the board, it also consists of three independent members. The representatives of owners and workers separately propose minimum wage for the workers respectively and place it to the board in black and white. The government of Bangladesh gazette the new wage structure for RMG workers setting Tk 8,000 or USD 97.71¹⁷ as the minimum monthly pay in 2019.

Table 5-3 Revised wage structure by GoB in 2019

Grade	Basic Wage in 2018 Gazette (BDT)	Revised Basic Wage in 2019 (BDT)	Increase in basic wage (%)	Gross wage in 2018 (BDT)	Gross wage in 2019 (BDT)	Increase in gross wage (%)
7	4100	4100	0.0%	8000	8000	0.0%
6	4370	4380	0.2%	8405	8420	0.2%
5	4670	4683	0.3%	8855	8875	0.2%
4	4930	4998	1.4%	9245	9347	1.1%
3	5160	5330	3.3%	9590	9845	2.7%
2	8520	9044	6.2%	14630	15416	5.4%
1	10440	10938	4.8%	17510	18257	4.3%

¹⁷ Considering USD 1= BDT 81.17 on 1 January 2019

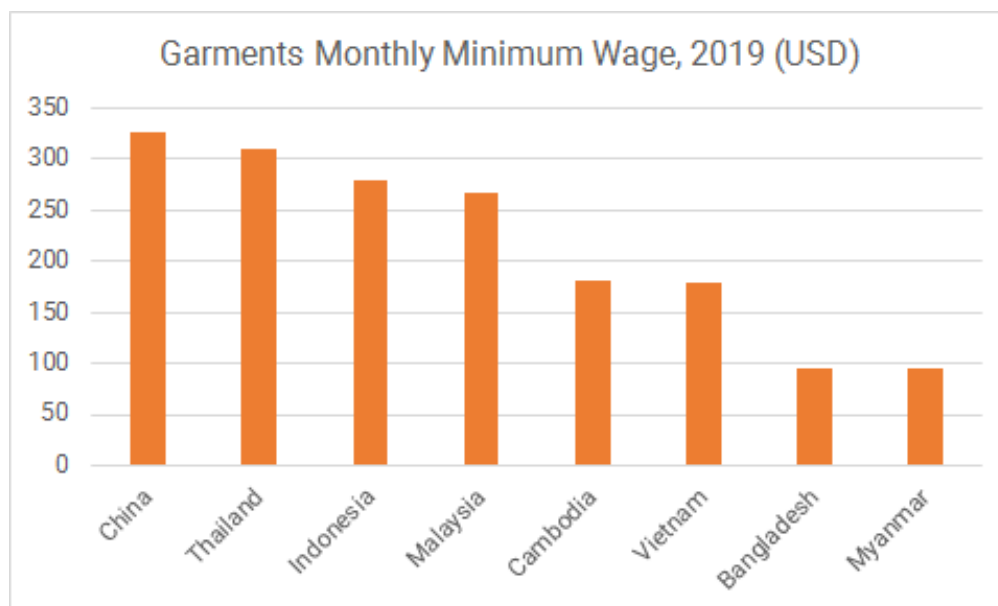


Figure 5-8 Country-wise Garments Monthly Minimum Wage 2019 / Source: NYU Stern Center for Business and Human Rights

The survey results showed that in RMG sector there is no gender-based discrimination in terms of wages and other benefits. In terms of gender equality, the sector has very low wage discrimination at similar work-levels although some works are exclusive terrains of male workers. Also, there is fair compensation provided to workers as per their pay grades. Most of the workers generally get festival bonus. Women workers get lunch break. However, except lunch break workers are not provided with opportunity to take rest.

However, knowledge among the women workers about calculation of overtime compensation is poor. Three in every four women do not know the calculation method. So, the authority can manipulate them easily. Also, there is a discrimination in disbursement of salary on time. According to government policy, salary of the workers should be cleared within seven days after the due date. But in case of female workers there is often a delay in salary disbursement. Moreover, on terms of trainings and other opportunities women are less promoted. As a result, they get less promotions compared to their male colleague due lack of skill and technical knowledge. This is one of the reasons why there are few women in the supervisor and managerial positions.

5.8 Workplace Environment

The study also revealed that, overall working conditions for women is favorable - wage discrimination on the basis of gender in the sector is largely absent, special benefits are provided for women with special needs (like infrastructure for lactating mothers, maternity leaves etc.). The incidence of workplace harassment does exist and in case any such cases are reported, there is a robust redress mechanism in place. Also, examples of child labor are also limited.

However, the scenario is only present in large scale factories and mostly absent in medium and small-scale factories. The workers from medium and small factories still suffer from unhealthy working condition, not having standard working hours, inadequate breaks during working hours and inadequate worker safety. Workplaces of the women workers are not free

from harassment. In other words, women workers experience different forms of harassment at workplace. Of the different forms, verbal abuse/harassment is the most common one. The reason behind this contrasting level of compliance between large scale and medium/small scale units is the growing concern of the buyers in social, health, safety and gender considerations after the Rana Plaza incident.

Sexual harassment has decreased in the garments sectors in recent years due to strict rules and monitoring at the factories. Most of the large enterprises are highly compliant ensuring safety and security of their workers and there is functional sexual harassment complaint committee. In case of small and medium enterprises, there are sexual harassment complaint committee on paper, but their functionality and transparency are questionable. However, in all cases the workers have less trust on the committees and most of the time do not file any complaint. So, this issue can be considered as a moderate risk factor as there is already existing committees for sexual harassment and the rate of sexual harassment is decreasing.

Post Rana Plaza, there was a severe market and media pressure on the buyers i.e. large apparel brands to fulfil their social responsibility and ensure better working conditions for the workers employed in these manufacturing units. Accordingly, the buyers demanded stringent compliance with safety and health norms from the manufacturing units and would withdraw business in case these norms are not met. The Accord¹⁸ and Alliance¹⁹ was developed and strictly implemented for this purpose – audits were conducted, and the unit were notified on the areas of improvement. This prompted the large industries, who are in direct business with the buyers, to comply with these norms. However, the trickledown effect on small and medium scale units was limited since they rarely have forward linkages with the final buyers and mostly supply inputs for these large-scale units. Hence, there is a large scope of improving the social, health and safety conditions for the medium and small-scale units.

Women workers are mostly deprived of social protection related facilities at workplace. High prevalence of ignorance/unawareness among workers concerning the social protection facilities are a striking feature in this regard for a long time. However, the situation is changing now. At present the factories try to comply with existing regulations and provide facilities to female workers.

¹⁸ The Accord is an independent, legally binding agreement between brands and trade unions to work towards a safe and healthy garment and textile industry in Bangladesh.

¹⁹ The Alliance for Bangladesh Worker Safety (Alliance) is a legally binding, five-year commitment to improve safety in Bangladeshi ready-made garment (RMG) factories.

From this study it was also observed that most factories try to provide maternity leave and special provisions for pregnant and lactating mothers as shown in the graph below.

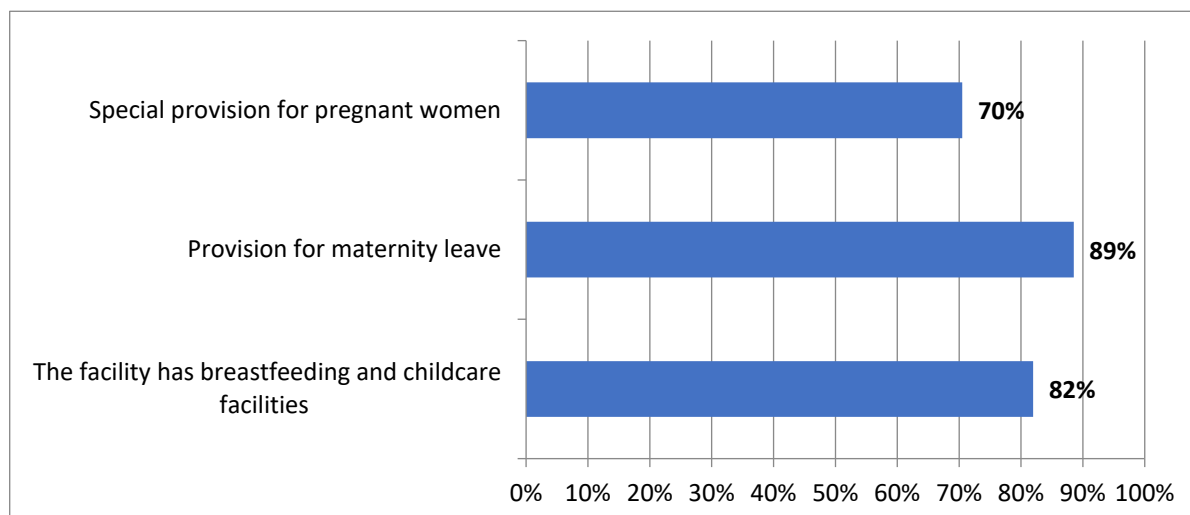


Figure 5-9: Workers' Responses on Maternity Care

In terms of safety issues, there is a risk of fire hazard as most of the materials and chemicals used in RMG are flammable. So proper fire safety needs to be ensured. In terms of health and safety issues this sector has improved a lot, especially after the Rana Plaza incident. However, they are still not up to the standard. The study found that among the total respondents, 48% said that they received trainings on chemical handling and storage of which 15% were female and 33% male workers. On the other hand, 52% said that they did not receive any training of which 21% were female and 31% were male (Figure 6-11).

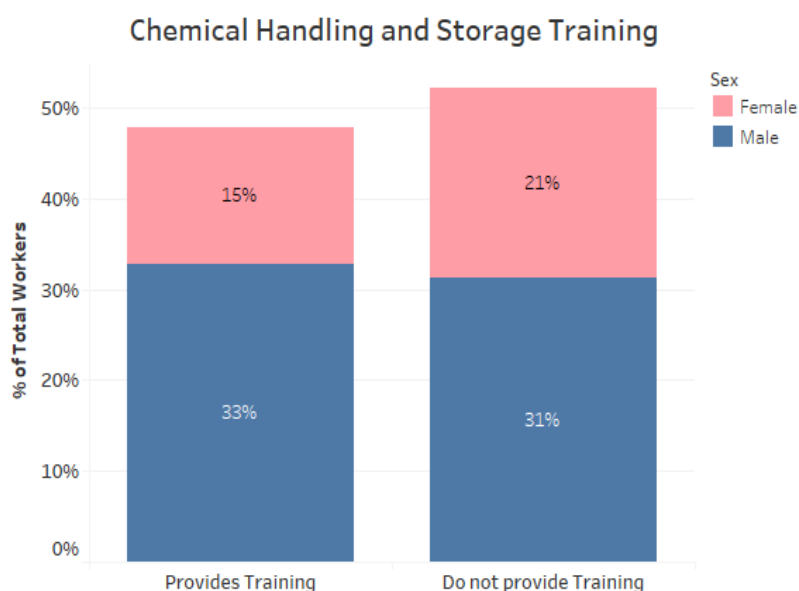


Figure 5-10 Respondents receiving training on chemical handling and storage

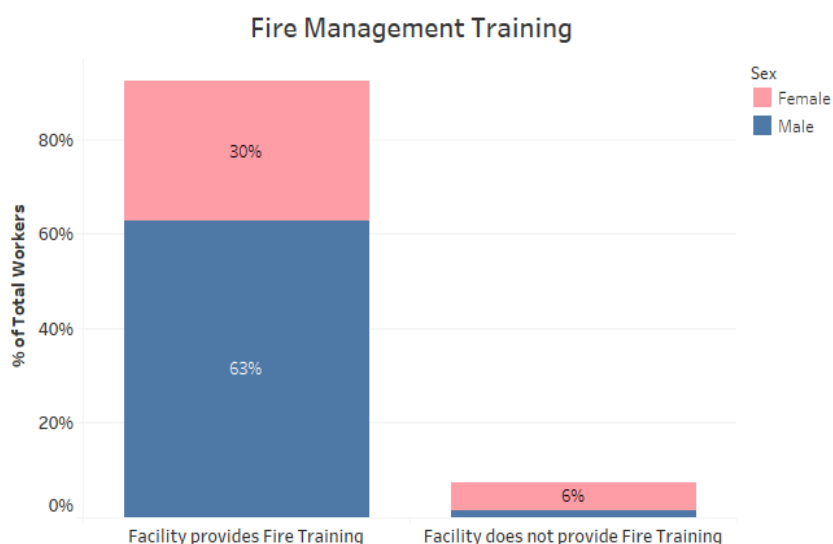


Figure 5-11 Respondents receiving training on fire management

The study findings also showed that most of the facilities that were surveyed provided fire safety trainings to the workers.

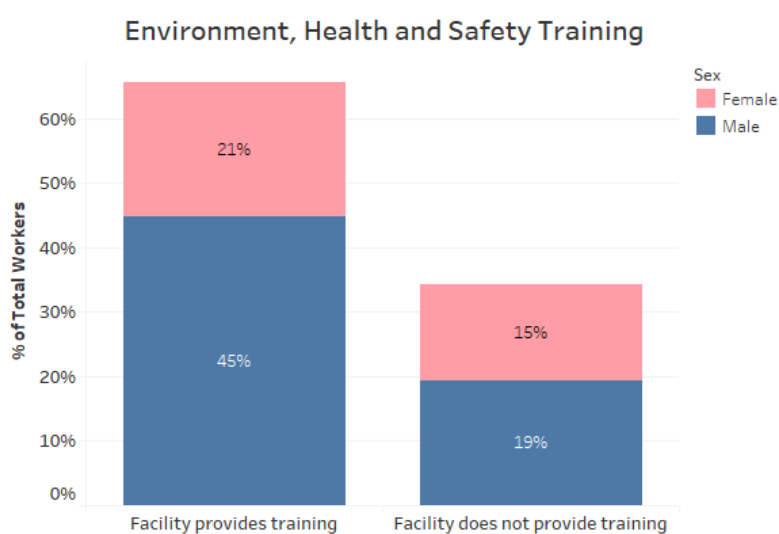


Figure 5-12 Respondents receiving environment, health and safety training

The study revealed that among the respondents 66% received training on environment, health and safety issues of which 21% were female and 45% were male. Rest of the 44% respondents said that they did not receive any sorts of training related to this issue from their facility.

The effectiveness of initiatives undertaken by Government to address environment, social and gender related issues in RMG sector is high. But the owners are not always interested to follow those unless the initiatives are imposed on them. However, the RMG sector is doing comparatively better than other industries. Monitoring bodies like Accord, Alliance, ILO push the owners to follow the standards for which the current situation in terms of compliance have

improved. But there is still a lack of awareness among the workers about their rights in terms of wages and other facilities. So, they need to sensitize more so that they can be aware of their rights.

5.9 Social beliefs, perceptions, and stereotypes

Participation of women is increasing in the Bangladeshi labor market, but socio-culturally constructed norms envisage that women and men's position is not equal. Women tend to be concentrated in certain low-paying, low-skilled sectors such as the garment industry, and within it, are often stuck in lower paying occupations such as sewing machine operators and helpers. Due to stereotypes regarding women's and men's aspirations, preferences and capabilities, employers' perceptions of women's and men's skills and attitudes are affected. This trend has been strengthened over the last two decades: as garment production becomes more technology-intensive, occupations become increasingly skilled and better paid, employing a higher proportion of men. Similar scenario was observed from the KII that female workers are decreased in Bangladesh in the RMG sector mainly because of –

- Increased wage structure attracts male workers to this job and owners prefer male over female.
- Due to automation and lack of skill in female workers.
- Female workers need more breaks after marriage or after having children. This is also a reason why owner prefer male workers more.
- Safety issues need to be assured for female workers and mostly they do not want to stay for night shifts.

Moreover, most of the female labors in the garment industries are mainly involved in cutting, sewing, hemming, stitching, ironing, buttoning, and folding²⁰. This is obviously seen as a gender-based skills compared to men whereas men mainly occupy supervisor or managerial positions. At the same time, men are firstly introduced to formal training, new technologies, and fashions. Women workers whom called “unskilled” are identified as marginal and low-paid labors and excluded from organizational and managerial training. Thus, women are marginalized in managerial roles because gender stereotypical beliefs and practices.

Like many other countries, the participation of women in market economy especially in the managerial position of any kind of business firms in Bangladesh is seriously affected on limits by gender nature or stereotypes. Nevertheless, when men and women are in similar professions, clearly men's earning, wages and profit is higher than women. In Bangladesh although government job's payment system is equal for men and women, but non-government jobs salary is varied for men and women, especially in managerial position. Male than female mainly occupies educational institutions, bank, and insurance companies' senior positions, which represent that strong gender stereotypes exist in the organizations. Besides these, women's pregnancy and child rearing are further barrier to achieving stable or top most

²⁰ Habiba Z (2001) Paid Work and Socio-Political Consciousness of Garment Workers in Bangladesh. *Journal of Contemporary Asia* 31: 145-160.

position of an organization²¹. Trade Union is still dominated by male, whereas women's position is as a recipient^{22,23}.

It was observed from the KII and field survey that there is less participation of women in management and technical positions. From social perspective, male ratio is higher in management positions and roles. It will be solved by creating awareness and providing training programs. There are some barriers like mind-set or cognitive biases against women, lack of equity, lack of awareness, etc. These are expected to be reduced over a period of time.

5.10 Access to Information

There is wide variation of in practice of leave provisions. Among several types of leave weekly holiday is the most enjoyed. Many workers are even unaware about the practice of leave at their workplaces. Night duty is mandatory for many women workers. In other words, when night duty is scheduled these workers are left with not options but to do night duty. There are workers who lack awareness in this regard. Lack of awareness has also been among many workers regarding practice of taking written consent from workers for night duty. However, largest segment of the workers claim that employers never take written consent from them.

Not all employers generally do provide workers with information regarding risks at workplace. One in every four workers claims that their employers do not inform workers in this regard. RMG workers who have faced/observed accidents (pricking of needle during work, cutting finger or hand, fracturing leg and fracturing hand, fire) at workplaces are quite high. 40.4 percent workers have this experience. There are also workers, though few, who don't know whether any accident took place at their workplaces. Most workplaces maintain/have made available different facilities/arrangement for ensuring workplace safety. However, it also true that still some factories have not made enough arrangement. Many workers do not know whether organizations/committees exist at workplace.

Presently, most of the garments have their own sexual harassment policies and any issues regarding sexual harassment is dealt with strictness. So, number of cases of sexual harassment has decreased in recent years. However, in many cases the victims do not file complaint due to lack of trust for getting justice or they feel embarrassed. Also, they do not know whether there is a system to file complaint or whom to approach first and. What are the procedures. To address these issues, sexual harassment should be added during the grievance redressal. Also, awareness of the workers should be increased about the committees and the complaint filing process.

²¹ Sarah S, Sonia J, Shahana R (2005) Women's Employment in Urban Bangladesh: A Challenge to Gender Identity? Development and Change 36: 317-349.

²² Nidhi K (2009) The Ready-Made Garments Industry in Bangladesh: A Means to Reducing Gender-Based Social Exclusion of Women? Journal of International Women's Studies 11: 289-303.

²³ Naila K (2004) Globalization Labor standards and women's rights: dilemmas of collective (in) action in an interdependent world Feminist Economics 10: 03-35.

5.11 Women and Health

According to BLA, employers must ensure basic standards for health and safety in the workplace. In the Bangladesh labour Law-2006, specified the rights of health and hygiene in the section (51, 52, 53, 54, 56, 57, 59, and 60). Despite the specific code in the sections, health and safety violations are common in garment factories. Women in Bangladesh are more disadvantaged group than men in terms of health and nutrition. In most cases, due to closed windows, light and air is not sufficient in RMG factories. Toilets are not sufficient for the crowd. So, toilet remain dirty all the time proper water supply can't properly supplied harmful gas emission from the toilet for the huge crowd and the environment of factory remains stinky most of the time.²⁴

Most of the female RMG workers live in urban slums or semi slums where clean drinking water, hygiene and sanitation health care are inaccessible due to scarcity and their low wages. Also, study identified that, community diseases are common in the workers live in slums. The female workers working in fabrics section (stitching, coloring, dye etc) are exposed to toxic elements and inhale dust of fabric which cause different breathing and respiratory diseases and those who are working. The female RMG workers mostly suffer from malnutrition, anemia, gastric, dysentery, hepatitis, diarrhea, respiratory problem, gynecological problems, tuberculosis and urine infection etc. Other than these diseases as they are sitting for a long working hour lead them to suffer from bones such as neck pain, joint pain, musculoskeletal pain, neural problem and problem in body muscles, joint, tendons, ligaments and bones. Also, the noisy environment from sewing machines cause hearing disability.²⁵ The female RMG workers receive only BDT 200 as medical allowance as per wages boards, 2010 but their average monthly medicine expense is BDT 497.5, which is more than double what they are provided.²⁶

Few RMG workers get the day care and in many garment factories, mothers do not have space or time to breastfeed. Breastfeeding rates by working women in garment factories can be as low as 10 percent (UNICEF 2015). Workplaces of these workers are not gender sensitive— there are insufficient numbers of toilets for women; no arrangement of sanitary napkins at workplace; and most of the factories also do not arrange women doctors.²⁷ Unwanted physical contact and sexual harassment by lineman, supervisor and other male workers lead to sexual diseases as well as hamper mental wellbeing. Moreover, female workers abort babies to retain their job due to their poor economic condition.

5.12 Access to Financial Services by RMG Workers

Even though the majority of the RMG workers are female, their wages are not up to mark to satisfy them even after working for a long time (including overtime). Consequently, they do not

²⁴ Rights of Female Garments Workers in Bangladesh: A Case of Chittagong.

²⁵ Ahmed, S., & Raihan, M. Z. (2014). Health status of the female workers in the garment sector of Bangladesh. *J Faculty Econom Administr Sci*, 4(1), 43-58.

²⁶ Mustafa, A., Islam, M. S., Islam, S., & Khatun, M. (2016). Impact of RMG sector on livelihood change of women employees of Bangladesh. *Social and Economic Geography*, 2(1), 1-10.

²⁷ Monitoring Work and Working Condition of Women Employed in Ready-Made Garment Industries of Bangladesh. March 2019. Karmojibi Nari. Dhaka.

have access to basic needs such as nutritious food or better clothing, medications etc. A study conducted on female Garment workers in Chittagong Metropolitan area identified the insecure condition of the female workers who are still trapped in the vicious cycle of poverty.²⁸

A recent study conducted by Care Bangladesh indicates that female RMG workers want to use financial services, for benefits that include loans, savings, their own bank/Bkash²⁹ accounts, etc.³⁰ According to the IFC data, there is a stark gender gap in access to financial services in Bangladesh, where only 36% of women have a formal bank account, compared to 65 percent of men.³¹ The study also indicated that many female RMG workers may not have full control over their earnings. Thus, financial services that enables a female RMG worker to exercise greater management of her finances may empower her and facilitate greater savings and more financial control. Furthermore, due to the limited amount of wages, they cannot save much for the future.

Given the economic significance of the readymade garment sector, which is dominated by female workers, digitizing wage payments through MFS can have a significant impact on women's financial inclusion. Almost about 90% of factories pay their workers in cash, which is a missed opportunity for both the financial and RMG sectors. When garments workers open MFS accounts to receive their salaries, they can avail a broad range of financial services. Through IFC's initiative, over 70,000 female workers benefitted from digital wage payments to their mobile financial services accounts between 2016 and 2018, and they can now use their accounts to send money, build savings and make payments securely. However, non-availability of national ID of workers, lack of financial literacy and awareness about accounts, and difficulty in covering fixed costs incurred by MFS providers are among major constraints in wage digitization in readymade garment industry in Bangladesh.³²

5.13 Leadership Opportunities for Women in RMG Sector

In Bangladesh, textile and garments sector is entirely male dominated. Even though the present elected president of BGMEA is a woman, there is no existing database on women enterprises or women involved in the higher management positions in the textile and garments industry.

According to the Global Gender Gap Report 2020, companies with female-majority ownership in Bangladesh are only 1.70%, whereas firms with female top managers are 4.80%. During the gender assessment, no women led enterprise could be reached. According to a recent report, women in leadership and in organizational decision-making in businesses as well as in the company's board are very few, because of social stigma towards women. Male

²⁸Mustafa, A., Islam, M. S., Islam, S., & Khatun, M. (2016). Impact of RMG sector on livelihood change of women employees of Bangladesh. *Social and Economic Geography*, 2(1), 1-10.

²⁹ bKash is a mobile financial service in Bangladesh operating under the authority of Bangladesh Bank as a subsidiary of BRAC Bank Limited.

³⁰ http://www.carebangladesh.org/publication/Publication_8660144.pdf

³¹ <https://www.dhakatribune.com/business/2019/06/26/lack-of-financial-literacy-awareness-major-challenges-for-wage-digitizing>

³² <https://www.dhakatribune.com/business/2019/06/26/lack-of-financial-literacy-awareness-major-challenges-for-wage-digitizing>

dominance continues to persist in managerial or decision-making positions at corporate or in the boards of companies. To engage more women enterprises in the garments sector, government and BGMEA should create enabling environment and facilities. To get more women enterprises interested in the sector, gender stereotyping should be eliminated.

Women do not have the right skill for promotion from blue collar type work to a more white-collar job profile and capacity building is required in solving this. Perception of owners is an important determinant of the extent to which gender issues get resolved. Focusing on improving about the male-female ratio may not necessarily resolve gender issues, however efforts on improving overall working conditions might help more. Nowadays workers are getting different types of training for skill development (sometimes free of cost training). BKMEA, BGMEA and other organizations provide some training free of cost (If needed), and additionally provide some money and refreshments, as additional motivators to increase participation in trainings. This has led to increased shift of female workers from supervisor or operator level. While development of skills in female workers require time, upskilling to counter advancement in technology can surely be undertaken. Hence, the technology factor is expected not to be a major issue in limiting job prospects of females.

6 Gender Inclusion Practices in Bangladesh

Equal rights for workers can be best assured by a combination of legislative measures as well as awareness of the business owners and their management about the advantages to be gained through employment of workers based on their capabilities rather than gender or other personal attributes. Recent government initiatives including amendments to the Labour Law in 2013 and Labour Rules in 2015 have made inroads into ensuring equality in the workplace, and the introduction of training initiatives including those related to effective workplace cooperation and Essentials of Occupational Safety and Health (EOSH) conducted by the BGMEA, BKMEA and BEF, with support from ILO-ITC since 2014, to establish a culture of occupational safety and health and respect at the factory level, go some way to improving the situation.³³

Equal responsibility within the RMG industry has been greatly helped by initiatives launched by the Bangladesh government and other bodies, which include the recently formed Tripartite Consultative Committee (TCC) that addresses labour and workplace related issues in the RMG sector and is a substantial step towards forming a collaborative and all-inclusive entity.³⁴ It was established specifically to deal with all unresolved or disputed issues within the workplace. Initiatives such as these, together with the work undertaken with the guidance of Accord and Alliance and other agencies, have created an environment in which ethical, environmental and sustainable standards are not only considered the norm, but also the minimum level that any RMG entity today has to abide by in order to gain necessary compliance certification from the customers.

Promoting equal participation of men and women in trade unions can play a crucial role in ensuring rights and interests of workers especially raising women workers' issues such as sexual reproductive health rights, sexual harassment, adopting a gender approach to OSH, leadership and communication skills.³⁵ In addition, it also helps to mainstream gender issues in all policies and programmes, including collective bargaining to promote women workers' rights. However, Trade unions' activities must be focused, rights based, efficient and aligned with regulatory measures provided in the Bangladesh provisions provided in the Bangladesh Labour Act (2006) and the Labour Rules (2016).

Bangladesh RMG industry is subject to the harshest scrutiny regarding all aspects of health and safety at work, compliance and structural standards following the tragic events of April 23, 2013 at the Rana Plaza. The Accord, Alliance, and National Action Plan (NAP) initiatives, implemented since that fateful day, have ensured that the RMG sector has greatly improved and now has 7 of the 10 highest rated LEED factories in the world, complying with the highest ethical and environmental standards throughout the industry. Level of compliance is high for those industries which have realized the business need of complying with safety regulations. In this respect, Rana Plaza incident has completely transformed the market by highlighting the disastrous effect the lack of safety can have on not only the workers but also on overall business prospects of the RMG units. The key reasons for non-compliance with applicable

³³ <https://www.thedailystar.net/opinion/perspective/enhancing-equality-within-rmg-industry-1573354>

³⁴ <https://www.dhakatribune.com/business/2017/03/12/govt-forms-tripartite-council-rmg-sector>

³⁵ https://www.ilo.org/wcmsp5/groups/public/---asia/---ro-bangkok/---ilo-dhaka/documents/publication/wcms_615874.pdf

regulations would extra cost of compliance which in some cases are an added burden to the limited financial capacity. In current market scenario, with business margins getting constrained, this added cost is leading to disparity in competitive advantage – with non-compliant companies having spare resources for creating market outreach. However, awareness about the benefits of compliance is increasing and it is expected levels of compliance will further increase in the days ahead, with market pressure increasing as well.

However, the investment to achieve such improvements is not always recognized by the immediate customer, and there is an unhealthy culture that regards Bangladesh as a low-cost source, no matter what the price. Equality at this level surely demands that the buyers respect the investments and the advances made by the RMG industry in Bangladesh and also realize that these changes have come about through significant investment in both plant and machinery and that, in a world of escalating raw material costs, some price levels are artificially suppressed.

7 Legal and Administrative Framework Protecting and Promoting Gender Equality in Bangladesh

Bangladeshi women have been struggling to establish their rights in family, society and in the state. In reality, women are still facing discrimination, exclusion and injustice and have a negligible influence in decision-making processes in a male-dominated society of Bangladesh. Discriminatory laws and policies hinder formal equality and socio-political conditions prevent women from exercising their rights.³⁶

The implementation of women development policy and work plan is the basic responsibility of the Government. This responsibility can be discharged on a sound basis through building up a well-organized and broad-based institutional system. Despite having under-representation of women at different levels, the government of Bangladesh has been implementing different steps to ensure gender equality and women's empowerment in the country. Through the vision 2021 and 2041, a momentum has been created for taking forward Bangladesh to a middle and high-income level, respectively. The GoB has undertaken various programs to materialize Vision 2021 for establishing rights of women, their empowerment and mainstreaming them in the overall development. The 7th Five-Year Plan (2016-2020) of the Bangladesh government considers women's engagement in political and economic activities as a crosscutting issue and one of the main drivers of transformation. Realizing the essence of ensuring gender equality, the government has shown its commitment to attaining the Sustainable Development Goals (SDG) of gender equality and empowering women and implementing the Convention on the Elimination of All Forms of Discrimination against Women (CEDAW) and the Beijing Platform for Action.

There is a dedicated ministry, the Ministry of women and Children Affairs (MoWCA) for the development and welfare of women in the country. Also, the Ministry of Labour and Employment (MoLE) that deals with labor related issues have particular focus on ensuring gender equality in the employment sector and to achieve that they promote different types of capacity building activities for different sectors. The MoWCA has been implementing different programs to alleviate poverty of women, stop abuse of women, stop trafficking of women, and ensure safety in all areas including their workplace with a view to ensuring full and equal participation of women in socio-economic activities. ILO and Ministry of Labour and Employment have taken initiatives to provide capacity building trainings and leadership programs to women working in RMG sector.³⁷ BGMEA and BKMEA also have training and capacity development programs for both male and female workers. BGMEA also promotes potential female workers for obtaining higher education by providing them scholarships.

Bangladesh has already substantially achieved the MDGs as it has achieved gender parity in primary and secondary education at the national level, among other successes of the MDGs. Bangladesh has been working relentlessly to ensure women's overall development by ensuring their equal and active participation in the mainstream socio-economic activities and removing the various impediments to their empowerment. In order to attain the SDG goal of gender equality; the government of Bangladesh has been implementing different initiatives so

³⁶ <https://www.daily-sun.com/post/345296/SDGs-and-Gender-Equality-in-Bangladesh>

³⁷ https://www.ilo.org/dhaka/informationresources/WCMS_583272/lang--en/index.htm

that gender equality and women empowerment can be ensured in the country. The following are some important initiatives of the government.

Since 2009 the gender-responsive budget has been planned by the Government of Bangladesh for ensuring the participation of women in various programmes. At present, all the ministries have adopted gender-responsive budgeting and have made accountable towards spending their allocation for the economic and social empowerment of women. Allocation for women's development is too 7,908.7 million, which is 26.80 per cent of the total budget in the current fiscal year.

To ensure gender equality and protection of and security to women, the government of Bangladesh has formed several acts and policies. These include:

- **Mobile Court Act, 2009:** The executive magistrate was given power to take steps by linking Section 509 of the Bangladesh Penal Code in the schedule of Mobile Court Act. This would resist and prevent eve teasing and sexual harassment of the girls and women.
- **Citizenship Act (amended), 2009:** The provision for giving citizenship by mother to child was made by the national parliament by amending the citizenship act in 2009.
- **Dowry Prohibition Act, 1980:** Prevention of financial obligation to provide gifts and assets
- **Women and Children Repression Prevention Act, 2000:** To ensure the safety and protection to women at homes and their surroundings.
- **Domestic Violence (Prevention and Protection) Act, 2010:** Prevention of domestic violence, protection of women and children from domestic violence
- **Prevention and Suppression of Human Trafficking Act, 2012:** To prevent trafficking for slavery, sexual interests and bonded labour.
- **High Court (2009) ruled to include mother's name in all official documents,** school admission forms, either alone or with that of the father. Consequently, climate change and its impacts are enhancing vulnerabilities for women due to the lack of climate adaptive capacities.
- **Legal Aid Services Act, 2000:** An Act to provide legal aid for the litigants who are incapable of seeking justice due to financial insolvency, destitution, helplessness and for various socioeconomic conditions.
 - i. Acid Crime Prevention (Amended) Act, 2010;
- **Human Trafficking (Deterrence and Suppression) Act, 2012:** The Act aims to restrain and suppress human trafficking, to ensure the protection of victims of the offence of human trafficking and their rights, and to ensure safe migration. So, under this act, if any offence has been committed regarding human trafficking, anyone can file a complaint (following the provisions of the Code of Criminal Procedure 1898), to the police or Tribunal and his identity may be concealed for the security purpose.
 - ii. Children Act, 2013;
 - iii. National Education Policy 2010;
 - iv. Comprehensive Early Childhood Care and Development Policy 2013;
 - v. National Plan of Action to implement National Women Development Policy;
 - vi. National Action Plan to Prevent Violence Against Women and Children 2013-2025: The action plan aims to ensure the accountability and promoting the agencies for women empowerment and gender equality by developing and implementing the
 - vii. National Action Plan for Combating Human Trafficking 2015-2017.

Besides the above initiatives, the government has taken different initiatives in other aspects to ensure gender equality and women empowerment. These initiatives have been taken in different sectors, including (i) social security and social empowerment of women; (ii) rehabilitation and legal aid for women victims; (iii) prevention of women's oppression; (iv) institutional mechanism for women's development; (v) sports and women; and (vi) digital Bangladesh.

8 Alignment of the Project with GCF Gender Assessment Guideline

According to GCF guideline, gender analysis needs to take place early in the planning process so that an understanding of gender roles and power relations is built into the project. All projects/programs should include an analysis of gender and include data on gender issues within the overall situation analysis. In this project survey was conducted in selected 20 RMG factories among 70 workers and 20 facility managers. The survey questions were based on educational qualification, labor force participation, participation in decision making process, existing facilities on the factory, gender considerations etc.

The project reviewed the existing policies, strategies and frameworks related to labor participation especially considered gender aspects both at national and international level. The projects interventions will be designed considering the legal status of women as well. In addition, the project considered commonly held beliefs, perceptions, and stereotypes related to gender, the division of labor among women and men and the participation between women and men in the formal/informal economy in the project/program footprint area. Also, during the stakeholder engagement workshops, KII and factory surveys it was ensured that both male and female participants take part in the discussions and give their valuable opinion. However, RMG sector and associated sectors related to this has very little involvement of women in higher management, research or academic line. Still the study tried to ensure participation of female respondents as much as possible.

During the first and second workshop 15% and 11% female participants were present respectively. The workshops were designed to share ideas around a wide range of energy efficiency practitioners including regulators, policy makers, experts, academicians, researchers and garment sector entrepreneurs, as well as environment, social and gender experts. There were participants from different ministries and government agencies, BGMEA, BKMEA, garments units along with development partners, NGOs, academic and research organizations who have been working with different issues related to RMG sector in Bangladesh, especially environmental, social and gender concerns.

During KII, 25% respondents were women and 50% of the respondents had knowledge and/or experience on working with gender related issues in the RMG sector in Bangladesh. Especially, respondents from Ministry of Labour and employment (MoLE), Ministry of Environment, Forest and Climate Change (MoEFCC) and Economic Relations Division (ERD) under Ministry of Foreign Affairs shared their views and knowledge regarding gender consideration and its importance in the sustainability of RMG sector. Representatives from BGMEA and BKMEA had first-hand experience on gender issues and providing capacity and knowledge building training to both male and female workers, promote female workers with potential for higher education and training and provide trainings to the workers on health and safety related issues. The researchers and academicians were also directly involved in addressing gender issues in the RMG sector and working for gender equality within the sector.

During factory survey more than 40% respondents were female workers. However, during the survey with facility managers there were no female respondents as the selected RMG units did not have any female facility managers till the survey was conducted.

Moreover, the project considered the existing gender inequalities and situation of women and men in the interventions related energy efficiency and emphasize more on capacity building on energy efficiency for both male and female workers. Also, access to financial resources and control over these resources is considered while designing the interventions. In addition, it will address differential needs/priorities of women and men in the context of the project.

The project will ensure the following issues in the gender action plan:

- Services and technologies provided by the project/program be available and accessible to both women and men
- Equal access for the women and men from vulnerable communities to information and opportunities necessary to participate and benefit fully from the anticipated outcomes of the project
- Ensure capacity development activities of disadvantaged groups
- Encourage women entrepreneurship in RMG sector
- Encourage more educated women to enter RMG sector in higher management
- Promote leadership capacity of female workers

The information gathered from the gender analysis/assessment should be considered in all stages of the project cycle: design, formulation, implementation, and monitoring and evaluation. In each of these stages, project managers should keep a 'gender lens' in mind, looking at ways the project/program can address gender inequalities that emerge from the project; ensure the differential needs of women and men are addressed; ensure women and men have equal access to resources, services, and capacity development; ensure equal participation of women and men in management arrangements and as beneficiaries, partners and key stakeholders; and ensure women's equal participation in decision – making processes.

Moreover, the Ministry of Women and Children's Affair (MoWCA) and Ministry of Labour and Employment (MoLE) will be involved in the project activities. While developing the training modules for capacity building, consultations will be done with these respective ministries as well so that they can provide adequate guidelines. Also, experiences shared by the ministries regarding capacity building of industrial workers, especially underprivileged women will help to develop the proposed actions in a sustainable way.

9 Grievance Redressal System (GRS)

Grievance Redress Mechanism (GRM) is recognized worldwide as one of the important indicators of an organization's proficiency and efficiency. Grievance redressal is, in fact, a governance related management process. In this case, the GRM will help to identify whether the stakeholders as well as communities and households living near the factories are facing any negative impacts due the implementation and operation of the sub-projects.

Management staff in the RMG sector come from varied backgrounds and do not necessarily have professional management trainings to understand and address issues related to well-being of workers. So, the GRS need to easily manageable, easily accessible and user friendly.

In this project the End Borrower would ensure that grievance redress procedures are in place and would monitor those procedures to ensure that grievance are handled properly. The End Borrower will establish a procedure to answer sub-project-related queries and address complaints, disputes, and grievances about any aspect of the sub-project, including disagreements regarding the assessment and mitigation of environmental and social impacts. Generally, the grievance redress committees (GRC) are of two types (i) formal courts of appeal and (ii) a locally constitutes GRC for dispute resolution. The second may not totally avoid but may reduce the problem significantly. Grievance Redress Committee (GRC) will be formed. The GRC will ensure proper presentation of complaints and grievances, as well as impartial hearings and transparent decisions. The GRCs will meet periodically to discuss the merit of each case and fix a date for hearing and notify the aggrieved persons to submit necessary documents in proof of her/his claim/case; resolve grievances within one month of receipt of complaint. Additional details regarding the functioning of GRC is presented in the SMF.

A general guideline for the GRS will be developed by the LFI in consultation with the MoWCA and BGMEA. It should be ensured that the GRM within the RMG units are aligned with the GCF Independent Redress Mechanism as well as the Accredited Entity's Grievance Redress Mechanism (including contact details, accessibility and basic procedures of such mechanisms, etc.).

Grievance Redress Committee (GRC) will be formed and established at each project site. Grievance response focal point will be available at each shelter site for instant response to an aggrieved person, receiving written complaints or suggestions, produce them to the GRC for hearing and resolution. The GRC will be comprise of the members from the PIU (*Please refer Gender action Plan: Activity 3.1.1: Formulation of a project committee with women participating for monitoring of the activities under the program.*) and representatives from workers (one male and one female). This GRC will be responsible for reporting to the LFIs. Detailed structure and mechanism are described in the ESMF. The Grievance Redress Committee (GRC) will be same for ESMF & gender issue.

The following issues should be considered during for a successful GRM:

User manual: An easy to understand instruction document should be provided for the GRM for the general user wishing to avail the proposed central GRM.

User friendliness: The system should not be too complicated either in technical terms or in language for the general public irrespective of education background.

End-to-end tracking: Tracking of grievance both by the complainant and the authority for grievance redressal is very important.

Confidentiality, security and anonymity: Keeping complainant's identity a secret is very important for security reasons. Ensuring that the system provides an option to complainants for lodging their grievances anonymously would also ensure confidentiality and security.

Progress monitoring: Grievance should be addressed within 7 days of receiving. A grievance redress system is no good if the grievances are not addressed regularly. When the grievances are not addressed for a long time then the employees will lose faith from the system and stop sharing their issues.

Feedback option: There should be a feedback option to measure client satisfaction and collect opinions on the effectiveness of the system.

10 Recommendations

The issue of gender equality has immense significance as it is not only a fundamental human right but also a necessary foundation for a peaceful, prosperous and sustainable world.³⁸ Providing women and girls with equal access to education, health care, decent work, and representation in political and economic decision-making processes will fuel sustainable economies and benefit societies and humanity at large. Implementing new legal frameworks regarding female equality in the workplace and the eradication of harmful practices targeted at women is crucial to ending the gender-based discrimination prevalent in many countries around the world. Thus, several initiatives are required to be taken to augment the process of attainment gender equality in Bangladesh.

In order to create a gender equitable environment at the workplace, the following initiatives can be implemented:

- Create enabling environment for hiring workers considering their skills rather than gender.
- Ensure job security by issuing job contracts and ensure effective implementation of different laws and regulations in terms of wages and other facilities.
- Ensure trainings and capacity development workshops for both male and female workers.
- Awareness raising activities among communities and families so that no woman has to leave their carrier due to social pressure or responsibilities.
- Capacity development of female RMG workers to avail financial services and awareness on existing financial services.
- Advocacy needs to be done with the family members to engage women in the family and community level economic activities so that women can be financially independent and take their own decisions.
- Building a network among women, influential community and educational leaders is important so that the network would work as a catalyst by bringing changes in the society.
- Provide safe transportation facilities especially for woman so that they can join work in different shifts as well as men without getting concerned about security issues.
- Provide better medical and mental health facilities to workers for increasing their health which will eventually increase their performance.
- Ensure special provisions for pregnant and lactating mothers and maternity leave should be as per government rules.
- Provision of daycare facilities by the factory authority as well as community.
- Facilitate gender responsive and socially inclusive training in the RMG units.
- Initiatives to improve the women's access to social services.
- Take initiatives to foster equality in the decision-making process and encourage participation of both male and female workers in trade bodies or labour unions for developing better leadership skills.
- Implement different programmes those would facilitate women's access over resources.
- Organize capacity-building events for industry personnel on gender sensitivity and equal opportunities.
- Ensure proper occupational health and safety for all the workers irrespective of gender.
- Ensure harassment free working environment for the workers regardless of gender.

³⁸ <https://www.daily-sun.com/post/345296/SDGs-and-Gender-Equality-in-Bangladesh>

11 Gender Action Plan

11.1 Outline

A project/program-specific Gender Action Plan (GAP) is a tool used to ensure gender mainstreaming is clearly visible in project/program design and implementation. The project GAP is not a separate component. It mirrors the logical framework of the project and is an integral part of project/program design. GAPs include clear targets, gender design features and measurable performance indicators to ensure women's participation and benefits. Key aspects of the GAP are incorporated into project/program assurances to encourage buy-in from AEs and other partners.

The GAP presents:

- Work undertaken to address gender issues in the project/program;
- Targets and design features, included in the project/program to address gender concerns and ensures tangible benefits to women and men, especially from vulnerable communities;
- Mechanisms to ensure implementation of the gender design elements;
- Gender - sensitive monitoring and evaluation indicators.

Below is presented a GAP that contains impact, outcome and output statements, gender – responsive activities, gender – performance indicators and targets, timeline, responsibility lines, and approximate budgetary allocations against each activity.

Gender Action Plan

Activities	Indicators and Targets	Timeline	Responsibilities	Budget
Impact Statement: Enhanced livelihood opportunities and reduced inequalities at work through training and skill development to improve the productivity and competitiveness for both men and women from the implementation of energy efficient technologies and equipment in readymade garment (RMG) sector.				
Outcome Statement: Increased access of the garment sector to concessional financing for energy efficiency, leading to energy savings through large scale adoption of energy efficient technologies to reduce GHG emissions.				
Component 1: Concessional loans for energy efficiency in garment sector				
Output 1.1: Women entrepreneur successfully engaged in the program				
Activity 1.1.1: Encouragement of women entrepreneur to venture in the garment sector	Indicator Encourage loan application from garment units with significant participation of (Chairman/Any Director/MD/CEO) women entrepreneurs Target To approach minimum 06 (six) woman led garment business entities under the programme	Minimum 01 woman led garment business entity will be approached in each year. Total 06 (five) woman led garment business entities will be approached within 05 years of GCF loan availability period	Implementation responsibility by LFI	Funding Source: LFI Amount: XXXXX Within the "Project Management Cost" from LFI in line of regular business process
Output 1.2: Investment to comply with national standards for women welfare and recover costs from the energy savings from programme/project implementation				
Activity 1.2.1: Reducing the gender wage gap (1) Unbiased employment as per the "Labour rules (2015)" (2) Encourage similar wages for men and women for	Indicator Encouraging women participation by providing unbiased employment opportunities and providing similar wages for same jobs to men and women Target Submission of existing pay scale ensuring similar wages for men & women or proposed pay scale with a time bound implementation plan ensuring such similar wages	Number of such submission will depend on the number of end borrowers approaching LFI for their financing. There is no yearly target of how many end	LFI & End borrowers	Funding Source: End borrowers Amount: XXXXX The cost of any such implementation will be borne by end borrowers &

similar work type as per the “National Wage Board”		borrowers will be financed rather specific amount for each year & a total of entities have been mentioned in the log frame. So LFIs will ensure submission of existing pay scale ensuring similar wages for men & women or proposed pay scale with a time bound implementation plan ensuring such similar wages in line with their actual financing in each year.		this will not be part of GCF project cost
Activity 1.2.2: Facilitate women participation in the sector through provision of: (1) Medical Facilities (2) Child care (3) Sanitation facilities	Indicator Encouraging women participation by facilitating them with medical assistance, child care and sanitation as prescribed under Labour Rule 2015 & other regulatory compliances Target Share existing status of the mentioned parameters during loan application and ensure compliance with the national standards (if not already complied) within 12 months of the loan disbursement.	Number of such sharing will depend on the number of end borrowers approaching LFIs for their financing. There is no yearly target of how many end borrowers will be financed rather specific amount for each year & a total of	LFIs & End borrowers	Funding Source: End borrowers Amount: XXXXX The cost of any such implementation will be borne by end borrowers & this will not be part of GCF project cost

		entities have been mentioned in the log frame. So LFIs will ensure sharing of existing status of the mentioned parameters during loan application and ensure compliance with the national standards (if not already complied) within 12 months of the loan disbursement in line with their actual financing in each year within 05 years of GCF loan availability period.		
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Component 2: Large scale adoption of Energy Efficient technologies and equipment

Output 2.1: Energy Efficient technology upgrades in the garment units employing new programmed/automated processes

Activity 2.1.1: Training and skill development for machine use and operating on programming software (1) Development of training modules (2) Trainings and evaluation	Indicator Ensure participation in skill development trainings leading to women empowerment by providing skill development through the technical assistance workshops Target	Within 05 years of GCF loan availability period. As the LFIs will arrange this separately, so there is no year specific	Each LFI will ensure at least 01 training for their respective end borrowers & ensure targeted women participation	Funding Source: End borrowers & LFIs Amount: XXXXX The cost of any such implementation will be borne by end borrowers &
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	Ensure women employee participation a minimum of 25% in skill development and training workshop (Total 04 workshops)	time for arranging such training.		this will not be part of GCF project cost. The cost of LFIs will be within the “Project Management Cost” from LFIs in line of regular business process
Activity 2.1.2: Safety trainings while operating on new machinery/equipment (1) Development of training modules (2) Periodic trainings and evaluation	Indicator Reduction in accidents and safety issues rising from new certified machinery for which due trainings have been provided to the workers including women. Target Ensure women employee participation a minimum 20% in the safety training for workers (Total 04 safety trainings)	Within 05 years of GCF loan availability period As the LFIs will arrange this separately, so there is no year specific time for arranging such training.	Each LFI will ensure at least 01 training for their respective end borrowers & ensure targeted women participation	Funding Source: End borrowers, LFIs & GCF Amount: USD 24,710 from GCF The cost of any such implementation will be borne by end borrowers & this will not be part of GCF project cost. The cost of LFIs will be within the “Project Management Cost” from LFIs in line of regular business process. GCF will finance USD 24,710 under TA component Activity 4.3.3
Activity 2.1.3: Grievance Redressal (1) Formulation of a project committee for addressing grievance (2) Establishing grievance redressal process	Indicator Encouragement for establishment of grievance redressed system for enhanced women employee security Target An active grievance redressal system in place within 12 months of project operation	Number of such grievance redressal system will depend on the number of end borrowers receiving financing from this program. There is no yearly target of how	LFIs & end borrowers	Funding Source: End borrowers & LFIs Amount: XXXXX The cost of any such implementation will be borne by end borrowers &

		many end borrowers will be financed rather specific amount for each year & a total of entities have been mentioned in the log frame. So LFIs will ensure such active grievance redressal system in place within 12 months of project operation in line with their actual financing in each year within 05 years of GCF loan availability period		this will not be part of GCF project cost. The cost of LFIs will be within the “Project Management Cost” from LFIs in line of regular business process
Component 3: Gender balanced participation during the Implementation/operationalization of the project/programme				
Output 3.1: Women Participation during project implementation				
Activity 3.1.1: Formulation of a project committee with women participating for monitoring of the activities under the program.	Indicator Gender balanced project/programme committee in each LFI Target To ensure women participation in the Project Committee & ensure at least 01 female member participation	To form a Project committee within 06 months after signing the Funded Activity Agreement (FAA) with GCF	LFIs	Funding Source: LFIs Amount: XXXXX The cost of LFIs will be within the “Project Management Cost” from LFIs in line of regular business process
Activity 3.1.2: Training & skill development of relevant Business Units of LFIs	Indicator Increased women skills & participation in various management aspects of the program Target	02 workshop in Year-2 02 workshops in Year-6	IDCOL, along with LFIs, will arrange training program for Business	Funding Source: GCF Amount: USD 192,701 (For entire activities)

	To ensure women employee participation (from LFIs) in the relevant capacity building initiatives as beneficiary	(Please see project Implementation Time table)	Units of LFIs & ensure targeted women participation	Activity 4.2.1 Conduct capacity building trainings/workshops to develop capacity of corporate/head office level officials to effectively conduct banking due diligence of loan applications for implementation of energy efficient technologies. Activity 4.2.2: Conduct capacity building trainings/workshops to develop technical capacity of banking officials at branch/field level to efficiently monitor implementation of energy efficient technologies over the duration of repayment of loan amount.
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11.2 Monitoring of Gender Action Plan

Ensuring the sustainability of the gender action plan is a major concern and to achieve this a proper monitoring system should be in place. At the sub-project level, there will be Project Implementation Unit (PIU)³⁹ who will be responsible for the proper implementation of the gender related interventions. The PIU will submit periodic monitoring reports (yearly) to the project committee in *Activity 3.1.1: Formulation of a project committee with women participating for monitoring of the activities under the program*. The LFIs will then monitor whether the end borrowers (i.e., RMG units) are following the policies and guidelines regarding gender issues and the action plan is implemented ensuring gender equality and women empowerment within the sector.

The selected LFIs involved in the program have experience on monitoring gender related issues. Most of their RMG clients work with renowned international buyers who have certain conditions precedent on gender related matters. As a result, they have to maintain certain guidelines to ensure gender equality and women empowerment. Thus, the LFIs are well aware of the gender related concerns and are capable of monitoring the overall gender related activities during the 05 years of GCF loan availability period.

For the Gender Action Plan of this programme, the LFIs have agreed to conduct onsite as well as offsite periodic inspection for ensuring proper practice of gender related issues in the projects. In the appraisal of loan proposal, they will assess the client's willingness and ability to implement gender issues as required by GCF and other regulatory authorities. They are cooperative in terms of gender related concerns and they will establish a monitoring mechanism within their organization. The LFIs will conduct annual assessment after the project implementation throughout the 05 years of GCF loan availability period. During the assessment it will be assessed whether the end borrowers have implemented the gender issues mentioned in the gender action plan and created enabling environment of increased women empowerment.

³⁹ The structure of the PIU will be same as proposed in the ESMF.

Gender Action Plan

Activities	Indicators and Targets	Timeline	Responsibilities	Budget
Impact Statement: Enhanced livelihood opportunities and reduced inequalities at work through training and skill development to improve the productivity and competitiveness for both men and women from the implementation of energy efficient technologies and equipment in readymade garment (RMG) sector.				
Outcome Statement: Increased access of the garment sector to concessional financing for energy efficiency, leading to energy savings through large scale adoption of energy efficient technologies to reduce GHG emissions.				
Component 1: Concessional loans for energy efficiency in garment sector				
Output 1.1: Women entrepreneur successfully engaged in the program				
Activity 1.1.1: Encouragement of women entrepreneur to venture in the garment sector	Indicator Encourage loan application from garment units with significant participation of (Chairman/Any Director/MD/CEO) women entrepreneurs Target To approach minimum 06 (six) woman led garment business entities under the programme	Minimum 01 woman led garment business entity will be approached in each year. Total 06 (five) woman led garment business entities will be approached within 05 years of GCF loan availability period	Implementation responsibility by LFIs	Funding Source: LFIs Amount: XXXXX Within the “Project Management Cost” from LFIs in line of regular business process
Output 1.2: Investment to comply with national standards for women welfare and recover costs from the energy savings from programme/project implementation				
Activity 1.2.1: Reducing the gender wage gap (1) Unbiased employment as per the “Labour rules (2015)” (2) Encourage similar wages for men and women for	Indicator Encouraging women participation by providing unbiased employment opportunities and providing similar wages for same jobs to men and women Target Submission of existing pay scale ensuring similar wages for men & women or proposed pay scale with a time bound implementation plan ensuring such similar wages	Number of such submission will depend on the number of end borrowers approaching LFIs for their financing. There is no yearly target of how many	LFIs & End borrowers	Funding Source: End borrowers Amount: XXXXX The cost of any such implementation will be borne by end borrowers &

similar work type as per the “National Wage Board”		end borrowers will be financed rather specific amount for each year & a total of entities have been mentioned in the log frame. So LFIs will ensure submission of existing pay scale ensuring similar wages for men & women or proposed pay scale with a time bound implementation plan ensuring such similar wages in line with their actual financing in each year.		this will not be part of GCF project cost
Activity 1.2.2: Facilitate women participation in the sector through provision of: (1) Medical Facilities (2) Child care (3) Sanitation facilities	Indicator Encouraging women participation by facilitating them with medical assistance, child care and sanitation as prescribed under Labour Rule 2015 & other regulatory compliances Target Share existing status of the mentioned parameters during loan application and ensure compliance with the national standards (if not already complied) within 12 months of the loan disbursement.	Number of such sharing will depend on the number of end borrowers approaching LFIs for their financing. There is no yearly target of how many end borrowers will be financed rather specific amount for each year & a total	LFIs & End borrowers	Funding Source: End borrowers Amount: XXXXX The cost of any such implementation will be borne by end borrowers & this will not be part of GCF project cost

		of entities have been mentioned in the log frame. So LFIs will ensure sharing of existing status of the mentioned parameters during loan application and ensure compliance with the national standards (if not already complied) within 12 months of the loan disbursement in line with their actual financing in each year within 05 years of GCF loan availability period.		
Component 2: Large scale adoption of Energy Efficient technologies and equipment				
Output 2.1: Energy Efficient technology upgrades in the garment units employing new programmed/automated processes				
Activity 2.1.1: Training and skill development for machine use and operating on programming software (1) Development of training modules (2) Trainings and evaluation	Indicator Ensure participation in skill development trainings leading to women empowerment by providing skill development through the technical assistance workshops Target	Within 05 years of GCF loan availability period. As the LFIs will arrange this separately, so there is no year specific	Each LFI will ensure at least 01 training for their respective end borrowers & ensure targeted women participation	Funding Source: End borrowers & LFIs Amount: XXXXX The cost of any such implementation will be borne by end borrowers &

Annex VIII – Gender Action Plan

GREEN CLIMATE FUND FUNDING PROPOSAL



	Ensure women employee participation a minimum of 25% in skill development and training workshop (Total 04 workshops)	time for arranging such training.		this will not be part of GCF project cost. The cost of LFIs will be within the “Project Management Cost” from LFIs in line of regular business process
Activity 2.1.2: Safety trainings while operating on new machinery/equipment (1) Development of training modules (2) Periodic trainings and evaluation	Indicator Reduction in accidents and safety issues rising from new certified machinery for which due trainings have been provided to the workers including women. Target Ensure women employee participation a minimum 20% in the safety training for workers (Total 04 safety trainings)	Within 05 years of GCF loan availability period As the LFIs will arrange this separately, so there is no year specific time for arranging such training.	Each LFI will ensure at least 01 training for their respective end borrowers & ensure targeted women participation	Funding Source: End borrowers, LFIs & GCF Amount: USD 24,710 from GCF The cost of any such implementation will be borne by end borrowers & this will not be part of GCF project cost. The cost of LFIs will be within the “Project Management Cost” from LFIs in line of regular business process. GCF will finance USD 24,710 under TA component Activity 4.3.3
Activity 2.1.3: Grievance Redressal (1) Formulation of a project committee for addressing grievance (2) Establishing grievance redressal process	Indicator Encouragement for establishment of grievance redressed system for enhanced women employee security Target An active grievance redressal system in place within 12 months of project operation	Number of such grievance redressal system will depend on the number of end borrowers receiving financing from this program. There is no yearly	LFIs & end borrowers	Funding Source: End borrowers & LFIs Amount: XXXXX The cost of any such implementation will be borne by end borrowers &

		target of how many end borrowers will be financed rather specific amount for each year & a total of entities have been mentioned in the log frame. So LFIs will ensure such active grievance redressal system in place within 12 months of project operation in line with their actual financing in each year within 05 years of GCF loan availability period		this will not be part of GCF project cost. The cost of LFIs will be within the “Project Management Cost” from LFIs in line of regular business process
Component 3: Gender balanced participation during the Implementation/operationalization of the project/programme				
Output 3.1: Women Participation during project implementation				
Activity 3.1.1: Formulation of a project committee with women participating for monitoring of the activities under the program.	Indicator Gender balanced project/programme committee in each LFI Target To ensure women participation in the Project Committee & ensure at least 01 female member participation	To form a Project committee within 06 months after signing the Funded Activity Agreement (FAA) with GCF	LFIs	Funding Source: LFIs Amount: XXXXX The cost of LFIs will be within the “Project Management Cost” from LFIs in line of regular business process
Activity 3.1.2: Training & skill development of relevant Business Units of LFIs	Indicator Increased women skills & participation in various management aspects of the program Target	02 workshop in Year-2 02 workshops in Year-6	IDCOL, along with LFIs, will arrange training program for Business	Funding Source: GCF Amount: USD 192,701 (For entire activities)

Annex VIII – Gender Action Plan

GREEN CLIMATE FUND FUNDING PROPOSAL



	To ensure women employee participation (from LFIs) in the relevant capacity building initiatives as beneficiary	(Please see project Implementation Time table)	Units of LFIs & ensure targeted women participation	Activity 4.2.1 Conduct capacity building trainings/workshops to develop capacity of corporate/head office level officials to effectively conduct banking due diligence of loan applications for implementation of energy efficient technologies. Activity 4.2.2: Conduct capacity building trainings/workshops to develop technical capacity of banking officials at branch/field level to efficiently monitor implementation of energy efficient technologies over the duration of repayment of loan amount.
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GENDER ASSESSMENT REPORT

Promoting private sector investment through large scale adoption of
energy saving technologies and equipment for textile sector of
Bangladesh

Table of Contents

1. Introduction	4
1.1. Programme brief and relevance	5
1.2. GCF’s guideline on gender equality	7
1.2.1. Objective, Principles and Mandatory requirements	7
2. Methodology	8
3. A contextual snapshot of Gender sensitivity and equality in textile sector in Bangladesh	9
3.1. Gender and energy efficiency in textile sector	9
3.2. Perceptions and stereotypes	10
3.3. Institutional arrangements (Legal and administrative)	11
3.3.1. Policy	11
3.3.2. Laws	11
3.3.3. Institutions	12
3.3.4. Accords and Conventions	14
4. Situational analysis: understanding vulnerabilities through Gender Assessment	14
4.1. Socio-economic assessment	14
4.1.1. Average age of women in working in textile units	14
4.1.2. Marital status and average household size (adults and children)	15
4.1.3. Education Status	15
4.1.4. Skills and trainings (acquired personally and with support from employer)	16
4.1.5. Gender ratio in textile workforce	16
4.1.6. Average work hours	17
4.1.7. Sources of income, average income and expenditure	17
4.1.8. Female led households, house ownership and tenancy	17
4.1.9. Savings and loans	18
4.1.10. Provident fund/life Insurance	18
4.2. Health and Safety assessment	18
4.2.1. Health issues faced by men and women in textile units	18
4.2.2. Health camps and medical facilities (like dispensary) and health insurance	18
4.2.3. Maternity leaves	18
4.2.4. Violence and abuse at work	18
4.2.5. Injuries and Accidents	19
4.2.6. Safety precautions, trainings and tools	19
4.2.7. Compensation to victims	19
4.3. Sanitation and access to basic facilities	20

4.3.1. Access to drinking water, washrooms and sanitation facilities (soaps, sanitary napkins, etc)	20
5. Qualitative Gap Analysis	21
6. Bridging Gaps: gender sensitivity and applicability of the programme.....	28
6.1. Inclusivity in stakeholder approach (planning, management and implementation).....	28
6.1.1. First stakeholder workshop.....	28
6.1.2. Second stakeholder workshop.....	28
6.2. Monitoring interventions and Evaluation	28

Table of figures and tables

Figure 1: The Fundamental Principles of GCF's Gender Policy	8
Figure 2: Percentage of Textile workers belonging to various age groups	15
Figure 3: Marital status of men and women	15
Figure 4: Gender wise education status of textile worker group	16
Figure 5: Gender wise break-up of the monthly income of textile worker group.....	17
Figure 6: Elements of the Gender Action Plan.....	29
Table 1: Percentage of women participation in stakeholder engagement	5
Table 2: Project activities that are aligned with SDGs for social and gender inclusion	5
Table 3: Survey Statistics.....	8
Table 4: Institutional framework relevant to the programme	12
Table 5: Responses of the textile workers on trainings and skill development	16
Table 6: Responses of textile labour on respective issues.....	19
Table 7: Gender biases in textile sector in Bangladesh	21

List of acronyms

ADB	Asian Development Bank
AE	Accredited Entity
BTMA	Bangladesh Textile Mills Association
DAE	Direct Access Entity
DOE	Department of Environment
E&S	Environmental and social
ECA	Environment Conservation Act
ECA	Environment Court Act
ECC	Environment Clearance Certificate
ECR	Environment Conservation Rules
EE	Electrical Efficiency
EHS	Environment, Health and Safety
EIA	Environment Impact Assessment
ESMF	Environmental and Social Management Framework
ETP	Effluent Treatment Plant
GCF	Green Climate Fund
GHG	Green House Gas
GoB	Government of Bangladesh
GRC	Grievance Redressal Committee
IDCOL	Infrastructure Development Company Limited
IE	Implementing Entity
IFC	International Finance Corporation
INDC	Intended Nationally Determined Contributions
PS	Performance Standards

1. Introduction

Government of Bangladesh committed to various energy consumption reduction targets in the past few years and initiated EE&C programmes like Energy Management Program, EE Labelling Program and EE Buildings Program, targets at large energy consuming entities and equipment in the industrial, residential and commercial sectors. Bangladesh being an international hub for clothing (textile and readymade garments production¹), has one of the world's fastest growing economy, largely driven by its exports in textile and readymade garments. The textile sector alone contributes to above 12% in GDP with more than 81% of the export earning coming from textiles & textile related products².

To meet the demands from world over, the sector has been growing and expanding rapidly over the last few years. With concentration of more production units, the electricity consumption in the sector is increasing and so is the employment of the workforce. The textile labour market has its own set of challenges like unorganised, uneducated-unskilled labour, urban-rural and wide male-female worker ratio disparities. Besides, the industrial challenges for the workforce are low wages, exhaustive long shifts, compromised health & safety, hazardous working conditions, sexual harassment/discrimination and lack of human resource policies.

Evidently, deriving from the above-mentioned facts, gender inequalities affect more woman than men in this case, given the fact, women are more personally, socially, economically and politically challenged. The industrial units prefer women labourers, especially young women workers since they are in abundant supply at a very low cost. However, in textile units, women employment is less due to the manual, labour intensive processes³. Stereotypically it isn't favoured by owners to employ women due to their low competence, productivity and capacity to perform certain tasks in textile units. Therefore, presence of women is very less and the rest few that are working in textile units as directors, managers or executives face unequal access to authority and decision-making, human and legal rights, house/asset ownership, bank credits, education and skill development, economic resources, and health services.

Bangladesh's Third National Communication (TNC)⁴ to UNFCCC clearly states that Bangladesh is one of the most climate vulnerable countries in the world and the challenges posed by climate change make the lives and livelihoods of millions of people increasingly complex and vulnerable. Bangladesh is prone to climate disasters like floods, droughts and cyclones. Consequently, climate change and its impacts are enhancing vulnerabilities for women due to the lack of climate adaptive capacities. Struggles in coping up with the impacts of climate disasters, living in extreme unhealthy working conditions, inability to earn a living due to health issues and lack of education and skills, fully covered drapes and clothes that reduce the possibility of escaping the place of climate event and man-made disasters like fire, are a few to name.

To understand the delicate relation between energy efficiency and gender equality, there is a need to understand the capacity of energy efficient processes and equipment to influence the life of workers both socially and economically by deep diving into the socio-economic status of the textile workforce which determines the human and economic development in the past and the upon technology

¹ Raw material for RMG sector is fabrics produced locally. Hence, RMG is a textile dependent sector.

² As quoted by Bangladesh Textile Mills Association (BTMA). Available at: <https://www.btmadhaka.com/>

³ Primary data sources (field survey and interviews)

⁴ TNC

(https://unfccc.int/sites/default/files/resource/TNC%20Report%20%28Low%20Resolution%29%2003_01_2019.pdf)

upgrades. In this context, it's important to understand gender specific inequalities and vulnerabilities. The following sections will discuss the concern in detail.

1.1. Programme brief and relevance

The programme 'Promoting private sector investment through large scale adoption of energy saving technologies and equipment for textile sector of Bangladesh' aims to reach out to the business establishments and entrepreneurs in textile sector to provide them with financial and market resources to avail investment opportunities for energy saving technology upgrades. The programme objectives have clearly been laid out to ensure direct and indirect benefits to the vulnerable and underprivileged sections of the society which are dependent on this sector. Project activities consider encouraging gender equality throughout the phases which particularly involves women in consultations, decision making and embedding gender specific benefits during the implementation of the programme.

During the design and development of the programme, gender equality has been maintained at the stakeholder consultation and workshop, key informants at planning and management level (women in the ministries, relevant department and associations), executive and labour level. Women respondents have been included and preferred to answer the survey questions related to facilities and challenges at their workplaces (in different sections in textile units).

Table 1: Percentage of women participation in stakeholder engagement

Stakeholder Engagement	Stakeholder Type	% of women participants
Key Informants Interview (KII)	Primary Stakeholders: Ministries, Associations, Multilaterals, Research/Academia and Industry	27%
Workshop/Group consultation	Primary and Secondary stakeholders: Ministries, Associations, Multilaterals, Research/Academia and Industry	Workshop 1 23%
		Workshop 2 24%
Industry Survey	Industry: Women led, Workers and Community	Women led industries: 35% factories surveyed have women leaders in capacities of Directors/DGM/CEO
		Executives/workers: 30%

Table 2: Project activities that are aligned with SDGs for social and gender inclusion

Project Activities	SDGs and Gender specific considerations		
► Assess the feasibility of the programme by conducting an in-depth review of the			

Annex VIII – Gender Assessment Report

GREEN CLIMATE FUND FUNDING PROPOSAL

technologies and equipment's used in Bangladesh's textile sector and developing a baseline for energy consumption in textile sector in current scenario.	The assessment to consider the feasibility of the programme employs research and review of existing baselines in energy consumption of the textile units and the roles and responsibilities of the associated institutions. Gender equality has been ensured while conducting the studies to assess the status and during implementation, ensure provision of clean energy, industrial and infrastructural development and reduction in consumption of energy throughout the textile value chain.
► Assess environmental, social and gender gaps in the current scenario and impacts on introduction of new technology and equipment's.	Environmental, social and gender assessments to assess the attitudes of the management and executives towards the impact of their currently deployed processes and operations on environment, society and the underprivileged gender and bridge the observed gaps during the implementation of the proposed programme. To develop a Gender Action Plan from the findings of gender assessment based on primary and secondary data.
► Analyse and evaluate environmental, social, political, financial, operational risks associated with the sector.	Evaluation of potential risks that may occur during the implementation of programme which may hamper the benefits and mitigation measures to be adopted for a gender equal approach.
► Identify programme/programme-level indicators to assist development of a GCF proposal aimed at Promoting private sector investment through large scale adoption of energy saving technologies and	

equipment for textile sector of Bangladesh	Development of a GCF proposal which is gender sensitive and proposes a gender action plan to be incorporated for the implementation of the programme. Innovation and infrastructure development, economic growth, sustainable value chain and provision of affordable and clean energy is directly related to women empowerment and welfare, lowering their sectoral and climate related vulnerabilities.
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1.2. GCF's guideline on gender equality

The Green Climate Fund has a mandate to support a paradigm shift to low-emission and climate-resilient development and gender mainstreaming is central to its objectives and guiding principles, through engaging women and men of all ages as stakeholders in the design, development and implementation of strategies and activities to be financed⁵.

The Gender Policy and Action Plan was adopted in March 2015, by the GCF board. The policy states, that a gender-sensitive approach is therefore part of a paradigm shift; Impacts of climate change are borne differently by man and women and existing gender inequalities are likely to be exacerbated; greater vulnerability of women to climate change stems from gender norms and discrimination that result in the imbalanced division of labour, lower incomes, and lesser livelihood opportunities; less access and control over land and other productive assets; fewer legal rights; lesser mobility and lesser political and professional representation.

1.2.1. Objective, Principles and Mandatory requirements

The overall objective of the gender policy is to ensure that by adopting a gender-sensitive approach, the Fund will efficiently contribute to gender equality and achieve greater and more sustainable climate change results, outcomes and impacts⁶.

It is a mandatory requirement of the gender initiatives under the GCF programme to satisfy the basic requirements i.e. gender equality and equity, inclusiveness, accountability, country ownership through alignment with national policies and priorities in stakeholder inclusion, enhancing competencies through fund's institutional framework and equitable resource /allocation for equal benefits for men and women through fund's climate action.

⁵ Mainstreaming Gender in Green Climate Fund Projects: A practical manual to support the integration of gender equality in climate change interventions and climate finance (https://www.greenclimate.fund/documents/20182/194568/Guidelines_-_GCF_Toolkit_Mainstreaming_Gender.pdf/860d1d03-877d-4c64-9a49-c0160c794ca7)

⁶ Gender Policy and Action Plan (GCF/B.08/19 6 October 2014) Available at: https://www.greenclimate.fund/documents/20182/24946/GCF_B.08_19_-_Gender_Policy_and_Action_Plan.pdf/afd29fd9-3efa-41c3-8318-7d86587c7701

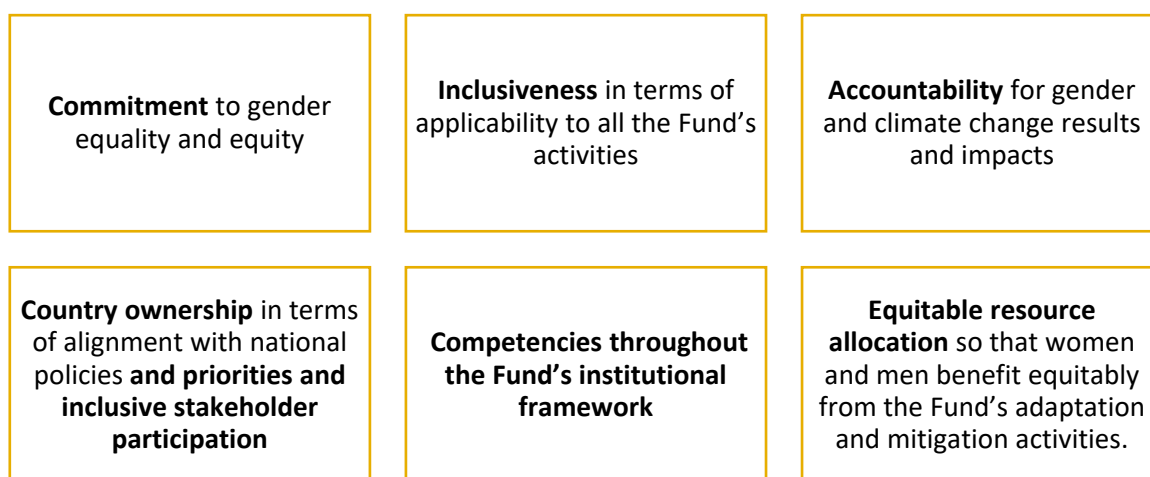


Figure 1: The Fundamental Principles of GCF's Gender Policy

Mainstreaming gender in GCF programmes can be synthesized into an approach which mandates a gender analysis to assess social, economic and political factors underlying climate change-exacerbated gender inequality, action plans with methods and tools to promote equality, and gender sensitivity of the monitoring and evaluation of the programme to measure and quantify the outcomes and impacts of programme activities on women and men's resilience to climate change.

In line with GCF's gender policy, IDCOL has conducted a gender assessment and has ensured gender balance during all activities conducted under the PPF for this programme. The gender specific needs and priorities of men and women have been assessed and taken up for an action plan as instructed by the GCF.

2. Methodology

Gender assessment is done to investigate gender roles and power relations in relation to the programme design and identify multiple causes of vulnerability, including gender inequality. It is a tool used to identify and build on the diverse knowledge and capacities within communities/households that can be used to make them more resilient to climate related shocks and risks.

This document is based on the review of national data and statistics, progress reports and government published documents; websites (ministries, departments, associations, international agencies and other active organisations) and resources from authentic sources; Meetings and Key informant interviews with the stakeholders and gender specialists. The data and information obtained from the secondary sources is equated from the primary sources (site visit, field survey, worker and community interviews, etc). The industrial survey has been done with the a) Facility lead/CEO/MD; b) Facility engineer; c) Facility workers (male and female) d) Community. The textile facilities were picked through random sampling. Once selected, these factories were visited for the survey and the workers were picked through stratified random sampling. The workers were split through the Strata 1: Female and Strata 2: Male and then picked randomly. The sample was stratified in order to maintain a gender balance.

Table 3: Survey Statistics

Total no. of Textile units surveyed: Each for Facility lead/CEO/MD and Facility engineer	20
Total no. of workers surveyed	201
Total no. of women workers	59

These sources provided with the status of the targeted/planned initiatives and the on-ground implementation of the actions. The huge amount of data obtained from this exercise is aligned with the nationally published data to highlight the gaps and the areas that require immediate attention to suggest the pathways and an action plan to bridge the gaps.

3. A contextual snapshot of Gender sensitivity and equality in textile sector in Bangladesh

Textile sector being the major recruiter in Bangladesh has a men-women ratio of 70:30⁷. Some of the textile mills don't even have a single female worker. The few employed women in the textile sector belong to the most vulnerable sections of the society. However, both women and men labour have increased vulnerability due to wage and income gaps, access to finance, lack of rights for land and property, access to information, health and safety issues⁸.

International statistics show that Bangladesh has a Gender Inequality Index Rank of 119⁹ and a Global Gender Gap Index Rank 72¹⁰. These are the function of reproductive health (maternal mortality ratio and adolescent birth rate), empowerment (share of parliamentary seats held by women and share of population with at least some secondary education), and labour market participation (labour force participation rate).

Hence, Bangladesh has a long way to go for which programmes that incorporate gender sensitive approach are vital. Development of this sector will have direct and indirect benefits for women currently employed and those that seek employment which will show up in a few years.

3.1. Gender and energy efficiency in textile sector

There exists a delicate relationship in gender and energy efficiency. It is essential to look at the trends of employment, which suggest, textile industry has more male workers than female. The employment of men is favoured in textile industry because of the common perceptions like processes in the textile units are labour intensive and manual, women can't perform tasks which include heavy lifting, shifting and loading. Women experience occupational segregation, as most of which are employed are given lighter, sitting jobs. This leads to insufficient skill development and possibly lower wages for women employed in Bangladesh's textile mills.

The equipment and processes currently being used in textile sector in Bangladesh are labour and energy intensive. In most cases, smaller and medium textile units do not undertake energy audits and continue to use decade-old technologies. Quite evidently, lack of skills and use of age-old, labour-intensive machinery are big constraints for job opportunities for women in this sector.

In the past, buyers to bring down their costs, often overlooked the working conditions of their manufacturing labour class and exploited their skills for low wages. The government of Bangladesh has been making stringent rules and amendments¹¹ in the existing laws to counter these challenges. Recently, it has been observed that the international buyers like M&S, H&M, Zara, Puma, etc., to meet

⁷ Calculated from Primary sources of data: Field survey and KII with the facility managers

⁸ International Labour Organisation (<https://www.ilo.org/infostories/en-GB/Stories/Employment/barriers-women#intro>)

⁹ UNDP, Human Development Report, 2016

¹⁰ UN Women. <http://evaw-global-database.unwomen.org/en/countries/asia/bangladesh>

¹¹ Labour Rules 2015 and amendment in Labour Law (2013), discussed in following sections of the report.

their sustainability standards/commitments and CSR targets¹², have started pushing the owner/suppliers to maintain certain standards in employee facilities and safety¹³. The textile mills have been pushed further to reduce the gender gaps in employment. To facilitate the mandate of their buyers, mills are undertaking initiatives like equal wages, child care centres, maternity and paternity leaves, medical facilities, safety precautions and trainings for skill development and awareness. Some of the reputed international brands demand ‘state of art’ machinery and technology upgrades to be introduced in the mills, manufacturing their products, to maintain brand competencies and quality. The interviews with the managers and CEOs of the large textile facilities indicate that the introduction of new and energy efficient, world-class technologies and equipment that are user friendly have opened more opportunities and jobs for men and women in textile sector. However, the textile mills who have integrated energy efficiency in their operations are only a few large industries, while many other due to lack of financial support and expertise, lag much behind. There’s a need of financial support to facilitate large-scale adoption of technology upgrades in textile sector to open new opportunities for men and women both.

3.2. Perceptions and stereotypes

Perceptions continue to enhance gender inequality in the sector. Most of the key informants and survey respondent shared their perceptions and views regarding the issue.

The major perceptions/misconceptions for women, that continue to exist are:

- **Inability to perform labour intense assignments.** The labour-intensive jobs are thus assigned to men, reducing the scope of work and wages for women.
- **Long, uncertain maternity leaves.** Major concern for hiring managers is the long leaves women take on account of pregnancy and child care.
- **Women are Technologically challenged.** Common perception is that the women can’t operate on complicated machinery due to their poor understanding of technology, mechanics and programming.
- **There’s no direct relationship between gender and energy efficiency.** Planners often tend to overlook the potential of the new, user-friendly energy efficient equipment which will facilitate more jobs for women.
- **Inefficient to be involved in decision making.** In a male dominated sector, participation of women in decision making isn’t encouraged. It’s very rare to find women entrepreneurs, MDs and CEOs that are actively involved in planning and management.
- **Restriction on mobility.** Most women in rural regions are discouraged to go beyond the boundaries of their village or travel for work. Their mobility should be within their permissive boundaries. Barriers on mobility have also reduced work opportunities.

Women both skilled and unskilled work in mills and perform domestic chores as well. Most of the existing misconceptions are led by the patriarchal structure of society which limit women empowerment. Training and skill development could break many barriers for women to be better able to compete with men and progress financially.

¹²Interviews with Facility lead, MDs and CEOs, informed that the brands are conscious of their social impact and influence manufacturers to adhere to their social and gender policies and standards.

¹³ The brands signed the ACCORD agreement and its obligatory for their manufacturing units to comply with the prescribed standards and agree for sudden site inspections/investigations.

3.3. Institutional arrangements (Legal and administrative)

Women holding the seats of Prime Minister, speaker of parliament, opposition leader and Foreign Minister, Bangladeshi women have experienced a great deal of political empowerment. The empowering policy, laws and institutions have been discussed below.

3.3.1. Policy

Bangladesh's National Women Development Policy of 2011¹⁴ mandates to incorporate gender perspective in matter of innovation, import and application of new technology. Further, to take initiatives to purge the harmful elements, if any, from the technology by conducting research on it if the application of such innovated technology affects adversely the interest of the women and make necessary legislations, in favour of the women in the field of technology.

Bangladesh Climate Change Strategy and Action Plan (BCCSAP), 2009¹⁵ is a 10-year program (2009 – 2018) of the country to meet the challenge of climate change. BCCSAP addresses the needs of the poor and vulnerable, including women and children to be mainstreamed in all the activities under the Action Plan.

Bangladesh Climate Change and Gender Action Plan (ccGAP), 2013 by Ministry of Environment and Forest discusses the climate vulnerabilities and gender action plan for four priority areas namely food security social protection and health; comprehensive disaster management plan; infrastructure; mitigation and low carbon development.

The **National Action Plan**¹⁶ to prevent violence against women and children (2013-2025) encompasses all strategic and practical issues involved in the four pillars of the Women, Peace and Security Agenda: Participation, Prevention, Protection, and Relief and Recovery.

3.3.2. Laws

The government has undertaken various programs to materialize Vision 2021¹⁷ for establishing rights of women, their empowerment and mainstreaming them in the overall development. Amendments have been made to the existing laws and new legislations have been made for protection of women and children.

- **Mobile Court Act, 2009:** The executive magistrate was given power to take steps by linking Section 509 of the Bangladesh Penal Code in the schedule of Mobile Court Act. This would resist and prevent eve teasing and sexual harassment of the girls and women.
- **Citizenship Act (amended), 2009:** The provision for giving citizenship by mother to child was made by the national parliament by amending the citizenship act in 2009.
- **Dowry Prohibition Act, 1980:** Prevention of financial obligation to provide gifts and assets
- **Women and Children Repression Prevention Act, 2000:** To ensure the safety and protection to women at homes and their surroundings.

¹⁴https://mowca.portal.gov.bd/sites/default/files/files/mowca.portal.gov.bd/policies/64238d39_0ecd_4a56_b00c_b834cc54f88d/National-Women-Policy-2011English.pdf

¹⁵ https://www.iucn.org/downloads/bangladesh_climate_change_strategy_and_action_plan_2009.pdf

¹⁶https://mowca.portal.gov.bd/sites/default/files/files/mowca.portal.gov.bd/page/bcf75e01_95e3_48ba_bfe4_3d88ea5f593c/Bangla-National-Action-Plan-2013-2025.pdf

¹⁷ The Government of Bangladesh has declared 'Vision 2021' with a target to make Bangladesh as a middle-income country using Information and Communication Technology (ICT) and development of favourable business environment for innovative companies. (http://bhtpa.portal.gov.bd/sites/default/files/files/bhtpa.portal.gov.bd/page/8f8ac427_fca2_4875_bf9a_65051666d011/Mission%20&%20Visson.pdf)

- **Domestic Violence (Prevention and Protection) Act, 2010:** Prevention of domestic violence, protection of women and children from domestic violence
- **Prevention and Suppression of Human Trafficking Act, 2012:** To prevent trafficking for slavery, sexual interests and bonded labour.
- **High Court (2009) ruled to include mother's name in all official documents,** school admission forms, either alone or with that of the father. Consequently, climate change and its impacts are enhancing vulnerabilities for women due to the lack of climate adaptive capacities.
- **The Bangladesh Labour Act, 2006 amended in 2013:** promotes equality of opportunity in employment and provides for equal pay amongst men and women.
- **Bangladesh's Labour Rule, 2015:** Mandate for strict rules for misconduct and punishments, provident fund, health and fire safety, wages, holiday and bonuses. Especially for women, there is a relaxation as the rules prohibits extreme jobs like lifting, carrying or discharging weights with hands or head outside the permitted quantity and if pregnant, any of these is completely prohibited; Inclusion in the apprenticeship program; provision of separate toilets for women; provision of child care or Crèches and consent form for women working in night shifts.

The Committee on the Elimination of Discrimination Against Women, 2011 in the Forty-eighth session indicated that the persistence of discrimination against women in the labour market, occupational segregation, has a wide gender wage gap and the exploitation of girls is also prevalent.¹⁸

3.3.3. Institutions

The country has significantly progressed by making and adopting integrated gender sensitive national policies and institutional arrangements. Since gender is a cross cutting issue, there are a number of institutions that are responsible for the recent development in women's status. Especially, the initiatives undertaken by the government institutions like MOWCA¹⁹ which is mandated by the government to act as a focal point and a catalyst in action for women equality and development through different government agencies.

Table 4: Institutional framework relevant to the programme

Organisation	Role
1. Ministry of Women and Child Affairs	Overall development of women, empowering the women and preserving their rights and socio-economic development of women by ➤ Awareness creation
2. Department of Women Welfare (MOWCA)	➤ Establishing the social, political and legal rights of women ➤ Creating equal opportunities for women ➤ Providing technical and vocational training
3. Jatiya Mohila Sangstha (MOWCA)	➤ Creating bridges for the interest of women with relevant government & non-government organizations at Home and abroad ➤ Initiating involvement of women in the national development activities, arranges conferences, seminars and workshops regarding the interest of women
4. Parliamentary Standing Committee (MOWCA)	➤ A Policy Leadership and Advocacy Unit (PLAU) provides technical support to MOWCA in coordinating women's advancement activities within the Government

¹⁸ BANGLADESH CLIMATE CHANGE AND GENDER ACTION PLAN
(http://cmsdata.iucn.org/downloads/iucn__bangladesh_climate_change__gender_action_plan__1.pdf)

¹⁹ Ministry of Women and Child Affairs, Bangladesh (<https://mowca.gov.bd/>)

Annex VIII – Gender Assessment Report

GREEN CLIMATE FUND FUNDING PROPOSAL

5.	The National Council for Women and Children Development (NCWCD)	Chaired by the Honourable Prime Minister to provide guidance through laws and rules for protection of women and children and monitor the implementation of critical policy decisions
6.	Women In Development Focal Point (WIDFP)	Focal points Mechanism established in 1990 has a designated focal point from each ministry to ensure women concerns are integrated in policy, plans and programmes, District and Thana level WID coordination committees. The WIDFPs of all ministries lead to address gender concerns within the policies and plans of their respective ministries through its three working legs ²⁰ : ➤ WIDFP Implementation and Evaluation Committee (WIDFPIEC) ➤ Women In Development Focal Point Coordination Committee (WIDFPCC) ➤ Women In Development Focal Point Network Committee (WIDFPNC)
7.	District and Upazilla Levels	District Women Affairs Officer and Upazilla Women Affairs Officer in coordination with other concerned officials shall discharge their responsibilities in implementation of all the programs undertaken for the development of women. -One Stop Crisis Centre and Cells -National Trauma Counselling centre
8.	Grassroots Level	In village and Unions, the women have been organized as self-sustained groups. To strengthen these groups, they are transformed into a registered organization under different government organizations.
9.	NGOs and Social Organizations	National Schemes and programs are adopted and implemented with assistance from these institutions to change social attitude towards women and expedite the process of women empowerment.
10.	International Co-operation	International, regional and sub-regional co-operation is encouraged through bilateral and multilateral financial and technical co-operation and exchanges of experience and technology aimed at empowerment of women ²¹ .
11.	Helpline Centres	National Helpline Centres for violence against women and children Helpline No. 10921

Ministry of Finance in 2003, established a monitoring and reporting facility named Recurrent, Capital, Gender, and Poverty model which assess the percentage of allocated funds that contribute to poverty reduction and gender equity. Women Abuse Prevention Cell and rehabilitation centres for abused women have been established to give legal assistance and counselling for prevention of women and children abuse.

Most recently, Bangladesh's women launched a resistance movement to combat SGBV²² after a series of brutal attacks in October 2016. Since the turn of the century, Bangladesh has weathered many conflicts with neighbouring states, India and Myanmar, as well as internal conflicts. In addition to these conflicts, women still face domestic violence and acid attacks.

²⁰ Report of the National Human Rights Commission, Bangladesh
(http://nhrc.portal.gov.bd/sites/default/files/files/nhrc.portal.gov.bd/page/348ec5eb_22f8_4754_bb62_6a0d15ba1513/NHRC%20Report%20on%20CEDAW%20Sate%20Report.pdf)

²¹ National Women Development Policy 2011

²² Sexual and Gender Based Violence

3.3.4. *Accords and Conventions*

Bangladesh has been actively participating in many international human rights conventions, treaties, agreements and instruments and endorsed the Beijing Platform for Action (BPFA). It also adopted the United Nations Millennium Declaration (2000) and committed to the goals and targets known as the Millennium Development Goals (MDGs). Few of the key relevant commitments are listed below.

Chittagong Hill Tracts Peace Accord

The struggle for promoting the role of women in conflict resolution was a challenge especially in case of the Chittagong Hill Tribes region in Bangladesh where many women were widowed during the conflicts. Through the peace accord, the regional council gave the opportunity to women to access decision-making bodies. The Chittagong Hill Tracts Peace Accord²³ also known as Chittagong Hill Tracts Treaty, 1997 mandated to have 3 (three) seats for women in every Hill District Council and one-third of the said seat for non-tribal women.

Convention on the Elimination of All Forms of Discrimination against Women (CEDAW)

Bangladesh ratified the Convention on the Elimination of All Forms of Discrimination against Women in 1984 and has submitted seven progress reports on its implementation and participated in meetings of the Committee on the Elimination of All Forms of Discrimination against Women.

ACCORD

The Accord on Fire and Building Safety in Bangladesh²⁴ helps ensure that no worker needs to fear fire, building collapses or other accidents that can be prevented with reasonable health and safety measures. This agreement was signed in the immediate aftermath to the Rana Plaza building collapse on 24 April 2013, which killed 1,133 workers and critically injured thousands more. The Accord protects workers' rights to:

- Refuse unsafe work
- Participate in the work of their factory Safety Committee
- File a complaint when they see a safety problem in their factory
- Protection against reprisal for reporting safety-related matters
- Freedom of Association in relation to protecting their own safety

4. *Situational analysis: understanding vulnerabilities through Gender Assessment*

The situational analysis is based on the real time data collected on field through survey interviews with the managers, workers and other stakeholders.

4.1. *Socio-economic assessment*

The socio-economic assessment will determine the way the proposed infrastructure development programme will affect or change the lives of current or future workforce in textile sector.

4.1.1. *Average age of women in working in textile units*

Approximately, 80 percent²⁵ women workers in textile fall under the age group of 18 to 45 years of age. Most of them tend to leave this sector during mid-40s or early 50s.

²³ https://peaceaccords.nd.edu/sites/default/files/accords/Chittagong_Hill_Tracts_Peace_Accord.pdf

²⁴ <https://bangladeshaccord.org/>

²⁵ Of the total women respondents

Annex VIII – Gender Assessment Report

GREEN CLIMATE FUND FUNDING PROPOSAL

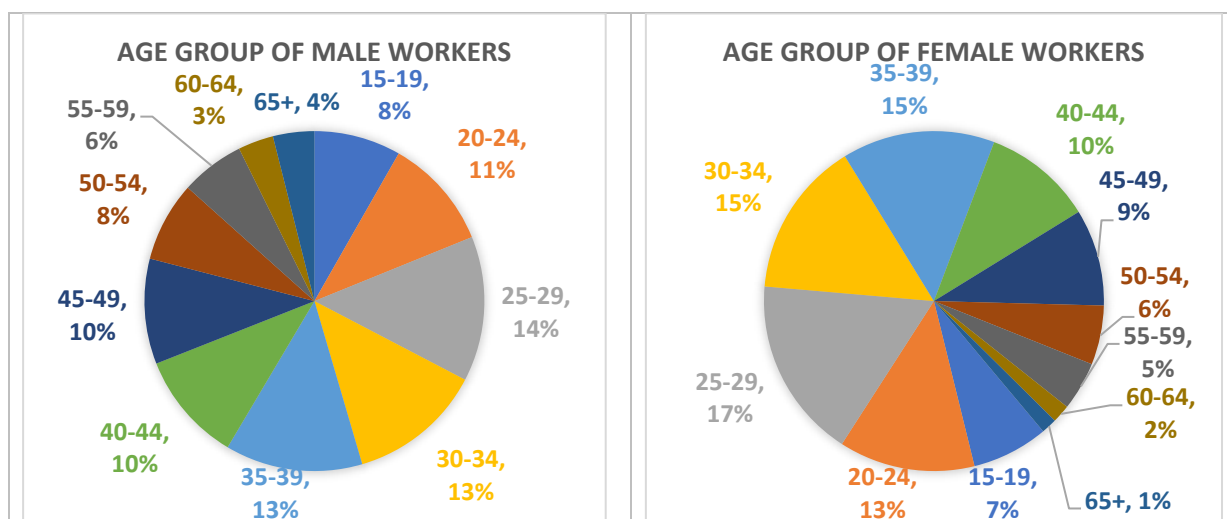


Figure 2: Percentage of Textile workers belonging to various age groups

4.1.2. Marital status and average household size (adults and children)

As per the field survey, above 60% of the men and women both in textile labour is married and has dependents. Forty five percent survey respondents have an average HH size above 4 members and the rest of the workers have less than 4 members in the households.

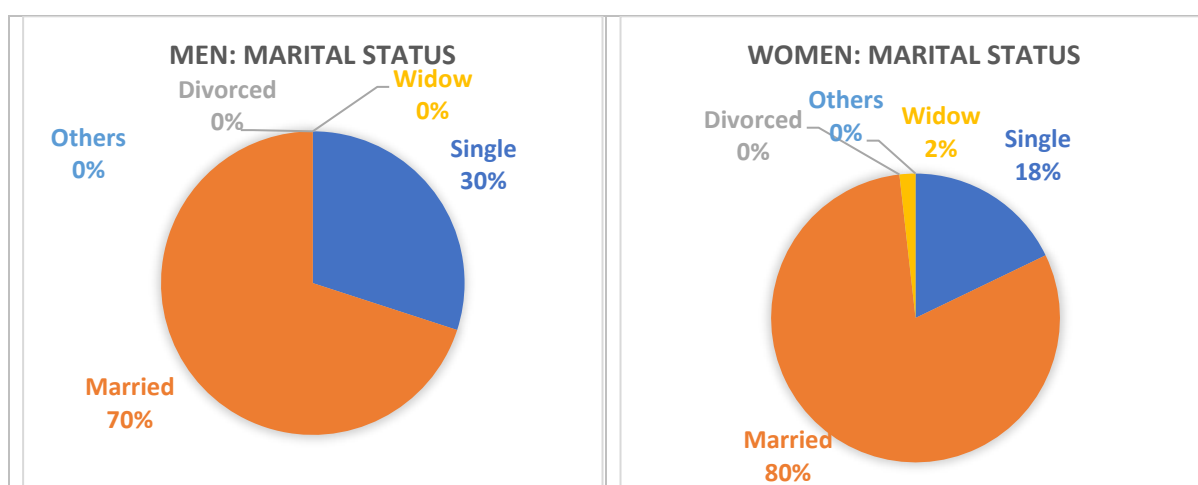


Figure 3: Marital status of men and women

4.1.3. Education Status

Women and men workers in textile units hold at least primary education and the highest being HSC as observed in the case of men. For women, Primary education is held by the most and very few have any education above SSC level (see figure 4).

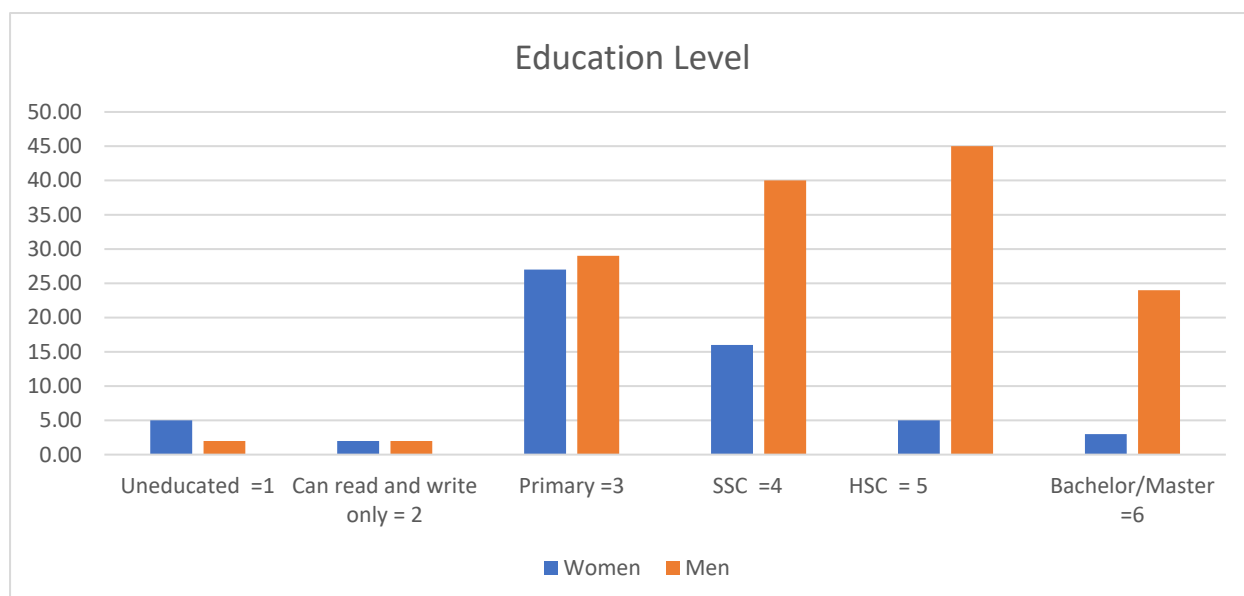


Figure 4: Gender wise education status of textile worker group

4.1.4. Skills and trainings (acquired personally and with support from employer)

Approximately 60% of the workforce admits that they are given training and orientation before they are given responsibility for a task or operations on a machine. Women are given training for skills and capacity development on the job as women rarely pursue vocational trainings. While below 10% men have undertaken higher/vocational trainings the others undertook trainings at work.

Table 5: Responses of the textile workers on trainings and skill development

Trainings undertaken by all workers		Women	Men
Vocational	5%	0	6%
On the job	87%	100%	83%
No training	8%	0	11%

4.1.5. Gender ratio in textile workforce

Survey reveals that the textile labour comprises of below 30% of women labour. The reasons for this disparity have already been discussed in the previous sections.

4.1.6. Average work hours

The survey suggests, there are three, 8 hours shift in medium and large textile units and small textile units usually have two, 12 hours shift. Women mostly serve 8 hours shift during day hours.

As per the Labour Act, No adult worker shall ordinarily work or be required to work in an establishment for more than 48 (forty-eight) hours in a week, however, any deviation may happen only as per the clauses mentioned.

4.1.7. Sources of income, average income and expenditure

Besides their main income source i.e. textile unit, workers that come from farming background make money through the agricultural produce, local shops in the area, etc. The analysis from the survey data speaks volumes about the disparities in income and wages for men and women (see figure 5).

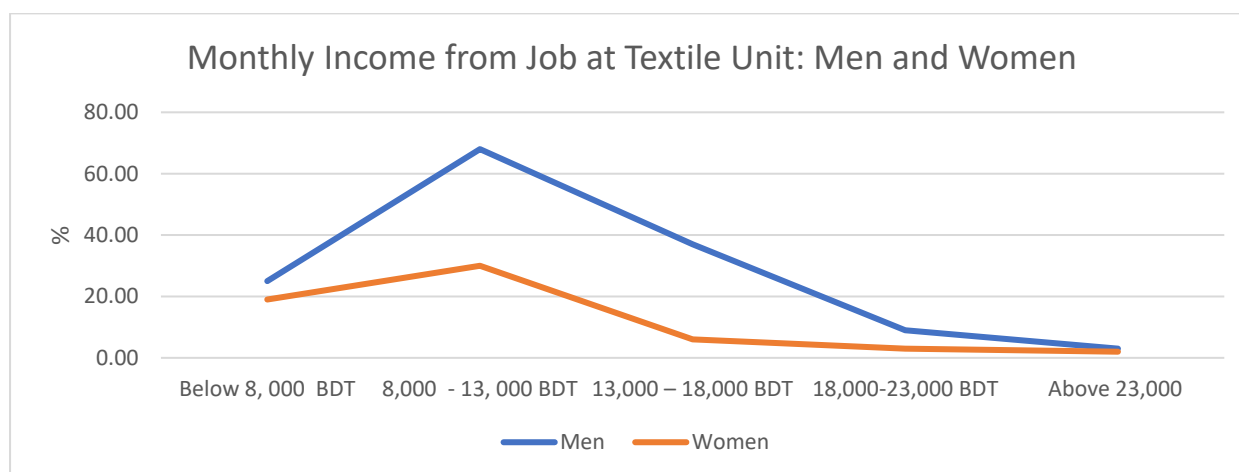


Figure 5: Gender wise break-up of the monthly income of textile worker group

The above graph reveals that women worker are paid around 8 to 10% less on average than their male compatriots. Lack of skills/technical knowledge and experience, attrition due to unplanned pregnancy, selective availability for single day shift, unwillingness to stretch for more hours at work are the most frequently listed reason, as per the field survey. Government of Bangladesh through the Labour Rules, Labour Law and Women Development Policy have initiated change, but the disparities continue to exist in the system.

4.1.8. Female led households, house ownership and tenancy

Female led house-holds are those where women provide for the basic needs of the family like buying food, clothes, house rents. Apparently, above 30% of the respondent (both men and women) stated that women provide for the basic needs of the family which is close to the national statistics²⁶. Supplemented saying men invest their income as savings, house maintenance and other agricultural farms. Most survey respondents reported that they are living in a rented accommodation away from their village. The men and women respondents gave an average estimate of their monthly rents which varies from 2,500 BDT to 4,500 BDT a month depending on their requirement (no. of family members and assets).

²⁶ Bureau of Statistics, Ministry of Planning, Bangladesh
http://bbs.portal.gov.bd/sites/default/files/files/bbs.portal.gov.bd/page/a1d32f13_8553_44f1_92e6_8ff80a4f82e/Bangladesh%20Statistics-2017.pdf

4.1.9. Savings and loans

According to the Global Findex Database 2017²⁷, Bangladesh has approx. 30% gender gap in account ownership. Above 90% women and men textile workers do not have a savings and resort to local lending in times of needs. The workers that do have savings, have it either in banks or local cooperatives. The ease of savings mobilization, availability of savings instruments and the presence and intensity of social safety nets determine the savings behaviour.²⁸ As per the responses, 80% men and 75% women do not have loans and those who do, have personal loans. Loans for house, vehicle and education hasn't been reported.

4.1.10. Provident fund/life Insurance

All the respondents stated that their employers do not give or facilitate provident fund opportunity. The respondents, either men or women do not have any kind of insurance either personal or provided by the employer. Recently, the labour rules have mandated the provision of Provident Fund by the employer and laid out clauses for the same and also for the obligatory group insurance policy by the employer.

4.2. Health and Safety assessment

4.2.1. Health issues faced by men and women in textile units

It is known that the textile units have a low air quality index and a high suspended particulate matter which is suffocating and in the long-term lead to bronchial diseases. Besides, there's heat and chemicals in the work premises that the employees work with.²⁹ As reported by the workers, there are frequent instances of burning from the heated machinery and effluents and chemical reactions by the dyes and colours.

4.2.2. Health camps and medical facilities (like dispensary) and health insurance

Men and women in large industries reported that their employers conduct health camps in association with government and private hospitals. Similarly, it was reported that there are 24*7 functional dispensaries in the large units that are looked after by government certified doctors. While the workers in small and medium textile units reported that there are no health camps or in-house medical facilities made available by their employer and they are directed to local medical practitioner in the vicinity.

In case of accidents and injuries at workplace in larger units, if the damage cannot be handled by a resident doctor, the victims are rushed to the nearby medical facility and the finances are covered by the employer.

The respondents shared there are no health insurance being provided by the employers.

4.2.3. Maternity leaves

Respondents shared that there is a provision of maternity leaves for 2 to 4 months and the Government also has legalised the maternity leave for 6 months and the female workers can also opt for maternity benefit during their absence. However, there isn't provision for paternity leaves.

4.2.4. Violence and abuse at work

Men and women responded of not having been abused at work in large and medium units, however, in smaller units a few instances of verbal abuse have been reported. Most of the instances of verbal

²⁷ World Bank Global Findex 2017 (<https://globalfindex.worldbank.org/>)

²⁸ Savings and Income Inequality in Bangladesh
(<https://www.bb.org.bd/pub/research/workingpaper/wp1617.pdf>)

²⁹ World Bank EHS Guideline for textile sector

abuse have been reported due to work related issues and incompetency to work on complex machine systems or carry out extreme tasks like carry, shift, load, the material.

The Labour Rules 2015 has prescribed a limited load quantity to be carried by both men and women and anything over and above that is an offence as per the rules.

4.2.5. Injuries and Accidents

Of the total respondents, 25% of women and men did mention of having experienced an injury at work while 15 % women and 25% men shared their experiences of having witnessed accidents and injuries due to fire or other occupational hazards like heated boilers, hot dyeing machines and hot dyed material taken off the dyeing machines for cooling and finishing, chemical dyes, generators and electric equipment, Switch boards while power on/off and much more. The cost of damage is mostly covered by the employer if the accident/injury takes place at workplace. The Labour Rules has set rules in favour of the workers where they/their dependents can seek money from the employer through the Labour Court. Also, The owner shall place Material Safety Data Sheet (MSDS) of dangerous materials in an easily noticeable place so that the employed worker can be well informed about the possible hazards.

4.2.6. Safety precautions, trainings and tools

Typical risks in a textile factory have previously been discussed. To prevent accidents and injuries and reduce the intensity of disaster, there are specific guidelines that needs to be followed while operating on machines, doing manual labour, dealing with electricity or heat.

There needs to be protection for respiratory health, eyes, fingers and limbs, for which the employer must mandatorily provide with necessary safety attire like safety shoes, helmets, goggles, masks, hand gloves, ear muffs, ear plugs, waist belts, aprons etc. and arrange training programs for the workers in using these materials and ensure their usage. Also, Employment of children, teenage laborers in the activities that are declared as dangerous from time to time by the Inspector General or his representative is prohibited³⁰.

. The following issues were discussed with the respondents and their responses were recorded.

Table 6: Responses of textile labour on respective issues

Specific issues	Yes: Male (% of total)	Yes: Female (% of total)	No (% of total)	Don't Know (% of total)
Trainings on health safety and security issues	47	34	10	9
Trainings on fire safety and rescue	25	19	33	23
Trainings on handling of heavy machines and chemicals	29	22	43	6
Awareness of the safety rules and policies in the work place	59	37	3	1

For safe machinery operation, refer to the Labour Rules 2015.

4.2.7. Compensation to victims

All women respondents and 80% men respondents agreed that their employer does provide compensation and medical support in times of loss that has happened during the work hours, at the

³⁰ Bangladesh Labour Rules 2015

workplace. Also, Labour Rule states 'All Owners, to whom Section 99, applies shall initiate group insurance policy for each worker.' Hence, under damage/death, relevant compensation would be provided.

4.3. Sanitation and access to basic facilities

4.3.1. *Access to drinking water, washrooms and sanitation facilities (soaps, sanitary napkins, etc)*

As per the site visits, the drinking water available for the workers are from the government's water pipeline being provided for the locality. The size of the textile units determines if the water is further processed by in-house water filters. Out of 20 units that were visited 17 units had water purification facility.

Similarly, hygiene and washroom facilities are dependent on the size of the textile units. The larger the set-up and workforce, the better facilities are provided. As observed, separate washrooms are provided if the facilities are medium to large scale and have a bigger workforce.

However, Labour Rules state 'there will be provision of pure drinking water in easily accessible and suitable place for all workers in each firm and the water should be stored in a hygienic way' and lays down rules for Toilets and washrooms- the number, location and cleanliness of the toilets and wash room should be in accordance and in case of female workers, one toilet for each 25 females up to the first 100 and one for each 50 next.

5. Qualitative Gap Analysis

Background research and field survey show similar picture of the status of the gender issues in the sector. Since the gap analysis is based on a comparative analysis of both genders, so, gender balance was essential in analysing gaps in this exercise. The respondents from both genders were interviewed from the worker and manager groups in textile sector to discuss about the differential needs and inequalities (accessibility/provision of services, opportunities, etc).

Table 7: Gender biases in textile sector in Bangladesh

Basic Rights (determining factor)	IFC Guidelines ³¹	National Guidelines/initiatives	Disparities/Inequalities as observed	
			Men	Women
Education	Education programs or improved infrastructure to facilitate increased access to education	Emphasis will be given on women's education to ensure women's comprehensive development and empowerment and women's participation in a balanced social advancement ³²	Male members of the society are more highly educated across all age groups.	Females have been facilitated by the different programmes and initiatives of the government and that has led to a change in trends and more educated young women workers. On field, the male workforce ranges from uneducated to higher degrees while women mostly belonged to the Primary to HSC range. This indicates that women workers usually have the basic skill set and the men have higher degrees. Besides, there are job opportunities for uneducated men as well and women have to fulfil a certain criterion to get onboard with the textile firms.
Skill training	Provide training opportunities to allow for more equal access	The arrangement of own training facilities on	Male worker group is observed to be skilled and better adapted to	Women have a low presence in textile labour force and they aren't

³¹ IFC Gender Toolkit: Unlocking Opportunities For Women And Business

³² Bangladesh's National Education Policy, 2010

	to alternative livelihood and employment opportunities	necessary efficiency for the positions for which the workers are supplied or agreement with any such approved training providing organization which can provide training.³³ The National Skills Development Policy (NSDP) 2011 emphasized on quality vocational and technical education for women considering market needs.³⁴	working on technologically complicated machinery.	usually trained and delegated to operate on machinery.
Jobs	The client will not make employment decisions on the basis of personal characteristics based on as gender, race, nationality, ethnic, social and indigenous origin, religion or belief, disability, age, or sexual orientation³⁵ The client will identify migrant workers and ensure that they are engaged on substantially equivalent terms and	The Bangladesh Labour Act, 2013 provides for equal pay for equal work, women friendly working environment, maternity leave, and redress against harassment at the workplace. To undertake all out initiative for employment of the educated and illiterate women labour.³⁶	More opportunities for men due to the nature of work and technological competence. No such evidence has been reported where migrant workers have experienced discrimination.	Less job opportunities due to lack of skills and labour intensive operations. Women often experience discrimination due to physical and technological incompetence, lack of relevant skills, uncertain maternity breaks, etc.

³³ Bangladesh Labour Rues 2015

³⁴ CEDAW: Eighth periodic report of States parties due in 2015

³⁵ IFC Performance Standard 2: Labour

³⁶ National Women Development Policy 2011

	conditions to non-migrant workers carrying out similar work			
Wages	Identify and close gender-based disparities in compensation including salary, benefits, and bonuses—awarded to men and women who hold equivalent jobs in the company.	In determining wages or fixing the minimum rate of wages for any worker, the principle of equal wages for male, female and handicapped workers for work of equal nature or standard or value shall be followed; and no discrimination shall be made in this respect on the ground of being male, female or handicapped. ³⁷	Competitive wages due to skilfulness and great competence gained over processes and equipment.	Despite government's initiatives to reduce wage gaps, women experience low wages due to negligible skills and low competence over equipment handling as compared to men.
Savings and loans	Employer may provide financial support, including no/low interest loans, salary advances, or support paying medical bills to women in need and experiencing Intimate Partner Violence	The banking system provides loan facilities to all, including women for housing, business, and personal purpose. The collateral requirements for higher amount of loan often make women ineligible for lending as they lack assets to mortgage. ³⁸	Given that textile labour has reported zero savings, it's only a few men who have savings either in banks or cooperatives. Very few men have reported loans and that only for personal use.	Women have informal saving pockets. Most women respondents have denied having a savings account in a bank.
House and Asset Ownership	Provide assistance to support equal access to Land titles: Divorcees and widows are in particular need of assistance	Limitation of resources is there in attaining the desired target of women development.	The house ownership is mostly in the name of the male member of the family.	Most women have been observed to have comparatively lower

³⁷ Bangladesh's Labour Act, 2013

³⁸ CEDAW: Eighth periodic report of States parties due in 2015

	here. They may face cultural, political, or legal difficulties in accessing their right to their land. Also note that in some contexts, women who hold the formal title to land may still face cultural, procedural, or customary law barriers to exercising this right. Support for women's land ownership and land titling. ³⁹	To give women the rights to wealth and resources earned through income, succession, loan/credit, land and market management. To include women perspective in the housing plan and shelter arrangement in the countryside and the urban areas ⁴⁰		ownership over land, house and other assets. ⁴¹
Dependents	Make sure that female-headed households receive the same benefits as male-headed house-holds. Make sure to include single, pregnant, elderly, female-headed households, and any other vulnerable groups in these consultations. Create a more inclusive, gender-supportive worksite culture and Support flexible work arrangements. ⁴²	Out of 40% population of Bangladesh living under the poverty line, two-thirds of them are women and female headed families are greater in number among them.	Traditionally, men have provided for the basic needs of the family like food and shelter.	Of late, the respondents have reported a change in ideologies and role play. With more women earning, the dependence of family members has been shared by them with their partners.

³⁹ IFC Gender Toolkit: Unlocking Opportunities For Women And Business

⁴⁰ National Women Development Policy, 2011

⁴¹ Khan, Issa et al. "The right of women in property sharing in Bangladesh: Can the Islamic inheritance system eliminate discrimination." SpringerPlus vol. 5, 1695. 3 Oct. 2016, doi:10.1186/s40064-016-3347-2 (Available at: <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5047860/>)

⁴² IFC Gender Toolkit: Unlocking Opportunities For Women And Business

Health	Assessment of health risks and provision of health care centres	The National Industrial Health and Safety Council has drafted an Occupational Safety and Health Policy	Allergies and Lung issues are common to both men and women.	Women experience unfavourable conditions during their menstruation, pregnancy, etc. Low air quality index and a high suspended particulate matter is suffocating and in the long-term lead to bronchial diseases ⁴³
Safety	The client will provide a safe and healthy work environment, taking into account inherent risks in its particular sector and specific classes of hazards in the client's work areas, including physical, chemical, biological, and radiological hazards, and specific threats to women. ⁴⁴	Every worker shall be made aware of the hazards of work through training in order to ensure the protection and safety of his professional health in the place of work. ⁴⁵	Men are more vulnerable to accidents during operations and disasters	Women are equally vulnerable to safety risks and hazards in the workstation. The major risks from handling of heat and chemicals can be reduced only upon trainings on safety management. However, survey reveals, women have had less opportunities for trainings on safety and health precautions.
Violence and Abuse	The client will take measures to prevent and address harassment, intimidation, and/or exploitation, especially in regard to women.	MOWCA has formulated National Action Plan to prevent Violence Against Women and Children 2013-2025 to protect and ensure women's human rights. ⁴⁶ The Multi-Sectoral Program on Violence Against Women	Men have not reported any cases of violence and abuse.	Women have reported verbal abuse in smaller textile units.

⁴³ World Bank EHS Guideline for textile sector

⁴⁴ IFC Performance standards 2

⁴⁵ Bangladesh's Labour Act, 2013

⁴⁶ CEDAW: Eighth periodic report of States parties due in 2015

		(MSP-VAW) provides health care, police assistance, DNA test, social services, legal assistance, psychological counselling and shelter for the woman victims through 8 One-stop Crisis Centres (OCCs).		
Sanitation facilities (Clean washroom, soap, etc)	Guarantee equal or improved access to civic infrastructure, including water, sanitation (facilities such as toilets and bathing facilities) and fuel resources.	In case of female workers, separate bathrooms must be arranged for them. Additionally, washing facilities must be covered in such a way so that the male workers cannot see the inside area of the bathrooms while working or moving. ⁴⁷	Men have provided neutral responses towards provision of sanitation facilities.	Women reported to have experienced absence of women washrooms in smaller units and bad sanitation experienced due to sharing toilets with men. Also, women require and expect to have basic facilities like soap, sanitary napkins from their employer which in most cases isn't taken care of.
Planning, management and decision-making	Enable men's and women's voices to be heard in decision making and accountability mechanisms	Women's participation in the National Parliament and local government institutions has been ensured by reserving quota, in addition to their eligibility for the regular seats. Women are visible on boards of private corporations, in senior management, in	Male dominated sector from planning and management to execution.	Women are present at the planning and management, however, their engagement in decision-making is still unsure. Women at execution level also are very less.

⁴⁷ Labour Rules, 2015

		diplomatic missions and in armed forces. There are more women in the departments and directorates engaged in service delivery, such as teachers, health and family planning assistants, nurses, etc. There are 23 per cent of women in the judiciary.		
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Evidently, presence of women in the sector is very less. Only a few women have been reported to be leading the textile units as Managing Directors and CEOs. There is a dire need to address inequalities being experienced by the respective genders in different cases. The development programmes in textile sector in Bangladesh must address gender issues for growth and social welfare.

6. Bridging Gaps: gender sensitivity and applicability of the programme

As per the GCF's Gender Policy, the following must be considered for the Gender Action Plan intended to

6.1. Inclusivity in stakeholder approach (planning, management and implementation)

Infrastructure development programmes with the GCF have a mandate to have a gender sensitive approach and gender inclusion in all activities including stakeholder engagement and consultations to operationalise fair and balanced programme actions. Since, there has been a proactive engagement of the stakeholders both men and women in planning and designing (consultations and surveys) of the programme, the existing issues in textile sector could surface.

6.1.1. First stakeholder workshop

A gender balanced scoping workshop on '**Importance of energy efficiency in textile sector and its influence on environment, society and gender**' was conducted at Dhaka on 29th May 2019 to engage with stakeholders from ministries, associations, industry and civil society organisations and discuss issues related to gender parity in the sector. Respective stakeholders could communicate their concerns about the status and the impact of programme through various phases (planning, designing, and implementation). The discussions on average male female ratio, roles of gender in management and decision-making, the trends in perceptions and attitudes and enabling change for a balanced approached, paved way for an integrated programme that's socially benefitting to the vulnerable group.

6.1.2. Second stakeholder workshop

A validation cum consultation workshop for dissemination of the findings of the gender assessment done through various stakeholder engagement methods (field survey, worker interviews, manager interviews, policy makers interviews, etc) was held titled '**Impacts of Energy Efficiency on Environment, Social and Gender in Textile Sector of Bangladesh**' on 29th July 2019 in Dhaka. Most of the distinguished stakeholders from ministries, associations, trade unions, industries, academicians and development professionals participated in the workshop to state views and comment on the observations and conclusions of the exercise.

6.2. Monitoring interventions and Evaluation

Gender sensitive vision and objectives has led to specific log-frame indicators to embed gender inclusion in programme design. The M&E log frame will measure the performance of the gender-based activities. The action plan will present:

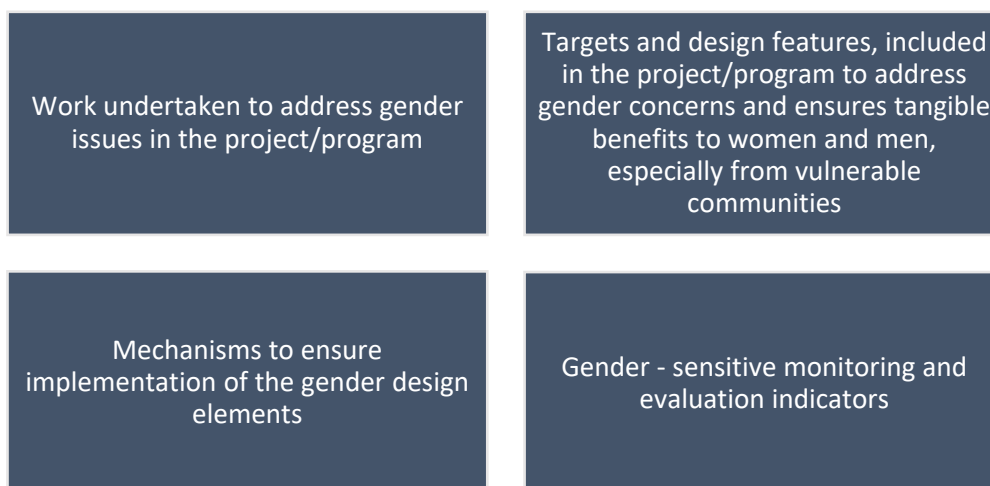


Figure 6: Elements of the Gender Action Plan

Specialised monitoring framework to measure outputs and track progress of gender specific elements while evaluation framework to measure impacts of the gender specific activities in addressing gender biases. The monitoring and evaluation will feed a detailed gender action plan which will lay base for a gender-sensitive programme/programme.

Gender Action Plan

Activities	Indicators and Targets	Timeline	Responsibilities	Budget
Impact Statement: Enhanced livelihood opportunities and reduced inequalities at work through training and skill development to improve the productivity and competitiveness for both men and women from the implementation of energy efficient technologies and equipment in textile sector.				
Outcome Statement: Increased access of the textile sector to concessional financing for energy efficiency, leading to energy savings through large scale adoption of energy efficient technologies to reduce GHG emissions.				
Component 1: Concessional loans for energy efficiency in textile sector				
Output 1.1: Women entrepreneur successfully engaged in the program				
Activity 1.1.1: Encouragement of women entrepreneur to venture in the textile sector	Indicator Encourage loan application from textile units with significant participation of (Chairman/Any Director/MD/CEO) women entrepreneurs Target To approach minimum 05 (five) woman led textile business entities under the programme	Minimum 01 woman led garment business entity will be approached in each year.	Respective Business Units of IDCOL	Funding Source: IDCOL Amount: XXXXX Within the “Project Management Cost” from IDCOL in line of regular business process
Output 1.2: Investment to comply with national standards for women welfare and recover costs from the energy savings from programme/project implementation				
Activity 1.2.1: Reducing the gender wage gap (1) Unbiased employment as per the “Labour rules (2015)” (2) Encourage similar wages for men and women for similar work type as per the “National Wage Board”	Indicator Encouraging women participation by providing unbiased employment opportunities and providing similar wages for same jobs to men and women Target Submission of existing pay scale ensuring similar wages for men & women or proposed pay scale with a time bound implementation plan ensuring such similar wages	Number of such submission will depend on the number of end borrowers approaching IDCOL for their financing. There is no yearly target of how many end borrowers will	End borrowers	Funding Source: End borrowers Amount: XXXXX The cost of any such implementation will be borne by end borrowers & this will not be

Annex VIII – Gender Action Plan

GREEN CLIMATE FUND FUNDING PROPOSAL

		be financed rather specific amount for each year & a total of entities have been mentioned in the log frame. So IDCOL will ensure submission of existing pay scale ensuring similar wages for men & women or proposed pay scale with a time bound implementation plan ensuring such similar wages in line with their actual financing in each year.		part of GCF project cost
Activity 1.2.2: Facilitate women participation in the sector through provision of: (1) Medical Facilities (2) Child care (3) Sanitation facilities	Indicator Encouraging women participation by facilitating them with medical assistance, child care and sanitation as prescribed under Labour Rule 2015 & other regulatory compliances Target Share existing status of the mentioned parameters during loan application and ensure compliance with the national standards (if not already complied) within 12 months of the loan disbursement.	Number of such sharing will depend on the number of end borrowers approaching IDCOL for their financing. There is no yearly target of how many end borrowers will be financed rather specific amount for each year & a total of entities have been	End borrowers	Funding Source: End borrowers Amount: XXXXX The cost of any such implementation will be borne by end borrowers & this will not be part of GCF project cost

		mentioned in the log frame. So IDCOL will ensure sharing of existing status of the mentioned parameters during loan application and ensure compliance with the national standards (if not already complied) within 12 months of the loan disbursement in line with their actual financing in each year within 05 years of GCF loan availability period.		
Component 2: Large scale adoption of Energy Efficient technologies and equipment				
Output 2.1: Energy Efficient technology upgrades in the textile units employing new programmed/automated processes				
Activity 2.1.1: Training and skill development for machine use and operating on programming software (1) Development of training modules (2) Periodic trainings and evaluation	Indicator Ensure participation in skill development trainings leading to women empowerment by providing skill development through the technical assistance workshops Target Ensure women employee participation a minimum of 25% in skill development and training workshop	In Year 3 within 05 years of GCF loan availability period	IDCOL will ensure at least 01 training for their respective end borrowers & ensure targeted women participation	Funding Source: End borrowers & IDCOL Amount: XXXXX The cost of any such implementation will be borne by end borrowers & this will not be

				part of GCF project cost. The cost of IDCOL will be within the “Project Management Cost” from IDCOL in line of regular business process
Activity 2.1.2: Safety trainings while operating on new machinery/equipment (1) Development of training modules (2) Periodic trainings and evaluation	Indicator Reduction in accidents and safety issues rising from new certified machinery for which due trainings have been provided to the workers including women. Target Ensure women employee participation a minimum 25% in the safety training for workers	In Year 4 within 05 years of GCF loan availability period	IDCOL will ensure at least 01 training for their respective end borrowers & ensure targeted women participation	Funding Source: End borrowers & IDCOL Amount: XXXXX The cost of any such implementation will be borne by end borrowers & this will not be part of GCF project cost. The cost of IDCOL will be within the “Project Management Cost” from IDCOL in line of regular business process

Annex VIII – Gender Action Plan

GREEN CLIMATE FUND FUNDING PROPOSAL

Activity 2.1.3: Grievance Redressal (1) Formulation of a project committee for addressing grievance (2) Establishing grievance redressal process	Indicator Encouragement for establishment of grievance redressed system for enhanced women employee security Target An active grievance redressal system in place within 12 months of project operation	Number of such grievance redressal system will depend on the number of end borrowers receiving financing from this program. There is no yearly target of how many end borrowers will be financed rather specific amount for each year & a total of entities have been mentioned in the log frame. So IDCOL will ensure such active grievance redressal system in place within 12 months of project operation in line with their actual financing in each year within 05 years of GCF loan availability period	IDCOL & End borrower	Funding Source: End borrowers & IDCOL Amount: XXXXX The cost of any such implementation will be borne by end borrowers & this will not be part of GCF project cost. The cost of IDCOL will be within the “Project Management Cost” from IDCOL in line of regular business process
Component 3: Gender balanced participation during the Implementation/operationalization of the project/programme				
Output 3.1: Women Participation during project implementation				
Activity 3.1.1: Formulation of a project committee with women participating for monitoring of	Indicator Gender balanced project/programme committee in IDCOL Target	To form a Project committee within 06 months after signing the Funded Activity	IDCOL	Funding Source: IDCOL Amount: XXXXX

the activities under the program.	To ensure women participation in the Project Committee & ensure at least 01 female member participation	Agreement (FAA) with GCF		The cost of IDCOL will be within the “Project Management Cost” from IDCOL in line of regular business process
Activity 3.1.2: Training & skill development of relevant Business Units of IDCOL	Indicator Increased women skills & participation in various management aspects of the program Target To ensure women employee participation (from IDCOL) in the relevant capacity building initiatives as beneficiary	• 1st workshop in Year-2 • 2nd workshops in Year-5 (Please see project Implementation Time table)	IDCOL	Funding Source: GCF Amount: USD 78,133.00 (for entire activity) Activity 2.3.2: Conduct capacity building workshops and training sessions for IDCOL, stakeholders and end borrowers (participants from the textile industry irrespective of being eligible for component 01) who are the applicants for the loan on the tools and requirements to be fulfilled for



				successfully securing the loan
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